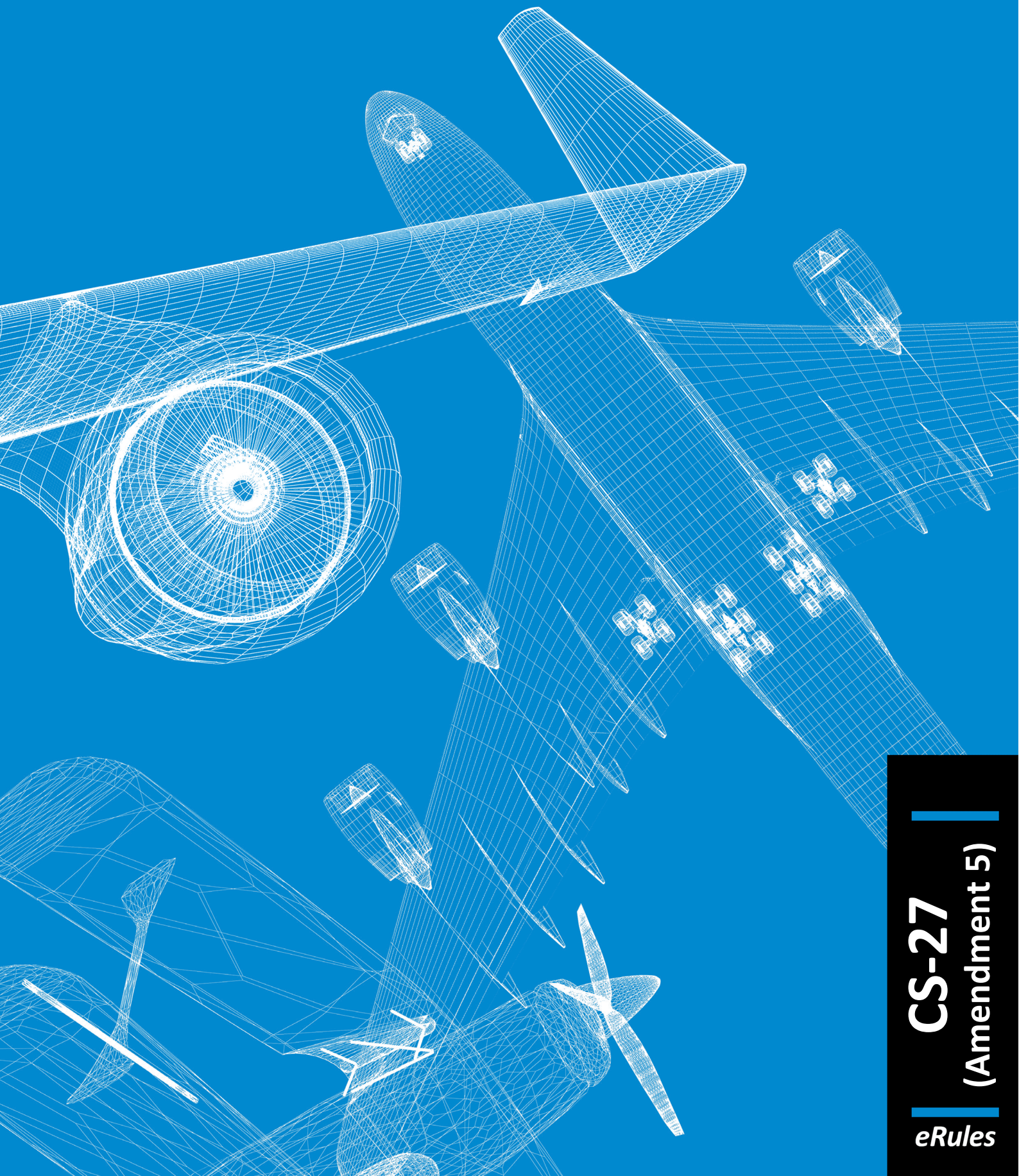




Easy Access Rules for Small Rotorcraft (CS-27) (Amendment 5)

EASA eRules: aviation rules for the 21st century

Rules and regulations are the core of the European Union civil aviation system. The aim of the **EASA eRules** project is to make them **accessible** in an efficient and reliable way to stakeholders.

EASA eRules will be a comprehensive, single system for the drafting, sharing and storing of rules. It will be the single source for all aviation safety rules applicable to European airspace users. It will offer easy (online) access to all rules and regulations as well as new and innovative applications such as rulemaking process automation, stakeholder consultation, cross-referencing, and comparison with ICAO and third countries' standards.

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Published June 2019¹

¹ The published date represents the date when the consolidated version of the document was generated.

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NOTE FROM THE EDITOR

The content of this document is arranged as follows: the certification specifications (CS) are followed by the related acceptable means of compliance (AMC) and guidance material (GM) paragraph(s).

All elements (i.e. CS, AMC and GM) are colour-coded and can be identified according to the illustration below. The EASA Executive Director (ED) decision through which the point or paragraph was introduced or last amended is indicated below the paragraph title(s) *in italics*.

Certification specification

ED decision

Acceptable means of compliance

ED decision

Guidance material

ED decision

The format of this document has been adjusted to make it user-friendly and for reference purposes. Any comments should be sent to erules@easa.europa.eu.

INCORPORATED AMENDMENTS

CS/AMC/GM (ED DECISIONS)

Incorporated ED Decision	CS/AMC/GM Issue No, Amendment No	Applicability date
ED Decision 2003/15/RM	CS-27/ Initial issue	14/11/2003
ED Decision 2007/013/R	CS-27/ Amendment 1	30/11/2007
ED Decision 2008/009/R	CS-27/ Amendment 2	17/11/2008
ED Decision 2012/021/R	CS-27/ Amendment 3	18/12/2012
ED Decision 2016/024/R	CS-27/ Amendment 4	2/12/2016
ED Decision 2018/007/R	CS-27/ Amendment 5	26/6/2018

Note: To access the official versions, please click on the hyperlinks provided above.

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PREAMBLE

ED Decision 2018/007/R

CS-27 Amendment 5

The following is a list of paragraphs affected by this amendment:

Subpart A	
CS-27 Appendix C	Amended (NPA 2016-01 , NPA 2017-07)
Subpart C	
CS 27.563	Amended (NPA 2016-01)
AMC 27.563	Created (NPA 2016-01)
Subpart D	
CS 27.783	Amended (NPA 2016-01)
AMC 27.783	Created (NPA 2016-01)
CS 27.801	Amended (NPA 2016-01)
AMC 27.801	Created (NPA 2016-01)
AMC to CS 27.801(e) and CS 27.802(c)	Created (NPA 2016-01)
CS 27.802	Created (NPA 2016-01)
AMC 27.802	Created (NPA 2016-01)
CS 27.805	Amended (NPA 2016-01)
AMC 27.805(c)	Created (NPA 2016-01)
CS 27.807	Amended (NPA 2016-01)
AMC 27.807(d)	Created (NPA 2016-01)
CS 27.865	Amended (Article 16 consultation with the ABs)
AMC 27.865	Created (Article 16 consultation with the ABs)
AMC No 1 to CS 27.865	Amended (Article 16 consultation with the ABs)
AMC No 2 to CS 27.865	Created (Article 16 consultation with the ABs)
AMC No 3 27.865	Created (Article 16 consultation with the ABs)
Subpart F	
CS 27.1411	Amended (NPA 2016-01)
AMC 27.1411	Created (NPA 2016-01)
CS 27.1415	Amended (NPA 2016-01)
AMC 27.1415	Created (NPA 2016-01)
CS 27.1470	Created (NPA 2016-01)
AMC 27.1470	Created (NPA 2016-01)
Subpart G	
CS 27.1555	Amended (NPA 2016-01)
AMC 27.1555	Created (NPA 2016-01)
CS 27.1557	Amended (NPA 2016-01)
CS 27.1561	Amended (NPA 2016-01)
AMC 27.1561	Created (NPA 2016-01)
CS 27.1587	Amended (NPA 2016-01)
AMC 27.1587(b)(3)	Created (NPA 2016-01)

CS-27 Amendment 4

The following is a list of paragraphs affected by this amendment:

Subpart A	
AMC 27 General	Amended (NPA 2013-04)
CS 27.1	Amended (editorial change)
CS-27 Appendix C	Amended (NPA 2013-04)
Subpart C	
AMC No 1 to CS 27.351	Created (NPA 2013-21)
AMC No 2 to CS 27.351	Renamed and amended (NPA 2013-21)
Subpart D	
CS 27.610	Amended (NPA 2014-16)
Subpart F	
CS 27.1309	Amended (NPA 2014-16)
CS 27.1316	Created (NPA 2014-16)
CS 27.1317	Created (NPA 2014-16)
CS-27 Appendix D	Created (NPA 2014-16)
Subpart G	
CS 27.1501	Amended (NPA 2011-17)
CS 27.1593	Created (NPA 2011-17)
AMC 27.1593	Created (NPA 2011-17)
Miscellaneous guidance	
MG5	Created (NPA 2013-04)
MG6	Created (NPA 2013-04)

CS-27 Amendment 3

The following is a list of paragraphs affected by this amendment:

Subpart A	
CS 27.2	Editorial change
Subpart C	
CS 27.547	Editorial change
CS 27.549	Editorial change
CS 27.573	Amended (NPA 2010-04)
Subpart D	
CS 27.865	Editorial change
Subpart F	
CS 27.1401	Editorial change
Subpart G	
CS 27.1521	Editorial change
CS-27 Appendix A	Amended (NPA 2010-04)

ED Decision 2008/009/R

CS-27 Amendment 2

The following is a list of paragraphs affected by this amendment:

Subpart A	
AMC 27 General	Amended (NPA 2007-17)
Appendix C	Amended (NPA 2007-17)
Subpart C	
AMC 27.351	Created (NPA 2007-17)
Subpart D	
AMC 27.602	Deleted (NPA 2007-17)
AMC 27.865	Created (NPA 2007-17)
Subpart F	
CS 27.1305	Amended (NPA 2007-17)
AMC 27.1305(t) and (u)	Deleted (NPA 2007-17)
Miscellaneous guidance	
MG4	Created (NPA 2007-17)

ED Decision 2007/013/R

CS-27 Amendment 1

The following is a list of paragraphs affected by this amendment:

Preamble	Preamble added
Subpart B	
CS 27.25	Amended (NPA 11/2006)
CS 27.49	Created by renaming CS 27.73 (NPA 11/2006)
CS 27.51	Amended (NPA 11/2006)
CS 27.73	Deleted and moved to CS 27.49 (NPA 11/2006)
CS 27.75	Amended (NPA 11/2006)
CS 27.79	Amended (NPA 11/2006)
CS-27 Appendix B	Amended (NPA 11/2006)
CS 27.143	Amended (NPA 11/2006)
CS 27.173	Amended (NPA 11/2006)
CS 27.175	Amended (NPA 11/2006)
CS 27.177	Amended (NPA 11/2006)
Subpart E	
CS 27.903	Amended (NPA 11/2006)
Subpart G	
CS 27.1587	Amended (NPA 11/2006)

SUBPART A – GENERAL

AMC 27 General

ED Decision 2016/024/R

1. The AMC to CS-27 consists of FAA AC 27-1B Change 4 dated 1 May 2014 with the changes/additions given in this Book 2 of CS-27.
2. The primary reference for each of these AMCs is the CS-27 paragraph. Where there is an appropriate paragraph in FAA AC 27-1B Change 4 dated 1 May 2014 this is added as a secondary reference.

[Amdt 27/2]

[Amdt 27/4]

CS 27.1 Applicability

ED Decision 2016/024/R

- (a) These certification specifications are applicable to small rotorcraft with maximum weights of 3 175 kg (7 000 lbs) or less and nine or less passenger seats.
- (b) reserved
- (c) Multi-engine rotorcraft may be type certificated as Category A provided the requirements referenced in [Appendix C](#) are met.

[Amdt 27/4]

Appendix C – Criteria for Category A

ED Decision 2018/007/R

C27.1 General. A small multi-engine rotorcraft may not be type certificated for category A operation unless it meets the design installation and performance provisions contained in this appendix in addition to the provisions of this CS-27.

C27.2 Applicable CS-29 paragraphs. The following paragraphs of CS-29 must be met in addition to the requirements of this CS:

- | | |
|---------------------|--|
| 29.45(a) and (b)(2) | — General. |
| 29.49(a) | — Performance at minimum operating speed. |
| 29.51 | — Take-off data: General. |
| 29.53 | — Take-off: Category A. |
| 29.55 | — Take-off decision point: Category A. |
| 29.59 | — Take-off path: Category A. |
| 29.60 | — Elevated heliport take-off path: Category A. |
| 29.61 | — Take-off distance: Category A. |
| 29.62 | — Rejected take-off: Category A. |
| 29.64 | — Climb: General. |
| 29.65(a) | — Climb: AEO. |
| 29.67(a) | — Climb: OEI. |
| 29.75 | — Landing: General. |
| 29.77 | — Landing decision point: Category A. |
| 29.79 | — Landing: Category A. |
| 29.81 | — Landing distance (ground level sites): Category A. |
| 29.85 | — Balked landing: Category A. |

29.87(a)	—	Height-velocity envelope.
29.547(a) and (b)	—	Main and tail rotor structure.
(29.571	—	Fatigue evaluation of structure.) AC Material only: AC 29-2C Change 4 dated 1 May 2014, Paragraph AC29.571A.b(2).
29.861(a)	—	Fire protection of structure, controls and other parts.
29.901(c)	—	Powerplant: Installation.
29.903(b), (c) and (e)	—	Engines.
29.908(a)	—	Cooling fans.
29.917(a), (b) and (c)(1)	—	Rotor drive system: Design. (29.917(a) replaces 27.917(d))
29.927(c)(1) and (c)(2)	—	Additional tests.
29.953(a)	—	Fuel system independence.
29.1027(a)	—	Transmission and gearboxes: General.
29.1027(a)	—	Transmission and gearboxes: General.
29.1045(a)(1), (b), (c), (d) and (f)	—	Climb cooling test procedures.
29.1047(a)	—	Take-off cooling test procedures.
29.1181(a)	—	Designated fire zones: Regions included.
29.1187(e)	—	Drainage and ventilation of fire zones.
29.1189(c)	—	Shutoff means.
29.1191(a)(l)	—	Firewalls.
29.1193(e)	—	Cowling and engine compartment covering.
29.1195(a) and (d)	—	Fire extinguishing systems (one shot).
29.1197	—	Fire extinguishing agents.
29.1199	—	Extinguishing agent containers.
29.1201	—	Fire extinguishing system materials.
29.1305(a)(6) and (b)	—	Powerplant instruments.
29.1309(b)(2)(i) and (d)	—	Equipment, systems and installations.
29.1323(c)(1)	—	Airspeed indicating system.
29.1331(b)	—	Instruments using a power supply.
29.1351(d)(2)	—	Additional requirements for Category A rotorcraft (Operation with the normal electrical power generating system inoperative.)
29.1585(h)	—	Operating Procedures.
29.1587(a)	—	Performance information.

If certification with ditching provisions is requested by the applicant, the following requirements of CS-29 must also be met in addition to the ones of this CS:

29.801(c) and (g)	—	Ditching.
29.803(c)	—	Emergency evacuation.
29.809(j)(2)	—	Emergency exit arrangement.
29.811(h)(1)	—	Emergency exit marking.
29.1415(d)	—	Ditching equipment.

If certification of an emergency flotation system alone is requested by the applicant, the following requirements of CS 29 must also be met in addition to the ones of this CS:

29.801(g)	—	Ditching.
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(See AC 29-2C Change 4 dated 1 May 2014 and AMC material to CS-29)

[Amdt No: 27/2]

[Amdt No: 27/4]

[Amdt No: 27/5]

CS 27.2 Special Retroactive Requirements

ED Decision 2012/021/R

- (a) reserved
- (b) For rotorcraft with a certification basis established prior to 1 May 2001
 - (1) The maximum passenger seat capacity may be increased to eight or nine provided compliance is shown with all the airworthiness requirements in effect from the initial issue of CS-27.
 - (2) The maximum weight may be increased to greater than 2 722 kg (6 000 lbs) provided -
 - (i) The number of passenger seats is not increased above the maximum number previously certificated; or
 - (ii) Compliance is shown with all of the airworthiness requirements in effect from the issue of CS-27.

[Amdt 27/3]

SUBPART B – FLIGHT

GENERAL

CS 27.21 Proof of compliance

ED Decision 2003/15/RM

Each requirement of this Subpart must be met at each appropriate combination of weight and centre of gravity within the range of loading conditions for which certification is requested. This must be shown:

- (a) By tests upon a rotorcraft of the type for which certification is requested or by calculations based on, and equal in accuracy to, the results of testing; and
- (b) By systematic investigation of each required combination of weight and centre of gravity if compliance cannot be reasonably inferred from combinations investigated.

CS 27.25 Weight limits

ED Decision 2007/013/R

- (a) *Maximum weight.* The maximum weight, the highest weight at which compliance with each applicable requirement of this CS-27 is shown, must be established so that it is:
 - (1) Not more than:
 - (i) The highest weight selected by the applicant;
 - (ii) The design maximum weight, the highest weight at which compliance with each applicable structural loading condition of this CS-27 is shown; or
 - (iii) The highest weight at which compliance with each applicable flight requirement of this CS-27 is shown; or
 - (iv) The highest weight, as a function of altitude and temperature, in which the provisions of [CS 27.79](#) and/or [CS 27.143\(c\)\(1\)](#) are demonstrated if the operating conditions (altitude and temperature) prescribed by those requirements can not be met; and
 - (2) Not less than the sum of:
 - (i) The empty weight determined under [CS 27.29](#);
 - (ii) The weight of usable fuel appropriate to the intended operation with full payload;
 - (iii) The weight of full oil capacity; and
 - (iv) For each seat, an occupant weight of 77 kg (170 lbs) or any lower weight for which certification is requested.
- (b) *Minimum weight.* The minimum weight, the lowest weight at which compliance with each applicable requirement of this CS-27 is shown, must be established so that it is:
 - (1) Not more than the sum of:
 - (i) The empty weight determined under [CS 27.29](#); and

- (ii) The weight of the minimum crew necessary to operate the rotorcraft, assuming for each crew member a weight no more than 77 kg (170 lbs), or any lower weight selected by the applicant or included in the loading instructions; and
- (2) Not less than:
 - (i) The lowest weight selected by the applicant;
 - (ii) The design minimum weight, the lowest weight at which compliance with each applicable structural loading condition of this CS-27 is shown; or
 - (iii) The lowest weight at which compliance with each applicable flight requirement of this CS-27 is shown.
- (c) *Total weight with jettisonable external load.* A total weight for the rotorcraft with a jettisonable external load attached that is greater than the maximum weight established under sub-paragraph (a) may be established for any rotorcraft-load combination if
 - (1) The rotorcraft-load combination does not include human external cargo,
 - (2) Structural component approval for external load operations under either [CS 27.865](#), or under equivalent operational standards is obtained,
 - (3) The portion of the total weight that is greater than the maximum weight established under sub-paragraph (a) is made up only of the weight of all or part of the jettisonable external load,
 - (4) Structural components of the rotorcraft are shown to comply with the applicable structural requirements of this CS-27 under the increased loads and stresses caused by the weight increase over that established under sub-paragraph (a), and
 - (5) Operation of the rotorcraft at a total weight greater than the maximum certificated weight established under sub-paragraph (a) is limited by appropriate operating limitations under [CS 27.865\(a\) and \(d\)](#).

[Amdt. No.: 27/1]

CS 27.27 Centre of gravity limits

ED Decision 2003/15/RM

The extreme forward and aft centres of gravity and, where critical, the extreme lateral centres of gravity must be established for each weight established under [CS 27.25](#). Such an extreme may not lie beyond:

- (a) The extremes selected by the applicant;
- (b) The extremes within which the structure is proven; or
- (c) The extremes within which compliance with the applicable flight requirements is shown.

CS 27.29 Empty weight and corresponding centre of gravity

ED Decision 2003/15/RM

- (a) The empty weight and corresponding centre of gravity must be determined by weighing the rotorcraft without the crew and payload but with:
- (1) Fixed ballast;
 - (2) Unusable fuel; and
 - (3) Full operating fluids, including:
 - (i) Oil;
 - (ii) Hydraulic fluid; and
 - (iii) Other fluids required for normal operation of rotorcraft systems, except water intended for injection in the engines.
- (b) The condition of the rotorcraft at the time of determining empty weight must be one that is well defined and can be easily repeated, particularly with respect to the weights of fuel, oil, coolant, and installed equipment.

CS 27.31 Removable ballast

ED Decision 2003/15/RM

Removable ballast may be used in showing compliance with the flight requirements of this Subpart.

CS 27.33 Main rotor speed and pitch limits

ED Decision 2003/15/RM

- (a) *Main rotor speed limits.* A range of main rotor speeds must be established that:
- (1) With power-on, provides adequate margin to accommodate the variations in rotor speed occurring in any appropriate manoeuvre, and is consistent with the kind of governor or synchroniser used; and
 - (2) With power-off, allows each appropriate autorotative manoeuvre to be performed throughout the ranges of airspeed and weight for which certification is requested.
- (b) *Normal main rotor high pitch limits (power-on).* For rotorcraft, except helicopters required to have a main rotor low speed warning under sub-paragraph (e). It must be shown, with power-on and without exceeding approved engine maximum limitations, that main rotor speeds substantially less than the minimum approved main rotor speed will not occur under any sustained flight condition. This must be met by:
- (1) Appropriate setting of the main rotor high pitch stop;
 - (2) Inherent rotorcraft characteristics that make unsafe low main rotor speeds unlikely; or
 - (3) Adequate means to warn the pilot of unsafe main rotor speeds.
- (c) *Normal main rotor low pitch limits (power-off).* It must be shown, with power-off, that:
- (1) The normal main rotor low pitch limit provides sufficient rotor speed, in any autorotative condition, under the most critical combinations of weight and airspeed; and
 - (2) It is possible to prevent overspeeding of the rotor without exceptional piloting skill.

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- (d) *Emergency high pitch.* If the main rotor high pitch stop is set to meet sub-paragraph (b)(1), and if that stop cannot be exceeded inadvertently, additional pitch may be made available for emergency use.
- (e) *Main rotor low speed warning for helicopters.* For each single engine helicopter, and each multi-engine helicopter that does not have an approved device that automatically increases power on the operating engines when one engine fails, there must be a main rotor low speed warning which meets the following requirements:
- (1) The warning must be furnished to the pilot in all flight conditions, including power-on and power-off flight, when the speed of a main rotor approaches a value that can jeopardise safe flight.
 - (2) The warning may be furnished either through the inherent aerodynamic qualities of the helicopter or by a device.
 - (3) The warning must be clear and distinct under all conditions, and must be clearly distinguishable from all other warnings. A visual device that requires the attention of the crew within the cockpit is not acceptable by itself.
 - (4) If a warning device is used, the device must automatically de-activate and reset when the low-speed condition is corrected. If the device has an audible warning, it must also be equipped with a means for the pilot to manually silence the audible warning before the low-speed condition is corrected.

PERFORMANCE

CS 27.45 General

ED Decision 2003/15/RM

- (a) Unless otherwise prescribed, the performance requirements of this Subpart must be met for still air and a standard atmosphere.
- (b) The performance must correspond to the engine power available under the particular ambient atmospheric conditions, the particular flight condition, and the relative humidity specified in sub-paragraphs (d) or (e), as appropriate.
- (c) The available power must correspond to engine power, not exceeding the approved power, less:
 - (1) Installation losses; and
 - (2) The power absorbed by the accessories and services appropriate to the particular ambient atmospheric conditions and the particular flight condition.
- (d) For reciprocating engine-powered rotorcraft, the performance, as affected by engine power, must be based on a relative humidity of 80% in a standard atmosphere.
- (e) For turbine engine-powered rotorcraft, the performance, as affected by engine power, must be based on a relative humidity of:
 - (1) 80%, at and below standard temperature; and
 - (2) 34%, at and above standard temperature plus 28°C (50°F) between these two temperatures, the relative humidity must vary linearly.
- (f) For turbine engine-powered rotorcraft, a means must be provided to permit the pilot to determine prior to take-off that each engine is capable of developing the power necessary to achieve the applicable rotorcraft performance prescribed in this Subpart.

CS 27.49 Performance at minimum operating speed

ED Decision 2007/013/R

- (a) For helicopters:
 - (1) The hovering ceiling must be determined over the ranges of weight, altitude, and temperature for which certification is requested, with:
 - (i) Take-off power;
 - (ii) The landing gear extended; and
 - (iii) The helicopter in ground effect at a height consistent with normal take-off procedures; and
 - (2) The hovering ceiling determined in sub-paragraph (a)(1) of this paragraph must be at least:
 - (i) For reciprocating engine-powered helicopters, 1219 m (4 000 ft) at maximum weight with a standard atmosphere; or
 - (ii) For turbine engine-powered helicopters, 762 m (2 500 ft) pressure altitude at maximum weight at a temperature of standard +22°C (+40°F).

- (3) The out-of-ground effect hovering performance must be determined over the ranges of weight, altitude, and temperature for which certification is requested, using take-off power.
- (b) For rotorcraft other than helicopters, the steady rate of climb at the minimum operating speed must be determined, over the ranges of weight, altitude, and temperature for which certification is requested, with:
 - (1) Take-off power; and
 - (2) The landing gear extended.

[Amdt. No.: 27/1]

CS 27.51 Take-off

ED Decision 2007/013/R

The take-off, with take-off power and r.p.m. at the most critical center of gravity, and with weight from the maximum weight at sea-level to the weight for which take-off certification is requested for each altitude covered by this paragraph:

- (a) May not require exceptional piloting skill or exceptionally favourable conditions throughout the ranges of altitude from standard sea-level conditions to the maximum altitude for which take-off and landing certification is requested, and
- (b) Must be made in such a manner that a landing can be made safely at any point along the flight path if an engine fails. This must be demonstrated up to the maximum altitude for which take-off and landing certification is requested or 2134 m (7,000 ft) density altitude, whichever is less.

[Amdt. No.: 27/1]

CS 27.65 Climb: all-engines-operating

ED Decision 2003/15/RM

- (a) For rotorcraft other than helicopters –
 - (1) The steady rate of climb, at V_Y must be determined:
 - (i) With maximum continuous power on each engine;
 - (ii) With the landing gear retracted; and
 - (iii) For the weights, altitudes, and temperatures for which certification is requested; and
 - (2) The climb gradient, at the rate of climb determined in accordance with subparagraph (a)(1), must be either:
 - (i) At least 1:10 if the horizontal distance required to take off and climb over a 15 m (50 ft) obstacle is determined for each weight, altitude, and temperature within the range for which certification is requested; or
 - (ii) At least 1:6 under standard sea-level conditions.
- (b) Each helicopter must meet the following requirements:
 - (1) V_Y must be determined:
 - (i) For standard sea-level conditions;

- (ii) At maximum weight; and
- (iii) With maximum continuous power on each engine.
- (2) The steady rate of climb must be determined:
 - (i) At the climb speed selected by the applicant at or below V_{NE} ;
 - (ii) Within the range from sea-level up to the maximum altitude for which certification is requested;
 - (iii) For the weights and temperatures that correspond to the altitude range set forth in sub-paragraph (b)(2)(ii) and for which certification is requested; and
 - (iv) With maximum continuous power on each engine.

CS 27.67 Climb: one-engine-inoperative

ED Decision 2003/15/RM

For multi-engine helicopters, the steady rate of climb (or descent), at V_Y (or at the speed for minimum rate of descent), must be determined with:

- (a) Maximum weight;
- (b) The critical engine inoperative and the remaining engines at either –
 - (1) Maximum continuous power and, for helicopters for which certification for the use of 30-minute one engine inoperative (OEI) power is requested, at 30-minute OEI power; or
 - (2) Continuous OEI power for helicopters for which certification for the use of continuous OEI power is requested.

CS 27.71 Glide performance

ED Decision 2003/15/RM

For single-engine helicopters and multi-engine helicopters that do not meet the category A engine isolation requirements of CS-27, the minimum rate of descent airspeed and the best angle-of-glide airspeed must be determined in autorotation at:

- (a) Maximum weight; and
- (b) Rotor speed(s) selected by the applicant.

CS 27.75 Landing

ED Decision 2007/013/R

- (a) The rotorcraft must be able to be landed with no excessive vertical acceleration, no tendency to bounce, nose over, ground loop, porpoise, or water loop, and without exceptional piloting skill or exceptionally favourable conditions, with:
 - (1) Approach or autorotation speeds appropriate to the type of rotorcraft and selected by the applicant;
 - (2) The approach and landing made with:
 - (i) Power off, for single-engine rotorcraft and entered from steady state autorotation; or

- (ii) One-engine inoperative (OEI) for multi-engine rotorcraft, with each operating engine within approved operating limitations, and entered from an established OEI approach.
- (b) Multi-engine rotorcraft must be able to be landed safely after complete power failure under normal operating conditions.

[Amdt. No.: 27/1]

CS 27.79 Limiting height-speed envelope

ED Decision 2007/013/R

- (a) If there is any combination of height and forward speed, including hover, under which a safe landing cannot be made under the applicable power failure condition in sub-paragraph (b), a limiting height-speed envelope must be established, including all pertinent information, for that condition, throughout the ranges of:
 - (1) Altitude, from standard sea-level conditions to the maximum altitude capability of the rotorcraft, or 2134 m (7 000 ft) density altitude, whichever is less; and
 - (2) Weight from the maximum weight at sea-level to the weight selected by the applicant for each altitude covered by sub-paragraph (a)(1) of this paragraph. For helicopters, the weight at altitudes above sea-level may not be less than the maximum weight or the highest weight allowing hovering out of ground effect whichever is lower.
- (b) The applicable power failure conditions are:
 - (1) For single-engine helicopters, full autorotation;
 - (2) For multi-engine helicopters, OEI, where engine isolation features ensure continued operation of the remaining engines, and the remaining engine(s) within approved limits and at the minimum installed specification power available for the most critical combination of approved ambient temperature and pressure altitude resulting in 2134 m (7000 ft) density altitude or the maximum altitude capability of the helicopter, whichever is less, and
 - (3) For other rotorcraft, conditions appropriate to the type.

[Amdt. No.: 27/1]

FLIGHT CHARACTERISTICS

CS 27.141 General

ED Decision 2003/15/RM

The rotorcraft must:

- (a) Except as specifically required in the applicable paragraph, meet the flight characteristics requirements of this Subpart –
 - (1) At the altitudes and temperatures expected in operation;
 - (2) Under any critical loading condition within the range of weights and centres of gravity for which certification is requested;
 - (3) For power-on operations, under any condition of speed, power, and rotor rpm for which certification is requested; and
 - (4) For power-off operations, under any condition of speed and rotor rpm for which certification is requested that is attainable with the controls rigged in accordance with the approved rigging instructions and tolerances;
- (b) Be able to maintain any required flight condition and make a smooth transition from any flight condition to any other flight condition without exceptional piloting skill, alertness, or strength, and without danger of exceeding the limit load factor under any operating condition probable for the type, including:
 - (1) Sudden failure of one engine, for multi-engine rotorcraft meeting category A engine isolation requirements of CS-29;
 - (2) Sudden, complete power failure for other rotorcraft; and
 - (3) Sudden, complete control system failures specified in [CS 27.695](#); and
- (c) Have any additional characteristic required for night or instrument operation, if certification for those kinds of operation is requested. Requirements for helicopter instrument flight are contained in [appendix B](#).

Appendix B – Airworthiness Criteria for Helicopter Instrument Flight

ED Decision 2007/013/R

- I. *General.* A small helicopter may not be type certificated for operation under the instrument flight rules (IFR) unless it meets the design and installation requirements contained in this appendix.
- II. *Definitions*
 - (a) V_{YI} means instrument climb speed, utilised instead of V_Y for compliance with the climb requirements for instrument flight.
 - (b) V_{NEI} means instrument flight never exceed speed, utilised instead of V_{NE} for compliance with maximum limit speed requirements for instrument flight.
 - (c) V_{MINI} means instrument flight minimum speed, utilised in complying with minimum limit speed requirements for instrument flight.
- III. *Trim.* It must be possible to trim the cyclic, collective, and directional control forces to zero at all approved IFR airspeeds, power settings, and configurations appropriate to the type.
- IV. *Static longitudinal stability*
 - (a) *General.* The helicopter must possess positive static longitudinal control force stability at critical combinations of weight and centre of gravity at the conditions specified in paragraphs IV (b) or (c) of this Appendix. The stick force must vary with speed so that any substantial speed change results in a stick force clearly perceptible to the pilot. For single pilot approval the airspeed must return to within 10% of the trim speed when the control force is slowly released for each trim condition specified in paragraph IV(b) of this Appendix.
 - (b) *For single-pilot approval*
 - (1) *Climb.* Stability must be shown in climb throughout the speed range 37 km/h (20 knots) either side of trim with –
 - (i) The helicopter trimmed at V_{YI} ;
 - (ii) Landing gear retracted (if retractable); and
 - (iii) Power required for limit climb rate (at least 5 m/s (1000 fpm)) at V_{YI} or maximum continuous power, whichever is less.
 - (2) *Cruise.* Stability must be shown throughout the speed range from 0.7 to 1.1 V_H or V_{NEI} , whichever is lower, not to exceed ± 37 km/h (± 20 knots) from trim with –
 - (i) The helicopter trimmed and power adjusted for level flight at 0.9 V_H or 0.9 V_{NEI} , whichever is lower; and
 - (ii) Landing gear retracted (if retractable).
 - (3) *Slow cruise.* Stability must be shown throughout the speed range from 0.9 V_{MINI} to 1.3 V_{MINI} or 37 km/h (20 knots) above trim speed, whichever is greater, with –
 - (i) The helicopter trimmed and power adjusted for level flight at 1.1 V_{MINI} ; and
 - (ii) Landing gear retracted (if retractable).

-
- (4) *Descent.* Stability must be shown throughout the speed range 37 km/h (20 knots) either side of trim with –
 - (i) The helicopter trimmed at $0.8 V_H$ or $0.8 V_{NEI}$ (or $0.8 V_{LE}$ for the landing gear extended case), whichever is lower;
 - (ii) Power required for 1000 fpm descent at trim speed; and
 - (iii) Landing gear extended and retracted, if applicable.
 - (5) *Approach.* Stability must be shown throughout the speed range from 0.7 times the minimum recommended approach speed to 37 km/h (20 knots) above the maximum recommended approach speed with –
 - (i) The helicopter trimmed at the recommended approach speed or speeds;
 - (ii) Landing gear extended and retracted, if applicable; and
 - (iii) Power required to maintain a 3° glide path and power required to maintain the steepest approach gradient for which approval is requested.
 - (c) Helicopters approved for a minimum crew of two pilots must comply with the provisions of paragraphs IV(b)(2) and IV(b)(5) of this Appendix.
- V. Static lateral-directional stability
- (a) Static directional stability must be positive throughout the approved ranges of airspeed, power, and vertical speed. In straight and steady sideslips up to $\pm 10^\circ$ from trim, directional control position must increase without discontinuity with the angle of sideslip, except for a small range of sideslip angles around trim. At greater angles up to the maximum sideslip angle appropriate to the type, increased directional control position must produce increased angle of sideslip. It must be possible to maintain balanced flight without exceptional pilot skill or alertness.
 - (b) During sideslips up to $\pm 10^\circ$ from trim throughout the approved ranges of airspeed, power, and vertical speed there must be no negative dihedral stability perceptible to the pilot through lateral control motion or force. Longitudinal cyclic movement with sideslip must not be excessive.
- VI. Dynamic stability
- (a) For single-pilot approval –
 - (1) Any oscillation having a period of less than 5 seconds must damp to $\frac{1}{2}$ amplitude in not more than one cycle.
 - (2) Any oscillation having a period of 5 seconds or more but less than 10 seconds must damp to $\frac{1}{2}$ amplitude in not more than two cycles.
 - (3) Any oscillation having a period of 10 seconds or more but less than 20 seconds must be damped.
 - (4) Any oscillation having a period of 20 seconds or more may not achieve double amplitude in less than 20 seconds.
 - (5) Any a periodic response may not achieve double amplitude in less than 6 seconds.
 - (b) For helicopters approved with a minimum crew of two pilots –
 - (1) Any oscillation having a period of less than 5 seconds must damp to $\frac{1}{2}$ amplitude in not more than two cycles.

- (2) Any oscillation having a period of 5 seconds or more but less than 10 seconds must be damped.
- (3) Any oscillation having a period of 10 seconds or more may not achieve double amplitude in less than 10 seconds.

VII. *Stability augmentation system (SAS)*

- (a) If a SAS is used, the reliability of the SAS must be related to the effects of its failure. Any SAS failure condition that would prevent continued safe flight and landing must be extremely improbable. It must be shown that, for any failure condition of the SAS which is not shown to be extremely improbable –
 - (1) The helicopter is safely controllable when the failure or malfunction occurs at any speed or altitude within the approved IFR operating limitations; and
 - (2) The overall flight characteristics of the helicopter allow for prolonged instrument flight without undue pilot effort. Additional unrelated probable failures affecting the control system must be considered. In addition:
 - (i) The controllability and manoeuvrability requirements in Subpart B of CS-27 must be met throughout a practical flight envelope;
 - (ii) The flight control, trim, and dynamic stability characteristics must not be impaired below a level needed to allow continued safe flight and landing; and
 - (iii) The static longitudinal and static directional stability requirements of Subpart B of CS-27 must be met throughout a practical flight envelope.
- (b) The SAS must be designed so that it cannot create a hazardous deviation in flight path or produce hazardous loads on the helicopter during normal operation or in the event of malfunction or failure, assuming corrective action begins within an appropriate period of time. Where multiple systems are installed, subsequent malfunction conditions must be considered in sequence unless their occurrence is shown to be improbable.

VIII. *Equipment, systems, and installation.* The basic equipment and installation must comply with CS 29.1303, 29.1431 and 29.1433, with the following exceptions and additions:

- (a) Flight and navigation instruments
 - (1) A magnetic gyro-stabilised direction indicator instead of the gyroscopic direction indicator required by CS 29.1303(h); and
 - (2) A standby attitude indicator which meets the requirements of CS 29.1303(g)(1) to (7), instead of a rate-of-turn indicator required by CS 29.1303(g). For two-pilot configurations, one pilot's primary indicator may be designated for this purpose. If standby batteries are provided they may be charged from the aircraft electrical system if adequate isolation is incorporated.
- (b) Miscellaneous requirements
 - (1) Instrument systems and other systems essential for IFR flight that could be adversely affected by icing must be adequately protected when exposed to the continuous and intermittent maximum icing conditions defined in appendix C of CS-29, whether or not the rotorcraft is certificated for operation in icing conditions.

- (2) There must be means in the generating system to automatically de-energise and disconnect from the main bus any power source developing hazardous overvoltage.
 - (3) Each required flight instrument using a power supply (electric, vacuum, etc.) must have a visual means integral with the instrument to indicate the adequacy of the power being supplied.
 - (4) When multiple systems performing like functions are required, each system must be grouped, routed, and spaced so that physical separation between systems is provided to ensure that a single malfunction will not adversely affect more than one system.
 - (5) For systems that operate the required flight instruments at each pilot's station –
 - (i) Only the required flight instruments for the first pilot may be connected to that operating system;
 - (ii) Additional instruments, systems, or equipment may not be connected to an operating system for a second pilot unless provisions are made to ensure the continued normal functioning of the required instruments in the event of any malfunction of the additional instruments, systems, or equipment which is not shown to be extremely improbable;
 - (iii) The equipment, systems, and installations must be designed so that one display of the information essential to the safety of flight which is provided by the instruments will remain available to a pilot, without additional crewmember action, after any single failure or combination of failures that is not shown to be extremely improbable; and For single-pilot configurations, instruments which require a static source must be provided with a means of selecting an alternate source and that source must be calibrated.
- IX. *Rotorcraft flight manual.* A rotorcraft flight manual or rotorcraft flight manual IFR supplement must be provided and must contain –
- (a) *Limitations.* The approved IFR flight envelope, the IFR flight crew composition, the revised kinds of operation, and the steepest IFR precision approach gradient for which the helicopter is approved;
 - (b) *Procedures.* Required information for proper operation of IFR systems and the recommended procedures in the event of stability augmentation or electrical system failures; and
 - (c) *Performance.* If V_{YI} differs from V_Y , climb performance at V_{YI} and with maximum continuous power throughout the ranges of weight, altitude, and temperature for which approval is requested.

[Amdt. No.: 27/1]

CS 27.143 Controllability and manoeuvrability

ED Decision 2007/013/R

- (a) The rotorcraft must be safely controllable and manoeuvrable:
- (1) During steady flight; and
 - (2) During any manoeuvre appropriate to the type, including:
 - (i) Take-off;
 - (ii) Climb;
 - (iii) Level flight;
 - (iv) Turning flight;
 - (v) Autorotation;
 - (vi) Landing (power-on and power-off); and
 - (vii) Recovery to power-on flight from a bailed autorotative approach.
- (b) The margin of cyclic control must allow satisfactory roll and pitch control at V_{NE} with:
- (1) Critical weight;
 - (2) Critical centre of gravity;
 - (3) Critical rotor rpm; and
 - (4) Power off, except for helicopters demonstrating compliance with sub-paragraph (f), and power on.
- (c) Wind velocities from zero to at least 31 km/h (17 knots), from all azimuths, must be established in which the rotorcraft can be operated without loss of control on or near the ground in any manoeuvre appropriate to the type, such as crosswind take-offs, sideward flight and rearward flight:
- (1) With altitude, from standard sea-level conditions to the maximum take-off and landing altitude capability of the rotorcraft or 2134 m (7000 ft) density altitude, whichever is less; with:
 - (i) Critical weight;
 - (ii) Critical centre of gravity; and
 - (iii) Critical rotor rpm.
 - (2) For take-off and landing altitudes above 2134 m (7000 ft) density altitude with:
 - (i) Weight selected by the applicant;
 - (ii) Critical center of gravity; and
 - (iii) Critical rotor rpm.
- (d) Wind velocities from zero to at least 31 km/h (17 knots), from all azimuths, must be established in which the rotorcraft can be operated without loss of control out-of-ground effect, with:
- (1) Weight selected by the applicant;
 - (2) Critical center of gravity;
 - (3) Rotor rpm selected by the applicant; and

- (4) Altitude, from standard sea-level conditions to the maximum take-off and landing altitude capability of the rotorcraft.
- (e) The rotorcraft, after
 - (1) failure of one engine in the case of multi-engine rotorcraft that meet Category A engine isolation requirements, or
 - (2) complete engine failure in the case of other rotorcraft, must be controllable over the range of speeds and altitudes for which certification is requested when such power failure occurs with maximum continuous power and critical weight. No corrective action time delay for any condition following power failure may be less than:
 - (i) For the cruise condition, one second, or normal pilot reaction time (whichever is greater); and
 - (ii) For any other condition, normal pilot reaction time.
- (f) For helicopters for which a V_{NE} (power-off) is established under [CS 27.1505\(c\)](#), compliance must be demonstrated with the following requirements with critical weight, critical centre of gravity, and critical rotor rpm:
 - (1) The helicopter must be safely slowed to V_{NE} (power-off), without exceptional pilot skill, after the last operating engine is made inoperative at power-on V_{NE} ;
 - (2) At a speed of $1.1 V_{NE}$ (power-off), the margin of cyclic control must allow satisfactory roll and pitch control with power off.

[Amdt. No.: 27/1]

CS 27.151 Flight controls

ED Decision 2003/15/RM

- (a) Longitudinal, lateral, directional, and collective controls may not exhibit excessive breakout force, friction or preload.
- (b) Control system forces and free play may not inhibit a smooth, direct rotorcraft response to control system input.

CS 27.161 Trim control

ED Decision 2003/15/RM

The trim control:

- (a) Must trim any steady longitudinal, lateral, and collective control forces to zero in level flight at any appropriate speed; and
- (b) May not introduce any undesirable discontinuities in control force gradients.

CS 27.171 Stability: general

ED Decision 2003/15/RM

The rotorcraft must be able to be flown, without undue pilot fatigue or strain, in any normal manoeuvre for a period of time as long as that expected in normal operation. At least three landings and take-offs must be made during this demonstration.

CS 27.173 Static longitudinal stability

ED Decision 2007/013/R

- (a) The longitudinal control must be designed so that a rearward movement of the control is necessary to obtain an airspeed less than the trim speed, and a forward movement of the control is necessary to obtain an airspeed more than the trim speed.
- (b) Throughout the full range of altitude for which certification is requested, with the throttle and collective pitch held constant during the manoeuvres specified in [CS 27.175\(a\) through \(d\)](#), the slope of the control position versus airspeed curve must be positive. However, in limited flight conditions or modes of operation determined by the Agency to be acceptable, the slope of the control position versus airspeed curve may be neutral or negative if the rotorcraft possesses flight characteristics that allow the pilot to maintain airspeed within ± 9 km/h (± 5 knots) of the desired trim airspeed without exceptional piloting skill or alertness.

[Amdt. No.: 27/1]

CS 27.175 Demonstration of static longitudinal stability

ED Decision 2007/013/R

- (a) *Climb.* Static longitudinal stability must be shown in the climb condition at speeds from $V_Y - 19$ km/h (10 knots) to $V_Y + 19$ km/h (10 knots), with:
 - (1) Critical weight;
 - (2) Critical centre of gravity;
 - (3) Maximum continuous power;
 - (4) The landing gear retracted; and
 - (5) The rotorcraft trimmed at V_Y .
- (b) *Cruise.* Static longitudinal stability must be shown in the cruise condition at speeds from $0.8 V_{NE} - 19$ km/h (10 knots) to $0.8 V_{NE} + 19$ km/h (10 knots) or, if V_H is less than $0.8 V_{NE}$, from $V_H - 19$ km/h (10 knots) to $V_H + 19$ km/h (10 knots), with:
 - (1) Critical weight;
 - (2) Critical centre of gravity;
 - (3) Power for level flight at $0.8 V_{NE}$ or V_H , whichever is less;
 - (4) The landing gear retracted; and
 - (5) The rotorcraft trimmed at $0.8 V_{NE}$ or V_H , whichever is less.
- (c) V_{NE} . Static longitudinal stability must be shown at speeds from $V_{NE} - 28$ km/h (20 knots) to V_{NE} with:
 - (1) Critical weight;
 - (2) Critical center of gravity;
 - (3) Power required for level flight at $V_{NE} - 19$ km/h (10 knots) or maximum continuous power, whichever is less;
 - (4) The landing gear retracted; and
 - (5) The rotorcraft trimmed at $V_{NE} - 19$ km/h (10 knots)

- (d) *Autorotation*. Static longitudinal stability must be shown in autorotation at:
- (1) Airspeeds from the minimum rate of descent airspeed – 19 km/h (10 knots) to the minimum rate of descent airspeed + 19 km/h (10 knots), with:
 - (i) Critical weight;
 - (ii) Critical center of gravity;
 - (iii) The landing gear extended; and
 - (iv) The rotorcraft trimmed at the minimum rate of descent airspeed.
 - (2) Airspeeds from the best angle-of-glide airspeed – 19 km/h (10 knots) to the best angle-of-glide airspeed + 19 km/h (10 knots), with:
 - (i) Critical weight;
 - (ii) Critical center of gravity;
 - (iii) The landing gear retracted; and
 - (iv) The rotorcraft trimmed at the best angle-of-glide airspeed.

[Amdt. No.: 27/1]

CS 27.177 Static directional stability

ED Decision 2007/013/R

- (a) The directional controls must operate in such a manner that the sense and direction of motion of the rotorcraft following control displacement are in the direction of the pedal motion with throttle and collective controls held constant at the trim conditions specified in [CS 27.175\(a\), \(b\), and \(c\)](#). Sideslip angles must increase with steadily increasing directional control deflection for sideslip angles up to the lesser of:
- (1) ± 25 degrees from trim at a speed of 28 km/h (15 knots) less than the speed for minimum rate of descent varying linearly to ± 10 degrees from trim at V_{NE} ;
 - (2) The steady state sideslip angles established by [CS 27.351](#);
 - (3) A sideslip angle selected by the applicant which corresponds to a sideforce of at least 0.1 g; or,
 - (4) The sideslip angle attained by maximum directional control input.
- (b) Sufficient cues must accompany the sideslip to alert the pilot when the aircraft is approaching the sideslip limits.
- (c) During the manoeuvre specified in sub-paragraph (a) of this paragraph, the sideslip angle versus directional control position curve may have a negative slope within a small range of angles around trim, provided the desired heading can be maintained without exceptional piloting skill or alertness.

[Amdt. No.: 27/1]

GROUND AND WATER HANDLING CHARACTERISTICS

CS 27.231 General

ED Decision 2003/15/RM

The rotorcraft must have satisfactory ground and water handling characteristics, including freedom from uncontrollable tendencies in any condition expected in operation.

CS 27.235 Taxying condition

ED Decision 2003/15/RM

The rotorcraft must be designed to withstand the loads that would occur when the rotorcraft is taxied over the roughest ground that may reasonably be expected in normal operation.

CS 27.239 Spray characteristics

ED Decision 2003/15/RM

If certification for water operation is requested, no spray characteristics during taxying, take-off, or landing may obscure the vision of the pilot or damage the rotors, propellers, or other parts of the rotorcraft.

CS 27.241 Ground resonance

ED Decision 2003/15/RM

The rotorcraft may have no dangerous tendency to oscillate on the ground with the rotor turning.

MISCELLANEOUS FLIGHT REQUIREMENTS

CS 27.251 Vibration

ED Decision 2003/15/RM

Each part of the rotorcraft must be free from excessive vibration under each appropriate speed and power condition.

SUBPART C – STRENGTH REQUIREMENTS

GENERAL

CS 27.301 Loads

ED Decision 2003/15/RM

- (a) Strength requirements are specified in terms of limit loads (the maximum loads to be expected in service) and ultimate loads (limit loads multiplied by prescribed factors of safety). Unless otherwise provided, prescribed loads are limit loads.
- (b) Unless otherwise provided, the specified air, ground, and water loads must be placed in equilibrium with inertia forces, considering each item of mass in the rotorcraft. These loads must be distributed to closely approximate or conservatively represent actual conditions.
- (c) If deflections under load would significantly change the distribution of external or internal loads, this redistribution must be taken into account.

CS 27.303 Factor of safety

ED Decision 2003/15/RM

Unless otherwise provided, a factor of safety of 1.5 must be used. This factor applies to external and inertia loads unless its application to the resulting internal stresses is more conservative.

CS 27.305 Strength and deformation

ED Decision 2003/15/RM

- (a) The structure must be able to support limit loads without detrimental or permanent deformation. At any load up to limit loads, the deformation may not interfere with safe operation.
- (b) The structure must be able to support ultimate loads without failure. This must be shown by:
 - (1) Applying ultimate loads to the structure in a static test for at least 3 seconds; or
 - (2) Dynamic tests simulating actual load application.

CS 27.307 Proof of structure

ED Decision 2003/15/RM

- (a) Compliance with the strength and deformation requirements of this Subpart must be shown for each critical loading condition accounting for the environment to which the structure will be exposed in operation. Structural analysis (static or fatigue) may be used only if the structure conforms to those structures for which experience has shown this method to be reliable.

In other cases, substantiating load tests must be made.

- (b) Proof of compliance with the strength requirements of this Subpart must include:
 - (1) Dynamic and endurance tests of rotors, rotor drives, and rotor controls;
 - (2) Limit load tests of the control system, including control surfaces;
 - (3) Operation tests of the control system;
 - (4) Flight stress measurement tests;

- (5) Landing gear drop tests; and
- (6) Any additional tests required for new or unusual design features.

CS 27.309 Design limitations

ED Decision 2003/15/RM

The following values and limitations must be established to show compliance with the structural requirements of this Subpart:

- (a) The design maximum weight.
- (b) The main rotor rpm ranges power on and power off.
- (c) The maximum forward speeds for each main rotor rpm within the ranges determined in sub-paragraph (b).
- (d) The maximum rearward and sideward flight speeds.
- (e) The centre of gravity limits corresponding to the limitations determined under sub-paragraphs (b), (c), and (d).
- (f) The rotational speed ratios between each powerplant and each connected rotating component.
- (g) The positive and negative limit manoeuvring load factors.

FLIGHT LOADS

CS 27.321 General

ED Decision 2003/15/RM

- (a) The flight load factor must be assumed to act normal to the longitudinal axis of the rotorcraft, and to be equal in magnitude and opposite in direction to the rotorcraft inertia load factor at the centre of gravity.
- (b) Compliance with the flight load requirements of this Subpart must be shown:
 - (1) At each weight from the design minimum weight to the design maximum weight; and
 - (2) With any practical distribution of disposable load within the operating limitations in the Rotorcraft Flight Manual.

CS 27.337 Limit manoeuvring load factor

ED Decision 2003/15/RM

The rotorcraft must be designed for:

- (a) A limit manoeuvring load factor ranging from a positive limit of 3.5 to a negative limit of – 1.0; or
- (b) Any positive limit manoeuvring load factor not less than 2.0 and any negative limit manoeuvring load factor of not less than – 0.5 for which:
 - (1) The probability of being exceeded is shown by analysis and flight tests to be extremely remote; and
 - (2) The selected values are appropriate to each weight condition between the design maximum and design minimum weights.

CS 27.339 Resultant limit manoeuvring loads

ED Decision 2003/15/RM

The loads resulting from the application of limit manoeuvring load factors are assumed to act at the centre of each rotor hub and at each auxiliary lifting surface, and to act in directions, and with distributions of load among the rotors and auxiliary lifting surfaces, so as to represent each critical manoeuvring condition, including power-on and power-off flight with the maximum design rotor tip speed ratio. The rotor tip speed ratio is the ratio of the rotorcraft flight velocity component in the plane of the rotor disc to the rotational tip speed of the rotor blades, and is expressed as follows:

$$\mu = \frac{V \cos a}{\Omega R}$$

where:

- V = The airspeed along the flight path;
- a = The angle between the projection, in the plane of symmetry, of the axis of no feathering and a line perpendicular to the flight path (positive when the axis is pointing aft);
- Ω = The angular velocity of rotor; and
- R = The rotor radius.

CS 27.341 Gust loads

ED Decision 2003/15/RM

The rotorcraft must be designed to withstand, at each critical airspeed including hovering, the loads resulting from a vertical gust of 9.1 m/s (30 ft/s).

CS 27.351 Yawing conditions

ED Decision 2003/15/RM

- (a) Each rotorcraft must be designed for the loads resulting from the manoeuvres specified in subparagraphs (b) and (c) with:
 - (1) Unbalanced aerodynamic moments about the centre of gravity which the aircraft reacts to in a rational or conservative manner considering the principal masses furnishing the reacting inertia forces; and
 - (2) Maximum main rotor speed.
- (b) To produce the load required in subparagraph (a), in unaccelerated flight with zero yaw, at forward speeds from zero up to $0.6 V_{NE}$:
 - (1) Displace the cockpit directional control suddenly to the maximum deflection limited by the control stops or by the maximum pilot force specified in [CS 27.397\(a\)](#);
 - (2) Attain a resulting sideslip angle or 90° , whichever is less; and
 - (3) Return the directional control suddenly to neutral.
- (c) To produce the load required in sub-paragraph (a), in unaccelerated flight with zero yaw, at forward speeds from $0.6 V_{NE}$ up to V_{NE} or V_H , whichever is less:
 - (1) Displace the cockpit directional control suddenly to the maximum deflection limited by the control stops or by the maximum pilot force specified in [CS 27.397\(a\)](#);
 - (2) Attain a resulting sideslip angle or 15° , whichever is less, at the lesser speed of V_{NE} or V_H ;
 - (3) Vary the sideslip angles of subparagraphs (b)(2) and (c)(2) directly with speed; and
 - (4) Return the directional control suddenly to neutral.

AMC No 1 to CS 27.351 Yawing conditions

ED Decision 2016/024/R

- (a) Definitions:
 - (1) Suddenly. For the purpose of this AMC, 'suddenly' is defined as an interval not to exceed 0.2 seconds for a complete control input. A rational analysis may be used to substantiate an alternative value.
 - (2) Initial Trim Condition. Steady, 1G level flight condition with zero bank angle or zero sideslip.
 - (3) 'Line'. The rotorcraft's sideslip envelope, defined by the rule, between 90° at $0.6V_{NE}$ and 15° at V_{NE} or V_H whichever is less (see Figure 1).
 - (4) Resulting Sideslip Angle. The rotorcraft's stabilised sideslip angle that results from a sustained maximum cockpit directional control deflection or as limited by pilot effort in the initial level flight power conditions.

- (b) **Explanation:** The rule requires a rotorcraft's 'structural' yaw or sideslip design envelope that must cover a minimum forward speed or hover to V_{NE} or V_H whichever is less. The scope of the rule is intended to cover structural components that are primarily designed for the critical combinations of tail rotor thrust, inertial and aerodynamic forces. This may include but is not limited to fuselage, tailboom and attachments, vertical control surfaces, tail rotor and tail rotor support structure.
- (1) The rotorcraft's structure must be designed to withstand the loads in the specified yawing conditions. The standard does not require a structural flight demonstration. It is a structural design standard.
 - (2) The standard applies only to power-on conditions. Autorotation need not be considered.
 - (3) This standard requires the maximum allowable rotor revolutions per minute (RPM) consistent with each flight condition for which certification is requested.
 - (4) For the purpose of this AMC, the analysis may be performed in international standard atmosphere (ISA) sea level conditions.
 - (5) Maximum displacement of the directional control, except as limited by pilot effort ([27.397\(a\)](#)), is required for the conditions cited in the rule. A control-system-limiting device may be used, however the probability of failure or malfunction of these system(s) should be considered (See [AMC No 2 to CS 27.351](#) Interaction of System and Structure).
 - (6) Both right and left yaw conditions should be evaluated.
 - (7) The air loads on the vertical stabilisers may be assumed independent of the tail rotor thrust.
 - (8) Loads associated with sideslip angles exceeding the values of the 'line', defined in Figure 1, do not need to be considered. The corresponding points of the manoeuvre may be deleted.
- (c) **Procedure:** The design loads should be evaluated within the limits of Figure 1 or the maximum yaw capability of the rotorcraft whichever is less at speeds from zero to V_H or V_{NE} whichever is less for the following phases of the manoeuvre (see Note 1):
- (1) With the rotorcraft at an initial trim condition, the cockpit directional control is suddenly displaced to the maximum deflection limited by the control stops or by the maximum pilot force specified in [27.397\(a\)](#). This is intended to generate a high tail rotor thrust.
 - (2) While maintaining maximum cockpit directional control deflection, within the limitation specified in (c)(1) of this AMC allow the rotorcraft to yaw to the maximum transient sideslip angle. This is intended to generate high aerodynamic loads that are determined based on the maximum transient sideslip angle or the value defined by the 'line' in Figure 1 whichever is less (see Note 1).
 - (3) Allow the rotorcraft to attain the resulting sideslip angle. In the event that the resulting sideslip angle is greater than the value defined by the 'line' in Figure 1, the rotorcraft should be trimmed to that value of the angle using less than maximum cockpit directional-control deflection by taking into consideration the manoeuvre's entry airspeed (see Note 2).
 - (4) With the rotorcraft yawed to the resulting sideslip angle specified in (c)(3) of this AMC, the cockpit control is suddenly returned to its initial trim position. This is intended to combine a high tail rotor thrust and high aerodynamic restoring forces.

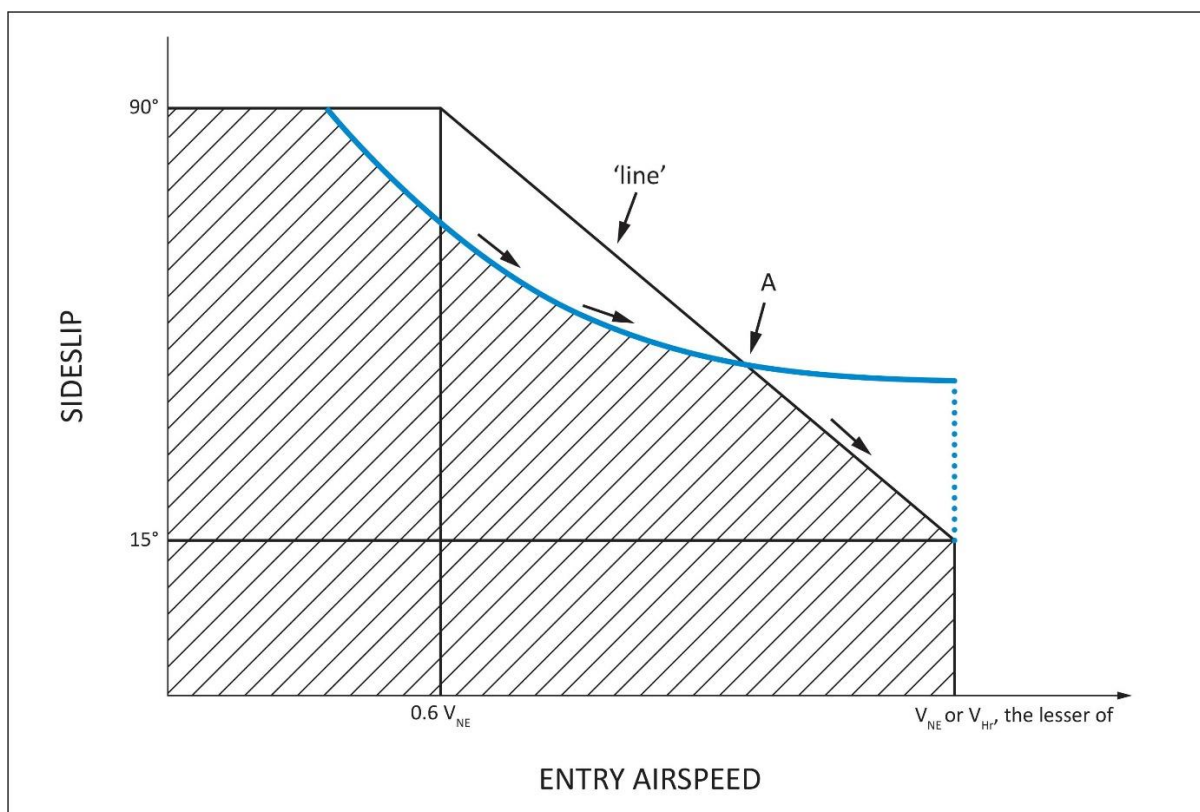


Figure 1 — YAW/FORWARD SPEED DIAGRAM

NOTE:

- (1) When comparing the rotorcraft's sideslip angle against the 'line' of Figure 1, the entry airspeed of the manoeuvre should be used.
- (2) When evaluating the yawing condition against the 'line' of Figure 1, sufficient points should be investigated in order to determine the critical design conditions. This investigation should include the loads that result from the manoeuvre, specifically initiated at the intermediate airspeed which is coincident with the intersection of the 'line' and the resultant sideslip angle (point A in Figure 1).
- (d) Another method of compliance may be used with a rational analysis (dynamic simulation), acceptable to the Agency/Authority, performed up to V_H or V_{NE} whichever is less, to the maximum yaw capability of the rotorcraft with recovery initiated at the resulting sideslip angle at its associated airspeed. Loads should be considered for all portions of the manoeuvre.

[Amdt 27/4]

AMC No 2 to 27.351 Yaw manoeuvre conditions

ED Decision 2016/024/R

1. Introduction

This AMC provides further guidance and acceptable means of compliance to supplement FAA AC 27-1B § AC 27.351. § 27.351 to meet the Agency's interpretation of [CS 27.351](#). As such it should be used in conjunction with the FAA AC but take precedence over it, where stipulated, in the showing of compliance.

Specifically, this AMC addresses two areas where the FAA AC has been deemed by the Agency as being unclear or at variance to the Agency's interpretation. These areas are as follows:

a. Aerodynamic Loads

The certification specification [CS 27.351](#) provides a minimum safety standard for the design of rotorcraft structural components that are subjected in flight to critical loads combinations of anti-torque system thrust (e.g. tail rotor), inertia and aerodynamics. A typical example of these structural components is the tailboom.

However, compliance with this standard according to FAA AC may not necessarily be adequate for the design of rotorcraft structural components that are principally subjected in flight to significant aerodynamic loads (e.g. vertical empennage, fins, cowlings and doors).

For these components and their supporting structure, suitable design criteria should be developed by the Applicant and agreed with the Agency.

In lieu of acceptable design criteria developed by the applicant, a suitable combination of sideslip angle and airspeed for the design of rotorcraft components subjected to aerodynamic loads may be obtained from a simulation of the yaw manoeuvre of [CS 27.351](#), starting from the initial directional control input specified in [CS 27.351\(b\)\(1\) and \(c\)\(1\)](#), until the rotorcraft reaches the maximum transient sideslip angle (overswing) resulting from its motion around the yaw axis.

b. Interaction of System and Structure

Maximum displacement of the directional control, except as limited by pilot effort ([CS 27.397\(a\)](#)), is required for the conditions cited in the certification specification. In the load evaluation credit may be taken for consideration of the effects of control system limiting devices.

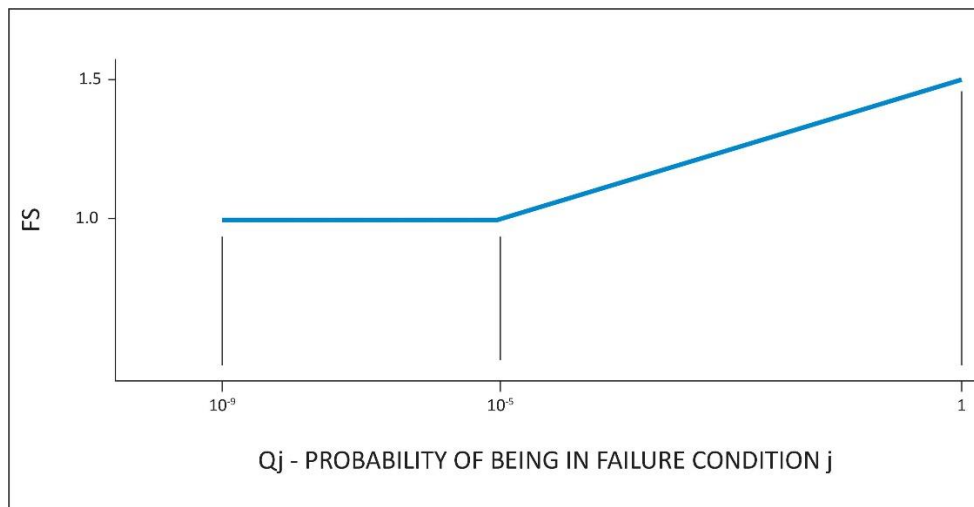
However, the probability of failure or malfunction of these system(s) should also be considered and if it is shown not to be extremely improbable then further load conditions with the system in the failed state should be evaluated. This evaluation may include Flight Manual Limitations, if failure of the system is reliably indicated to the crew.

A yaw limiting device is a typical example of a system whose failed condition should be investigated in the assessment of the loads requested by [CS 27.351](#).

An acceptable methodology to investigate the effects of all system failures not shown to be extremely improbable on the loading conditions of [CS 27.351](#) is as follows:

- (i) With the system in the failed state and considering any appropriate reconfiguration and flight limitations, it should be shown that the rotorcraft structure can withstand without failure the loading conditions of [CS 27.351](#), when the manoeuvre is performed in accordance with the provisions of this AMC.

- (ii) The factor of safety to apply to the above specified loading conditions to comply with [CS 27.305](#) is defined in the figure below.



$$Q_j = (T_j)(P_j)$$

where:

T_j = Average flight time spent with a failed limiting system j (in hours)

P_j = Probability of occurrence of failure of control limiting system j (per hour)

Note: If P_j is greater than 1×10^{-3} per flight hour then a 1.5 factor of safety should be applied to all limit load conditions evaluated for the system failure under consideration.

[Amdt 27/2]

[Amdt 27/4]

CS 27.361 Engine torque

ED Decision 2003/15/RM

- (a) For turbine engines, the limit torque may not be less than the highest of:
- (1) The mean torque for maximum continuous power multiplied by 1.25;
 - (2) The torque required by [CS 27.923](#);
 - (3) The torque required by [CS 27.927](#);
- or
- (4) The torque imposed by sudden engine stoppage due to malfunction or structural failure (such as compressor jamming).
- (b) For reciprocating engines, the limit torque may not be less than the mean torque for maximum continuous power multiplied by:
- (1) 1.33, for engines with five or more cylinders; and
 - (2) Two, three, and four, for engines with four, three, and two cylinders, respectively.

CONTROL SURFACE AND SYSTEM LOADS

CS 27.391 General

ED Decision 2003/15/RM

Each auxiliary rotor, each fixed or movable stabilising or control surface, and each system operating any flight control must meet the requirements of [CS 27.395](#), [27.397](#), [27.399](#), [27.411](#) and [27.427](#).

CS 27.395 Control system

ED Decision 2003/15/RM

- (a) The part of each control system from the pilot's controls to the control stops must be designed to withstand pilot forces of not less than –
 - (1) The forces specified in [CS 27.397](#); or
 - (2) If the system prevents the pilot from applying the limit pilot forces to the system, the maximum forces that the system allows the pilot to apply, but not less than 0.60 times the forces specified in [CS 27.397](#).
- (b) Each primary control system including its supporting structure, must be designed as follows:
 - (1) The system must withstand loads resulting from the limit pilot forces prescribed in [CS 27.397](#).
 - (2) Notwithstanding sub-paragraph (b)(3), when power-operated actuator controls or power boost controls are used, the system must also withstand the loads resulting from the force output of each normally energised power device, including any single power boost or actuator system failure.
 - (3) If the system design or the normal operating loads are such that a part of the system cannot react to the limit forces prescribed in [CS 27.397](#), that part of the system must be designed to withstand the maximum loads that can be obtained in normal operation. The minimum design loads must, in any case, provide a rugged system for service use, including consideration of fatigue, jamming, ground gusts, control inertia and friction loads. In the absence of rational analysis, the design loads resulting from 0.60 of the specified limit pilot forces are acceptable minimum design loads.
 - (4) If operational loads may be exceeded through jamming, ground gusts, control inertia, or friction, the system must withstand the limit pilot forces specified in [CS 27.397](#), without yielding.

CS 27.397 Limit pilot forces and torques

ED Decision 2003/15/RM

- (a) Except as provided in sub-paragraph (b) the limit pilot forces are as follows:
 - (1) For foot controls, 578 N (130 lbs).
 - (2) For stick controls, 445 N (100 lbs) fore and aft, and 298 N (67 lbs) laterally.
- (b) For flap, tab, stabiliser, rotor brake, and landing gear operating controls, the following apply:

- (1) Crank, wheel, and lever controls, $(25.4 + R) \times 2.919$ N, where R = radius in millimetres ($\left[\frac{1+R}{3}\right] \times 50$ lbs, where R = radius in inches), but not less than 222 N (50 lbs) nor more than 445 N (100 lbs) for hand-operated controls or 578 N (130 lbs) for foot-operated controls, applied at any angle within 20° of the plane of motion of the control.
- (2) Twist controls, $356 \times R$ Newton-millimetres, where R = radius in millimetres ($80 \times R$ inch-pounds where R = radius in inches).

CS 27.399 Dual control system

ED Decision 2003/15/RM

Each dual primary flight control system must be designed to withstand the loads that result when pilot forces of 0.75 times those obtained under [CS 27.395](#) are applied –

- (a) In opposition; and
- (b) In the same direction.

CS 27.411 Ground clearance: tail rotor guard

ED Decision 2003/15/RM

- (a) It must be impossible for the tail rotor to contact the landing surface during a normal landing.
- (b) If a tail rotor guard is required to show compliance with sub-paragraph (a):
 - (1) Suitable design loads must be established for the guard; and
 - (2) The guard and its supporting structure must be designed to withstand those loads.

CS 27.427 Unsymmetrical loads

ED Decision 2003/15/RM

- (a) Horizontal tail surfaces and their supporting structure must be designed for unsymmetrical loads arising from yawing and rotor wake effects in combination with the prescribed flight conditions.
- (b) To meet the design criteria of sub-paragraph (a), in the absence of more rational data, both of the following must be met:
 - (1) 100% of the maximum loading from the symmetrical flight conditions acts on the surface on one side of the plane of symmetry and no loading acts on the other side.
 - (2) 50% of the maximum loading from the symmetrical flight conditions acts on the surface on each side of the plane of symmetry but in opposite directions.
- (c) For empennage arrangements where the horizontal tail surfaces are supported by the vertical tail surfaces, the vertical tail surfaces and supporting structure must be designed for the combined vertical and horizontal surface loads resulting from each prescribed flight condition, considered separately. The flight conditions must be selected so the maximum design loads are obtained on each surface. In the absence of more rational data, the unsymmetrical horizontal tail surface loading distributions described in this paragraph must be assumed.

GROUND LOADS

CS 27.471 General

ED Decision 2003/15/RM

- (a) *Loads and equilibrium.* For limit ground loads –
- (1) The limit ground loads obtained in the landing conditions in this Subpart must be considered to be external loads that would occur in the rotorcraft structure if it were acting as a rigid body; and
 - (2) In each specified landing condition, the external loads must be placed in equilibrium with linear and angular inertia loads in a rational or conservative manner.
- (b) *Critical centres of gravity.* The critical centres of gravity within the range for which certification is requested must be selected so that the maximum design loads are obtained in each landing gear element.

CS 27.473 Ground loading conditions and assumptions

ED Decision 2003/15/RM

- (a) For specified landing conditions, a design maximum weight must be used that is not less than the maximum weight. A rotor lift may be assumed to act through the centre of gravity throughout the landing impact. This lift may not exceed two-thirds of the design maximum weight.
- (b) Unless otherwise prescribed, for each specified landing condition, the rotorcraft must be designed for a limit load factor of not less than the limit inertia load factor substantiated under [CS 27.725](#).

CS 27.475 Tyres and shock absorbers

ED Decision 2003/15/RM

Unless otherwise prescribed, for each specified landing condition, the tyres must be assumed to be in their static position and the shock absorbers to be in their most critical position.

CS 27.477 Landing gear arrangement

ED Decision 2003/15/RM

Paragraphs [CS 27.235](#), [27.479](#) to [27.485](#), and [CS 27.493](#) apply to landing gear with two wheels aft, and one or more wheels forward, of the centre of gravity.

CS 27.479 Level landing conditions

ED Decision 2003/15/RM

- (a) *Attitudes.* Under each of the loading conditions prescribed in sub-paragraph (b), the rotorcraft is assumed to be in each of the following level landing attitudes:
- (1) An attitude in which all wheels contact the ground simultaneously.
 - (2) An attitude in which the aft wheels contact the ground with the forward wheels just clear of the ground.

- (b) *Loading conditions.* The rotorcraft must be designed for the following landing loading conditions:
- (1) Vertical loads applied under [CS 27.471](#).
 - (2) The loads resulting from a combination of the loads applied under sub-paragraph (b)(1) with drag loads at each wheel of not less than 25% of the vertical load at that wheel.
 - (3) If there are two wheels forward, a distribution of the loads applied to those wheels under sub-paragraphs (b)(1) and (2) in a ratio of 40:60.
- (c) *Pitching moments.* Pitching moments are assumed to be resisted by:
- (1) In the case of the attitude in sub-paragraph (a)(1), the forward landing gear, and
 - (2) In the case of the attitude in sub-paragraph (a)(2), the angular inertia forces.

CS 27.481 Tail-down landing conditions

ED Decision 2003/15/RM

- (a) The rotorcraft is assumed to be in the maximum nose-up attitude allowing ground clearance by each part of the rotorcraft.
- (b) In this attitude, ground loads are assumed to act perpendicular to the ground.

CS 27.483 One-wheel landing conditions

ED Decision 2003/15/RM

For the one-wheel landing condition, the rotorcraft is assumed to be in the level attitude and to contact the ground on one aft wheel. In this attitude:

- (a) The vertical load must be the same as that obtained on that side under [CS 27.479\(b\)\(1\)](#); and
- (b) The unbalanced external loads must be reacted by rotorcraft inertia.

CS 27.485 Lateral drift landing conditions

ED Decision 2003/15/RM

- (a) The rotorcraft is assumed to be in the level landing attitude, with:
- (1) Side loads combined with one-half of the maximum ground reactions obtained in the level landing conditions of [CS 27.479\(b\)\(1\)](#); and
 - (2) The loads obtained under sub-paragraph (a)(1) applied:
 - (i) At the ground contact point; or
 - (ii) For full-swivelling gear, at the centre of the axle.
- (b) The rotorcraft must be designed to withstand, at ground contact –
- (1) When only the aft wheels contact the ground, side loads of 0.8 times the vertical reaction acting inward on one side, and 0,6 times the vertical reaction acting outward on the other side, all combined with the vertical loads specified in sub-paragraph (a) ; and
 - (2) When all wheels contact the ground simultaneously:
 - (i) For the aft wheels, the side loads specified in sub-paragraph (b)(1); and

- (ii) For the forward wheels, a side load of 0.8 times the vertical reaction combined with the vertical load specified in sub-paragraph (a).

CS 27.493 Braked roll conditions

ED Decision 2003/15/RM

Under braked roll conditions with the shock absorbers in their static positions:

- (a) The limit vertical load must be based on a load factor of at least:
 - (1) 1.33, for the attitude specified in [CS 27.479\(a\)\(1\)](#); and
 - (2) 1.0 for the attitude specified in [CS 27.479\(a\)\(2\)](#); and
- (b) The structure must be designed to withstand at the ground contact point of each wheel with brakes, a drag load at least the lesser of:
 - (1) The vertical load multiplied by a coefficient of friction of 0.8; and
 - (2) The maximum value based on limiting brake torque.

CS 27.497 Ground loading conditions: landing gear with tail wheels

ED Decision 2003/15/RM

- (a) *General.* Rotorcraft with landing gear with two wheels forward, and one wheel aft, of the centre of gravity must be designed for loading conditions as prescribed in this paragraph.
- (b) *Level landing attitude with only the forward wheels contacting the ground.* In this attitude:
 - (1) The vertical loads must be applied under [CS 27.471](#) to [27.475](#);
 - (2) The vertical load at each axle must be combined with a drag load at that axle of not less than 25% of that vertical load; and
 - (3) Unbalanced pitching moments are assumed to be resisted by angular inertia forces.
- (c) *Level landing attitude with all wheels contacting the ground simultaneously.* In this attitude, the rotorcraft must be designed for landing loading conditions as prescribed in sub-paragraph (b).
- (d) *Maximum nose-up attitude with only the rear wheel contacting the ground.* The attitude for this condition must be the maximum nose-up attitude expected in normal operation, including autorotative landings. In this attitude:
 - (1) The appropriate ground loads specified in sub-paragraphs (b)(1) and (2) must be determined and applied, using a rational method to account for the moment arm between the rear wheel ground reaction and the rotorcraft centre of gravity; or
 - (2) The probability of landing with initial contact on the rear wheel must be shown to be extremely remote.
- (e) *Level landing attitude with only one forward wheel contacting the ground.* In this attitude, the rotorcraft must be designed for ground loads as specified in sub-paragraphs (b) (1) and (3).
- (f) *Side loads in the level landing attitude.* In the attitudes specified in sub-paragraphs (b) and (c) the following apply:
 - (1) The side loads must be combined at each wheel with one-half of the maximum vertical ground reactions obtained for that wheel under sub-paragraphs (b) and (c). In this condition the side loads must be:

- (i) For the forward wheels, 0.8 times the vertical reaction (on one side) acting inward, and 0.6 times the vertical reaction (on the other side) acting outward; and
 - (ii) For the rear wheel, 0.8 times the vertical reaction.
- (2) The loads specified in sub-paragraph (f)(1) must be applied:
 - (i) At the ground contact point with the wheel in the trailing position (for non-full swivelling landing gear or for full-swivelling landing gear with a lock, steering device, or shimmy damper to keep the wheel in the trailing position); or (ii) At the centre of the axle (for full swivelling landing gear without a lock, steering device, or shimmy damper).
- (g) *Braked roll conditions in the level landing attitude.* In the attitudes specified in sub-paragraphs (b) and (c), and with shock absorbers in their static positions, the rotorcraft must be designed for braked roll loads as follows:
 - (1) The limit vertical load must be based on a limit vertical load factor of not less than:
 - (i) 1.0 for the attitude specified in sub-paragraph (b); and
 - (ii) 1.33, for the attitude specified in sub-paragraph (c).
 - (2) For each wheel with brakes, a drag load must be applied, at the ground contact point, of not less than the lesser of:
 - (i) 0.8 times the vertical load; and
 - (ii) The maximum based on limiting brake torque.
- (h) *Rear wheel turning loads in the static ground attitude.* In the static ground attitude, and with the shock absorbers and tyres in their static positions, the rotorcraft must be designed for rear wheel turning loads as follows:
 - (1) A vertical ground reaction equal to the static load on the rear wheel must be combined with an equal sideload.
 - (2) The load specified in sub-paragraph (h)(1) must be applied to the rear landing gear:
 - (i) Through the axle, if there is a swivel (the rear wheel being assumed to be swivelled 90° to the longitudinal axis of the rotorcraft); or
 - (ii) At the ground contact point, if there is a lock, steering device or shimmy damper (the rear wheel being assumed to be in the trailing position).
- (i) *Taxying condition.* The rotorcraft and its landing gear must be designed for loads that would occur when the rotorcraft is taxied over the roughest ground that may reasonably be expected in normal operation.

CS 27.501 Ground loading conditions: landing gear with skids

ED Decision 2003/15/RM

- (a) *General.* Rotorcraft with landing gear with skids must be designed for the loading conditions specified in this paragraph. In showing compliance with this paragraph, the following apply:
 - (1) The design maximum weight, centre of gravity, and load factor must be determined under [CS 27.471](#) to [27.475](#).
 - (2) Structural yielding of elastic spring members under limit loads is acceptable.

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- (3) Design ultimate loads for elastic spring members need not exceed those obtained in a drop test of the gear with:
- (i) A drop height of 1.5 times that specified in [CS 27.725](#); and
 - (ii) An assumed rotor lift of not more than 1.5 times that used in the limit drop tests prescribed in [CS 27.725](#).
- (4) Compliance with sub-paragraphs (b) to (e) must be shown with:
- (i) The gear in its most critically deflected position for the landing condition being considered; and
 - (ii) The ground reactions rationally distributed along the bottom of the skid tube.
- (b) *Vertical reactions in the level landing attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of both skids, the vertical reactions must be applied as prescribed in sub-paragraph (a).
- (c) *Drag reactions in the level landing attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of both skids, the following apply:
- (1) The vertical reactions must be combined with horizontal drag reactions of 50% of the vertical reaction applied at the ground.
 - (2) The resultant ground loads must equal the vertical load specified in sub-paragraph (b).
- (d) *Side loads in the level landing attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of both skids, the following apply:
- (1) The vertical ground reaction must be:
 - (i) Equal to the vertical loads obtained in the condition specified in sub-paragraph (b); and
 - (ii) Divided equally among the skids.
 - (2) The vertical ground reactions must be combined with a horizontal sideload of 25% of their value.
 - (3) The total sideload must be applied equally between the skids and along the length of the skids.
 - (4) The unbalanced moments are assumed to be resisted by angular inertia.
 - (5) The skid gear must be investigated for:
 - (i) Inward acting sideloads; and
 - (ii) Outward acting sideloads.
- (e) *One-skid landing loads in the level attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of one skid only, the following apply:
- (1) The vertical load on the ground contact side must be the same as that obtained on that side in the condition specified in subparagraph (b).
 - (2) The unbalanced moments are assumed to be resisted by angular inertia.
- (f) *Special conditions.* In addition to the conditions specified in sub-paragraphs (b) and (c), the rotorcraft must be designed for the following ground reactions:

- (1) A ground reaction load acting up and aft at an angle of 45° to the longitudinal axis of the rotorcraft. This load must be:
 - (i) Equal to 1.33 times the maximum weight;
 - (ii) Distributed symmetrically among the skids;
 - (iii) Concentrated at the forward end of the straight part of the skid tube; and
 - (iv) Applied only to the forward end of the skid tube and its attachment to the rotorcraft.
- (2) With the rotorcraft in the level landing attitude, a vertical ground reaction load equal to one-half of the vertical load determined in sub-paragraph (b). This load must be –
 - (i) Applied only to the skid tube and its attachment to the rotorcraft; and
 - (ii) Distributed equally over 33.3% of the length between the skid tube attachments and centrally located midway between the skid tube attachments.

CS 27.505 Ski landing conditions

ED Decision 2003/15/RM

If certification for ski operation is requested, the rotorcraft, with skis, must be designed to withstand the following loading conditions (where P is the maximum static weight on each ski with the rotorcraft at design maximum weight, and n is the limit load factor determined under [CS 27.473\(b\)](#)).

- (a) Up-load conditions in which:
 - (1) A vertical load of Pn and a horizontal load of $Pn/4$ are simultaneously applied at the pedestal bearings; and
 - (2) A vertical load of $1.33 P$ is applied at the pedestal bearings.
- (b) A side-load condition in which a side load of $0.35 Pn$ is applied at the pedestal bearings in a horizontal plane perpendicular to the centreline of the rotorcraft.
- (c) A torque-load condition in which a torque load of $1.33 P$ (in foot pounds) is applied to the ski about the vertical axis through the centreline of the pedestal bearings.

WATER LOADS

CS 27.521 Float landing conditions

ED Decision 2003/15/RM

If certification for float operation is requested, the rotorcraft, with floats, must be designed to withstand the following loading conditions (where the limit load factor is determined under [CS 27.473\(b\)](#) or assumed to be equal to that determined for wheel landing gear):

- (a) Up-load conditions in which:
 - (1) A load is applied so that, with the rotorcraft in the static level attitude, the resultant water reaction passes vertically through the centre of gravity; and
 - (2) The vertical load prescribed in sub-paragraph (a)(1) is applied simultaneously with an aft component of 0.25 times the vertical component.
- (b) A side-load condition in which:
 - (1) A vertical load of 0.75 times the total vertical load specified in sub-paragraph (a)(1) is divided equally among the floats; and
 - (2) For each float, the load share determined under sub-paragraph (b)(1), combined with a total sideload of 0.25 times the total vertical load specified in sub-paragraph (b)(1), is applied to the float only.

MAIN COMPONENT REQUIREMENTS

CS 27.547 Main rotor structure

ED Decision 2012/021/R

- (a) Each main rotor assembly (including rotor hubs and blades) must be designed as prescribed in this paragraph.
- (b) The main rotor structure must be designed to withstand the following loads prescribed in [CS 27.337](#) to [27.341](#):
 - (1) Critical flight loads.
 - (2) Limit loads occurring under normal conditions of autorotation. For this condition, the rotor rpm must be selected to include the effects of altitude.
- (c) The main rotor structure must be designed to withstand loads simulating:
 - (1) For the rotor blades, hubs, and flapping hinges, the impact force of each blade against its stop during ground operation; and
 - (2) Any other critical condition expected in normal operation.
- (d) The main rotor structure must be designed to withstand the limit torque at any rotational speed, including zero. In addition:
 - (1) The limit torque need not be greater than the torque defined by a torque limiting device (where provided), and may not be less than the greater of:
 - (i) The maximum torque likely to be transmitted to the rotor structure in either direction; and
 - (ii) The limit engine torque specified in [CS 27.361](#).
 - (2) The limit torque must be distributed to the rotor blades in a rational manner.

[Amdt 27/3]

CS 27.549 Fuselage, landing gear, and rotor pylon structures

ED Decision 2012/021/R

- (a) Each fuselage, landing gear, and rotor pylon structure must be designed as prescribed in this paragraph. Resultant rotor forces may be represented as a single force applied at the rotor hub attachment point.
- (b) Each structure must be designed to withstand:
 - (1) The critical loads prescribed in [CS 27.337](#) to [27.341](#);
 - (2) The applicable ground loads prescribed in [CS 27.235](#), [27.471](#) to [27.485](#), [CS 27.493](#), [27.497](#), [27.501](#), [27.505](#), and [27.521](#); and
 - (3) The loads prescribed in [CS 27.547\(c\)\(2\) and \(d\)](#).
- (c) Auxiliary rotor thrust, and the balancing air and inertia loads occurring under accelerated flight conditions, must be considered.
- (d) Each engine mount and adjacent fuselage structure must be designed to withstand the loads occurring under accelerated flight and landing conditions, including engine torque.

[Amdt 27/3]

EMERGENCY LANDING CONDITIONS

CS 27.561 General

ED Decision 2003/15/RM

- (a) The rotorcraft, although it may be damaged in emergency landing conditions on land or water, must be designed as prescribed in this paragraph to protect the occupants under those conditions.
- (b) The structure must be designed to give each occupant every reasonable chance of escaping serious injury in a crash landing when:
 - (1) Proper use is made of seats, belts, and other safety design provisions;
 - (2) The wheels are retracted (where applicable); and
 - (3) Each occupant and each item of mass inside the cabin that could injure an occupant is restrained when subjected to the following ultimate inertial load factors relative to the surrounding structure:
 - (i) Upward – 4 g
 - (ii) Forward – 16 g
 - (iii) Sideward – 8 g
 - (iv) Downward – 20 g, after the intended displacement of the seat device
 - (v) Rearward – 1.5 g
- (c) The supporting structure must be designed to restrain, under any ultimate inertial load up to those specified in this paragraph, any item of mass above and/or behind the crew and passenger compartment that could injure an occupant if it came loose in an emergency landing. Items of mass to be considered include, but are not limited to, rotors, transmissions, and engines. The items of mass must be restrained for the following ultimate inertial load factors:
 - (1) Upward – 1.5 g
 - (2) Forward – 12 g
 - (3) Sideward – 6 g
 - (4) Downward – 12 g
 - (5) Rearward – 1.5 g
- (d) Any fuselage structure in the area of internal fuel tanks below the passenger floor level must be designed to resist the following ultimate inertial factors and loads and to protect the fuel tanks from rupture when those loads are applied to that area:
 - (1) Upward – 1.5 g
 - (2) Forward – 4.0 g
 - (3) Sideward – 2.0 g
 - (4) Downward – 4.0 g

CS 27.562 Emergency landing dynamic conditions

ED Decision 2003/15/RM

- (a) The rotorcraft, although it may be damaged in an emergency crash landing, must be designed to reasonably protect each occupant when:
 - (1) The occupant properly uses the seats, safety belts, and shoulder harnesses provided in the design; and
 - (2) The occupant is exposed to the loads resulting from the conditions prescribed in this paragraph.
- (b) Each seat type design or other seating device approved for crew or passenger occupancy during take-off and landing must successfully complete dynamic tests or be demonstrated by rational analysis based on dynamic tests of a similar type seat in accordance with the following criteria. The tests must be conducted with an occupant, simulated by a 77 kg (170-pound) anthropomorphic test dummy (ATD), sitting in the normal upright position.
 - (1) A change in downward velocity of not less than 9.1 m/s (30 ft/s) when the seat or other seating device is oriented in its nominal position with respect to the rotorcraft's reference system, the rotorcraft's longitudinal axis is canted upward 60° with respect to the impact velocity vector, and the rotorcraft's lateral axis is perpendicular to a vertical plane containing the impact velocity vector and the rotorcraft's longitudinal axis. Peak floor deceleration must occur in not more than 0.031 seconds after impact and must reach a minimum of 30 g.
 - (2) A change in forward velocity of not less than 12.8 m/s (42 ft/s) when the seat or other seating device is oriented in its nominal position with respect to the rotorcraft's reference system, the rotorcraft's longitudinal axis is yawed 10° either right or left of the impact velocity vector (whichever would cause the greatest load on the shoulder harness), the rotorcraft's lateral axis is contained in a horizontal plane containing the impact velocity vector, and the rotorcraft's vertical axis is perpendicular to a horizontal plane containing the impact velocity vector. Peak floor deceleration must occur in not more than 0.071 seconds after impact and must reach a minimum of 18.4 g.
 - (3) Where floor rails or floor or sidewall attachment devices are used to attach the seating devices to the airframe structure for the conditions of this paragraph, the rails or devices must be misaligned with respect to each other by at least 10° vertically (i.e. pitch out of parallel) and by at least a 10° lateral roll, with the directions optional, to account for possible floor warp.
- (c) Compliance with the following must be shown:
 - (1) The seating device system must remain intact although it may experience separation intended as part of its design.
 - (2) The attachment between the seating device and the airframe structure must remain intact, although the structure may have exceeded its limit load.
 - (3) The ATD's shoulder harness strap or straps must remain on or in the immediate vicinity of the ATD's shoulder during the impact.
 - (4) The safety belt must remain on the ATD's pelvis during the impact.

- (5) The ATD's head either does not contact any portion of the crew or passenger compartment, or if contact is made, the head impact does not exceed a head injury criteria (HIC) of 1000 as determined by this equation.

$$HIC = (t_2 - t_1) \left[\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} a(t) dt \right]^{2.5}$$

Where: $a(t)$ is the resultant acceleration at the centre of gravity of the head form expressed as a multiple of g (the acceleration of gravity) and $t_2 - t_1$ is the time duration, in seconds, of major head impact, not to exceed 0.05 seconds.

- (6) Loads in individual upper torso harness straps must not exceed 7784 N (1750 lbs). If dual straps are used for retaining the upper torso, the total harness strap loads must not exceed 8896 N (2000 lbs).
- (7) The maximum compressive load measured between the pelvis and the lumbar column of the ATD must not exceed 6674 N (1500 lbs).
- (d) An alternate approach that achieves an equivalent or greater level of occupant protection, as required by this paragraph, must be substantiated on a rational basis.

CS 27.563 Structural ditching and emergency flotation provisions

ED Decision 2018/007/R

If certification with ditching provisions or if certification with emergency flotation provisions is requested by the applicant, structural strength must meet the requirements of this CS. If certification with ditching provisions is requested by the applicant, the requirements of [CS 27.801\(f\)](#) must also be met. The loading conditions apply to all parts of the rotorcraft, unless otherwise stated by this CS and [CS 27.802\(b\)](#).

- (a) *Landing conditions.* The conditions considered must be those resulting from an emergency landing into the most severe sea conditions for which certification is requested by the applicant, at a forward ground speed not less than 15.4 m/s (30 knots), and a vertical speed not less than 1.5 m/s (5 ft/s), in likely pitch, roll and yaw attitudes. Rotor lift may be assumed to act through the centre of gravity during water entry. This lift may not exceed two-thirds of the design maximum weight.

- (b) *Loads.*

- (1) *Floats fixed or intended to be deployed before initial water contact.* The loads to be considered are those resulting from the rotorcraft entering the water, in the conditions defined in (a), and in accordance with flight manual procedures. In addition, each float, and its support and attaching structure, must be designed for the loads developed by a fully immersed float unless it can be shown that full immersion is unlikely. If full immersion is unlikely, the highest likely float buoyancy load must be applied. Appropriate air loads shall be used in substantiation of the floats and their attachment to the rotorcraft. For this purpose, the design airspeed for limit load is the maximum operating airspeed limit with fixed or deployed floats multiplied by 1.11.

In the case of approval with ditching provisions, water entry with deployable floats in the unintended stowed position must also be accounted for. It must be established that in such a case, damage to the un-deployed floats, attachments or surrounding structure, that would prevent proper deployment and functioning of the floats, will not occur.

- (2) *Floats intended to be deployed after initial water contact.* The loads to be considered are those resulting from the rotorcraft entering the water, in the conditions defined in (a), and in accordance with flight manual procedures. In addition, each float and its support and attaching structure must be designed for combined vertical and drag loads. The vertical load must be that developed by a fully immersed float, unless it can be shown that full immersion is unlikely. If full immersion is unlikely, the highest likely float buoyancy load must be applied. The drag load must be determined assuming a relative speed of 10.3 m/s (20 knots) between the rotorcraft and the water.

[Amdt No: 27/5]

AMC 27.563 Structural ditching and emergency flotation provisions

ED Decision 2018/007/R

This AMC replaces FAA AC 27.563 and AC 27.563A.

(a) Explanation.

This AMC contains specific structural conditions to be considered to support the ditching requirements of [CS 27.801](#), and the emergency flotation requirements of [CS 27.802](#).

For rotorcraft for which certification with ditching provisions is requested by the applicant, in accordance with [CS 27.801\(a\)](#), the structural conditions apply to the complete rotorcraft.

For rotorcraft for which certification with emergency flotation provisions is requested by the applicant, in accordance with [CS 27.802\(b\)](#), the structural conditions apply only to the flotation units and their attachments to the rotorcraft.

At Amendment 5, the requirement for flotation stability on waves was appreciably changed. A requirement for the substantiation of acceptable stability by means of scale model testing in irregular waves was introduced at this amendment. This change made the usage of Sea State (World Meteorological Organization) no longer appropriate. The sea conditions are now defined in terms of significant wave height (H_s) and mean wave period (T_z). These terms are therefore also used in this AMC when defining sea conditions.

- (1) The landing conditions specified in [CS 27.563\(a\)](#) may be considered as follows:
- (i) The rotorcraft contacts the most severe sea conditions for which certification with ditching or emergency flotation provisions is requested by the applicant, selected in accordance with Table 1 of [AMC to CS 27.801\(e\) and 27.802\(c\)](#) and as illustrated in Figure 1a). These conditions may be simulated considering the rotorcraft contacting a plane of stationary water as illustrated in Figure 1b), inclined with a range of steepness from zero to the significant steepness given by $S_s = 2\pi H_s / (g T_z^2)$. Values of S_s are given in Table 1 of [AMC to 27.801\(e\) and 27.802\(c\)](#). The rotorcraft contacts the inclined plane of stationary water with a flight direction contained in a vertical plane. This vertical plane is perpendicular to the inclined plane, as illustrated in Figure 1 b). Likely rotorcraft pitch, roll and yaw attitudes at water entry that would reasonably be expected to occur in service, should also be considered. Autorotation, run-on landing, or one-engine-inoperative flight tests, or a validated simulation should be used to confirm the attitudes selected.
 - (ii) The forward ground speed should not be less than 15.4 m/s (30 kt), and the vertical speed not less than 1.5 m/s (5 ft/s).
 - (iii) A rotor lift of not more than two-thirds of the design maximum weight may be assumed to act through the rotorcraft's centre of gravity during water entry.

- (iv) The above conditions may be simulated or tested using a calm horizontal water surface with an equivalent impact angle and speed relative to the water surface as illustrated in Figure 1 c).
- (2) For floats that are fixed or intended to be deployed before water contact, [CS 27.563\(b\)\(1\)](#) defines the applicable load condition for entry into water, with the floats in their intended configuration.

[CS 27.563\(b\)\(1\)](#) also requires consideration of the following cases:

- The floats and their attachments to the rotorcraft should be designed for the loads resulting from a fully immersed float unless it is shown that full immersion is unlikely. If full immersion is shown to be unlikely, the determination of the highest likely buoyancy load should include consideration of a partially immersed float creating restoring moments to compensate for the upsetting moments caused by the side wind, unsymmetrical rotorcraft loading, water wave action, rotorcraft inertia, and probable structural damage and leakage considered under CS 27.801(e) or 27.802(c). The maximum roll and pitch angles established during compliance with CS 27.801(e) or 27.802(c) may be used, to determine the extent of immersion of each float. When determining this, damage to the rotorcraft that could be reasonably expected should be accounted for.
 - To mitigate the case when the crew is unable to, or omits to, deploy a normally stowed emergency flotation system before entering the water, if approval with ditching provisions is sought, it should be substantiated that the floats will survive and function properly. The floats in their un-deployed condition, their attachments to the rotorcraft and the local structure should be designed to withstand the water entry loads without damage that would prevent the floats inflating as intended. Risks such as the splintering of surrounding components in a way that might damage the un-deployed or deploying floats should be considered. There is, however, no requirement to assess the expected loading on other parts of the rotorcraft when entering the water, with unintended un-deployed floats.
 - The floats and their attachments to the rotorcraft should be substantiated as capable of withstanding the loads generated in flight. The airspeed chosen for assessment of the loads should be the appropriate operating limitation multiplied by 1.11. For fixed floats, the operating limitation should be the rotorcraft V_{NE} . For deployable floats, if an operating limitation for the deployment of floats and/or flight with floats deployed is given, the highest such limitation should be used, otherwise the rotorcraft V_{NE} should be used.
- (3) For floats intended to be deployed after water contact, [CS 27.563\(b\)\(2\)](#) requires the floats and their attachments to the rotorcraft to be designed to withstand the loads generated when entering the water with the floats in their intended condition.

Simultaneous vertical and drag loading on the floats and their attachments should be considered to account for the rotorcraft moving forward through the water during float deployment.

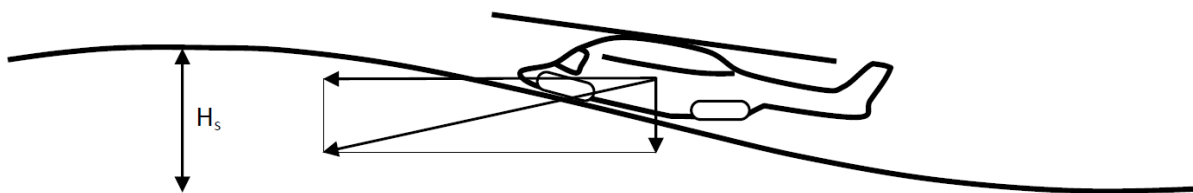
The vertical loads should be those resulting from fully immersed floats unless it is shown that full immersion is unlikely. If full immersion is shown to be unlikely, the determination of the highest likely buoyancy load should include consideration of a partially immersed float creating restoring moments to compensate for the upsetting moments caused by side wind, unsymmetrical rotorcraft loading, water wave action, rotorcraft inertia, and probable structural damage and leakage considered under CS 27.801(e) or 27.802(c). The

maximum roll and pitch angles established during compliance with CS 27.801(e) or 27.802(c) may be used, if significant, to determine the extent of immersion of each float. When determining this, damage to the rotorcraft that could be reasonably expected should be accounted for.

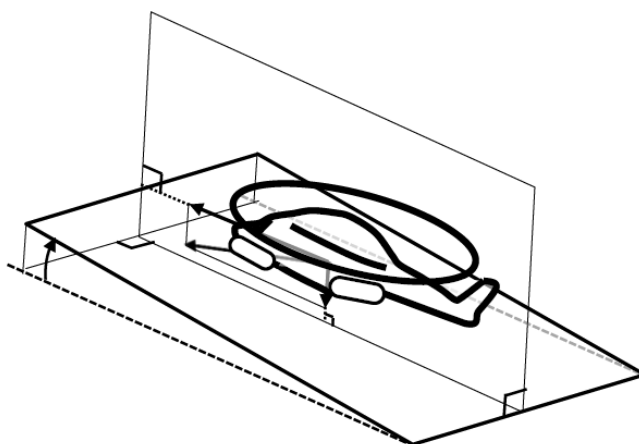
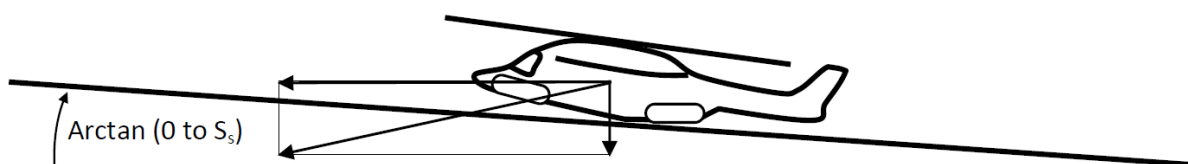
The drag loads should be those resulting from movement of the rotorcraft through the water at 10.3 m/s (20 knots).

(b) Procedures

- (1) The floats and the float attachment structure should be substantiated for rational limit and ultimate loads.
- (2) The most severe sea conditions for which certification with ditching or emergency flotation provisions is requested by the applicant are to be considered. The sea conditions should be selected in accordance with the [AMC to CS 27.801\(e\) and 27.802\(c\)](#).
- (3) Landing load factors and the water load distribution may be determined by water drop tests or validated analysis.

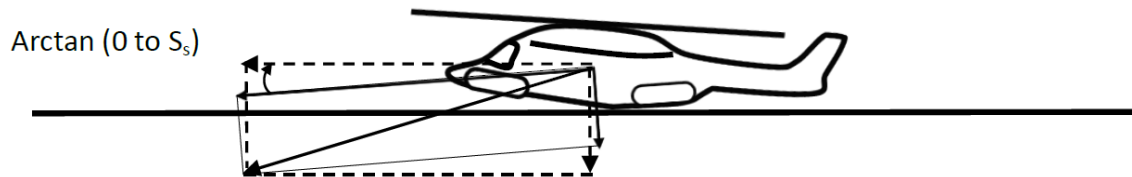


a) Water entry into wave



- b) Water entry into inclined plane of stationary water, steepness range - zero to significant steepness (S_s)

$$S_s = 2\pi H_s / (g T_z^2)$$



- c) Water entry into a stationary horizontal water surface using an equivalent water entry angle and velocity relative to the water surface

(Dashed arrows show required horizontal and vertical speeds)

Figure 1 – Illustration of water entry test or simulation conditions which may be considered for structural provisions assessment

[Amdt No: 27/5]

FATIGUE EVALUATION

CS 27.571 Fatigue evaluation of flight structure

ED Decision 2003/15/RM

- (a) *General.* Each portion of the flight structure (the flight structure includes rotors, rotor drive systems between the engines and the rotor hubs, controls, fuselage, landing gear, and their related primary attachments) the failure of which could be catastrophic, must be identified and must be evaluated under sub-paragraph (b), (c), (d), or (e). The following apply to each fatigue evaluation:
- (1) The procedure for the evaluation must be approved.
 - (2) The locations of probable failure must be determined.
 - (3) In-flight measurement must be included in determining the following:
 - (i) Loads or stresses in all critical conditions throughout the range of limitations in [CS 27.309](#), except that manoeuvring load factors need not exceed the maximum values expected in operation.
 - (ii) The effect of altitude upon these loads or stresses.
 - (4) The loading spectra must be as severe as those expected in operation including, but not limited to, external cargo operations, if applicable, and ground-air-ground cycles. The loading spectra must be based on loads or stresses determined under sub-paragraph (a)(3).
- (b) *Fatigue tolerance evaluation.* It must be shown that the fatigue tolerance of the structure ensures that the probability of catastrophic fatigue failure is extremely remote without establishing replacement times, inspection intervals or other procedures under paragraph A27.4 of [appendix A](#).
- (c) *Replacement time evaluation.* It must be shown that the probability of catastrophic fatigue failure is extremely remote within a replacement time furnished under paragraph A27.4 of [appendix A](#).
- (d) *Fail-safe evaluation.* The following apply to fail-safe evaluation:
- (1) It must be shown that all partial failures will become readily detectable under inspection procedures furnished under paragraph A27.4 of [appendix A](#).
 - (2) The interval between the time when any partial failure becomes readily detectable under sub-paragraph (d)(1), and the time when any such failure is expected to reduce the remaining strength of the structure to limit or maximum attainable loads (whichever is less), must be determined.
 - (3) It must be shown that the interval determined under sub-paragraph (d)(2) is long enough, in relation to the inspection intervals and related procedures furnished under paragraph A27.4 of [appendix A](#), to provide a probability of detection great enough to ensure that the probability of catastrophic failure is extremely remote.
- (e) *Combination of replacement time and fail-safe evaluations.* A component may be evaluated under a combination of sub-paragraphs (c) and (d). For such component it must be shown that the probability of catastrophic failure is extremely remote with an approved combination of replacement time, inspection intervals, and related procedures furnished under paragraph A27.4 of [appendix A](#).

CS 27.573 Damage Tolerance and Fatigue Evaluation of Composite Rotorcraft Structures

ED Decision 2012/021/R

- (a) Composite rotorcraft structure must be evaluated under the damage tolerance requirements of sub-paragraph (d) unless the applicant establishes that a damage tolerance evaluation is impractical within the limits of geometry, inspectability, and good design practice. In such a case, the composite rotorcraft structure must undergo a fatigue evaluation in accordance with sub-paragraph (e).
- (b) Reserved
- (c) Reserved
- (d) Damage Tolerance Evaluation:
 - (1) Damage tolerance evaluations of composite structures must show that Catastrophic Failure due to static and fatigue loads is avoided throughout the operational life or prescribed inspection intervals of the rotorcraft.
 - (2) The damage tolerance evaluation must include PSEs of the airframe, main and tail rotor drive systems, main and tail rotor blades and hubs, rotor controls, fixed and movable control surfaces, engine and transmission mountings, landing gear, and any other detail design points or parts whose failure or detachment could prevent continued safe flight and landing.
 - (3) Each damage tolerance evaluation must include:
 - (i) The identification of the structure being evaluated;
 - (ii) A determination of the structural loads or stresses for all critical conditions throughout the range of limits in [CS 27.309](#) (including altitude effects), supported by in-flight and ground measurements, except that manoeuvring load factors need not exceed the maximum values expected in service;
 - (iii) The loading spectra as severe as those expected in service based on loads or stresses determined under sub-paragraph (d)(3)(ii), including external load operations, if applicable, and other operations including high torque events;
 - (iv) A Threat Assessment for all structure being evaluated that specifies the locations, types, and sizes of damage, considering fatigue, environmental effects, intrinsic and discrete flaws, and impact or other accidental damage (including the discrete source of the accidental damage) that may occur during manufacture or operation;
 - (v) An assessment of the residual strength and fatigue characteristics of all structure being evaluated that supports the replacement times and inspection intervals established under sub-paragraph (d)(4); and
 - (vi) allowances for the detrimental effects of material, fabrication techniques, and process variability.
 - (4) Replacement times, inspections, or other procedures must be established to require the repair or replacement of damaged parts to prevent Catastrophic Failure. These replacement times, inspections, or other procedures must be included in the Airworthiness Limitations Section of the Instructions for Continued Airworthiness required by [CS 27.1529](#).

- (i) Replacement times must be determined by tests, or by analysis supported by tests to show that throughout its life the structure is able to withstand the repeated loads of variable magnitude expected in-service. In establishing these replacement times, the following items must be considered:
 - (A) Damage identified in the Threat Assessment required by sub-paragraph (d)(3)(iv);
 - (B) Maximum acceptable manufacturing defects and in-service damage (i.e., those that do not lower the residual strength below ultimate design loads and those that can be repaired to restore ultimate strength); and
 - (C) Ultimate load strength capability after applying repeated loads.
 - (ii) Inspection intervals must be established to reveal any damage identified in the Threat Assessment required by sub-paragraph (d)(3)(iv) that may occur from fatigue or other in-service causes before such damage has grown to the extent that the component cannot sustain the required residual strength capability. In establishing these inspection intervals, the following items must be considered:
 - (A) The growth rate, including no-growth, of the damage under the repeated loads expected in-service determined by tests or analysis supported by tests; and
 - (B) The required residual strength for the assumed damage established after considering the damage type, inspection interval, detectability of damage, and the techniques adopted for damage detection. The minimum required residual strength is limit load.
 - (5) The effects of damage on stiffness, dynamic behaviour, loads and functional performance must be taken into account when substantiating the maximum assumed damage size and inspection interval.
- (e) Fatigue Evaluation:
- If an applicant establishes that the damage tolerance evaluation described in sub-paragraph (d) is impractical within the limits of geometry, inspectability, or good design practice, the applicant must do a fatigue evaluation of the particular composite rotorcraft structure and:
- (1) Identify structure considered in the fatigue evaluation;
 - (2) Identify the types of damage considered in the fatigue evaluation;
 - (3) Establish supplemental procedures to minimise the risk of Catastrophic Failure associated with damage identified in sub-paragraph (e)(2); and
 - (4) Include these supplemental procedures in the Airworthiness Limitations section of the Instructions for Continued Airworthiness required by [CS 27.1529](#).

[Amdt 27/3]

SUBPART D – DESIGN AND CONSTRUCTION

GENERAL

CS 27.601 Design

ED Decision 2003/15/RM

- (a) The rotorcraft may have no design features or details that experience has shown to be hazardous or unreliable.
- (b) The suitability of each questionable design detail and part must be established by tests.

CS 27.602 Critical parts

ED Decision 2003/15/RM

- (a) Critical part - A critical part is a part, the failure of which could have a catastrophic effect upon the rotorcraft, and for which critical characteristics have been identified which must be controlled to ensure the required level of integrity.
- (b) If the type design includes critical parts, a critical parts list shall be established. Procedures shall be established to define the critical design characteristics, identify processes that affect those characteristics, and identify the design change and process change controls necessary for showing compliance with the quality assurance requirements of Part-21.

CS 27.603 Materials

ED Decision 2003/15/RM

The suitability and durability of materials used for parts, the failure of which could adversely affect safety, must:

- (a) Be established on the basis of experience or tests;
- (b) Meet approved specifications that ensure their having the strength and other properties assumed in the design data; and
- (c) Take into account the effects of environmental conditions, such as temperature and humidity, expected in service.

CS 27.605 Fabrication methods

ED Decision 2003/15/RM

- (a) The methods of fabrication used must produce consistently sound structures. If a fabrication process (such as gluing, spot welding, or heat-treating) requires close control to reach this objective, the process must be performed according to an approved process specification.
- (b) Each new aircraft fabrication method must be substantiated by a test program.

CS 27.607 Fasteners

ED Decision 2003/15/RM

- (a) Each removable bolt, screw, nut, pin, or other fastener whose loss could jeopardise the safe operation of the rotorcraft must incorporate two separate locking devices. The fastener and its locking devices may not be adversely affected by the environmental conditions associated with the particular installation.
- (b) No self-locking nut may be used on any bolt subject to rotation in operation unless a non-friction locking device is used in addition to the self-locking device.

CS 27.609 Protection of structure

ED Decision 2003/15/RM

Each part of the structure must:

- (a) Be suitably protected against deterioration or loss of strength in service due to any cause, including:
 - (1) Weathering;
 - (2) Corrosion; and
 - (3) Abrasion; and
- (b) Have provisions for ventilation and drainage where necessary to prevent the accumulation of corrosive, flammable, or noxious fluids.

CS 27.610 Lightning and static electricity protection

ED Decision 2016/024/R

- (a) The rotorcraft must be protected against catastrophic effects from lightning.
- (b) For metallic components, compliance with sub-paragraph (a) may be shown by:
 - (1) Electrically bonding the components properly to the airframe; or
 - (2) Designing the components so that a strike will not endanger the rotorcraft.
- (c) For non-metallic components, compliance with sub-paragraph (a) may be shown by:
 - (1) Designing the components to minimise the effect of a strike; or
 - (2) Incorporating acceptable means of diverting the resulting electrical current so as not to endanger the rotorcraft.
- (d) The electrical bonding and protection against lightning and static electricity must:
 - (1) Minimise the accumulation of electrostatic charge;
 - (2) Minimise the risk of electric shock to crew, passengers, and service and maintenance personnel using normal precautions;
 - (3) Provide an electrical return path, under both normal and fault conditions, on rotorcraft having grounded electrical systems; and
 - (4) Reduce to an acceptable level the effects of static electricity on the functioning of essential electrical and electronic equipment.

[Amdt 27/4]

CS 27.611 Inspection provisions

ED Decision 2003/15/RM

There must be means to allow the close examination of each part that requires:

- (a) Recurring inspection;
- (b) Adjustment for proper alignment and functioning; or
- (c) Lubrication.

CS 27.613 Material strength properties and design values

ED Decision 2003/15/RM

- (a) Material strength properties must be based on enough tests of material meeting specifications to establish design values on a statistical basis.
- (b) Design values must be chosen to minimise the probability of structural failure due to material variability. Except as provided in sub-paragraphs (d) and (e), compliance with this paragraph must be shown by selecting design values that assure material strength with the following probability:
 - (1) Where applied loads are eventually distributed through a single member within an assembly, the failure of which would result in loss of structural integrity of the component, 99% probability with 95% confidence; and
 - (2) For redundant structure, those in which the failure of individual elements would result in applied loads being safely distributed to other load carrying members, 90% probability with 95% confidence.
- (c) The strength, detail design, and fabrication of the structure must minimise the probability of disastrous fatigue failure, particularly at points of stress concentration.
- (d) Material specifications must be those contained in documents accepted by the Agency.
- (e) Other design values may be used if a selection of the material is made in which a specimen of each individual item is tested before use and it is determined that the actual strength properties of that particular item will equal or exceed those used in design.

CS 27.619 Special factors

ED Decision 2003/15/RM

- (a) The special factors prescribed in [CS 27.621](#) to [27.625](#) apply to each part of the structure whose strength is:
 - (1) Uncertain;
 - (2) Likely to deteriorate in service before normal replacement; or
 - (3) Subject to appreciable variability due to:
 - (i) Uncertainties in manufacturing processes; or
 - (ii) Uncertainties in inspection methods.
- (b) For each part to which [CS 27.621](#) to [27.625](#) apply, the factor of safety prescribed in [CS 27.303](#) must be multiplied by a special factor equal to:
 - (1) The applicable special factors prescribed in [CS 27.621](#) to [27.625](#); or

- (2) Any other factor great enough to ensure that the probability of the part being understrength because of the uncertainties specified in sub-paragraph (a) is extremely remote.

CS 27.621 Casting factors

ED Decision 2003/15/RM

- (a) *General.* The factors, tests, and inspections specified in sub-paragraphs (b) and (c) must be applied in addition to those necessary to establish foundry quality control. The inspections must meet approved specifications. Sub-paragraphs (c) and (d) apply to structural castings except castings that are pressure tested as parts of hydraulic or other fluid systems and do not support structural loads.
- (b) *Bearing stresses and surfaces.* The casting factors specified in sub-paragraphs (c) and (d):
- (1) Need not exceed 1.25 with respect to bearing stresses regardless of the method of inspection used; and
 - (2) Need not be used with respect to the bearing surfaces of a part whose bearing factor is larger than the applicable casting factor.
- (c) *Critical castings.* For each casting whose failure would preclude continued safe flight and landing of the rotorcraft or result in serious injury to any occupant, the following apply:
- (1) Each critical casting must –
 - (i) Have a casting factor of not less than 1.25; and
 - (ii) Receive 100% inspection by visual, radiographic, and magnetic particle (for ferromagnetic materials) or penetrant (for non-ferromagnetic materials) inspection methods or approved equivalent inspection methods.
 - (2) For each critical casting with a casting factor less than 1.50, three sample castings must be static tested and shown to meet –
 - (i) The strength requirements of [CS 27.305](#) at an ultimate load corresponding to a casting factor of 1.25; and
 - (ii) The deformation requirements of [CS 27.305](#) at a load of 1.15 times the limit load.
- (d) *Non-critical castings.* For each casting other than those specified in sub-paragraph (c), the following apply:
- (1) Except as provided in sub-paragraphs (d)(2) and (3), the casting factors and corresponding inspections must meet the following table:

Casting factor	Inspection
2.0 or greater.....	100% visual
Less than 2.0 greater than 1.5	100% visual and magnetic particle (ferromagnetic materials), penetrant (non-ferromagnetic materials), or approved equivalent inspection methods.
1.25 through 1.50.....	100% visual, and magnetic particle (ferromagnetic materials), penetrant non-ferromagnetic materials), and radiographic or approved equivalent inspection methods

- (2) The percentage of castings inspected by nonvisual methods may be reduced below that specified in sub-paragraph (d)(1) when an approved quality control procedure is established.
- (3) For castings procured to a specification that guarantees the mechanical properties of the material in the casting and provides for demonstration of these properties by test of coupons cut from the castings on a sampling basis:
 - (i) A casting factor of 1.0 may be used; and
 - (ii) The castings must be inspected as provided in sub-paragraph (d)(1) for casting factors of 1.25 to 1.50 and tested under sub-paragraph (c)(2).

CS 27.623 Bearing factors

ED Decision 2003/15/RM

- (a) Except as provided in sub-paragraph (b), each part that has clearance (free fit), and that is subject to pounding or vibration, must have a bearing factor large enough to provide for the effects of normal relative motion.
- (b) No bearing factor need be used on a part for which any larger special factor is prescribed.

CS 27.625 Fitting factors

ED Decision 2003/15/RM

For each fitting (part or terminal used to join one structural member to another) the following apply:

- (a) For each fitting whose strength is not proven by limit and ultimate load tests in which actual stress conditions are simulated in the fitting and surrounding structures, a fitting factor of at least 1.15 must be applied to each part of:
 - (1) The fitting;
 - (2) The means of attachment; and
 - (3) The bearing on the joined members.
- (b) No fitting factor need be used:
 - (1) For joints made under approved practices and based on comprehensive test data (such as continuous joints in metal plating, welded joints, and scarf joints in wood); and
 - (2) With respect to any bearing surface for which a larger special factor is used.
- (c) For each integral fitting, the part must be treated as a fitting up to the point at which the paragraph properties become typical of the member.
- (d) Each seat, berth, litter, safety belt, and harness attachment to the structure must be shown by analysis, tests, or both, to be able to withstand the inertia forces prescribed in [CS 27.561\(b\)\(3\)](#) multiplied by a fitting factor of 1.33.

CS 27.629 Flutter

ED Decision 2003/15/RM

Each aerodynamic surface of the rotorcraft must be free from flutter under each appropriate speed and power condition.

ROTORS

CS 27.653 Pressure venting and drainage of rotor blades

ED Decision 2003/15/RM

- (a) For each rotor blade:
 - (1) There must be means for venting the internal pressure of the blade;
 - (2) Drainage holes must be provided for the blade; and
 - (3) The blade must be designed to prevent water from becoming trapped in it.
- (b) Sub-paragraphs (a)(1) and (2) do not apply to sealed rotor blades capable of withstanding the maximum pressure differentials expected in service.

CS 27.659 Mass balance

ED Decision 2003/15/RM

- (a) The rotors and blades must be mass balanced as necessary to –
 - (1) Prevent excessive vibration; and
 - (2) Prevent flutter at any speed up to the maximum forward speed.
- (b) The structural integrity of the mass balance installation must be substantiated.

CS 27.661 Rotor blade clearance

ED Decision 2003/15/RM

There must be enough clearance between the rotor blades and other parts of the structure to prevent the blades from striking any part of the structure during any operating condition.

CS 27.663 Ground resonance prevention means

ED Decision 2003/15/RM

- (a) The reliability of the means for preventing ground resonance must be shown either by analysis and tests, or reliable service experience, or by showing through analysis or tests that malfunction or failure of a single means will not cause ground resonance.
- (b) The probable range of variations, during service, of the damping action of the ground resonance prevention means must be established and must be investigated during the test required by [CS 27.241](#).

CONTROL SYSTEMS

CS 27.671 General

ED Decision 2003/15/RM

- (a) Each control and control system must operate with the ease, smoothness, and positiveness appropriate to its function.
- (b) Each element of each flight control system must be designed, or distinctively and permanently marked, to minimise the probability of any incorrect assembly that could result in the malfunction of the system.

CS 27.672 Stability augmentation, automatic, and power-operated systems

ED Decision 2003/15/RM

If the functioning of stability augmentation or other automatic or power-operated systems is necessary to show compliance with the flight characteristics requirements of this CS-27, such systems must comply with [CS 27.671](#) and the following:

- (a) A warning which is clearly distinguishable to the pilot under expected flight conditions without requiring the pilot's attention must be provided for any failure in the stability augmentation system or in any other automatic or power-operated system which could result in an unsafe condition if the pilot is unaware of the failure. Warning systems must not activate the control systems.
- (b) The design of the stability augmentation system or of any other automatic or power-operated system must allow initial counteraction of failures without requiring exceptional pilot skill or strength by overriding the failure by movement of the flight controls in the normal sense and deactivating the failed system.
- (c) It must be shown that after any single failure of the stability augmentation system or any other automatic or power-operated system:
 - (1) The rotorcraft is safely controllable when the failure or malfunction occurs at any speed or altitude within the approved operating limitations;
 - (2) The controllability and manoeuvrability requirements of this CS-27 are met within a practical operational flight envelope (for example, speed, altitude, normal acceleration, and rotorcraft configurations) which is described in the Rotorcraft Flight Manual; and
 - (3) The trim and stability characteristics are not impaired below a level needed to permit continued safe flight and landing.

CS 27.673 Primary flight control

ED Decision 2003/15/RM

Primary flight controls are those used by the pilot for immediate control of pitch, roll, yaw, and vertical motion of the rotorcraft.

CS 27.674 Interconnected controls

ED Decision 2003/15/RM

Each primary flight control system must provide for safe flight and landing and operate independently after a malfunction, failure, or jam of any auxiliary interconnected control.

CS 27.675 Stops

ED Decision 2003/15/RM

- (a) Each control system must have stops that positively limit the range of motion of the pilot's controls.
- (b) Each stop must be located in the system so that the range of travel of its control is not appreciably affected by:
 - (1) Wear;
 - (2) Slackness; or
 - (3) Take-up adjustments.
- (c) Each stop must be able to withstand the loads corresponding to the design conditions for the system.
- (d) For each main rotor blade:
 - (1) Stops that are appropriate to the blade design must be provided to limit travel of the blade about its hinge points; and
 - (2) There must be means to keep the blade from hitting the droop stops during any operation other than starting and stopping the rotor.

CS 27.679 Control system locks

ED Decision 2003/15/RM

If there is a device to lock the control system with the rotorcraft on the ground or water, there must be means to:

- (a) Give unmistakable warning to the pilot when the lock is engaged; and
- (b) Prevent the lock from engaging in flight.

CS 27.681 Limit load static tests

ED Decision 2003/15/RM

- (a) Compliance with the limit load requirements of this CS-27 must be shown by tests in which:
 - (1) The direction of the test loads produces the most severe loading in the control system; and
 - (2) Each fitting, pulley, and bracket used in attaching the system to the main structure is included.
- (b) Compliance must be shown (by analyses or individual load tests) with the special factor requirements for control system joints subject to angular motion.

CS 27.683 Operation tests

ED Decision 2003/15/RM

It must be shown by operation tests that, when the controls are operated from the pilot compartment with the control system loaded to correspond with loads specified for the system, the system is free from:

- (a) Jamming;
- (b) Excessive friction; and
- (c) Excessive deflection.

CS 27.685 Control system details

ED Decision 2003/15/RM

- (a) Each detail of each control system must be designed to prevent jamming, chafing, and interference from cargo, passengers, loose objects or the freezing of moisture.
- (b) There must be means in the cockpit to prevent the entry of foreign objects into places where they would jam the system.
- (c) There must be means to prevent the slapping of cables or tubes against other parts.
- (d) Cable systems must be designed as follows:
 - (1) Cables, cable fittings, turnbuckles, splices and pulleys must be of an acceptable kind.
 - (2) The design of the cable systems must prevent any hazardous change in cable tension throughout the range of travel under any operating conditions and temperature variations.
 - (3) No cable smaller than 2.4 mm (3/32 inch) diameter may be used in any primary control system.
 - (4) Pulley kinds and sizes must correspond to the cables with which they are used.
 - (5) Pulleys must have close fitting guards to prevent the cables from being displaced or fouled.
 - (6) Pulleys must lie close enough to the plane passing through the cable to prevent the cable from rubbing against the pulley flange.
 - (7) No fairlead may cause a change in cable direction of more than 3°.
 - (8) No clevis pin subject to load or motion and retained only by cotter pins may be used in the control system.
 - (9) Turnbuckles attached to parts having angular motion must be installed to prevent binding throughout the range of travel.
 - (10) There must be means for visual inspection at each fairlead, pulley, terminal and turnbuckle.
- (e) Control system joints subject to angular motion must incorporate the following special factors with respect to the ultimate bearing strength of the softest material used as a bearing:
 - (1) 3.33 for push-pull systems other than ball and roller bearing systems.
 - (2) 2.0 for cable systems.

- (f) For control system joints, the manufacturer's static, non-Brinell rating of ball and roller bearings must not be exceeded.

CS 27.687 Spring devices

ED Decision 2003/15/RM

- (a) Each control system spring device where failure could cause flutter or other unsafe characteristics must be reliable.
- (b) Compliance with sub-paragraph (a) must be shown by tests simulating service conditions.

CS 27.691 Autorotation control mechanism

ED Decision 2003/15/RM

Each main rotor blade pitch control mechanism must allow rapid entry into autorotation after power failure.

CS 27.695 Power boost and power-operated control system

ED Decision 2003/15/RM

- (a) If a power boost or power-operated control system is used, an alternate system must be immediately available that allows continued safe flight and landing in the event of:
- (1) Any single failure in the power portion of the system; or
 - (2) The failure of all engines.
- (b) Each alternate system may be a duplicate power portion or a manually operated mechanical system. The power portion includes the power source (such as hydraulic pumps), and such items as valves, lines, and actuators.
- (c) The failure of mechanical parts (such as piston rods and links), and the jamming of power cylinders, must be considered unless they are extremely improbable.

LANDING GEAR

CS 27.723 Shock absorption tests

ED Decision 2003/15/RM

The landing inertia load factor and the reserve energy absorption capacity of the landing gear must be substantiated by the tests prescribed in [CS 27.725](#) and [27.727](#), respectively. These tests must be conducted on the complete rotorcraft or on units consisting of wheel, tyre, and shock absorber in their proper relation.

CS 27.725 Limit drop test

ED Decision 2003/15/RM

The limit drop test must be conducted as follows:

- (a) The drop height must be –
 - (1) 0.33 m (13 inches) from the lowest point of the landing gear to the ground; or
 - (2) Any lesser height, not less than 0.20 m (8 in), resulting in a drop contact velocity equal to the greatest probable sinking speed likely to occur at ground contact in normal power-off landings.
- (b) If considered, the rotor lift specified in [CS 27.473\(a\)](#) must be introduced into the drop test by appropriate energy absorbing devices or by the use of an effective mass.
- (c) Each landing gear unit must be tested in the attitude simulating the landing condition that is most critical from the standpoint of the energy to be absorbed by it.
- (d) When an effective mass is used in showing compliance with sub-paragraph (b) the following formula may be used instead of more rational computations:

$$W_e = W \frac{h + (1 - L)d}{h + d} : \text{and } n = n_j \frac{W_e}{W} + L$$

where:

W_e = the effective weight to be used in the drop test.

$W = W_M$ for main gear units, equal to the static reaction on the particular unit with the rotorcraft in the most critical attitude. A rational method may be used in computing a main gear static reaction, taking into consideration the moment arm between the main wheel reaction and the rotorcraft centre of gravity.

$W = W_N$ for nose gear units, equal to the vertical component of the static reaction that would exist at the nose wheel, assuming that the mass of the rotorcraft acts at the centre of gravity and exerts a force of 1.0 g downward and 0.25 g forward.

$W = W_T$ for tailwheel units equal to whichever of the following is critical:

- (1) The static weight on the tailwheel with the rotorcraft resting on all wheels; or
- (2) The vertical component of the ground reaction that would occur at the tailwheel, assuming that the mass of the rotorcraft acts at the centre of gravity and exerts a force of 1 g downward with the rotorcraft in the maximum nose-up attitude considered in the nose-up landing conditions.

- h = specified free drop height.
- L = ratio of assumed rotor lift to the rotorcraft weight.
- d = deflection under impact of the tyre (at the proper inflation pressure) plus the vertical component of the axle travel relative to the drop mass.
- n = limit inertia load factor.
- n_j = the load factor developed, during impact, on the mass used in the drop test (i.e., the acceleration dv/dt in g recorded in the drop test plus 1.0).

CS 27.727 Reserve energy absorption drop test

ED Decision 2003/15/RM

The reserve energy absorption drop test must be conducted as follows:

- (a) The drop height must be 1.5 times that specified in [CS 27.725\(a\)](#).
- (b) Rotor lift, where considered in a manner similar to that prescribed in [CS 27.725\(b\)](#), may not exceed 1.5 times the lift allowed under that paragraph.
- (c) The landing gear must withstand this test without collapsing. Collapse of the landing gear occurs when a member of the nose, tail, or main gear will not support the rotorcraft in the proper attitude or allows the rotorcraft structure, other than the landing gear and external accessories, to impact the landing surface.

CS 27.729 Retracting mechanism

ED Decision 2003/15/RM

For rotorcraft with retractable landing gear, the following apply:

- (a) *Loads.* The landing gear, retracting mechanism, wheel-well doors, and supporting structure must be designed for –
 - (1) The loads occurring in any manoeuvring condition with the gear retracted;
 - (2) The combined friction, inertia, and air loads occurring during retraction and extension at any airspeed up to the design maximum landing gear operating speed; and
 - (3) The flight loads, including those in yawed flight, occurring with the gear extended at any airspeed up to the design maximum landing gear extended speed.
- (b) *Landing gear lock.* A positive means must be provided to keep the gear extended.
- (c) *Emergency operation.* When other than manual power is used to operate the gear, emergency means must be provided for extending the gear in the event of –
 - (1) Any reasonably probable failure in the normal retraction system; or
 - (2) The failure of any single source of hydraulic, electric, or equivalent energy.
- (d) *Operation tests.* The proper functioning of the retracting mechanism must be shown by operation tests.
- (e) *Position indicator.* There must be a means to indicate to the pilot when the gear is secured in the extreme positions.
- (f) *Control.* The location and the operation of the retraction control must meet the requirements of [CS 27.777](#) and [27.779](#).

- (g) **Landing gear warning.** An aural or equally effective landing gear warning device must be provided that functions continuously when the rotorcraft is in a normal landing mode and the landing gear is not fully extended and locked. A manual shut-off capability must be provided for the warning device and the warning system must automatically reset when the rotorcraft is no longer in the landing mode.

CS 27.731 Wheels

ED Decision 2003/15/RM

- (a) Each landing gear wheel must be approved.
- (b) The maximum static load rating of each wheel may not be less than the corresponding static ground reaction with:
- (1) Maximum weight; and
 - (2) Critical centre of gravity.
- (c) The maximum limit load rating of each wheel must equal or exceed the maximum radial limit load determined under the applicable ground load requirements of this CS-27.

CS 27.733 Tyres

ED Decision 2003/15/RM

- (a) Each landing gear wheel must have a tyre:
- (1) That is a proper fit on the rim of the wheel; and
 - (2) Of the proper rating.
- (b) The maximum static load rating of each tyre must equal or exceed the static ground reaction obtained at its wheel, assuming:
- (1) The design maximum weight; and
 - (2) The most unfavourable centre of gravity.
- (c) Each tyre installed on a retractable landing gear system must, at the maximum size of the tyre type expected in service, have a clearance to surrounding structure and systems that is adequate to prevent contact between the tyre and any part of the structure or systems.

CS 27.735 Brakes

ED Decision 2003/15/RM

For rotorcraft with wheel-type landing gear, a braking device must be installed that is:

- (a) Controllable by the pilot;
- (b) Usable during power-off landings; and
- (c) Adequate to:
- (1) Counteract any normal unbalanced torque when starting or stopping the rotor; and
 - (2) Hold the rotorcraft parked on a 10° slope on a dry, smooth pavement.

CS 27.737 Skis

ED Decision 2003/15/RM

The maximum limit load rating of each ski must equal or exceed the maximum limit load determined under the applicable ground load requirements of this CS-27.

FLOATS AND HULLS

CS 27.751 Main float buoyancy

ED Decision 2003/15/RM

- (a) For main floats, the buoyancy necessary to support the maximum weight of the rotorcraft in fresh water must be exceeded by:
 - (1) 50%, for single floats; and
 - (2) 60%, for multiple floats.
- (b) Each main float must have enough watertight compartments so that, with any single main float compartment flooded, the main floats will provide a margin of positive stability great enough to minimise the probability of capsizing.

CS 27.753 Main float design

ED Decision 2003/15/RM

- (a) *Bag floats*. Each bag float must be designed to withstand:
 - (1) The maximum pressure differential that might be developed at the maximum altitude for which certification with that float is requested; and
 - (2) The vertical loads prescribed in [CS 27.521\(a\)](#), distributed along the length of the bag over three-quarters of its projected area.
- (b) *Rigid floats*. Each rigid float must be able to withstand the vertical, horizontal, and side loads prescribed in [CS 27.521](#). These loads may be distributed along the length of the float.

CS 27.755 Hulls

ED Decision 2003/15/RM

For each rotorcraft, with a hull and auxiliary floats, that is to be approved for both taking off from and landing on water, the hull and auxiliary floats must have enough watertight compartments so that, with any single compartment flooded, the buoyancy of the hull and auxiliary floats (and wheel tyres if used) provides a margin of positive stability great enough to minimise the probability of capsizing.

PERSONNEL AND CARGO ACCOMMODATIONS

CS 27.771 Pilot compartment

ED Decision 2003/15/RM

For each pilot compartment:

- (a) The compartment and its equipment must allow each pilot to perform his duties without unreasonable concentration or fatigue;
- (b) If there is provision for a second pilot, the rotorcraft must be controllable with equal safety from either pilot seat; and
- (c) The vibration and noise characteristics of cockpit appurtenances may not interfere with safe operation.

CS 27.773 Pilot compartment view

ED Decision 2003/15/RM

- (a) Each pilot compartment must be free from glare and reflections that could interfere with the pilot's view, and designed so that:
 - (1) Each pilot's view is sufficiently extensive, clear, and undistorted for safe operation; and
 - (2) Each pilot is protected from the elements so that moderate rain conditions do not unduly impair his view of the flight path in normal flight and while landing.
- (b) If certification for night operation is requested, compliance with sub-paragraph (a) must be shown in night flight tests.

CS 27.775 Windshields and windows

ED Decision 2003/15/RM

Windshields and windows must be made of material that will not break into dangerous fragments.

CS 27.777 Cockpit controls

ED Decision 2003/15/RM

Cockpit controls must be:

- (a) Located to provide convenient operation and to prevent confusion and inadvertent operation; and
- (b) Located and arranged with respect to the pilots' seats so that there is full and unrestricted movement of each control without interference from the cockpit structure or the pilot's clothing when pilots from 1.57 m (5 ft 2 inches) to 1.83 m (6 ft) in height are seated.

CS 27.779 Motion and effect of cockpit controls

ED Decision 2003/15/RM

Cockpit controls must be designed so that they operate in accordance with the following movements and actuation:

- (a) Flight controls, including the collective pitch control, must operate with a sense of motion which corresponds to the effect on the rotorcraft.
- (b) Twist-grip engine power controls must be designed so that, for left-hand operation, the motion of the pilot's hand is clockwise to increase power when the hand is viewed from the edge containing the index finger. Other engine power controls, excluding the collective control, must operate with a forward motion to increase power.
- (c) Normal landing gear controls must operate downward to extend the landing gear.

CS 27.783 Doors

ED Decision 2018/007/R

- (a) Each closed cabin must have at least one adequate and easily accessible external door.
- (b) Each external door must be located where persons using it will not be endangered by the rotors, propellers, engine intakes and exhausts when appropriate operating procedures are used. If opening procedures are required, they must be the marked inside, on or adjacent to the door opening device.
- (c) If certification with ditching provisions is requested by the applicant, any non-jettisonable doors intended for use after a ditching must have means to enable them to be secured in the open position and remain secure for emergency egress in all sea conditions for which ditching capability is requested by the applicant.

[Amdt No: 27/5]

AMC 27.783 Doors

ED Decision 2018/007/R

This AMC provides further guidance and acceptable means of compliance to supplement FAA AC 27-1B AC 27.783 § 27.783 to meet EASA's interpretation of [CS 27.783](#). As such it should be used in conjunction with the FAA AC but take precedence over it, where stipulated, in the showing of compliance.

Specifically, this AMC addresses one area where the FAA AC has been deemed by EASA as being at variance to EASA's interpretation. This area is as follows:

- (a) Explanation
 - [...]
 - (4) Any means of egress (door, hatch, openable window) intended for use following ditching need not have a threshold above the waterline of the rotorcraft in calm water. However, the usability of the egress means should be substantiated in all sea conditions up to and including those chosen for showing compliance with [CS 27.801\(e\)](#) or [27.802\(c\)](#) as appropriate. See also [AMC 27.801](#) paragraph (b)(10) and [AMC 27.802](#) paragraph (b)(7).

[Amdt No: 27/5]

CS 27.785 Seats, berths, safety belts, and harnesses

ED Decision 2003/15/RM

- (a) Each seat, safety belt, harness, and adjacent part of the rotorcraft at each station designated for occupancy during take-off and landing must be free of potentially injurious objects, sharp edges, protuberances, and hard surfaces and must be designed so that a person making proper use of these facilities will not suffer serious injury in an emergency landing as a result of the static inertial load factors specified in [CS 27.561\(b\)](#) and dynamic conditions specified in [CS 27.562](#).
- (b) Each occupant must be protected from serious head injury by a safety belt plus a shoulder harness that will prevent the head from contacting any injurious object except as provided for in [CS 27.562\(c\)\(5\)](#). A shoulder harness (upper torso restraint), in combination with the safety belt, constitutes a torso restraint system as described in ETSO-C114.
- (c) Each occupant's seat must have a combined safety belt and shoulder harness with a single-point release. Each pilot's combined safety belt and shoulder harness must allow each pilot when seated with safety belt and shoulder harness fastened, to perform all functions necessary for flight operations. There must be a means to secure belts and harnesses when not in use, to prevent interference with the operation of the rotorcraft and with rapid egress in an emergency.
- (d) If seat backs do not have a firm handhold, there must be hand grips or rails along each aisle to enable the occupants to steady themselves while using the aisle in moderately rough air.
- (e) Each projecting object that could injure persons seated or moving about in the rotorcraft in normal flight must be padded.
- (f) Each seat and its supporting structure must be designed for an occupant weight of at least 77 kg (170 lbs) considering the maximum load factors, inertial forces, and reactions between the occupant, seat, and safety belt or harness corresponding with the applicable flight and ground-load conditions, including the emergency landing conditions of [CS 27.561\(b\)](#). In addition:
 - (1) Each pilot seat must be designed for the reactions resulting from the application of the pilot forces prescribed in [CS 27.397](#); and
 - (2) The inertial forces prescribed in [CS 27.561\(b\)](#) must be multiplied by a factor of 1.33 in determining the strength of the attachment of:
 - (i) Each seat to the structure; and
 - (ii) Each safety belt or harness to the seat or structure.
- (g) When the safety belt and shoulder harness are combined, the rated strength of the safety belt and shoulder harness may not be less than that corresponding to the inertial forces specified in [CS 27.561\(b\)](#), considering the occupant weight of at least 77 kg (170 lbs), considering the dimensional characteristics of the restraint system installation, and using a distribution of at least a 60% load to the safety belt and at least a 40% load to the shoulder harness. If the safety belt is capable of being used without the shoulder harness, the inertial forces specified must be met by the safety belt alone.
- (h) When a headrest is used, the headrest and its supporting structure must be designed to resist the inertia forces specified in [CS 27.561](#), with a 1.33 fitting factor and a head weight of at least 5.9 kg (13 lbs).

- (i) Each seating device system includes the device such as the seat, the cushions, the occupant restraint system, and attachment devices.
- (j) Each seating device system may use design features such as crushing or separation of certain parts of the seats to reduce occupant loads for the emergency landing dynamic conditions of [CS 27.562](#); otherwise, the system must remain intact and must not interfere with rapid evacuation of the rotorcraft.
- (k) For the purposes of this paragraph, a litter is defined as a device designed to carry a non-ambulatory person, primarily in a recumbent position, into and on the rotorcraft. Each berth or litter must be designed to withstand the load reaction of an occupant weight of at least 77 kg (170 lbs) when the occupant is subjected to the forward inertial factors specified in [CS 27.561\(b\)](#). A berth or litter installed within 15° or less of the longitudinal axis of the rotorcraft must be provided with a padded end-board, cloth diaphragm, or equivalent means that can withstand the forward load reaction. A berth or litter oriented greater than 15° with the longitudinal axis of the rotorcraft must be equipped with appropriate restraints, such as straps or safety belts, to withstand the forward load reaction. In addition –
 - (1) The berth or litter must have a restraint system and must not have corners or other protuberances likely to cause serious injury to a person occupying it during emergency landing conditions; and
 - (2) The berth or litter attachment and the occupant restraint system attachments to the structure must be designed to withstand the critical loads resulting from flight and ground load conditions and from the conditions prescribed in [CS 27.561\(b\)](#). The fitting factor required by [CS 27.625\(d\)](#) shall be applied.

CS 27.787 Cargo and baggage compartments

ED Decision 2003/15/RM

- (a) Each cargo and baggage compartment must be designed for its placarded maximum weight of contents and for the critical load distributions at the appropriate maximum load factors corresponding to the specified flight and ground load conditions, except the emergency landing conditions of [CS 27.561](#).
- (b) There must be means to prevent the contents of any compartment from becoming a hazard by shifting under the loads specified in sub-paragraph (a).
- (c) Under the emergency landing conditions of [CS 27.561](#), cargo and baggage compartments must:
 - (1) Be positioned so that if the contents break loose they are unlikely to cause injury to the occupants or restrict any of the escape facilities provided for use after an emergency landing; or
 - (2) Have sufficient strength to withstand the conditions specified in [CS 27.561](#) including the means of restraint, and their attachments, required by sub-paragraph (b). Sufficient strength must be provided for the maximum authorised weight of cargo and baggage at the critical loading distribution.
- (d) If cargo compartment lamps are installed, each lamp must be installed so as to prevent contact between lamp bulb and cargo.

CS 27.801 Ditching

ED Decision 2018/007/R

- (a) If certification with ditching provisions is requested by the applicant, the rotorcraft must meet the requirements of this CS and [CS 27.563](#), [CS 27.783\(c\)](#), [CS 27.805\(c\)](#), [CS 27.807\(d\)](#), [CS 27.1411](#), [CS 27.1415](#), [CS 27.1470](#), [CS 27.1555\(d\)](#) and [CS 27.1561](#).
- (b) Each practicable design measure, compatible with the general characteristics of the rotorcraft, must be taken to minimise the probability that when ditching, the behaviour of the rotorcraft would cause immediate injury to the occupants or would make it impossible for them to escape.
- (c) An emergency flotation system that is stowed in a deflated condition during normal flight must:
 - (1) be designed such that the effects of a water impact (i.e. crash) on the emergency flotation system are minimised;
 - (2) have a means of automatic deployment following water entry.
- (d) The probable behaviour of the rotorcraft during ditching water entry must be shown to exhibit no unsafe characteristics.
- (e) The rotorcraft must be shown to resist capsize in the sea conditions selected by the applicant. The probability of capsizing in a 5-minute exposure to the sea conditions must be substantiated to be less than or equal to 3.0 % with a fully serviceable emergency flotation system and 30.0 % with the critical float compartment failed, with 95 % confidence.

Allowances must be made for probable structural damage and leakage.
- (f) Unless the effects of the collapse of external doors and windows are accounted for in the investigation of the probable behaviour of the rotorcraft during ditching (as prescribed in (d) and (e)), the external doors and windows must be designed to withstand the probable maximum local pressures.

[Amdt No: 27/5]

AMC 27.801 Ditching

ED Decision 2018/007/R

This AMC replaces FAA AC 27.801.

- (a) Definitions
 - (1) Ditching: a controlled emergency landing on water, deliberately executed in accordance with rotorcraft flight manual (RFM) procedures, with the intent of abandoning the rotorcraft as soon as practicable.
 - (2) Emergency flotation system (EFS): a system of floats and any associated parts (e.g. gas cylinders, means of deployment, pipework and electrical connections) that is designed and installed on a rotorcraft to provide buoyancy and flotation stability in a ditching.
- (b) Explanation
 - (1) Ditching certification is performed only if requested by the applicant.
 - (2) For a rotorcraft to be certified for ditching, in addition to the other applicable requirements of CS-27, the rotorcraft must specifically satisfy [CS 27.801](#) together with the requirements referenced in [CS 27.801\(a\)](#).

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- (3) Ditching certification encompasses four primary areas of concern: rotorcraft water entry and flotation stability (including loads and flotation system design), occupant egress, and occupant survival. CS-27 Amendment 5 has developed enhanced standards in all of these areas.
 - (4) The scope of the ditching requirements is expanded at Amendment 5 through a change in the ditching definition. All potential failure conditions that could result in a controlled ‘land immediately’ action by the pilot are now included. This primarily relates to changes in water entry conditions. While the limiting conditions for water entry have been retained (15.4 m/s (30 kt), 1.5 m/s (5 ft/s)), the alleviation that previously allowed less than 15.4 m/s (30 kt) forward speed to be used as the maximum applicable value has been removed (also from [CS 27.563](#)).
 - (5) Flotation stability is enhanced through the introduction of a new standard based on a probabilistic approach to capsizes.
 - (6) Failure of the EFS to operate when required will lead to the rotorcraft rapidly capsizing and sinking. Operational experience has shown that localised damage or failure of a single component of an EFS, or the failure of the flight crew to activate or deploy the EFS, can lead to the loss of the complete system. Therefore, the design of the EFS needs careful consideration; automatic deployment has been shown to be practicable and to offer a significant safety benefit.
 - (7) The sea conditions, on which certification with ditching provisions is to be based, are selected by the applicant and should take into account the expected sea conditions in the intended areas of operation. The wave climate of the northern North Sea is adopted as the default wave climate as it represents a conservative condition. The applicant may select alternative/additional sea areas, with any associated certification then being limited to those geographical regions. The significant wave height, and any geographical limitations (if applicable – see the [AMC to CS 27.801\(e\) and 27.802\(c\)](#)) should be included in the RFM as performance information.
 - (8) During scale model testing, appropriate allowances should be made for probable structural damage and leakage. Previous model tests and other data from rotorcraft of similar configurations that have already been substantiated, based on equivalent test conditions, may be used to satisfy the ditching requirements. In regard to flotation stability, the test conditions should be equivalent to those defined in the [AMC to CS 27.801\(e\) and 27.802\(c\)](#).
 - (9) [CS 27.801](#) requires that after ditching in sea conditions for which certification with ditching provisions is requested by the applicant, the probability of capsizing in a 5 minute exposure is acceptably low in order to allow the occupants to leave the rotorcraft and enter life rafts. This should be interpreted to mean that up to and including the worst-case sea conditions for which certification with ditching provisions is requested by the applicant, the probability that the rotorcraft will capsize should be not higher than the target stated in [CS 27.801\(e\)](#). An acceptable means of demonstrating post-ditching flotation stability is through scale model testing using irregular waves. The [AMC to CS 27.801\(e\) and 27.802\(c\)](#) contains a test specification that has been developed for this purpose.
 - (10) Providing a ‘wet floor’ concept (water in the cabin) by positioning the floats higher on the fuselage sides and allowing the rotorcraft to float lower in the water can be a way of increasing the stability of a ditched rotorcraft (although this would need to be verified for

the individual rotorcraft type for all weight and loading conditions), or it may be desirable for other reasons. This is permissible provided that the mean static level of water in the cabin is limited to being lower than the upper surface of the seat cushion (for all rotorcraft mass and centre of gravity cases, with all flotation units intact), and that the presence of water will not unduly restrict the ability of occupants to evacuate the rotorcraft and enter the life raft.

- (11) The sea conditions approved for ditching should be stated in the performance information section of the RFM.
- (12) Current practices allow wide latitude in the design of cabin interiors and, consequently, of stowage provisions for safety and ditching equipment. Rotorcraft manufacturers may deliver aircraft with unfinished (green) interiors that are to be completed by a modifier.
 - (i) Segmented certification is permitted to accommodate this practice. That is, the rotorcraft manufacturer shows compliance with the flotation time, stability, and emergency exit requirements while a modifier shows compliance with the equipment requirements and egress requirements with the interior completed. This procedure requires close cooperation and coordination between the manufacturer, modifier, and EASA.
 - (ii) The rotorcraft manufacturer may elect to establish a token interior for ditching certification. This interior may subsequently be modified by a supplemental type certificate (STC). The ditching provisions should be shown to be compliant with the applicable requirements after any interior configuration or limitation change.
 - (iii) The RFM and any RFM supplements deserve special attention if a segmented certification procedure is pursued.
- (c) Procedures
 - (1) Flotation system design
 - (i) Structural integrity should be established in accordance with [CS 27.563](#).
 - (ii) Rotorcraft handling qualities should be verified to comply with the applicable certification specifications throughout the approved flight envelope with floats installed. Where floats are normally deflated, and deployed in flight, the handling qualities should be verified for the approved operating envelopes with the floats in:
 - (A) the deflated and stowed condition;
 - (B) the fully inflated condition; and
 - (C) the in-flight inflation condition; for float systems which may be inflated in flight, rotorcraft controllability should be verified by test or analysis taking into account all possible emergency flotation system inflation failures.
 - (iii) Reliability should be considered in the basic design to assure approximately equal inflation of the floats to preclude excessive yaw, roll, or pitch in flight or in the water:
 - (A) Maintenance procedures should not degrade the flotation system (e.g. by introducing contaminants that could affect normal operation, etc.).

- (B) The flotation system design should preclude inadvertent damage due to normal personnel traffic flow and wear and tear. Protection covers should be evaluated for function and reliability.
 - (C) The designs of the floats should provide means to minimise the likelihood of damage or tear propagation between compartments. Single compartment float designs should be avoided.
 - (D) When showing compliance with [CS 27.801\(c\)\(1\)](#), and where practicable, the design of the flotation system should consider the likely effects of water impact (i.e. crash) loads. For example:
 - (a) locate system components away from the major effects of structural deformation;
 - (b) use flexible pipes/hoses; and
 - (c) avoid passing pipes/hoses or electrical wires through bulkheads that could act as a ‘guillotine’ when the structure is subject to water impact loads.
 - (iv) The floats should be fabricated from highly conspicuous material of to assist in locating the rotorcraft following a ditching (and possible capsiz).
- (2) Flotation system inflation.
- Emergency flotation systems (EFSs) that are normally stowed in a deflated condition and are inflated either in flight or after contact with water should be evaluated as follows:
- (i) The emergency flotation system should include a means to verify its system integrity prior to each flight.
 - (ii) Means should be provided to automatically trigger the inflation of the EFS upon water entry, irrespective of whether or not inflation prior to water entry is the intended operation mode. If a manual means of inflation is provided, the float activation switch should be located on one of the primary flight controls and should be safeguarded against inadvertent actuation.
 - (iii) The inflation system should be safeguarded against spontaneous or inadvertent actuation in flight conditions for which float deployment has not been demonstrated to be safe.
 - (iv) The maximum airspeeds for intentional in-flight actuation of the emergency flotation system and for flight with the floats inflated should be established as limitations in the RFM unless in-flight actuation is prohibited by the RFM.
 - (v) Activation of the emergency flotation system upon water entry (irrespective of whether or not inflation prior to water entry is the intended operation mode) should result in an inflation time short enough to prevent the rotorcraft from becoming excessively submerged.
 - (vi) A means should be provided for checking the pressure of the gas stowage cylinders prior to take-off. A table of acceptable gas cylinder pressure variation with ambient temperature and altitude (if applicable) should be provided.
 - (vii) A means should be provided to minimise the possibility of over inflation of the flotation units under any reasonably probable actuation conditions.

(viii) The ability of the floats to inflate without puncturing when subjected to actual water pressures should be substantiated. A demonstration of a full-scale float immersion in a calm body of water is one acceptable method of substantiation. Precautions should also be taken to avoid floats being punctured due to the proximity of sharp objects, during inflation in flight and with the helicopter in the water, and during subsequent movement of the helicopter in waves. Examples of objects that need to be considered are aerials, probes, overboard vents, unprotected split-pin tails, guttering and any projections sharper than a three-dimensional right-angled corner.

(3) Injury prevention during and following water entry.

An assessment of the cabin and cockpit layouts should be undertaken to minimise the potential for injury to occupants in a ditching. This may be performed as part of the compliance with [CS 27.785](#). Attention should be given to the avoidance of injuries due to leg/arm flailing, as these can be a significant impediment to occupant egress and subsequent survivability. Practical steps that could be taken include:

- (i) locating potentially hazardous items away from the occupants;
- (ii) installing energy-absorbing padding onto interior components;
- (iii) using frangible materials; and
- (iv) designs that exclude hard or sharp edges.

(4) Water entry procedures.

Tests or simulations (or a combination of both) should be conducted to establish procedures and techniques to be used for water entry, based on the conditions given in (5). These tests/simulations should include determination of the optimum pitch attitude and forward velocity for ditching in a calm sea, as well as entry procedures for the most severe sea condition to be certified. Procedures for all failure conditions that may lead to a 'land immediately' action (e.g. one engine inoperative, all engines inoperative, tail rotor/drive failure) should be established. However, only the procedures for the most critical all-engines-inoperative condition need be verified by water entry test data.

(5) Water entry behaviour.

[CS 27.801\(d\)](#) requires the probable behaviour of the rotorcraft to be shown to exhibit no unsafe characteristics, e.g. that would lead to an inability to remain upright.

This should be demonstrated by means of scale model testing, based on the following conditions:

- (i) For entry into a calm sea:
 - (A) the optimum pitch, roll and yaw attitudes determined in (c)(4) above, with consideration for variations that would reasonably be expected to occur in service;
 - (B) ground speeds from 0 to 15.4 m/s (0 to 30 kt); and
 - (C) descent rate of 1.5 m/s (5 ft/s) or greater;
- (ii) For entry into the most severe sea condition:
 - (A) the optimum pitch attitude and entry procedure determined in (c)(4) above;

- (B) ground speed of 15.4 m/s (30 kt);
- (C) descent rate of 1.5 m/s (5 ft/s) or greater;
- (D) likely roll and yaw attitudes; and
- (E) sea conditions may be represented by regular waves having a height at least equal to the significant wave height (H_s), and a period no larger than the wave zero-crossing period (T_z) for the wave spectrum chosen for demonstration of rotorcraft flotation stability after water entry (see (c)(6) below and [AMC to 27.801\(e\) and 27.802\(c\)](#));

- (iii) Scoops, flaps, projections, and any other factors likely to affect the hydrodynamic characteristics of the rotorcraft must be considered.
- (iv) Probable damage to the structure due to water entry should be considered during the water entry evaluations (e.g. failure of windows, doors, skins, panels, etc.); and
- (v) Rotor lift does not have to be considered.

Alternatively, if scale model test data for a helicopter of a similar configuration has been previously successfully used to justify water entry behaviour, this data could form the basis for a comparative analytical approach.

(6) Flotation stability tests.

An acceptable means of flotation stability testing is contained in the [AMC to CS 27.801\(e\) and 27.802\(c\)](#). Note that model tests in a wave basin on a number of different rotorcraft types have indicated that an improvement in seakeeping performance can consistently be achieved by fitting float scoops.

(7) Occupant egress and survival.

The ability of the occupants to deploy life rafts, egress the rotorcraft, and board the life rafts should be evaluated. For configurations which are considered to have critical occupant egress capabilities due to the life raft locations or the emergency exit locations and the proximity of the float (or a combination of both), an actual demonstration of egress may be required. When a demonstration is required, it may be conducted on a full-scale rotorcraft actually immersed in a calm body of water or using any other rig or ground test facility shown to be representative. The demonstration should show that the floats do not impede a satisfactory evacuation. Service experience has shown that it is possible for occupants to have escaped from the cabin but to have not been able to board a life raft and to have had difficulty in finding handholds to stay afloat and together. Handholds or lifelines should be provided on appropriate parts of the rotorcraft. The normal attitude of the rotorcraft and the possibility of capsizing should be considered when positioning the handholds or lifelines.

[Amdt No: 27/5]

AMC to CS 27.801(e) and 27.802(c) Model test method for flotation stability

ED Decision 2018/007/R

This AMC should be used when showing compliance with [CS 27.801\(e\)](#) or [CS 27.802\(c\)](#) as introduced at Amendment 5.

(a) Explanation

(1) Model test objectives

The objective of the model tests described in the certification specification is to establish the performance of the rotorcraft in terms of its stability in waves. The wave conditions in which the rotorcraft is to be certified should be selected according to the desired level of operability (see (a)(2) below).

This will enable the overall performance of the rotorcraft to be established for inclusion in the rotorcraft flight manual (RFM) as required by [CS 27.1587\(b\)\(3\)](#). In the case of approval with ditching provisions, the wave conditions selected for substantiation of behaviour during the water entry phase must also be taken into account.

The rotorcraft design is to be tested, at each mass condition (see paragraph b(1)(ii) below), with its flotation system intact, and with its single most critical flotation compartment damaged (i.e. the single-puncture case which has the worst adverse effect on flotation stability).

(2) Model test wave conditions

The rotorcraft is to be tested in a single sea condition comprising a single combination of significant wave height (H_s) and zero-crossing period (T_z). The values of H_s and T_z should be no less than, and no more than, respectively, those chosen for certification, i.e. as selected from table 1. This approach is necessary in order to constrain the quantity of testing required within reasonable limits and is considered to be conservative. The justification is detailed in [Appendix 2](#).

The applicant is at liberty to certify the rotorcraft to any significant wave height H_s . This significant wave height will be noted as performance information in the RFM.

Using reliable wave climate data for an appropriate region of the ocean for the anticipated flight operations, a T_z is selected to accompany the H_s . This T_z should be typical of those occurring at H_s as determined in the wave scatter table for the region. The mode or median of the T_z distribution at H_s should be used.

It is considered that the northern North Sea represents a conservatively ‘hostile’ region of the ocean worldwide and should be adopted as the default wave climate for certification. However, this does not preclude an applicant from certifying a rotorcraft specifically for a different region. Such a certification for a specific region would require the geographical limits of that certification region to be noted as performance information in the RFM. Certification for the default northern North Sea wave climate does not require any geographical limits.

In the case of an approval with emergency flotation provisions, operational limitations may limit flight to ‘non-hostile’ sea areas. For simplicity, the northern North Sea may still be selected as the wave climate for certification, or alternatively a wave climate derived from a non-hostile region’s data may be used. If the latter approach is chosen, and it is

desired to avoid geographical limits, a ‘non-hostile’ default wave climate will need to be agreed with EASA.

Wave climate data for the northern North Sea were obtained from the United Kingdom Meteorological Office (UK Met Office) for a typical ‘hostile’ helicopter route. The route selected was from Aberdeen to Block 211/27 in the UK sector of the North Sea. Data tables were derived from a UK Met Office analysis of 34 years of 3-hourly wave data generated within an 8-km, resolved wave model hindcast for European waters. This data represents the default wave climate.

Table 1 below has been derived from this data and contains combinations of H_s and T_z . Table 1 also includes the probability of exceedance (P_e) of the H_s .

Table 1 — Northern North Sea wave climate

Spectrum shape: JONSWAP, peak enhancement factor $\gamma = 3.3$				
	Significant wave height H_s	Mean wave period T_z	Significant steepness $S_s = 2\pi H_s / (g T_z^2)$	H_s probability of exceedance P_e
Intact flotation system	6 m	7.9 s	1/16.2	1.2 %
	5.5 m	7.6 s	1/16.4	2 %
	5 m	7.3 s	1/16.6	3 %
	4.5 m	7.0 s	1/17.0	5 %
	4 m	6.7 s	1/17.5	8 %
	3.5 m	6.3 s	1/17.7	13 %
	3 m	5.9 s	1/18.1	20 %
	2.5 m	5.5 s	1/18.9	29 %
	2 m	5.1 s	1/20.3	43 %
	1.25 m	4.4 s	1/24.2	72 %

(3) Target probability of capsizing

Target probabilities of capsizing have been derived from a risk assessment. The target probabilities to be applied are as stated in [CS 27.801\(e\)](#) and [27.802\(c\)](#), as applicable.

For ditching, the intact flotation system probability of capsizing of 3 % is derived from a historic ditching rate of 3.32×10^{-6} per flight hour and an AMC 27.1309 consequence of hazardous, which implies a frequency of capsizing of less than 10^{-7} per flight hour. The damaged flotation system probability of capsizing is increased by a factor of 10 to 30 % on the assumption that the probability of failure of the critical float compartment is 0.1; this probability has been estimated, as there is insufficient data on flotation system failure rates.

For emergency flotation equipment, an increase of half an order ($\sqrt{10}$) is allowed on the assumption of a reduced exposure to the risk, resulting in a probability of capsizing of 10 %. The probability of a capsizing with a damaged flotation system is consequently increased to 100 %, hence no test is required.

(4) Intact flotation system

For the case of an intact flotation system, if the northern North Sea default wave climate has been chosen for certification, the rotorcraft should be shown to resist capsize in a sea condition selected from Table 1. The probability of capsizing in a 5-minute exposure to the selected sea condition is to be demonstrated to be less than or equal to the

appropriate value provided in [CS 27.801\(e\)](#) or [27.802\(c\)](#), as appropriate, with a confidence of 95 % or greater.

(5) Damaged flotation system

For the case of a damaged flotation compartment (see (1) above), the same sea condition may be used, but a 10-fold increased probability of capsizing is permitted. This is because it is assumed that flotation system damage will occur in approximately one out of ten emergency landings on water. Thus, the probability of capsizing in a 5-minute exposure to the sea condition is to be demonstrated to be less than or equal to 10 times the required probability for the intact flotation system case, with a confidence of 95 % or greater. Where a 10-times probability is equal to or greater than 100 %, it is not necessary to perform a model test to determine the capsize probability with a damaged flotation system.

Alternatively, the applicant may select a wave condition with 10 times the probability of exceedance P_e of the significant wave height (H_s) selected for the intact flotation condition. In this case, the probability of capsizing in a 5-minute exposure to the sea condition is to be demonstrated to be less than or equal to the required value (see [CS 27.801\(e\)](#) or [27.802\(c\)](#)), with a confidence of 95 % or greater.

(6) Long-crested waves

Whilst it is recognised that ocean waves are in general multidirectional (short-crested), the model tests are to be performed in unidirectional (long-crested) waves, this being regarded as a conservative approach to capsize probability.

(b) Procedures

(1) Rotorcraft model

(i) Construction and scale of the model

The rotorcraft model, including its emergency flotation, is to be constructed to be geometrically similar to the full-scale rotorcraft design at a scale that will permit the required wave conditions to be accurately represented in the model basin. It is recommended that the scale of the model should be not smaller than 1/15.

The construction of the model is to be sufficiently light to permit the model to be ballasted to achieve the desired weight and rotational inertias specified in the mass conditions (see (b)(1)(ii) below)¹.

Where it is likely that water may flood into the internal spaces following an emergency landing on water, for example through doors opened to permit escape, or any other opening, the model should represent these internal spaces and openings as realistically as possible.

It is permissible to omit the main rotor(s) from the model, but its (their) mass is to be represented in the mass and inertia conditions².

¹ It should be noted that rotorcraft tend to have a high centre of gravity due to the position of the engines and gearbox on top of the cabin. It therefore follows that most of the ballast is likely to be required to be installed in these high locations of the model.

² Rotors touching the waves can promote capsize, but they can also be a stabilising factor depending on the exact circumstances. Furthermore, rotor blades are often lost during the ditching due to contact with the sea. It is therefore considered acceptable to omit them from the model.

(ii) Mass conditions

As it is unlikely that the most critical condition can be determined reliably prior to testing, the model is to be tested in two mass conditions:

- (A) maximum mass condition, mid C of G; and
- (B) minimum mass condition, mid C of G.

(iii) Mass properties

The model is to be ballasted in order to achieve the required scale weight, centre of gravity, roll and yaw inertia for each of the mass conditions to be tested.

Once ballasted, the model's floating draft and trim in calm water is to be checked and compared with the design floating attitude.

The required mass properties and floating draft and trim, and those measured during model preparation, are to be fully documented and compared in the report.

(iv) Model restraint system

The primary method of testing is with a restrained model, but an alternative option is for a free-floating model (See (3)(iii) below).

For the primary restrained method, a flexible restraint or mooring system is to be provided to restrain the model in order for it to remain beam-on to the waves in the model basin¹.

This restraint system should fulfil the following criteria:

- (A) be attached to the model on the centre line at the front and rear of the fuselage in such a position that roll motion coupling is minimised; an attachment at or near the waterline is preferred; and
- (B) be sufficiently flexible that the natural frequencies of the model surging/swaying on this restraint system are much lower than the lowest wave frequencies in the spectrum.

(v) Sea anchor

Whether or not the rotorcraft is to be fitted with a sea anchor, such an anchor is not to be represented in these model tests².

(2) Test facility

The model test facility is to have the capability to generate realistic long non-repeating sequences of unidirectional (long-crested) irregular waves, as well as the characteristic wave condition at the chosen model scale. The facility is to be deep enough to ensure that the waves are not influenced by the depth (i.e. deep-water waves).

¹ In general the model cannot be permitted to float freely in the basin because in the necessarily long-wave test durations, the model would otherwise drift down the basin and out of the calibrated wave region. Constraining the model to remain beam-on to the waves and not float freely is regarded as a conservative approach to the capsize test. A free-floating test is optional after a specific capsize event, in order to investigate whether the restraint system contributed to the event. It may also be possible to perform a complete free-floating test campaign by combining many short exposures in a wave basin capable of demonstrating a large calibrated wave region.

² A sea anchor deployed from the rotorcraft nose is intended to improve stability by keeping the rotorcraft nose into the waves. However, such devices take a significant time to deploy and become effective, and so, their beneficial effect is to be ignored. The rotorcraft model will be restrained to remain beam-on to the waves.

The dimensions of the test facility are to be sufficiently large to avoid any significant reflection/refraction effects influencing the behaviour of the rotorcraft model.

The facility is to be fitted with a high-quality wave-absorbing system or beach.

The model basin is to provide full details of the performance of the wave maker and the wave absorption system prior to testing.

(3) Model test set-up

(i) General

The model is to be installed in the wave facility in a location sufficiently distant from the wave maker, tank walls and beach/absorber such that the wave conditions are repeatable and not influenced by the boundaries.

The model is to be attached to the model restraint system (see (b)(1)(iv) above).

(ii) Instrumentation and visual records

During wave calibration tests, three wave elevation probes are to be installed and their outputs continuously recorded. These probes are to be installed at the intended model location, a few metres to the side and a few metres ahead of this location.

The wave probe at the model location is to be removed during tests with the rotorcraft model present.

All tests are to be continuously recorded on digital video. It is required that at least two simultaneous views of the model are to be recorded. One is to be in line with the model axis (i.e. viewing along the wave crests), and the other is to be a three-quarter view of the model from the up-wave direction. Video records are to incorporate a time code to facilitate synchronisation with the wave elevation records in order to permit the investigation of the circumstances and details of a particular capsize event.

(iii) Wave conditions and calibration

Prior to the installation of the rotorcraft model in the test facility, the required wave conditions are to be pre-calibrated.

Wave elevation probes are to be installed at the model location, alongside and ahead of the intended model location.

The intended wave spectrum is to be run for the full exposure duration required to demonstrate the required probability of capsizing. The analysis of these wave calibration runs is to be used to:

- (A) confirm that the required wave spectrum has been obtained at the model location; and
- (B) verify that the wave spectrum does not deteriorate appreciably during the run in order to help establish the maximum duration test that can be run before the test facility must be allowed to become calm again.

It should be demonstrated that the wave spectrum measured at each of the three locations is the same.

If a free-floating model is to be used, then the waves are to be calibrated for a range of locations down the basin, and the spectrum measured in each of these locations should be shown to be the same. The length of the basin covered by this range will be the permitted test region for the free-floating model, and the model will be recovered when it drifts outside this region (See Section 4). It should be demonstrated that the time series of the waves measured at the model location does not repeat during the run. Furthermore, it should be demonstrated that one or more continuation runs can be performed using exactly the same wave spectrum and period, but with different wave time series. This is to permit a long exposure to the wave conditions to be built up from a number of separate runs without any unrealistic repetition of the time series.

No wind simulation is to be used¹.

(iv) Required wave run durations

The total duration of runs required to demonstrate that the required probability of capsizing has been achieved (or bettered) is dependent on that probability itself, and on the reliability or confidence of the capsize probability required to be demonstrated.

With the assumption that each 5-minute exposure to the wave conditions is independent, the equations provided in (b)(5) below can be used to determine the duration without a capsize that is required to demonstrate the required performance.² (See [Appendix 1](#) below for examples.)

(4) Test execution and results

Tests are to start with the model at rest and the wave basin calm.

Following the start of the wave maker, sufficient time is to elapse to permit the slowest (highest-frequency) wave components to arrive at the model, before data recording starts.

Wave runs are to continue for the maximum permitted duration determined in the wave calibration test, or in the free-floating option for as long as the model remains in the calibrated wave region. Following sufficient time to allow the basin to become calm again, additional runs are to be conducted until the necessary total exposure duration (T_{test}) has been achieved (see (b)(5) below).

In the case of the free-floating option, the model may be recovered and relaunched without stopping the wave maker, provided that the maximum permitted duration is not exceeded. See paragraph (4)(iv) for requirements regarding relaunching the free-floating model.

If and when a model capsize occurs, the time of the capsize from the start of the run is to be recorded, and the run stopped. The model is to be recovered, drained of any water, and reset in the basin for a continuation run to be performed.

¹ Wind generally has a tendency to redirect the rotorcraft nose into the wind/waves, thus reducing the likelihood of capsize. Therefore, this conservative testing approach does not include a wind simulation.

² Each 5-minute exposure might not be independent if, for example, there was flooding of the rotorcraft, progressively degrading its stability. However, in this context, it is considered that the assumption of independence is conservative.

There are a number of options that may be taken following a capsize event:

- (i) Continuing with the same model configuration.

If the test is to be continued with the same model configuration, the test can be restarted with a different wave time series, or continued from the point of capsizing in a pseudorandom time series.

- (ii) Reducing the wave severity to achieve certification at a lower significant wave height.

Provided that the same basic pseudorandom wave time series can be reproduced by the wave basin at a lower wave height and corresponding period, it is permitted to restart the wave maker time series at a point at least 5 minutes prior to the capsize event, and if the model is now seen to survive the wave sequence that caused a capsize in the more severe condition, then credit can then be taken for the run duration successfully achieved prior to the capsize. Clearly, such a restart is only possible with a model basin using pseudorandom wave generation.

This method is only permitted if the change in significant wave height and period is sufficiently small that the same sequence of capsizing waves, albeit at a lower amplitude, can be seen in the wave basin. If this is not the case, then credit cannot be taken for the exposure time prior to capsize, and the wave time series must be restarted from the beginning.

- (iii) Modifying the model with the intention of avoiding a capsize.

If it is decided to modify the model flotation with the intention of demonstrating that the modified model does not capsize in the wave condition, then the pseudorandom wave maker time series should be restarted at a point at least 5 minutes prior to the capsize event so that the model is seen to survive the wave that caused a capsize prior to the modification. Credit can then be taken for the duration of the run successfully achieved prior to the capsize.

- (iv) Repeating a restrained capsize event with a free-floating model.

If it is suspected that the model restraint system might have contributed to the capsize event, it is permitted to repeat that part of the pseudorandom time series with a free-floating model. The model is to be temporally restrained with light lines and then released beam-on to the waves such that the free-floating model is seen to experience the same wave time series that caused a capsize in exactly the same position in the basin. It is accepted that it might require several attempts to find the precise model release time and position to achieve this.

If the free-floating model, having been launched beam-on to the waves, is seen to yaw into a more beneficial heading once released, and seen to survive the wave that caused a capsize in the restrained model, then this is accepted as negating the capsize seen with the restrained model.

The test may then continue with a restrained model as with (i) above.

- (v) Special considerations regarding relaunching a free-floating model into the calibrated wave region.

If a free-floating model is being used for the tests, then it is accepted that the model will need to be recovered as it leaves the calibrated wave region, and then

relaunched at the top of that region. It is essential that this process does not introduce any statistical or other bias into the behaviour of the model. For example, there might be a natural tendency to wait for a spell of calmer waves into which to launch the model. This particular bias is to be avoided by strictly obeying a fixed time delay between recovery and relaunch.

Any water accumulated inside the model is not to be drained prior to the relaunch.

If the model has taken up a heading to the waves that is not beam-on, then it is permissible to relaunch the model at that same heading.

In all the above cases, continuation runs are to be performed until the total duration of exposure to the wave condition is sufficient to establish that the 5-minute probability of capsizing has been determined with the required confidence of 95 %.

(5) Results analysis

Given that it has been demonstrated that the wave time series are non-repeating and statistically random, the results of the tests may be analysed on the assumption that each 5-minute element of the total time series is independent.

If the model rotorcraft has not capsized during the total duration of the tests, the confidence that the probability of capsizing within 5 minutes is less than the target value of $P_{\text{capsize(target)}}$, as shown below:

$$C = 1 - (1 - P_{\text{capsize(target)}})^{[T_{\text{test}}/T_{\text{criterion}}]}$$

$$\approx 1 - \exp\left(-\frac{P_{\text{capsize(target)}}T_{\text{test}}}{T_{\text{criterion}}}\right)$$

and so the total duration of the model test required without capsizing is provided by:

$$T_{\text{test}} \approx -\frac{T_{\text{criterion}} \ln(1 - C)}{P_{\text{capsize(target)}}$$

where:

- (A) T_{test} is the required full-scale duration of the test (in seconds);
- (B) $P_{\text{capsize(target)}}$ is the required maximum probability of capsizing within 5 minutes;
- (C) $T_{\text{criterion}}$ is the duration (in seconds) in which the rotorcraft must meet the no-capsize probability (= 5 x 60 s), as defined in [CS 27.801\(e\)](#); and
- (D) C is the required confidence that the probability of capsizing has been achieved (0.95).

If the rotorcraft has capsized N_{capsize} times during the tests, the probability of capsizing within 5 minutes can be estimated as:

$$P_{\text{capsize}} = \frac{N_{\text{capsize}} T_{\text{criterion}}}{T_{\text{test}}}$$

and the confidence that the required capsiz criteria have been met is:

$$C = 1 - \sum_{k=0}^{N_{capsize}} \frac{([T_{test}/T_{criterion}]!)!}{([T_{test}/T_{criterion}] - k)!} \{ (P_{capsize(target)})^k (1 - P_{capsize(target)})^{([T_{test}/T_{criterion}] - k)} \}$$

$$\approx 1 - \left\{ \sum_{k=0}^{N_{capsize}} \frac{1}{k!} \left(\frac{P_{capsize(target)} T_{test}}{T_{criterion}} \right)^k \right\} \exp \left(- \frac{P_{capsize} T_{test}}{T_{criterion}} \right)$$

It should be noted that, if the rotorcraft is permitted to fly over sea conditions with significant wave heights (H_s) above the certification limit, then $P_{capsiz(target)}$ should be reduced by the probability of exceedance of the certification limit for the significant wave height (Pe) (see [Appendix 2](#) below).

(c) Deliverables

- (1) A comprehensive report describing the model tests, the facility they were performed in, the model properties, the wave conditions used, the results of the tests, and the method of analysis to demonstrate compliance with [CS 27.801\(d\) and \(e\)](#).
- (2) Conclusions in this report are to clarify the compliance (or otherwise) with those provisions.
- (3) Digital video and data records of all tests performed.
- (4) A specification for a certification model test should also be expected to include:
 - (i) an execution plan and timescale;
 - (ii) formal progress reports on content and frequency; and
 - (iii) quality assurance requirements.

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Appendix 1 — Worked example

ED Decision 2018/007/R

The target 5-minute capsiz probabilities for a rotorcraft certified to [CS 27.801](#) are:

Certification with ditching provisions:

Fully serviceable emergency flotation system (EFS)	– 3 %
Critical flotation compartment failed	– 30 %

Certification with emergency flotation provisions:

Fully serviceable emergency flotation system (EFS)	– 10 %
Critical flotation compartment failed	– no demonstration required

One option available to the rotorcraft designer is to test at the selected wave height and demonstrate a probability of capsizing no greater than these values. However, to enhance offshore helicopter

safety, some national aviation authorities (NAAs) have imposed restrictions that prevent normal operations (i.e. excluding emergencies, search and rescue (SAR), etc.) over sea conditions that are more severe than those for which performance has been demonstrated. In such cases, the helicopter may be operationally limited.

These operational restrictions may be avoided by accounting for the probability of exposure to sea conditions that exceed the selected wave height by certifying the rotorcraft for a lower probability of capsizing. Since it is conservatively assumed that the probability of capsizing in sea conditions that exceed the certified wave height is unity, the lower capsize probability required to be met is the target value minus the probability of the selected wave height being exceeded. However, it should also be noted that, in addition to restricting normal helicopter overwater operations to the demonstrated capability, i.e. the applicant's chosen significant wave height limit ($H_{s(\text{limit})}$), an NAA may declare a maximum limit above which all operations will be suspended due to the difficulty of rescuing persons from the sea in extreme conditions. There will, therefore, be no operational benefit in certifying a rotorcraft for sea conditions that exceed the national limits for rescue.

In the following examples, we shall use the three target probabilities of capsizing without any reduction to avoid operational restrictions. The test times quoted are full-scale times; to obtain the actual model test run time, these times should be divided by the square root of the model scale.

Certification with ditching provisions — fully serviceable EFS

Taking this first case, we need to demonstrate a $\leq 3\%$ probability of capsizing with a 95 % confidence. Applying equation (5)(i) above, this can be achieved with a 499-minute (full-scale time) exposure to the sea condition without a capsize.

Rearranging this equation, we have:

$$T_{\text{test}} \approx -\ln(1 - C) \frac{T_{\text{criterion}}}{P_{\text{capsize}(\text{target})}}$$

$$T_{\text{test}} \approx -\ln(1 - 0.95) \frac{5 \times 60}{0.03} = 29957 \text{ s} = 499 \text{ min}$$

Alternatively, applying equation (5)(ii) above, the criterion would also be met if the model were seen to capsize just three times (for example) in a total 21.5 hours of exposure to the sea condition, or four times (for example) in a total of 25.5 hours of exposure.

Equation (ii) cannot be readily rearranged to solve T_{test} , so the easiest way to solve it is by using a spreadsheet on a trial-and-error method. For the four-capsize case, we find that a 25.5-hour exposure gives a confidence of 0.95.

$$C \approx 1 - \left\{ \sum_{k=0}^{4_e} \frac{1}{k!} \left(\frac{0.03 \times 25.5 \times 60 \times 60}{5 \times 60} \right)^k \right\} \exp \left(-\frac{0.03 \times 25.5 \times 60 \times 60}{5 \times 60} \right) = 0.95$$

Certification with ditching provisions — critical flotation compartment failed

In this case, we need to demonstrate a $\leq 30\%$ probability of capsizing with a 95 % confidence. This can be achieved with a 50-minute (full-scale time) exposure to the sea condition without a capsize.

$$T_{\text{test}} \approx -\ln(1 - 0.95) \frac{5 \times 60}{0.30} = 2996 \text{ s} = 50 \text{ min}$$

As above, the criterion would also be met if the model were seen to capsize just three times (for example) in a total 2.2 hours of exposure to the sea condition, or four times (for example) in a total of 2.6 hours of exposure.

Solving by trial and error in a spreadsheet, we find that a 2.6-hour exposure with no more than four capsizes gives a confidence of 0.95.

$$C \approx 1 - \left\{ \sum_{k=0}^{4_e} \frac{1}{k!} \left(\frac{0.30 \times 2.6 \times 60 \times 60}{5 \times 60} \right)^k \right\} \exp \left(- \frac{0.30 \times 2.6 \times 60 \times 60}{5 \times 60} \right) = 0.95$$

Certification with emergency flotation provisions — fully serviceable EFS

In this case, we need to demonstrate a $\leq 10\%$ probability of capsizing with a 95 % confidence. By solving the equations as above, this can be achieved with a 150-minute (full-scale time) exposure to the sea condition without a capsize.

$$T_{test} \approx -\ln(1 - 0.95) \frac{5 \times 60}{0.10} = 8987 \text{ s} = 150 \text{ min}$$

As above, the criterion would also be met if the model were seen to capsize just three times (for example) in a total 6.5 hours of exposure to the sea condition, or four times (for example) in a total of 7.6 hours of exposure.

Solving by trial and error in a spreadsheet we find that a 7.6-hour exposure with no more than four capsizes gives a confidence of 0.95.

$$C \approx 1 - \left\{ \sum_{k=0}^{4_e} \frac{1}{k!} \left(\frac{0.10 \times 7.6 \times 60 \times 60}{5 \times 60} \right)^k \right\} \exp \left(- \frac{0.10 \times 7.6 \times 60 \times 60}{5 \times 60} \right) = 0.95$$

Certification with ditching provisions — critical flotation compartment failed

As stated in [CS 27.802\(c\)](#), no demonstration of capsize resistance is required for the case of the critical float compartment having failed.

This is because the allowed factor of ten increase in the probability of capsizing, as explained in (a)(3) above, results in a probability of 100 %.

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Appendix 2 — Test specification rationale

ED Decision 2018/007/R

(a) Introduction

The overall risk of capsizing within the 5-minute exposure period consists of two components: the probability of capsizing in a given wave condition, and the probability of experiencing that wave condition in an emergency landing on water.

If it is assumed that an emergency landing on water occurs at random and is not linked with weather conditions, the overall risk of a capsizing can be established by combining two pieces of information:

- (1) The wave climate scatter table, which shows the probability of meeting any particular combination of H_s and T_z . An example scatter table is shown below in **Figure 1 — Example of all-year wave scatter table**. Each cell of the table contains the probability of experiencing a wave condition with H_s and T_z in the range provided. Thus, the total of all cells in the table adds up to unity.
- (2) The probability of a capsizing in a 5-minute exposure for each of these height/period combinations. This probability of capsizing is different for each helicopter design and for each wave height/period combination, and is to be established through scale model testing using the method defined above.

In theory, a model test for the rotorcraft design should be performed in the full range of wave height/period combinations covering all the cells in the scatter table. Clearly, wave height/period combinations with zero or very low probabilities of occurrence might be ignored. It might also be justifiably assumed that the probability of capsizing at very high wave heights is unity, and at very low wave heights, it is zero. However, there would still remain a very large number of intermediate wave height/period combinations that would need to be investigated in model tests, and it is considered that such a test programme would be too lengthy and costly to be practicable.

The objective here is therefore to establish a justifiable method of estimating the overall 5-minute capsize probability using model test results for a single-wave condition. That is a single combination of H_s and T_z . Such a method can never be rigorously linked with the safety objective, but it is proposed that it may be regarded as a conservative approximation.

(b) Test methodology

The proposed test methodology is as follows:

The rotorcraft designer selects a desired significant wave height limit $H_s(\text{limit})$ for ditching or the emergency flotation certification of his helicopter. Model tests are performed in the sea condition $H_s(\text{limit})$ $T_z(\text{limit})$ (where $T_z(\text{limit})$ is the zero-crossing period most likely to accompany $H_s(\text{limit})$) with the selected spectrum shape using the method specified above, and the 5-minute probability of capsizing (P_{capsize}) established in this sea condition.

The way in which P_{capsize} varies for other values of H_s and T_z is not known because it is not proposed to perform model tests in all the other possible combinations. Furthermore, there is no theoretical method to translate a probability of capsizing from one sea condition to another.

However, it is known that the probability of capsizing is related to the exposure to breaking waves of sufficient height, and that this is in turn linked with wave steepness. Hence:

- (1) the probability of capsizing is likely to be higher for wave heights just less than $H_{s(\text{limit})}$ but with wave periods shorter than $T_{z(\text{limit})}$; and
- (2) the probability of capsizing will be lower for the larger population of wave conditions with wave heights less than $H_{s(\text{limit})}$ and with wave periods longer than $T_{z(\text{limit})}$.

So, a reasonable and conservative assumption is that on average, the same P_{capsize} holds good for all wave conditions with heights less than or equal to $H_{s(\text{limit})}$.

In jurisdictions where flying over waves greater than $Hs(limit)$ is permitted, the rotorcraft will have passed the certification criteria provided that: $P_{capsize} \leq P_{capsize}(target) - P_e$, where P_e is the probability of exceedance of $Hs(limit)$. Clearly, in this case, it can be seen that it would not be permissible for the rotorcraft designer to select an $Hs(limit)$ which has a probability of exceedance greater than $P_{capsize}(target)$.

Hs (m) >= Hs (m) <																							TOTALS		
12.5	13																						0.00000	0.00000	0.0000
12	12.5																								0.0000
11.5	12																								0.0000
11	11.5																								0.0000
10.5	11																								0.0000
10	10.5																								0.0000
9.5	10																								0.0000
9	9.5																								0.0000
8.5	9																								0.0000
8	8.5																								0.0000
7.5	8																								0.0000
7	7.5																								0.0000
6.5	7																								0.0000
6	6.5																								0.0000
5.5	6																								0.0000
5	5.5																								0.0000
4.5	5																								0.0000
4	4.5																								0.0000
3.5	4																								0.0000
3	3.5																								0.0000
2.5	3																								0.0000
2	2.5																								0.0000
1.5	2																								0.0000
1	1.5																								0.0000
0.5	1																								0.0000
0	0.5	0.00002	0.00100	0.00446	0.00513	0.00427	0.00196	0.00050	0.00012	0.00003	0.00001	0.00000													0.0000
Tz (s) >=	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12	1.0000		
Tz (s) <	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12				

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- (a) The rotorcraft must be equipped with an approved emergency flotation system.
- (b) The flotation units of the emergency flotation system, and their attachments to the rotorcraft, must comply with [CS 27.563](#).
- (c) The rotorcraft must be shown to resist capsizing in the sea conditions selected by the applicant. The probability of capsizing in a 5-minute exposure to the sea conditions must be demonstrated to be less than or equal to 10.0 % with a fully serviceable emergency flotation system, with 95 % confidence. No demonstration of capsizing resistance is required for the case of the critical float compartment having failed.

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AMC 27.802 Emergency Flotation

ED Decision 2018/007/R

This AMC replaces FAA AC 27 MG 10.

(a) Definitions

- (1) Ditching: a controlled emergency landing on the water, deliberately executed in accordance with rotorcraft flight manual (RFM) procedures, with the intent of abandoning the rotorcraft as soon as practicable.

NOTE: Although the term ‘ditching’ is most commonly associated with the design standards related to [CS 27.801](#), a rotorcraft equipped to the less demanding requirements of [CS 27.802](#), when performing an emergency landing on water, would nevertheless be commonly described as carrying out the process of ditching. The term ‘ditching’ is therefore used in this AMC in this general sense.

- (2) Emergency flotation system (EFS): a system of floats and any associated parts (e.g. gas cylinders, means of deployment, pipework and electrical connections) that is designed and installed on a rotorcraft to provide buoyancy and flotation stability during and after ditching.

(b) Explanation

- (1) Approval of emergency flotation equipment is performed only if requested by the applicant. Operational rules may accept that a helicopter conducts flights over certain sea areas provided it is fitted with approved emergency flotation equipment (i.e. an EFS), rather than being certified with full ditching provisions.
- (2) Emergency flotation certification encompasses emergency flotation system loads and design, and rotorcraft flotation stability.
- (3) Failure of the EFS to operate when required will lead to the rotorcraft rapidly capsizing and sinking. Operational experience has shown that localised damage or failure of a single component of an EFS can lead to the loss of the complete system. Therefore, the design of the EFS needs careful consideration.
- (4) The sea conditions, on which certification with emergency flotation is to be based, are selected by the applicant and should take into account the expected sea conditions in the intended areas of operation. Capsizing resistance is required to meet the same requirements as for full ditching approval but with the allowable capsizing probability being set at 10 %. The default wave climate specified in this requirement is that of the northern North Sea, as it represents a conservative condition. An applicant might consider this to be inappropriate, as it represents a hostile sea area. The applicant may therefore propose a different wave climate based on data from a non-hostile sea area. The associated certification will then be limited to the geographical region(s) thus represented. Alternatively, a non-hostile default wave climate might be agreed, with no associated need for geographical limits to the certification. The significant wave height, and any geographical limitations (if applicable, see the [AMC to 27.801\(e\) and 27.802\(c\)](#)) should be included in the RFM as performance information.
- (5) During scale model testing, appropriate allowances should be made for probable structural damage and leakage. Previous model tests and other data from rotorcraft of similar configurations that have already been substantiated based on equivalent test conditions may be used to satisfy the emergency flotation requirements. In regard to

flotation stability, test conditions should be equivalent to those defined in the [AMC to 27.801\(e\) and 27.802\(c\)](#).

- (6) [CS 27.802](#) requires that in sea conditions for which certification with emergency flotation is requested by the applicant, the probability of capsizing in a 5-minute exposure is acceptably low in order to allow the occupants to leave the rotorcraft and enter the life rafts. This should be interpreted to mean that up to and including the worst-case sea conditions for which certification with emergency flotation is requested by the applicant, the probability that the rotorcraft will capsize should be not higher than the target stated in [CS 27.802\(c\)](#). An acceptable means of demonstrating post-ditching flotation stability is through scale model testing using irregular waves. The [AMC to 27.801\(e\) and 27.802\(c\)](#) contains a test specification that has been developed for this purpose.
 - (7) Providing a ‘wet floor’ concept (water in the cabin) by positioning the floats higher on the fuselage sides and allowing the rotorcraft to float lower in the water can be a way of increasing the stability of a ditched rotorcraft (although this would need to be verified for the individual rotorcraft type for all weight and loading conditions), or it may be desirable for other reasons. This is permissible provided that the mean static level of water in the cabin is limited to being lower than the upper surface of the seat cushion (for all rotorcraft mass and centre of gravity cases, with all flotation units intact), and that the presence of water will not unduly restrict the ability of occupants to evacuate the rotorcraft and enter the life raft.
 - (8) The sea conditions approved for ditching should be stated in the performance information section of the RFM.
- (c) Procedures
- (1) Flotation system design
 - (i) Structural integrity should be established in accordance with [CS 27.563](#). [CS 27.802\(a\)](#) only requires the floats and their attachments to the rotorcraft to be designed to withstand the load conditions defined in [CS 27.563](#). Other parts of the rotorcraft (e.g. fuselage underside structure, chin windows, doors) do not need to be shown to be capable of withstanding these load conditions.
 - (ii) Rotorcraft handling qualities should be verified to comply with the applicable certification specifications throughout the approved flight envelope with floats installed. Where floats are normally deflated and deployed in flight, the handling qualities should be verified for the approved operating envelopes with the floats in:
 - (A) the deflated and stowed condition;
 - (B) the fully inflated condition; and
 - (C) the in-flight inflation condition; for float systems which may be inflated in flight, rotorcraft controllability should be verified by test or analysis taking into account all possible emergency flotation system inflation failures.
 - (iii) Reliability should be considered in the basic design to assure approximately equal inflation of the floats to preclude excessive yaw, roll, or pitch in flight or in the water:
 - (A) Maintenance procedures should not degrade the flotation system (e.g. introducing contaminants that could affect normal operation, etc.).

- (B) The flotation system design should preclude inadvertent damage due to normal personnel traffic flow and wear and tear. Protection covers should be evaluated for function and reliability.
- (C) The designs of the floats should provide means to minimise the likelihood of damage or tear propagation between compartments. Single compartment float designs should be avoided.
- (iv) The floats should be fabricated from highly conspicuous materials to assist in locating the rotorcraft following a ditching (and possible capsizing).

(2) Flotation system inflation

Emergency flotation systems (EFSs) which are normally stowed in a deflated condition and are inflated either in flight or after water contact should be evaluated as follows:

- (i) The emergency flotation system should include a means to verify system integrity prior to each flight.
 - (ii) If a manual means of inflation is provided, the float activation switch should be located on one of the primary flight controls and should be safeguarded against inadvertent actuation.
 - (iii) The inflation system should be safeguarded against spontaneous or inadvertent actuation in flight conditions for which float deployment has not been demonstrated to be safe.
 - (iv) The maximum airspeeds for intentional in-flight actuation of the emergency flotation system and for flight with the floats inflated should be established as limitations in the RFM unless in-flight actuation is prohibited by the RFM.
 - (v) Activation of the emergency flotation system upon water entry (irrespective of whether or not inflation prior to water entry is the intended operation mode) should result in an inflation time short enough to prevent the rotorcraft from becoming excessively submerged.
 - (vi) A means should be provided for checking the pressure of the gas stowage cylinders prior to take-off. A table of acceptable gas cylinder pressure variation with ambient temperature and altitude (if applicable) should be provided.
 - (vii) A means should be provided to minimise the possibility of over-inflation of the flotation units under any reasonably probable actuation conditions.
 - (viii) The ability of the floats to inflate without puncturing when subjected to actual water pressures should be substantiated. A demonstration of a full-scale float immersion in a calm body of water is one acceptable method of substantiation. Precautions should also be taken to avoid floats being punctured due to the proximity of sharp objects, during inflation in flight or with the helicopter in the water, and during subsequent movement of the helicopter in waves. Examples of objects that need to be considered are airdrops, probes, overboard vents, unprotected split-pin tails, guttering and any projections sharper than a three-dimensional right angled corner.
- (3) Injury prevention during and following water entry.

An assessment of the cabin and cockpit layouts should be undertaken to minimise the potential for injury to occupants in a ditching. This may be performed as part of the

compliance with [CS 27.785](#). Attention should be given to the avoidance of injuries due to leg/arm flailing, as these can be a significant impediment to occupant egress and subsequent survivability. Practical steps that could be taken include:

- (i) locating potentially hazardous items away from the occupants;
 - (ii) installing energy-absorbing padding onto interior components;
 - (iii) using frangible materials; and
 - (iv) designs that exclude hard or sharp edges.
- (4) Water entry procedures.

Tests or simulations (or a combination of both) should be conducted to establish procedures and techniques to be used for water entry. These tests/simulations should include determination of the optimum pitch attitude and forward velocity for ditching in a calm sea, as well as entry procedures for the most severe sea condition to be certified. Procedures for all failure conditions that may lead to a 'land immediately' action (e.g. one engine inoperative, all engines inoperative, tail rotor/drive failure) should be established.

- (5) Flotation stability tests.

An acceptable means of flotation stability testing is contained in [AMC to 27.801\(e\) and 27.802\(c\)](#). Note that model tests in a wave basin on a number of different rotorcraft types have indicated that an improvement in seakeeping performance can consistently be achieved by fitting float scoops.

- (6) Occupant egress and survival.

The ability of the occupants to deploy life rafts, egress the rotorcraft, and board the life rafts should be evaluated. For configurations which are considered to have critical occupant egress capabilities due to the life raft locations or the emergency exit locations and the proximity of the float (or a combination of both), an actual demonstration of egress may be required. When a demonstration is required, it may be conducted on a full-scale rotorcraft actually immersed in a calm body of water or using any other rig or ground test facility shown to be representative. The demonstration should show that floats do not impede a satisfactory evacuation. Service experience has shown that it is possible for occupants to have escaped from the cabin but to have not been able to board a life raft and to have had difficulty in finding handholds to stay afloat and together. Handholds or lifelines should be provided on appropriate parts of the rotorcraft. The normal attitude of the rotorcraft and the possibility of a capsized condition should be considered when positioning the handholds or lifelines.

[Amdt No: 27/5]

CS 27.805 Flight crew emergency exits

ED Decision 2018/007/R

- (a) For rotorcraft with passenger emergency exits that are not convenient to the flight crew, there must be flight crew emergency exits, on both sides of the rotorcraft or as a top hatch, in the flight crew area.
- (b) Each flight crew emergency exit must be of sufficient size and must be located so as to allow rapid evacuation of the flight crew and must be marked so as to be readily located and operated even in darkness. This must be shown by test.

- (c) Underwater emergency exits for flight crew. If certification with ditching provisions is requested by the applicant, none of the flight crew emergency exits required by (a) and (b) may be obstructed by water or flotation devices after an emergency landing on water and each exit must shown by test, demonstration, or analysis to provide for rapid escape with the rotorcraft in the upright floating position or capsized. Each operational device (pull tab(s), operating handle, 'push here' decal, etc.) must be marked with black and yellow stripes and must be shown to be accessible for the range of flight crew heights as required by [CS 27.777\(b\)](#) and for both the case of an un-deformed seat and a seat with any deformation resulting from the test conditions required by [CS 27.562](#). Flight crew emergency exits must be reasonably protected from becoming jammed as a result of fuselage deformation. The markings required by (b) must remain visible if the rotorcraft is capsized and the cabin is submerged.

[Amdt No: 27/5]

AMC 27.805(c) Flight crew emergency exits

ED Decision 2018/007/R

This AMC supplements FAA AC 27.805.

(a) Explanation

To facilitate a rapid escape, flight crew underwater emergency exits should be designed for use with the rotorcraft in both the upright position and in any foreseeable floating attitude. The flight crew underwater emergency exits should not be obstructed during their operation by water or floats to the extent that rapid escape would not be possible or that damage to the flotation system may occur. This should be substantiated for any rotorcraft floating attitude, upright or capsized, and with the emergency flotation system intact and with any single compartment failed. With the rotorcraft capsized and floating, the flight crew underwater emergency exits should be usable with the cabin flooded, and the markings required to enable occupants to escape in darkness should continue to function when the rotorcraft is capsized and the cabin is submerged.

(b) Procedures

- (1) It should be shown by test, demonstration or analysis that there is no interference with the flight crew underwater emergency exits from water or any stowed or deployed emergency flotation devices, with the rotorcraft in any foreseeable floating attitude.
- (2) Flight crew should be able to reach the operating device for their underwater emergency exit, whilst seated, with restraints fastened, with seat energy absorption features at any design position, and with the rotorcraft in any attitude.
- (3) Likely damage sustained during a ditching should be considered.
- (4) It is acceptable for the underwater emergency exit threshold to be below the waterline when the rotorcraft is floating upright, but in such a case, it should be substantiated that there is no obstruction to the use of the exit and that no excessive force (see FAA AC 29.809) is required to operate the exit.
- (5) It is permissible for flight crew to be unable to directly enter life rafts from the underwater flight crew emergency exits and to have to take a more indirect route, e.g. by climbing over a forward flotation unit. In such a case, the feasibility of the exit procedure should be assessed. Handholds may need to be provided on the rotorcraft.

- (6) [CS 27.807\(b\)\(3\)](#) requires emergency exit markings to be provided and enable the emergency exit to be located and operated in darkness. Furthermore, [CS 27.805\(c\)](#) requires these illuminated markings to continue to function if the cabin becomes submerged. This should be shown by test, demonstration or analysis.
- (7) To make it easier to recognise underwater, the operating device for the underwater emergency exit should have black and yellow markings with at least two bands of each colour of approximately equal widths. Any other operating feature, e.g. highlighted 'push here' decal(s) for openable windows, should also incorporate black-and-yellow-striped markings.

[Amdt No: 27/5]

CS 27.807 Passenger emergency exits

ED Decision 2018/007/R

- (a) *Number and location.*
 - (1) There must be at least one emergency exit on each side of the cabin readily accessible to each passenger. One of these exits must be usable in any probable attitude that may result from a crash;
 - (2) Doors intended for normal use may also serve as emergency exits, provided that they meet the requirements of this CS; and
 - (3) If emergency flotation devices are installed, there must be an emergency exit accessible to each passenger on each side of the cabin that is shown by test, demonstration, or analysis to open without interference from flotation devices, whether stowed or deployed, and with the rotorcraft floating either upright or capsized.
- (b) *Type and operation.* Each emergency exit prescribed by (a) or (d) must:
 - (1) Consist of a moveable window or panel, or additional external door, providing an unobstructed opening that will admit a 0.48 m by 0.66 m (19 inch by 26 inch) ellipse;
 - (2) Have simple and obvious methods of opening, from the inside and from the outside, which do not require exceptional effort;
 - (3) Be arranged and marked so as to be readily located and operated even in darkness; and
 - (4) Be reasonably protected from becoming jammed as result of fuselage deformation.
- (c) *Tests.* The proper functioning of each emergency exit must be shown by test.
- (d) *Underwater emergency exits for passengers.* If certification with ditching provisions is requested by the applicant, underwater emergency exits must be provided in accordance with the following requirements and must be proven by test, demonstration, or analysis to provide for rapid escape with the rotorcraft in the upright floating position or capsized:
 - (1) One underwater emergency exit, meeting the size requirements of (b) above, must be installed in each side of the rotorcraft for each unit (or part of a unit) of four passenger seats. However, the seat-to-exit ratio may be increased for underwater emergency exits large enough to permit the simultaneous egress of two passengers side by side. Passenger seats must be located in relation to the underwater emergency exits in a way to best facilitate escape with the rotorcraft capsized and the cabin flooded.

- (2) Underwater emergency exits, including their means of operation, markings, lighting and accessibility, must be designed for use in a flooded and capsized cabin.
- (3) Each underwater emergency exit must be provided with a suitable handhold, or handholds adjacently located inside the cabin, to assist occupants in locating and operating the exit, as well as in egressing through the underwater emergency exit.
- (4) The markings required by sub-paragraph (b)(3) must be designed to remain visible if the rotorcraft is capsized and the cabin is submerged.
- (5) Each operational marking (pull tab(s), operating handle, 'push here' decal, etc.) must be marked with black and yellow stripes.

[Amdt No: 27/5]

AMC 27.807(d) Underwater emergency exits for passengers

ED Decision 2018/007/R

This AMC replaces FAA AC 27.807, AC 27.807A and AC 27.807B.

(a) Explanation

CS-27 Amendment 5 re-evaluates the need for and the concept behind emergency exits for rotorcraft approved with ditching provisions. Prior to CS-27 Amendment 5, there were no additional ditching provisions for rotorcraft certified for ditching with regard to the number of emergency exits.

Operational experience has shown that in a ditching in which the rotorcraft remains upright, use of the passenger doors can be very beneficial in ensuring a rapid and orderly evacuation onto the life raft(s). However, when a rotorcraft capsizes, doors may be unusable and the number and availability of emergency exits that can be readily used underwater will be crucial to ensuring that passengers are able to escape in a timely manner. Experience has shown that the number of emergency exits required in the past by design requirements has been inadequate in a capsized situation, and a common design solution has been to use the passenger cabin windows as additional emergency egress means by including a jettison feature. The jettison feature has commonly been provided by modifying the elastomeric window seal such that its retention strength is either reduced, or can be reduced by providing a removable part of its cross section, i.e. the so called 'push out' window, although other design solutions have been employed. The provision of openable windows has been required by some air operations regulations.

In recognition of this identified need for an increased number of exits for underwater escape, Amendment 5 created a new set of exit terminology and [CS 27.807\(d\)\(1\)](#) was revised to require one pair of 'underwater emergency exits', i.e. one on each side of the rotorcraft, to be provided for each unit, or part of a unit, of four passenger seats, and passenger seats to be located relative to these exits in a way to best facilitate escape. This new terminology was seen as describing the real intent of this higher number of required emergency exits for rotorcraft approved with ditching provisions.

The objective is for no passenger to be in a worse position than the second person to egress through an exit. The size of each underwater emergency exit should at least meet the dimensional provisions of [CS 27.807\(b\)\(1\)](#), i.e. it should provide an unobstructed opening through which a 0.48 m x 0.66 m (19 in. x 26 in.) elliptical object could pass.

This provision is based on the need to facilitate egress in the case of a capsizing that occurs soon after the rotorcraft has alighted on the water or in the event of a survivable water impact in which the cabin will likely be immediately flooded. The time available for evacuation is very short in such situations, and therefore, CS-27 Amendment 5 has increased the safety level by mandating additional exits, in the form of underwater emergency exits, to both shorten available escape routes and to ensure that no occupant should need to wait for more than one other person to escape before being able to make their own escape. The provision of an underwater emergency exit in each side of the fuselage for each unit (or part of a unit) of four passenger seats will make this possible, provided that seats are positioned relative to the exits in a favourable manner.

Critical evacuation factors are the distance to an underwater emergency exit and how direct and obvious the exit route is, taking into account that the passengers are likely to be disoriented.

So called ‘push-out’ windows (see above) have some advantages in that they are not susceptible to jamming and may open by themselves in a water impact due to flexing of the fuselage upon water entry and/or external water pressure.

The risk of a capsizing during evacuation onto the life rafts can be mitigated to some extent by instructing passengers to open all the underwater emergency exits as a matter of course soon after the helicopter has alighted on the water, thus avoiding the delay due to opening the exits in the event that the exits are needed. Such advice should be considered for inclusion in the documentation provided to the helicopter operator.

(b) Procedures

- (1) The number and the size of underwater emergency exits should be as specified above.
- (2) Care should be taken regarding oversized exits to avoid them becoming blocked if more than one passenger attempts to use the same exit simultaneously.
- (3) A higher seat-to-exit ratio may be accepted if the exits are large enough to allow the simultaneous escape of more than one passenger. For example, a pair of exits may be approved for eight passengers if the size of each exit provides an unobstructed area that encompasses two ellipses of 0.48 m x 0.66 m (19 in. x 26 in.) side by side.
- (4) Test, demonstration, compliance inspection, or analysis is required to substantiate that an exit is free from interference from stowed or deployed emergency flotation devices. In the event that an analysis or inspection is insufficient or that a given design is questionable, a test or demonstration may be required. Such a test or demonstration would consist of an accurate, full-size replica (or true representation) of the rotorcraft and flotation devices, both when stowed and after their deployment.
- (5) Consideration should be given to reducing the potential confusion caused by the lack of standardisation of the location of the operating devices (pull tab, handle) for underwater emergency exits. For example, the operating device should be located next to the handhold (see (10) below). The occupant then has only to find the handhold to locate the operating device. Each adjacent occupant should be able to reach the handhold and operating device whilst seated, with restraints fastened, with seat energy absorption features at any design position, and with the rotorcraft in any attitude. If a single underwater emergency exit is designed for the simultaneous egress of two occupants side by side, a handhold and an operating device should be within reach of each occupant seated adjacent to the exit.

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- (6) Underwater emergency exits should be shown to be operable with the rotorcraft in any foreseeable attitude, including with the rotorcraft capsized.
 - (7) Underwater emergency exits should be designed so that they are optimised for use with the rotorcraft capsized. For example, the handhold(s) should be located close to the bottom of the window (top if inverted) to assist an occupant in overcoming the buoyancy loads of an immersion suit, and by ensuring that markings and lighting will help identify the exit(s) and readily assist in an escape.
 - (8) The means to open an underwater emergency exit should be simple and obvious and should not require any exceptional effort. Designs with any of the following characteristics (non-exhaustive list) are considered to be non-compliant:
 - (i) More than one hand is needed to operate the exit itself (use of the handhold may occupy the other hand);
 - (ii) Any part of the opening means, e.g. an operating handle or control, is located remotely from the exit such that it would be outside of a person's direct vision when looking directly at the exit, or that the person needs to move away from the immediate vicinity of the exit in order to reach it; and
 - (iii) The exit does not meet the opening effort limitations set by FAA AC 29.809.
 - (9) It should be possible to readily grasp and operate any operating handle or control using either a bare or a gloved hand.
 - (10) Handholds, as required by [CS 27.807\(d\)\(3\)](#), should be mounted close to the bottom of each underwater emergency exit such that they fall easily to hand for a normally seated occupant. In the case of exits between face-to-face seating, the provision of two handholds is required. Handholds should be designed such that the risk is low of escapees' clothing or emergency equipment snagging on them.
 - (11) To make it easier to recognise underwater, the operating device for the underwater emergency exit should have black and yellow markings with at least two bands of each colour of approximately equal widths. Any other operating features, e.g. highlighted 'push here' decal(s) for openable windows, should also incorporate black-and-yellow-striped markings.
 - (12) With regard to the location of seats relative to the exits, the most obvious layout that maximises achievement of the objective that no passenger is in a worse position than the second person to egress through an exit is a four-abreast arrangement with all the seats in each row located appropriately and directly next to the emergency exits. However, this might not be possible in all rotorcraft designs due to issues such as limited cabin width, the need to locate seats such as to accommodate normal boarding and egress, and the installation of items other than seats in the cabin. Notwithstanding this, an egress route necessitating movement such as along an aisle, around a cabin item, or in any way other than directly towards the nearest emergency exit, to escape the rotorcraft is not considered to be compliant with [CS 27.807\(d\)\(1\)](#).

[Amdt No: 27/5]

CS 27.831 Ventilation

ED Decision 2003/15/RM

- (a) The ventilating system for the pilot and passenger compartments must be designed to prevent the presence of excessive quantities of fuel fumes and carbon monoxide.
- (b) The concentration of carbon monoxide may not exceed one part in 20 000 parts of air during forward flight or hovering in still air. If the concentration exceeds this value under other conditions, there must be suitable operating restrictions.

CS 27.833 Heaters

ED Decision 2003/15/RM

Each combustion heater must be approved.

FIRE PROTECTION

CS 27.853 Compartment interiors

ED Decision 2003/15/RM

For each compartment to be used by the crew or passengers:

- (a) The materials must be at least flame resistant;
- (b) (Reserved)
- (c) If smoking is to be prohibited, there must be a placard so stating, and if smoking is to be allowed:
 - (1) There must be an adequate number of self-contained, removable ashtrays; and
 - (2) Where the crew compartment is separated from the passenger compartment, there must be at least one illuminated sign (using either letters or symbols) notifying all passengers when smoking is prohibited. Signs which notify when smoking is prohibited must:
 - (i) When illuminated, be legible to each passenger seated in the passenger cabin under all probable lighting conditions; and
 - (ii) Be so constructed that the crew can turn the illumination on and off.

CS 27.855 Cargo and baggage compartments

ED Decision 2003/15/RM

- (a) Each cargo and baggage compartment must be constructed of, or lined with, materials that are at least:
 - (1) Flame resistant, in the case of compartments that are readily accessible to a crew member in flight; and
 - (2) Fire resistant, in the case of other compartments.
- (b) No compartment may contain any controls, wiring, lines, equipment, or accessories whose damage or failure would affect safe operation, unless those items are protected so that:
 - (1) They cannot be damaged by the movement of cargo in the compartment; and
 - (2) Their breakage or failure will not create a fire hazard.

CS 27.859 Heating systems

ED Decision 2003/15/RM

- (a) *General.* For each heating system that involves the passage of cabin air over, or close to, the exhaust manifold, there must be means to prevent carbon monoxide from entering any cabin or pilot compartment.
- (b) *Heat exchangers.* Each heat exchanger must be:
 - (1) Of suitable materials;
 - (2) Adequately cooled under all conditions; and
 - (3) Easily disassembled for inspection.
- (c) *Combustion heater fire protection.* Except for heaters which incorporate designs to prevent hazards in the event of fuel leakage in the heater fuel system, fire within the ventilating air

passage, or any other heater malfunction, each heater zone must incorporate the fire protection features of the applicable requirements of [CS 27.1183](#), [27.1185](#), [27.1189](#), [27.1191](#), and be provided with –

- (1) Approved, quick-acting fire detectors in numbers and locations ensuring prompt detection of fire in the heater region.
 - (2) Fire extinguisher systems that provide at least one adequate discharge to all areas of the heater region.
 - (3) Complete drainage of each part of each zone to minimise the hazards resulting from failure or malfunction of any component containing flammable fluids. The drainage means must be:
 - (i) Effective under conditions expected to prevail when drainage is needed; and
 - (ii) Arranged so that no discharged fluid will cause an additional fire hazard.
 - (4) Ventilation, arranged so that no discharged vapours will cause an additional fire hazard.
- (d) *Ventilating air ducts.* Each ventilating air duct passing through any heater region must be fireproof.
- (1) Unless isolation is provided by fireproof valves or by equally effective means, the ventilating air duct downstream of each heater must be fireproof for a distance great enough to ensure that any fire originating in the heater can be contained in the duct.
 - (2) Each part of any ventilating duct passing through any region having a flammable fluid system must be so constructed or isolated from that system that the malfunctioning of any component of that system cannot introduce flammable fluids or vapours into the ventilating airstream.
- (e) *Combustion air ducts.* Each combustion air duct must be fireproof for a distance great enough to prevent damage from backfiring or reverse flame propagation.
- (1) No combustion air duct may connect with the ventilating airstream unless flames from back-fires or reverse burning cannot enter the ventilating airstream under any operating condition, including reverse flow or malfunction of the heater or its associated components.
 - (2) No combustion air duct may restrict the prompt relief of any backfire that, if so restricted, could cause heater failure.
- (f) *Heater control.* General. There must be means to prevent the hazardous accumulation of water or ice on or in any heater control component, control system tubing, or safety control.
- (g) *Heater safety controls.* For each combustion heater, safety control means must be provided as follows:
- (1) Means independent of the components provided for the normal continuous control of air temperature, airflow, and fuel flow must be provided for each heater to automatically shut off the ignition and fuel supply of that heater at a point remote from that heater when any of the following occurs:
 - (i) The heat exchanger temperature exceeds safe limits.
 - (ii) The ventilating air temperature exceeds safe limits.
 - (iii) The combustion airflow becomes inadequate for safe operation.

- (iv) The ventilating airflow becomes inadequate for safe operation.
- (2) The means of complying with sub-paragraph (g)(1) for any individual heater must:
 - (i) Be independent of components serving any other heater, the heat output of which is essential for safe operation; and
 - (ii) Keep the heater off until restarted by the crew.
- (3) There must be means to warn the crew when any heater, the heat output of which is essential for safe operation, has been shut off by the automatic means prescribed in sub-paragraph (g)(1).
- (h) *Air intakes.* Each combustion and heat-ventilating air intake must be located so that no flammable fluids or vapours can enter the heater system:
 - (1) During normal operation; or
 - (2) As a result of the malfunction of any other component.
- (i) *Heater exhaust.* Each heater exhaust system must meet the requirements of [CS 27.1121](#) and [27.1123](#).
 - (1) Each exhaust shroud must be sealed so that no flammable fluids or hazardous quantities of vapours can reach the exhaust system through joints.
 - (2) No exhaust system may restrict the prompt relief of any backfire that, if so restricted, could cause heater failure.
- (j) *Heater fuel systems.* Each heater fuel system must meet the powerplant fuel system requirements affecting safe heater operation. Each heater fuel system component in the ventilating airstream must be protected by shrouds so that no leakage from those components can enter the ventilating airstream.
- (k) *Drains.* There must be means for safe drainage of any fuel that might accumulate in the combustion chamber or the heat exchanger.
 - (1) Each part of any drain that operates at high temperatures must be protected in the same manner as heater exhausts.
 - (2) Each drain must be protected against hazardous ice accumulation under any operating condition.

CS 27.861 Fire protection of structure, controls, and other parts

ED Decision 2003/15/RM

Each part of the structure, controls, rotor mechanism, and other parts essential to a controlled landing that would be affected by powerplant fires must be fireproof or protected so they can perform their essential functions for at least 5 minutes under any foreseeable powerplant fire conditions.

CS 27.863 Flammable fluid fire protection

ED Decision 2003/15/RM

- (a) In each area where flammable fluids or vapours might escape by leakage of a fluid system, there must be means to minimise the probability of ignition of the fluids and vapours, and the resultant hazards if ignition does occur.

- (b) Compliance with sub-paragraph (a) must be shown by analysis or tests, and the following factors must be considered:
 - (1) Possible sources and paths of fluid leakage, and means of detecting leakage.
 - (2) Flammability characteristics of fluids, including effects of any combustible or absorbing materials.
 - (3) Possible ignition sources, including electrical faults, over-heating of equipment, and malfunctioning of protective devices.
 - (4) Means available for controlling or extinguishing a fire, such as stopping flow of fluids, shutting down equipment, fireproof containment, or use of extinguishing agents.
 - (5) Ability of rotorcraft components that are critical to safety of flight to withstand fire and heat.
- (c) If action by the flight crew is required to prevent or counteract a fluid fire (e.g. equipment shutdown or actuation of a fire extinguisher) quick acting means must be provided to alert the crew.
- (d) Each area where flammable fluids or vapours might escape by leakage of a fluid system must be identified and defined.

EXTERNAL LOADS

CS 27.865 External loads

ED Decision 2018/007/R

- (a) It must be shown by analysis, test, or both, that the rotorcraft external-load attaching means for rotorcraft-load combinations to be used for non-human external cargo applications can withstand a limit static load equal to 2.5, or some lower load factor approved under [CS 27.337](#) through [27.341](#), multiplied by the maximum external load for which authorisation is requested. It must be shown by analysis, test, or both that the rotorcraft external-load attaching means and any complex personnel-carrying device system for rotorcraft-load combinations to be used for human external cargo applications can withstand a limit static load equal to 3.5 or some lower load factor, not less than 2.5, approved under [CS 27.337](#) through [27.341](#), multiplied by the maximum external load for which authorisation is requested. The load for any rotorcraft-load combination class, for any external cargo type, must be applied in the vertical direction. For jettisonable rotorcraft-load combinations, for any applicable external cargo type, the load must also be applied in any direction making the maximum angle with the vertical that can be achieved in service but not less than 30°. However, the 30° angle may be reduced to a lesser angle if:
- (1) An operating limitation is established limiting external load operations to those angles for which compliance with this paragraph has been shown; or
 - (2) It is shown that the lesser angle can not be exceeded in service.
- (b) The external-load attaching means, for jettisonable rotorcraft-load combinations, must include a quick-release system (QRS) to enable the pilot to release the external load quickly during flight. The QRS must consist of a primary quick-release subsystem and a backup quick-release subsystem that are isolated from one another. The QRS, and the means by which it is controlled, must comply with the following:
- (1) A control for the primary quick-release subsystem must be installed either on one of the pilot's primary controls or in an equivalently accessible location and must be designed and located so that it may be operated by either the pilot or a crew member without hazardously limiting the ability to control the rotorcraft during an emergency situation.
 - (2) A control for the backup quick-release subsystem, readily accessible to either the pilot or another crew member, must be provided.
 - (3) Both the primary and backup quick-release subsystems must:
 - (i) Be reliable, durable, and function properly with all external loads up to and including the maximum external limit load for which authorisation is requested.
 - (ii) Be protected against electromagnetic interference (EMI) from external and internal sources and against lightning to prevent inadvertent load release.
 - (A) The minimum level of protection required for jettisonable rotorcraft-load combinations used for non-human external cargo is a radio frequency field strength of 20 volts per metre.
 - (B) The minimum level of protection required for jettisonable rotorcraft-load combinations used for human external cargo is a radio frequency field strength of 200 volts per metre.

- (iii) Be protected against any failure that could be induced by a failure mode of any other electrical or mechanical rotorcraft system.
- (c) For rotorcraft-load combinations to be used for human external cargo applications, the rotorcraft must:
 - (1) For jettisonable external loads, have a QRS that meets the requirements of sub-paragraph (b) and that:
 - (i) Provides a dual actuation device for the primary quick-release subsystem, and
 - (ii) Provides a separate dual actuation device for the backup quick-release subsystem.
 - (2) Enable the safe utilisation of complex personnel-carrying device systems to transport occupants external to the helicopter or to restrain occupants inside the cabin. A personnel-carrying device system is considered complex if:
 - (i) it does not meet a European Norm (EN) standard under Directive 89/686/EEC¹ or Regulation (EU) 2016/425², as applicable, or subsequent revision;
 - (ii) it is designed to restrain more than a single person (e.g. a hoist or cargo hook operator, photographer, etc.) inside the cabin, or to restrain more than two persons outside the cabin; or
 - (iii) it is a rigid structure such as a cage, a platform or a basket.

Complex personnel-carrying device systems shall be reliable and have the structural capability and personnel safety features essential for external occupant safety through compliance with the specific requirements of CS 27.865, CS 27.571 and other relevant requirements of CS-27 for the proposed operating envelope.
 - (3) Have placards and markings at all appropriate locations that clearly state the essential system operating instructions and, for complex personnel-carrying device systems, ingress and egress instructions.
 - (4) Have equipment to allow direct intercommunication among required crew members and external occupants.
 - (5) Have the appropriate limitations and procedures incorporated in the flight manual for conducting human external cargo operations.
 - (6) For human external cargo applications requiring use of Category A rotorcraft, have one-engine-inoperative hover performance data and procedures in the flight manual for the weights, altitudes, and temperatures for which external load approval is requested.
- (d) The critically configured jettisonable external loads must be shown by a combination of analysis, ground tests, and flight tests to be both transportable and releasable throughout the approved operational envelope without hazard to the rotorcraft during normal flight conditions. In addition, these external loads must be shown to be releasable without hazard to the rotorcraft during emergency flight conditions.
- (e) A placard or marking must be installed next to the external-load attaching means stating the maximum authorised external load as demonstrated under [CS 27.25](#) and this paragraph.

¹ Council Directive 89/686/EEC of 21 December 1989 on the approximation of the laws of the Member States relating to personal protective equipment (OJ L 399, 30.12.1989, p. 18).

² Regulation (EU) 2016/425 of the European Parliament and of the Council of 9 March 2016 on personal protective equipment and repealing Council Directive 89/686/EEC (OJ L 81, 31.3.2016, p. 51).

- (f) The fatigue evaluation of [CS 27.571](#) does not apply rotorcraft-load combinations to be used for non-human external cargo except for the failure of critical structural elements that would result in a hazard to the rotorcraft. For rotorcraft-load combinations to be used for human external cargo, the fatigue evaluation of [CS 27.571](#) applies to the entire quick-release and complex personnel-carrying device structural systems and their attachments.

[Amdt No: 27/3]

[Amdt No: 27/5]

AMC 27.865 External Loads

ED Decision 2018/007/R

This AMC provides further guidance and acceptable means of compliance to supplement FAA AC 27-1B Change 4 AC 27.865B § 27.865 EXTERNAL LOADS to meet EASA's interpretation of [CS 27.865](#). As such, it should be used in conjunction with the FAA AC but should take precedence over it, where stipulated, in the showing of compliance.

[AMC No 1](#) addresses the differences between the operational requirements within the USA and those in Europe for Class D rotorcraft-load combinations for the carriage of human external cargo.

[AMC No 2](#) addresses the specificities of complex personnel-carrying device systems for human external cargo applications. This AMC provides further guidance and acceptable means of compliance to supplement FAA AC 27-1B Change 4 AC 27.865B § 27.865 (Amendment 27-36) EXTERNAL LOADS to meet EASA's interpretation of [CS 27.865](#).

[AMC No 3](#) contains a recognised approach to the approval of simple personnel-carrying device systems if required by the applicable operating rule or if an applicant elects to include simple personnel-carrying device systems within the scope of type certification.

[Amdt No: 27/5]

AMC No 1 to CS 27.865 Class D (Human External Cargo) for Operations within Europe

ED Decision 2018/007/R

1. Introduction

This additional EASA AMC, used in conjunction with FAA guidance¹ on Human External Cargo (HEC), provides an acceptable means of compliance with [CS 27.865](#) for rotorcraft intended for Class D Rotorcraft-Load Combinations (RLC) for the carriage of HEC. For all other RLC classes, reference should be made directly to the adopted FAA AC material.

The addition of this AMC has been necessary due to a difference in operational requirements within the USA and Europe and the absence of dedicated material within the FAA AC.

2. Basic Definition and Intended Use

A Class D RLC is one where personnel are at some point in the operation transported external to the rotorcraft, and the operator receives compensation from or on behalf of the person(s) being transported, e.g. transfer of personnel to/from a ship.

¹ See reference in [AMC 27 General](#)

3. Certification Considerations

Class D HEC was originally envisaged for Part 29/CS-29 rotorcraft only. However, CS-27 rotorcraft which have been shown to comply with the engine isolation specifications of CS-27 [Appendix C](#) are also eligible.

The rotorcraft must be certified for one-engine-inoperative/out-of-ground effect (OEI/OGE) hover performance weight, altitude and temperature envelope. This becomes the maximum envelope that can be used for Class D HEC operations.

4. Compliance Procedures

4.1 The rotorcraft is required to meet the Category A engine isolation specifications of CS-27 [Appendix C](#), and have an OEI/OGE hover performance capability in its approved, jettisonable HEC weight, altitude, and temperature envelope.

- (i) In determining OEI hover performance, dynamic engine failures should be considered. Each hover verification test should begin from a stabilised hover at the maximum OEI hover weight, at the requested in-ground-effect (IGE) or OGE skid or wheel height, and with all engines operating. At this point the critical engine should be failed and the aircraft should remain in a stabilized hover condition without exceeding any rotor limits or engine limits for the operating engine(s). As with all performance testing, engine power should be limited to minimum specification power. Engine failures may be simulated by rapidly moving the throttle to idle provided a 'needle split' is obtained between the rotor and engine RPM.
- (ii) Normal pilot reaction time should be used following the engine failure to maintain the stabilised hover flight condition. When hovering OGE or IGE at maximum OEI hover weight, an engine failure should not result in an altitude loss of more than 10 per cent or 4 feet, whichever is greater, of the altitude established at the time of engine failure. In either case, a sufficient power margin should be available from the operating engine(s) to regain the altitude lost during the dynamic engine failure and to transition to forward flight.
- (iii) Consideration should also be given to the time required to recover or manoeuvre the Class D external load and to transition into forward flight. For example, to winch up and bring aboard personnel in hoisting operations or manoeuvre clear of power lines for fixed strop/basket operations. The time necessary to perform such actions may exceed the short duration OEI power ratings. For example, for a helicopter with a 30-second/2-minute rating structure that sustains an engine failure at a height of 40 feet, the time required to re-stabilise in a hover, recover the external load (given the hoist speed limitations), and then transition to forward flight (with minimal altitude loss) would likely exceed 30 seconds and a power reduction into the 2-minute rating would be necessary.
- (iv) The rotorcraft flight manual (RFM) should contain information that describes the expected altitude loss, any special recovery techniques, and the time increment used for recovery of the external load when establishing maximum weights and wheel or skid heights. The OEI hover chart should be placed in the performance section of the RFM or RFM supplement. Allowable altitude extrapolation for the hover data should not exceed 2 000 feet.

4.2 For helicopters that incorporate engine-driven generators, the hoist should remain operational following an engine or generator failure. A hoist should not be powered from a bus that is automatically shed following the loss of an engine or generator. Maximum

two-engine generator loads should be established so that when one engine or generator fails, the remaining generator can assume the entire rotorcraft electrical load (including the maximum hoist electrical load) without exceeding approved limitations.

- 4.3 The external load attachment means and the complex personnel-carrying device should be shown to meet the specifications of [CS 27.865\(a\)](#) for the proposed operating envelope.
- 4.4 The rotorcraft is required to be equipped for, or otherwise allow, direct intercommunication under any operational conditions among crew members and the HEC. For Class D RLC operations, two-way radios or intercoms should be employed.

[Amdt No: 27/2]

[Amdt No: 27/5]

AMC No 2 to CS 27.865 External loads

ED Decision 2018/007/R

a. Explanation

- (1) This advisory material contains guidance for the certification of helicopter external-load attaching means and load-carrying systems to be used in conjunction with operating rules, such as Regulation (EU) No 965/2012 on Air Operations¹. The three RLC classes (and their eligibility for CS-27 use based on operational requirements) are summarised in Figure AMC 27.865-1 and discussed in paragraph d. Under the operating rules, RLC Classes A, B, and C are eligible, under specific restrictions, for both human external cargo (HEC) and non-human external cargo (NHEC) operations. However, under the operating rules, only Class D RLC is eligible for transporting HEC for remuneration (see Figure AC 27.865-1). Also, paragraph AC 27.25 (ref.: [CS 27.25](#)) also concerns, in part, jettisonable external cargo.
- (2) [CS 27.865](#) provides a minimum level of safety for small rotorcraft designs to be used with operating rules, such as Regulation (EU) No 965/2012 on Air Operations. Certain aspects of operations, such as microwave tower and high-line wirework, may also be regulated separately by other agencies or entities. For applications that could come under the regulations of more than one agency or entity, special certification emphasis will be required by both the applicant and the approving authority to assure all relevant safety requirements are identified and met. Potential additional requirements, where thought to exist, are noted herein.

b. Definitions

- (1) Applicable cargo type: the cargo type (i.e. non-human external cargo (NHEC), human external cargo (HEC), or both) that each RLC class is eligible to use by regulation.
- (2) Backup quick-release subsystem (BQRS): the secondary or 'second choice' subsystem used to perform a normal or emergency jettison of external cargo.
- (3) Cargo: the part of any rotorcraft-load combination that is removable, changeable, and is attached to the rotorcraft by an approved means.
- (4) Cargo hook: a hook that can be rated for both HEC and NHEC. It is typically used by being fixed directly to a designated hard point on the rotorcraft.

¹ Commission Regulation (EU) No 965/2012 of 5 October 2012 laying down technical requirements and administrative procedures related to air operations pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council (OJ L 296, 25.10.2012, p. 1).

- (5) Dual actuation device (DAD): this is a sequential control that requires two distinct actions in series for actuation. One example is the removal of a lock pin followed by the activation of a ‘then free’ switch or lever for load release to occur (in this scenario, a load release switch protected only by an uncovered switch guard is not acceptable). For jettisonable HEC applications, a simple, covered switch does not qualify as a DAD. Familiarity with covered switches allows the pilot to both open and activate the switch in one motion. This has led to inadvertent load release.
- (6) Emergency jettison (or complete load release): the intentional, instantaneous release of NHEC or HEC in a preset sequence by the quick-release system (QRS) that is normally performed to achieve safer aircraft operation in an emergency.
- (7) External fixture: a structure external to and in addition to the basic airframe that does not have true jettison capability and has no significant payload capability in addition to its own weight. An example is an agricultural spray boom. These configurations are not approvable as ‘External Loads’ under [CS 27.865](#).
- (8) Hoist: a hoist is a device that exerts a vertical pull, usually through a cable and drum system (i.e. a pull that does not typically exceed a 30-degree cone measured around the z-rotorcraft axis).
- (9) Hoist demonstration cycle (or ‘one cycle’): the complete extension and retraction of at least 95 % of the actual cable length, or 100 % of the cable length capable of being used in service (i.e. that would activate any extension or retraction limiting devices), whichever is greater.
- (10) Hoist load-speed combinations: some hoists are designed so that the extension and retraction speed slows as the load increases or nears the end of a cable extension. Other hoist designs maintain a constant speed as the load is varied. In the latter designs, the load-speed combination simply means the variation in load at the constant design speed of the hoist.
- (11) Human external cargo (HEC): a person (or persons) who, at some point in the operation, is (are) carried external to the rotorcraft. (Figure AC 27.865-1 contains explicit definitions for U.S. Part 133 Operations.) See non-human external cargo (NHEC).
- (12) Non-human external cargo (NHEC): any external cargo operation that does not at any time involve a person (or persons) carried external to the rotorcraft (Figure AC 27.865-1 contains explicit definitions for U.S. Part 133 Operations).
- (13) Normal jettison (or selective load release): the intentional release, normally at optimum jettison conditions, of NHEC.
- (14) Personnel-carrying device system (PCDS) is a device that has the structural capability and features needed to transport occupants external to the helicopter during HEC or helicopter hoist operations. A PCDS includes but is not limited to life safety harnesses (including, if applicable, a quick-release and stop with a connector ring), rigid baskets and cages that are either attached to a hoist or cargo hook or mounted to the rotorcraft airframe.
- (15) Primary quick-release subsystem (PQRS): the primary or ‘first choice’ subsystem used to perform a normal or emergency jettison of external cargo.
- (16) Quick-release system (QRS): the entire release system for jettisonable external cargo (i.e. the sum total of both the primary and backup quick-release subsystem). The QRS consists

of all the components including the controls, the release devices, and everything in between.

- (17) Rescue hook (or hook): a hook that can be rated for both HEC and NHEC. It is typically used in conjunction with a hoist or equivalent system.
- (18) Rotorcraft-load combination (RLC): the combination of a rotorcraft and an external load, including the external-load attaching means. RLCs are designated as Class A, Class B, Class C and Class D as follows:
 - (i) Class A RLC means one in which the external load cannot move freely, cannot be jettisoned, and does not extend below the landing gear.
 - (ii) Class B RLC means one in which the external load is jettisonable and is lifted free of land or water during the rotorcraft operation.
 - (iii) Class C RLC means one in which the external load is jettisonable and remains in contact with land or water during the rotorcraft operation.
 - (iv) Class D RLC means one in which the external load is other than a Class A, B or C and has been specifically approved by the relevant authority for that operation. Class D RLC operations are not allowed for Part 27 rotorcraft (Ref.: 133.45(e) restriction).
- (19) Spider: a spider is a system of attaching a lowering cable or rope or a harness to an NHEC (or HEC) RLC to eliminate undesirable flight dynamics during operations. A spider usually has four or more legs (or load paths) that connect to various points of a PCDS to equalise loading and prevent spinning, twisting, or other undesirable flight dynamics.
- (20) True jettison capability: the ability to safely release an external load using an approved QRS in 30 seconds or less.

NOTE: In all cases, a PQRS should release the external load in less than 5 seconds. Many PQRSSs will release the external load in milliseconds, once the activation device is triggered. However, a manual BQRS, such as a set of cable cutters, could take as much as 30 seconds to release the external load. The 30 seconds would be measured starting from the time the release command was given and ending when the external load was cut loose.
- (21) True payload capability: the ability of an external device or tank to carry a significant payload in addition to its own weight. If little or no payload can be carried, the external device or tank is an external fixture (see definition above).
- (22) Winch: a winch is a device that can employ a cable and drum or other means to exert a horizontal (i.e. x-rotorcraft axis) pull. However, since a winch can be used to perform a hoist function by use of a 90-degree cable direction change device (such as a pulley or pulley system), a winch system may be considered to be a hoist.

c. Procedures

The following certification procedures are provided in the most general form. Where there are significant differences between the cargo types, these differences are highlighted.

- (1) General Compliance Procedures for [CS 27.865](#): The applicant should clearly identify both the RLC and the applicable cargo types (NHEC or HEC) for which application is being made. The structural loads and operating envelopes for each RLC class and applicable cargo type should be determined and used to formulate the flight manual supplement and basic loads report. The applicant should show by analysis, test, or both, that the rotorcraft structure, the external-load attaching means, and the complex PCDS, if applicable, meet

the specific requirements of [CS 27.865](#) and any other relevant requirements of CS-27 for the proposed operating envelope.

NOTE: It is possible, if approved, to carry both HEC and NHEC externally, simultaneously as two separate external loads. However, in no case is it intended that the approved maximum internal gross weight should be exceeded for any approved HEC configuration (or combined NHEC and HEC configuration) in normal operations.

Reliability of the external load system. A failure of the external load system, including the complex PCDS where applicable, and its attachments to the rotorcraft should be shown to be extremely improbable (i.e. 1×10^{-9} failures per flight) for all failure modes that could cause a catastrophic failure, serious injury or a fatality anywhere in the total airborne system. All significant failure modes of lesser consequence should be shown to be improbable (i.e. 1×10^{-5} failures per flight). An acceptable method of achieving this goal is to submit the following for subsequent approval:

- (i) a failure modes and effects analysis (FMEA) showing that all potential failure modes of the airborne system that may result in catastrophic failures, serious injuries or fatalities are extremely improbable and any less significant failures are improbable;
- (ii) a repetitive test of all the functional devices that cycles these devices at least 30 times under critical structural conditions, operational conditions, or a combination of both;
- (iii) an environmental qualification review covering the proposed operating environment.

Figure AMC 27.865B-1

Rotorcraft-Load Combination Versus Applicable Cargo Type Data And Definition Summary				
Possible RLCs and cargo types		Category 'A' rating and one-engine-inoperative (OEI) hover capability	Notes	Direct two-way voice communications required See paragraph (d)(10)
HEC	RLC A	No	Note 2	No
NHEC	RLC A	No		N/A
HEC	RLC B	No	Note 2	No
NHEC	RLC B	No		N/A
HEC	RLC C	No	Note 2	No
NHEC	RLC C	N/A		N/A
HEC	RLC D	Note 1	Note 1	Note 1

NOTES:

- Class D RLC operations are not allowed for Part 27 rotorcraft (ref.: 133.45(e) restriction).
- A person (or persons) who is (are) not being carried or transported for remuneration is (are) knowledgeable of the risks involved, and at some point is (are) required to be outside the rotorcraft in order to fulfil the mission. This (these) person (persons) is (are) considered to be RLC Class A, B, or C HEC as appropriate to the operation.

- (2) [CS 27.865\(a\)](#) Static Structural Substantiation Procedures: The following static structural substantiation methods should be used:

- (i) Critical Basic Load Determination. The critical basic loads and corresponding flight envelope are determined by statically substantiating the gross weight range limits, the corresponding vertical limit load factors (N_{ZW}) and the safety factors applicable for the type of external load for which the application is being made.

NOTE: In cases where NHEC or HEC can have more than one shape, centre of gravity, centre of lift, or be carried at more than one distance in-flight from the rotorcraft attachment, a critical configuration for certification purposes may not be determinable. If such a critical configuration can be determined, it may be examined for approval as a ‘worst case’ to satisfy a particular certification criterion or several criteria, as appropriate. If such a critical configuration cannot be determined, the extreme points of the operational external load configuration envelope should be examined, with consideration given to any other points within the envelope that experience or any other rationale indicates as points that need to be investigated.

- (ii) Vertical Limit and Ultimate Load Factors. The basic N_{ZW} is converted to the ultimate load by multiplying the maximum vertical limit load by the appropriate safety factor (for restricted category approvals, see the guidance in paragraph AC 27 MG 5). This ultimate load is used to substantiate all the existing structure affected by, and all the added structure associated with, the load-carrying device, its attachments and its cargo. Casting factors, fitting factors, and other dynamic load factors should be applied where appropriate.

- (A) NHEC applications. In most cases, it is acceptable to perform a standard static analysis to show compliance. A vertical limit load factor (N_{ZW}) of 2.5 g is typical for heavy gross weight NHEC hauling configurations (ref.: [CS 27.337](#)). This vertical load factor should be applied to the maximum external load for which the application is being made, together with a minimum safety factor of 1.5.
- (B) HEC applications. If a safety factor of 3.0 or more is used, it is acceptable to perform a standard static analysis to show compliance. The safety factor should be applied to the yield strength of the weakest component in the system (QRS, complex PCDS, and attachment load path). If a safety factor of less than 3.0 is used, both an analysis and a full-scale ultimate load test of the relevant parts of the system should be performed.

Since HEC applications typically involve lower gross weight configurations, a higher vertical limit load factor is required to assure that the limit load is not exceeded in service. The applicant should use either the conservative value of 3.5 g or an analytically derived maximum vertical limit load factor for the requested operating envelope. Linear interpolation between the vertical load factors of the maximum and minimum design weights may be used. However, in no case may the vertical limit load factor be less than 2.5 g for any RLC application for HEC.

For the purpose of structural analysis or test, applicants should assume a 101.2-kg (223-pound) man as the minimum weight of each occupant carried as HEC.

NOTE: If the HEC is engaged in work tasks that employ devices of significant added weight (e.g. heavy backpacks, tools, fire extinguishers, etc.), the total weight of the 101.2-kg (223-pound) man and their equipment should be assumed in the structural analysis or test.

- (iii) Critical Structural Case. For applications involving more than one RLC class or cargo type, the structural substantiation is required only for the most critical case. The most critical case should be determined by rational analysis.
- (iv) Jettisonable Loads. For the substantiating analyses or tests of all jettisonable RLC external loads, including HEC, the maximum external load should be applied at the maximum angle that can be achieved in service, but not less than 30 degrees. The angle should be measured from the sling-load-line to the rotorcraft vertical axis (z axis) and may be in any direction that can be achieved in service. The 30-degree angle may be reduced in some or all directions if it is impossible to obtain due to physical constraints or operating limitations. The maximum allowable cable angle should be determined and approved. The angle approved should be based on structural requirements, mechanical interference limits, and flight-handling characteristics over the most critical conditions and combinations of conditions in the approved flight envelope.
- (v) Hoist System Limit Load.

NOTE: If a hoist cable or a long-line cable is utilised, a new dynamic system is established. The characteristics of the system should be evaluated to assure that either no hazardous failure modes exist or that they are acceptably minimised. For example, the hoist cable or long-line cable may exhibit a natural frequency that could be excited by sources internal to the overall structural system (i.e. the rotorcraft) or by sources external to the system. Another example is the loading effect of the cable acting as a spring between the rotorcraft and the suspended external load.

- (A) Determine the basic loads that would result in the failure or unspooling of the hoist or its installation, respectively.

NOTE: This determination should be based on static strength and any significant dynamic load magnification factors.

- (B) Select the lower of the two values as the ultimate load of the hoist system installation.
- (C) Divide the selected ultimate load by 1.5 to determine the true structural limit load of the system.
- (D) Determine the manufacturer's approved 'limit design safety factor' (or that which the applicant has applied for). Divide this factor into the true structural limit load (from (C) above) to determine the hoist system's working (or placarded) limit load.
- (E) Compare the system's derived limit load to that applied for one 'g' payload multiplied by the maximum downward vertical load factor (N_{ZWMAX}) to determine the critical payload's limit value.
- (F) The critical payload limit should be equal to or less than the system's derived limit load for the installation to be approvable.

- (3) [CS 27.865\(b\)](#) and CS 27.865(c) Procedures for Quick-Release Systems and Cargo Hooks: for jettisonable RLCs of any applicable cargo type, both a primary quick-release system (PQRS) and a backup quick-release system (BQRS) are required. Features that should be considered are:
- (i) The PQRS, BQRS and their load-release devices and subsystems (such as electronically actuated guillotines) should be separate (i.e. physically, systematically, and functionally redundant).
 - (ii) The controls for the PQRS should be installed on one of the pilot's primary controls, or in an equivalently accessible location. The use of an 'equivalent accessible location' should be reviewed on a case-by-case basis and utilised only where equivalent safety is clearly maintained.
 - (iii) The controls for the BQRS may be less sophisticated than those of the PQRS. For instance, manual cable cutters are acceptable provided they are listed in the flight manual as a required device and have a dedicated, placarded storage location.
 - (iv) The PQRS should release the external load in less than 5 seconds. The BQRS should release the external load in less than 30 seconds. This time interval begins the moment an emergency is declared and ends when the load is released.
 - (v) Each quick-release device should be designed and located to allow the pilot or a crew member to accomplish the release of the external cargo release without hazardously limiting the ability to control the rotorcraft during emergency situations. The flight manual should reflect the requirement for a crew member and their related functions.
 - (vi) Other Load Release Types. In some current configurations, such as those used for high line operations, a load release may be present that is not on the rotorcraft but is on the complex PCDS itself. Examples are a tension release device that lets out line under an operationally induced load or a personal rope cutter. These devices are acceptable if:
 - (A) the off-rotorcraft release is considered to be a 'third release'. This type of release is not a substitute for a required release (i.e. PQRS or BQRS);
 - (B) the release meets all other relevant requirements of [CS 27.865](#) and the methods of this AMC or equivalent methods; and
 - (C) the release has no operational or failure modes that would affect continued safe flight and landing under any operations, critical failure modes, conditions, or combination of either.
 - (vii) Cargo Hooks or Equivalent Devices and their Related Systems. All cargo hooks or equivalent devices should be approved to acceptable aircraft industry standards. The applicant should present these standards, and any related manufacturer's certificates of production or qualification, as part of the approval package.
 - (A) General. Cargo hook systems should have the same reliability goals and should be functionally demonstrated under the critical loads for NHEC and HEC, as appropriate. All engagement and release modes should be demonstrated. If the hook is used as a quick-release device, then the release of critical loads should be demonstrated under conditions that simulate the maximum allowable bank angles and speeds and any other critical operating conditions. Demonstration of any re-latching features and any safety or

warning devices should also be conducted. Demonstration of actual in-flight emergency quick-release capability may not be necessary if the quick-release capability can be acceptably simulated by other means.

NOTE 1: Cargo hook manufacturers specify particular shapes, sizes, and cross sections for lifting eyes to assure compatibility with their hook design (e.g. Breeze Eastern Service Bulletin CAB-100-41). Experience has shown that, under certain conditions, a load may inadvertently hang up because of improper geometry at the hook-to-eye interface that will not allow the eye to slide off an open hook as intended.

NOTE 2: For both NHEC and HEC designs, the phenomenon of hook dynamic roll-out (inadvertent opening of the hook latch and subsequent release of the load) should be considered to assure that QRS reliability goals are not compromised. This is of particular concern for HEC applications. Hook dynamic roll-out occurs during certain ground-handling and flight conditions that may allow the lifting eye to work its way out of the hook.

Hook dynamic roll-out typically occurs when either the RLC's sling or harness is not properly attached to the hook, is blown by down draft, is dragged along the ground or through water, or is otherwise placed into a dangerous hook-to-eye configuration.

The potential for hook dynamic roll-out can be minimised in design by specifying particular hook-and-eye shape and cross-section combinations. For non-jettisonable RLCs, a pin can be used to lock the hook-keeper in place during operations.

NOTE: Some cargo hook systems may employ two or more cargo hooks for safety. These systems are approvable. However, a loss of any load by a single hook should be shown to not result in a loss of control of the rotorcraft. In a dual hook system, if the hook itself is the quick-release device (i.e. if a single release point does not exist in the load path between the rotorcraft and the dual hooks), the pilot should have a dual PQRS that includes selectable, co-located individual quick releases that are independent for each hook used. A BQRS should also be present for each hook. For cargo hook systems with more than two hooks, either a single release point should be present in the load path between the rotorcraft and the multiple hook system, or multiple PQRSs and BQRSs should be present.

- (B) Jettisonable Cargo Hook Systems. For jettisonable applications, each cargo hook:
- (1) should have a sufficient amount of slack in the control cable to permit cargo hook movement without tripping the hook release;
 - (2) should be shown to be reliable (see paragraph c(1));
 - (3) for HEC systems, unless the cargo hook is to be the primary quick-release device, each cargo hook should be designed so that operationally induced loads cannot inadvertently release the load. For example, a simple cargo hook should have a one-way, spring-loaded gate (i.e. 'snap hook') that allows load attachment going into the gate but does not allow the gate to open (and subsequently lose the HEC) when an operationally induced load is applied in the opposite

direction. For HEC applications, cargo hooks that also serve as quick-release devices should be carefully reviewed to assure they are reliable.

- (4) CS 27.865(b)(3) Reliability Determination for QRSs and Devices: QRSs are required to be reliable. The primary electrical and mechanical failure modes that should be identified and minimised are: (1) load release by any means, and (2) loss of continued safe flight and landing capability due to a QRS failure. However, any failure that could result in catastrophic failure modes, serious injuries or fatalities should also be identified and shown to be extremely improbable. All other failure modes should be shown to be improbable. The reliability of each QRS system should be demonstrated by completion and approval of all of the following:
- (i) An FMEA showing that all potential failure modes of the QRS which may result in catastrophic failures, serious injuries or fatalities are extremely improbable and any less-significant failures are improbable.
 - (ii) A repetitive test of all functioning devices that affect or comprise the QRS, which tests all the critical conditions or combinations of critical conditions at least 10 times each for NHEC and 30 times each for HEC, using both the primary and backup quick-release subsystems.
 - (iii) An environmental qualification programme that includes consideration of high and low temperatures (typically – 40 °C (– 40 °F) to + 65.6 °C (+ 150 °F)), altitudes up to 12 000 feet, humidity, salt spray, sand and dust, vibration, shock, rain, fungus, and acceleration. Testing should be conducted in accordance with RTCA/DO-160 or MIL-STD-810 for high- and low-temperature tests and for vibrations.
 - (iv) Using the methods of compliance in other relevant paragraphs of AC 27-1B included where supplemented and amended by CS-27 Book 2 or equivalent methods.
- (5) Functional Reliability and Durability Compliance Procedures for Hoist Systems under CS 27.865(b)(3)(i) and (c)(2): hoist systems and their installations in the rotorcraft should be designed, approved, and demonstrated as follows:
- (i) Reserved
 - (ii) Reserved
 - (iii) It is assumed that only one hoist cycle will typically occur per flight. This rationale has been used to determine the requirement for 10 demonstration cycles for NHEC applications and 30 demonstration cycles for HEC applications. However, if a particular application requires more than one hoist cycle per flight, then the number of demonstration cycles should be increased accordingly.
 - (iv) The hoist or rescue hook system should be reliable for the phases of flight in which it is operable, unstowed, partially unstowed, or in which cargo is carried. The hoist should be disabled (or an overriding, fail-safe mechanical safety device such as either a flagged removable shear pin or a load-lowering brake should be utilised) to prevent inadvertent load unspooling or release during any extended flight phases in which hoist operation is not intended. Loss of hoist operational control should also be considered. The reliability of the system should be demonstrated by completion and approval of all of the following:

- (A) An FMEA showing that all potential failure modes of the hoist or rescue hook system which may result in catastrophic failures, serious injuries or fatalities are extremely improbable and any less-significant failures are improbable.
- (B) Unless a more rational test method is presented and approved, at least 10 repetitive tests of all functional devices, which exercise the entire system's functional parameters, should be conducted. These repetitive tests may be conducted on the rotorcraft, or by using a bench simulation that accurately replicates the rotorcraft installation.
- (C) A hoist unit environmental qualification programme that includes consideration of high and low temperatures (typically – 40 °C (– 40 °F) to + 65.6 °C (+ 150 °F)), altitudes up to 12 000 feet, humidity, salt spray, sand and dust, vibration, shock, rain, fungus, and acceleration. Testing in accordance with RTCA/DO-160 or MIL-STD-810 for high- and low-temperature tests and for vibrations. Hoist manufacturers should submit a test plan and follow-on test reports to the applicant and the authority following the completion of the qualification. It is intended that the hoist itself either be prequalified to the EMI and lightning threat levels specified for NHEC or HEC, as applicable for the requested operation, or that it be qualified as part of the entire on-board QRS to these threat levels.
- (D) All instructions and documents necessary for continued airworthiness, normal operations, and emergency operations (see paragraph c(17)).
- (v) Cable Attachment. Either the cable should be positively attached to the hoist drum and the attachment should have ultimate load capability, or equivalent means should be provided to minimise the possibility of inadvertent, complete cable unspooling.
- (vi) Cable Length and Marking. A length of the cable nearest to the cable's attachment to the hoist drum should be visually marked to indicate to the operator that the cable is near to its full extension. The length of cable to be marked is a function of the maximum extension speed of the system and the operator's reaction time needed to prevent cable run-out. It should be determined during certification demonstration tests. In no case should the length be less than 3.5 drum circumferences.
- (vii) Cable Stops. Means should be present to automatically stop cable movement quickly when the system's extension and retraction operational limits are reached.
- (viii) Hoist System Load-Speed Combination Ground Tests. The load versus speed combinations of the hoist should be demonstrated on the ground (either using an accurate engineering mock-up or a rotorcraft) by showing the repeatability of the no load-speed combination, the 50 per cent load-speed combination, the 75 per cent load-speed combination, and the 100 per cent (i.e. system-rated limit) load-speed combination. If more than one operational speed range exists, the preceding tests should be performed at either all speeds or at the most critical speed.
 - (A) At least 1/10 of the demonstration cycles (see definition) should include the maximum aft angular displacement of the load from the drum, applied for under [CS 27.865\(a\)](#).

- (B) A minimum of 6 consecutive, complete operation cycles should be conducted at the system's 100 per cent (i.e. system limit rated) load-speed combination.
- (C) In addition, the demonstration should cover all normal and emergency modes of intended operation and should include operation of all control devices such as limit switches, braking devices, and overload sensors in the system.
- (D) All quick-release devices and cable cutters should be demonstrated at 0, 25, 50, 75 and 100 per cent of the system limit load or at the most critical percentage value.

NOTE: Some hoist designs have built-in cable-tensioning devices that function at the no load-speed combination, as well as at other load-speed combinations. These devices should be shown to work during the no load-speed and other load-speed cable-cutting demonstrations.

- (E) All electrical and mechanical systems and load-release devices for any jettisonable NHEC or HEC RLC should be shown to be reliable by both analysis and testing.
- (F) Any devices or methods used to increase the mechanical advantage of the hoist should also be demonstrated.
- (G) During a portion of each demonstration cycle, the hoist should be operated from each station from which it can be controlled.

NOTE: A reasonable amount of starting and stopping during demonstration cycles is acceptable.

- (ix) Hoist System Continued Airworthiness. The design life of the hoist system and any life-limited components should be clearly identified, and the Airworthiness Limitations Section of the maintenance manual should include these requirements. For STCs, a maintenance manual supplement should be provided that includes these requirements.

NOTE: Design lives of hoist and cable systems are typically between 5 000 and 8 000 cycles. Some hoist systems have usage time meters installed. Others may have cycle counters installed. Cycle counters should be considered for HEC operations and high-load or other operations that may cause low-cycle fatigue failures.

- (x) Hoist System Flight Tests. An in-flight demonstration test of the hoist system should be conducted for helicopters designed to carry NHEC or HEC. The rotorcraft should be flown to the extremes of the applicable manoeuvre flight envelope and to all conditions that are critical to strength, manoeuvrability, stability, and control, or any other factor affecting airworthiness. Unless a lesser load is determined to be more critical for either dynamic stability or other reasons, the maximum hoist system rated load or, if less, the maximum load requested for approval (and the associated limit load data placards) should be used for these tests. The minimum hoist system load (or zero load) should also be demonstrated in these tests.

- (6) CS 27.865(b)(3)(ii) Electromagnetic Interference: protection of the QRS against potential internal and external sources of electromagnetic interference (EMI) and lightning is

required. This is necessary to prevent inadvertent load releases from sources such as lightning strikes, stray electromagnetic signals, and static electricity.

- (i) Jettisonable NHEC systems should be able to absorb a minimum of 20 volts per metre (i.e. CAT U) radio frequency (RF) field strength per RTCA/DO-160.
- (ii) Jettisonable HEC systems should be able to absorb a minimum of 200 volts per metre (i.e. CAT Y) RF field strength per RTCA/DO-160.

NOTE 1: These RF field threat levels may need to be increased for certain special applications such as microwave tower and high-voltage high line repairs. Separate criteria for special applications under the regulations of more than one agency or entity (such as the Institute of Electrical and Electronics Engineers (IEEE) or Occupational Safety and Health Administration (OSHA) standards) should also be addressed, as applicable, during certification. When necessary, the issue paper process can be used to establish a practicable level of safety for specific high-voltage or other special application conditions. For any devices or means added to meet the regulations of more than one agency or entity, their failure modes should not have an adverse effect on flight safety. Other certification authorities may require higher RF field threat levels than those required by CS 27.865 (e.g. CS-27 Appendix E).

NOTE 2: An approved standard rotorcraft test that includes the full HIRF frequency and amplitude external and internal environments on the QRS and complex PCDS (or the entire rotorcraft including the QRS and complex PCDS) could be substituted for the jettisonable NHEC and HEC systems tests defined by c(6)(i) and c(6)(ii) respectively, as long as the RF field strengths directly on the QRS and complex PCDS are shown to equal or exceed those of c(6)(i) and c(6)(ii).

NOTE 3: The EMI levels specified in c(6)(i) and c(6)(ii) are total EMI levels to be applied to the QRS (and affected QRS component) boundary. The total EMI level applied should include the effects of both external and internal EMI sources. All aspects of internally generated EMI should be carefully considered including peaks that could occur from time to time due to any combination of on-board systems being operated. For example, special attention should be given to EMI from hoist operations that involve the switching of very high currents. Those currents can generate significant voltages in closely spaced wiring that, if allowed to reach some squib designs, could activate the device. Shielding, bonding and grounding of wiring associated with the operation of the hoist and the quick-release mechanism should be clearly and adequately evaluated during design and certification. This evaluation may require testing. One acceptable test method to demonstrate the adequacy of QRS shielding, bonding and grounding would be to actuate the hoist under maximum load together with likely critical combinations of other aircraft electrical loads and demonstrate that the test squibs (which are more EMI sensitive than the squibs specified for use in the QRS) do not inadvertently operate during the test.

- (7) CS 27.865(c)(1) QRS Requirements for Jettisonable HEC Operations: For jettisonable HEC operations, both the PQRS and BQRS are required to have a dual actuation device (DAD) for external cargo release. Two distinct actions are required to minimise inadvertent jettison of HEC. The DAD is intended for emergency use during the phases of flight that the HEC is carried or retrieved. The DAD can be used for both NHEC and HEC operations. However, because it can be used for HEC, the Instructions for Continued Airworthiness should be carefully reviewed and documented. The DAD can be operated by the pilot from a primary control or, after a command is given by the pilot, by a crew member from

a remote location. If the backup DAD is a cable cutter, it should be properly secured, placarded and readily accessible to the crew member intended to use it.

- (8) CS 27.865(c)(2) PCDS: for all HEC applications that use complex PCDSs, an approval is required. The complex PCDS may be either previously approved or is required to be approved during certification. In either case, its installation should be approved. The failure of the complex PCDS, and its attachments to the rotorcraft, should be shown to be extremely improbable (i.e. 1×10^{-9} failures per flight) for all failure modes that could cause a catastrophic failure, serious injury or fatality. All significant failure modes of lesser consequence should be shown to be improbable (i.e. 1×10^{-5} failures per flight). An acceptable method of achieving this goal is to submit the following for subsequent approval:

- (i) an FMEA showing that all the potential failure modes of the complex PCDS that may result in catastrophic failures, serious injury or fatality, are extremely improbable and any less-significant failures are improbable;
- (ii) a repetitive test of all functional devices that cycles these devices at least 30 times under critical structural conditions, operational conditions, or a combination;
- (iii) an environmental qualification review of the proposed operating environment.

NOTE: Complex PCDS designs can include relatively complex devices such as multiple occupant cages or gondolas. The purpose of the complex PCDS is to provide a minimum acceptable level of safety for personnel being transported outside the rotorcraft. The personnel being transported may be healthy or injured, conscious or unconscious.

- (iv) Regulation (EU) No 965/2012 on Air Operations contains the minimum performance specifications and standards for simple PCDSs, such as HEC body harnesses.
- (v) Static Strength. The complex PCDS should be substantiated for the allowable ultimate load and loading conditions as determined under paragraph c(2).
- (vi) Fatigue. CS 27.865(f) requires the metallic components of the complex PCDSs to be substantiated for fatigue in accordance with [CS 27.571](#) (ref.: c(14)).
- (vii) Personnel Safety. For each complex PCDS design, the applicant should submit a design evaluation that assures the necessary level of personnel safety is provided. As a minimum, the following should be evaluated:
 - (A) The complex PCDS should be easily and readily entered or exited.
 - (B) It should be placarded with its proper capacity, the internal arrangement and location of occupants, and ingress and egress instructions.
 - (C) For door latch fail-safety, more than one fastener or closure device should be used. The latch device design should provide direct visual inspectability to assure it is fastened and secured.
 - (D) Any fabric used should be durable and should be at least flame-resistant.
 - (E) Reserved
 - (F) Occupant retention devices and the related design safety features should be used as necessary. In simple designs, rounded corners and edges with adequate strapping (or other means of HEC retention relative to the complex

PCDS) and head supports or pads may be all the safety features that are necessary. Complex PCDS designs may require safety features such as seat belts, handholds, shoulder harnesses, placards, or other personnel safety standards.

- (viii) EMI and Lightning Protection. All essential, affected components of the complex PCDS, such as intercommunication equipment, should be protected against RF field strengths to a minimum of RTCA/DO-160 CAT Y.
 - (ix) Instructions for Continued Airworthiness. All instructions and documents necessary for continued airworthiness, normal operations and emergency operations should be completed, reviewed and approved during the certification process (see paragraph c(17)).
 - (x) Flotation Devices. Complex PCDSs that are intended to have a dual role as flotation devices or life preservers should meet the relevant requirements for 'Life Preservers.' Also, any PCDS design to be used in the water should have a flotation kit. The flotation kit should support the weight of the maximum number of occupants and the complex PCDS in the water and minimise the possibility of the occupants floating face down.
 - (xi) Aerodynamic Considerations. Some complex PCDS designs may spin, twist or otherwise respond unacceptably in flight. Each of these designs should be structurally restrained with a device such as a spider, a harness, or an equivalent device to minimise undesirable flight dynamics.
 - (xii) Medical Design Considerations. Complex PCDSs should be designed to the maximum practicable extent and placarded to maximise the HEC's protection from medical considerations such as blocked air passages induced by improper body configurations and excessive losses of body heat during operations. Injured or water-soaked persons may be exposed to high body heat losses from sources such as rotor washes and the airstreams. The safety of occupants of complex PCDSs from transit-induced medical considerations can be greatly increased by proper design.
- (9) CS 27.865(c)(3) QRS Design, Installation and Placarding: for jettisonable HEC applications, the QRS design, installation and associated placarding should be given special consideration to assure the proper level of occupant safety.
 - (10) CS 27.865(c)(4) Intercom Systems for HEC Operations: for all HEC operations, the rotorcraft is required to be equipped for, or otherwise allow, direct intercommunication under any operational conditions among crew members and the HEC. For some systems, voice or hand signals to PCDS occupants may be acceptable. For other systems, more sophisticated devices such as two-way radios or intercoms should be employed.
 - (11) CS 27.865(c)(5) Flight Manual Procedures: appropriate flight manual procedures and limitations for all HEC operations should be presented. All limitations are required to be approved for all RLCs of Class A, B, or C that employ HEC. The flight manual should clearly define the method of communication between the flight crew and the HEC. These instructions and manuals should be validated during flight testing.
 - (12) CS 27.865(d) Flight Test Verification Work: flight test verification work (or an equivalent combination of analysis and ground testing, either in conjunction with or in addition to operating rules, such as Regulation (EU) No 965/2012 on Air Operations) that thoroughly examines the operational envelope should be conducted with the external cargo carriage

device for which approval is requested (especially those that involve HEC). The flight test programme should show that all aspects of the operations applied for are safe, uncomplicated, and can be conducted by a qualified flight crew under the most critical service environment and, in the case of HEC, under emergency conditions. Flight tests should be conducted for the simulated representative NHEC and HEC loads to demonstrate their in-flight handling and separation characteristics. Each placard, marking and flight manual supplement should be validated during flight testing.

- (i) General. Flight testing (or an equivalent combination of analysis and testing) should be conducted under the critical combinations of configurations and operating conditions for which basic type certification approval is sought. Additional combinations of external loads and operating conditions may be subsequently approved under the relevant operational requirements as long as the structural limits and reliability considerations of the basic certification approval are not exceeded (i.e. equivalent safety is maintained). The qualification flight test work of this subparagraph is intended to be accomplished primarily by analysis or bench testing. However, at least one in-flight limit load drop test should be conducted for the critical load case. If one critical load case cannot be clearly identified, then more than one drop test might be necessary. Also, in-flight tests for the minimum load case (i.e. typically the cable hook itself) with the load trailing both in the minimum and maximum cable length configurations should be conducted. Any safety-of-flight limitations should be documented and placed in the rotorcraft flight manual (RFM). In certain low gross weight, jettisonable HEC configurations, the complex PCDS may act as a trailing aerofoil that could result in entangling the complex PCDS and the rotorcraft. These configurations should be assessed on a case-by-case basis by analysis or flight test to assure that any safety-of-flight limitations are clearly identified and placed in the RFM.
- (ii) Separation Characteristics of Jettisonable External Loads. For all jettisonable RLCs of any applicable cargo type, the satisfactory post-jettison separation characteristics of all loads should meet the following minimum criteria:
 - (A) Immediate ‘clean’ operation of the QRS, including ‘clean’ separate functioning of the PQRS and BQRS.
 - (B) No damage to the helicopter during or following actuation of the QRS and load jettisoning.
 - (C) A jettison trajectory clear of the helicopter.
 - (D) No inherent instability of the jettisonable (or just jettisoned) HEC or NHEC while in proximity to the helicopter.
 - (E) No adverse or uncontrollable helicopter reactions at the time of jettison.
 - (F) Stability and control characteristics after jettison should be within the originally approved limits.
 - (G) No unacceptable degradation of the helicopter performance characteristics after jettison.
- (iii) Jettison Requirements for Jettisonable External Loads. For representative cargo types (low-, medium- and high-density loads on long and short lines), emergency and normal jettison procedures should be demonstrated (by a combination of

- analysis, ground tests, and flight tests) at sufficient combinations of flight conditions to establish a jettison envelope which should be placed in the RFM.
- (iv) QRS Demonstration. Repetitive jettison demonstrations should be conducted that use the PQRS. The BQRS should be utilised at least once.
 - (v) QRS Reliability (i.e. failure modes) Affecting Flight Performance. The FMEA of the QRS (ref.: c(4)) should show that no single system failure will result in unsatisfactory flight characteristics, including any QRS failures that result in asymmetric loading conditions.
 - (vi) Flight Test Weight and CG Locations. All flight tests should be conducted at the extreme or critical combinations of weight and longitudinal and lateral CG conditions within the flight envelope that is applied for. The rotorcraft should remain within the approved weight and CG limits both with the external load applied and after jettison of the load.
 - (vii) Jettison Envelopes. Emergency and normal jettison demonstrations should be performed at sufficient airspeeds and decent rates to establish any restrictions for satisfactory separation characteristics. Both the maximum and minimum airspeed limits and maximum descent rate for safe separation should be determined. The sideslip envelope as a function of airspeed should be determined.
 - (viii) Altitude. Emergency and normal jettison demonstrations should be performed at altitudes consistent with the approvable operational envelope and with the manoeuvring requirements necessary to overcome any adverse effects of the jettison.
 - (ix) Attitude. Emergency and normal jettison demonstrations should be performed from all attitudes appropriate to normal and emergency operational usage. Where the attitudes of HEC or NHEC with respect to the helicopter may vary the most critical attitude should be demonstrated. This demonstration would normally be accomplished by bench testing.
 - (x) Hoist and Rescue Hook Systems and Cargo Hook Systems. An in-flight demonstration test of the hoist system should be conducted for helicopters designed to carry NHEC or HEC. The rotorcraft should be flown to the extremes of the applicable manoeuvre flight envelope and to all conditions that are critical to strength, manoeuvrability, stability, and control, or any other factor affecting its airworthiness. Unless a lesser load is determined to be more critical for either dynamic stability or other reasons, the maximum hoist system rated load or, if less, the maximum load requested for approval (and the associated limit load data placards) should be used for these tests. The minimum hoist system load (or zero load) should also be demonstrated in these tests.
- (13) CS 27.865(e) External Loads Placards and Markings: placards and markings should be installed next to the external-load attaching means, in a clearly noticeable location, that state the primary operational limitations — specifically including the maximum authorised external load. Not all operational limitations need be stated on the placard (or equivalent markings); only those that are clearly necessary for immediate reference in operations. Other more detailed operational limitations of lesser immediate importance should be stated either directly in the RFM or in an RFM supplement.

- (14) CS 27.865(f) Fatigue Substantiation: the fatigue evaluation of [CS 27.571](#) should be applied as follows:

NOTE: The term ‘hazard to the rotorcraft’ is defined to include all hazards to either the rotorcraft, to the occupants thereof, or both.

- (i) Fatigue Evaluation of NHEC Applications. Any critical components of the suspended system and their attachments (such as the cargo hook or bolted or pinned truss attachments), the failure of which could result in a hazard to the rotorcraft, should undergo an acceptable fatigue analysis in accordance with AC 27 MG 11, paragraph e.
 - (ii) Fatigue Evaluation of HEC Applications. The entire complex PCDS and its attachments should be reviewed on a component-by-component basis to determine which components, if any, are fatigue-critical or damage-intolerant. These components should be analysed or tested (per AC 27 MG 11, AC 27 MG 11, or other equivalent methods) to assure their fatigue life limits are properly determined and placed in the limited life section of the maintenance manual.
- (15) Other Considerations
- (i) Agricultural Installations (AIs): AIs can be approved for either jettisonable or non-jettisonable NHEC or HEC operations as long as they meet relevant certification and operations requirements and follow appropriate compliance methods. However, most current AI designs are external fixtures (see definition), not external loads. External fixtures are not approvable as jettisonable external cargo because they do not have a true payload (see definition), true jettison capability (see definition), or a complete QRS. Many AI designs can dump their solid or liquid chemical loads by use of a ‘purge port’ release over a relatively long time period (i.e. greater than 30 seconds). This is not considered to be a true jettison capability (see definition) since the external load is not released by a QRS and since the release time span is typically greater than 30 seconds (ref.: b(20) and c(7)). Thus, these types of AIs should be approved as non-jettisonable external loads. However, other designs that have the entire AI (or significant portions thereof) attached to the rotorcraft, that have short time frame jettison (or release) capabilities provided by QRSs that meet the definitions herein and that have no post-jettison characteristics that would endanger continued safe flight and landing may be approved as jettisonable external loads. For example, if all the relevant criteria are properly met, a jettisonable fluid load can be approved as an NHEC external cargo. AC 27 MG 5 discusses other AI certification methodologies.
 - (ii) External Tanks: external tank configurations that have true payload (see definition) and true jettison capabilities (see definition) should be approved as jettisonable NHEC. External tank configurations that have true payload capabilities but do not have true jettison capabilities should be approved as non-jettisonable NHEC. An external tank that has neither a true payload capability nor true jettison capability is an external fixture; it should not be approved as an external load under [CS 27.865](#). If an external tank is to be jettisoned in flight, it should have a QRS that is approved for the maximum jettisonable external tank payload and is either inoperable or is otherwise rendered reliable to minimise inadvertent jettisons above the maximum jettisonable external tank payload.
 - (iii) Logging Operations: These operations are very susceptible to low-cycle fatigue because of the large loads and relatively high load cycles that are common to this

industry. It is recommended that load-measuring devices (such as load cells) be used to assure that no unrecorded overloads occur and to assure that cycles producing high fatigue damage are properly considered. Cycle counters are recommended to assure that acceptable cumulative fatigue damage levels are identifiable and are not exceeded. As either a supplementary method or an alternate method, maintenance instructions should be considered to assure proper cycle counting and load recording during operations.

- (16) Reserved
- (17) Instructions for Continued Airworthiness. Maintenance manuals (and RFM supplements) developed by applicants for external load applications should be presented for approval and should include all appropriate inspection and maintenance procedures. The applicant should provide sufficient data and other information to establish the frequency, extent, and methods of inspection of critical structures, systems and components. This information is required by [CS 27.1529](#) to be included in the maintenance manual. For example, maintenance requirements for sensitive QRS squibs should be carefully determined, documented, approved during certification, and included as specific mandatory scheduled maintenance requirements that may require either 'daily' or 'pre-flight' checks (especially for HEC applications).

[Amdt No: 27/5]

AMC No 3 27.865 External loads operations using simple personnel-carrying device systems

ED Decision 2018/007/R

If required by the applicable operating rule or if an applicant elects to, this AMC provides a means of compliance for the airworthiness certification of a simple personnel-carrying device system (PCDS) and attaching means to the hook, providing safety factors and consideration of calendar life replacement limits in lieu of a dedicated fatigue analysis and test.

A PCDS is considered to be simple if:

- (a) it meets an EN standard under Directive 89/686/EEC, or Regulation (EU) 2016/425, as applicable, or subsequent revision;
- (b) it is designed to restrain no more than a single person (e.g. hoist or cargo hook operator, photographer, etc.) inside the cabin, or to restrain no more than two persons outside the cabin;
- (c) it is not a rigid structure such as a cage, a platform or a basket.

PCDSs that cannot be considered to be simple are considered to be complex.

Note 1: EASA or the relevant Authority should be contacted to confirm the classification in the event that:

- a PCDS includes new or novel features;
- a PCDS has not been proven by appreciable and satisfactory service experience; or
- there is any doubt in the classification.

Approval of Simple PCDSs

If the approval of a simple PCDS is requested, then Directive 89/686/EEC, or Regulation (EU) 2016/425 or subsequent revision are an acceptable basis for the certification of a simple PCDS provided that:

- (a) the applicable Directive 89/686/EEC, or Regulation (EU) 2016/425, as applicable, or subsequent revision and corresponding EN standards for the respective components are complied with (EC Type Examination Certificate);
- (b) the applicant for the minor change has obtained from the manufacturer and keeps on record the applicable EC Conformity Certificate(s).

Note 2: A simple PCDS has an EC Type Examination Certificate (similar to an STC), issued by a Notified Certification Body and, for the production and marketing, an EC Conformity Certificate (similar to an EASA Form 1) issued by the manufacturer.

Note 3: In cases where ropes or elements connect simple PCDSs to the hoist/cargo hook or internal helicopter cabin, the EN certification can be achieved by a body meeting the transposition into national law of the applicable EC/EU regulation.

The EC-certified components are appropriately qualified for the intended use and the environmental conditions.

Note 4: The intended use and corresponding risks must be considered when selecting EN standards. For example hoist operators and rescuers that have to work at the edge of the cabin or outside should have full body harnesses to address the risk of inversion. Litters and the corresponding restraint systems should be adequately designed for the loads that can be generated during spinning.

Note 5: The assembly of the different components should also consider the intended use. For example the attachment of the tethering strap to the harness of a hoist operator should be of a quick-release type to allow quick detachment from the aircraft following a ditching or emergency landing. The tethering strap should also be adjustable to take up slack and avoid shock loads being transmitted to other components.

- (c) The maximum load applied to each component between the HEC and the hook is conservatively estimated. This is particularly important when more than one person is attached by a single system to the cargo hook/ hoist. Appendix 1 defines the appropriate minimum ultimate load (UL_{min}). If UL_{min} is above the static strength currently declared by the supplier of the PCDS or of a component of the attachments, through compliance with an EN standard, then proof of sufficient strength is to be provided by static tests. All possible service load cases (including asymmetric load distribution) are to be considered. In this case, the PCDS and/or the attaching means (e.g. rope, carabineer, shackles, etc.) must be capable of supporting UL_{min} for a minimum of 3 minutes without failure. There should be no deformation of components that could allow the release of the HEC. Components and details added to the EN-approved equipment (such as splicing, knots, stitching, seams, press fits, etc.) or the materials used (textiles, composites, etc.) that might reduce the strength of a product or could (in combination) have other detrimental effects have been investigated by the applicant and accounted for in the substantiation.
- (d) The effects of ageing (due to sunlight, temperature, water immersion, etc.) and other operational factors that may affect the strength of the PCDS are accounted for through appropriate inspections and the application of a calendar life limit as appropriate. The PCDS and the related attachment elements are limited to the carriage of HEC.
- (e) The risk of fatigue failure is minimised. See section below for further details.
- (f) Instructions for Continued Airworthiness (ICA) should be provided. Typically, the ICA would comprise an inspection programme and maintenance instructions based on the applicable manufacturer's data. The ICA should ensure that specific operational uses of the system that

might affect its strength are accounted for. A calendar life limit should be applied when appropriate.

- (g) When the harness is not designed to transport an incapacitated or untrained person, then the labelling and/or the user/flight manual should include a specific limitation of use as applicable.

Note 6: The following considerations and corresponding instructions/limitations should be taken for EN 1498 Type A and C rescue loops due to their potential detrimental physiological effects and the risk of falling out:

- (a) whether life is in imminent risk;
- (b) the physical condition of the person to be hoisted, particularly whether the rescuee will remain conscious and coherent during the hoist process;
- (c) the potential for the person to remain compliant with the brief given prior to hoisting;
- (d) alternative methods and devices to recover the person; and
- (e) whether the risk of falling from the device would result in further serious injury or death.

Simple PCDS Helicopter Compatibility

The ingress/egress of the simple PCDS in the cabin should be verified on the specific rotorcraft by means of a test. The compatibility with the hoist hook, unless the ring is already specified in the RFM, should also be verified by means of a test.

The verification of the hook and simple PCDS compatibility should also verify the absence of any roll-out/jamming phenomenon in order to:

- (a) prevent any inadvertent release of the load from the cargo hook; and/or
- (b) prevent the ring from jamming on the load beam during the release.

Manufacturing and Identification

Simple PCDSs that comply with Directive 89/686/EEC, or Regulation (EU) 2016/425, as applicable, or subsequent revision and the corresponding EN standards for the respective components are labelled by the manufacturer according to the applicable standard. If not already contained in the manufacturer labelling, the following additional information, as applicable, should be made visible on labelling on simple PCDSs:

- (a) manufacturing date;
- (b) life-limit date (if different from any existing one marked on the personal protective equipment (PPE));
- (c) manufacturer's identification;
- (d) part number;
- (e) serial number or unique identification of the single PCDS;
- (f) STC/minor change approval number (if applicable);
- (g) authorised load in kg;
- (h) authorised number of persons;
- (i) any other limitation not recorded in the manufacturer labelling.

Simple PCDS Static Strength

The PCDS should be substantiated for the loading conditions determined under the applicable paragraphs of FAA AC 27.865. For a PCDS to be certified separately from the hoist, using the guidance of this certification memo, the minimum ultimate load (UL_{min}) to be substantiated is defined as follows:

$$UL_{min} = M \times n \times j \times jf \times K \times g \text{ (units are Newtons)}$$

Where:

M is the total mass of the PCDS equipment/component and persons restrained by the part being substantiated (this is equivalent to the working load rating of an EN). The mass of each person should be assumed to be 100 kg.

NOTE: If the person(s) or their task requires the personal carriage of heavy items (backpacks, tools, fire extinguishers, etc.), these must be accounted for in the total mass M, in addition to the person's mass of 100 kg.

n is the helicopter manoeuvring limit load factor and must be assumed = 3.5 ([CS 27.337](#) and [27.865](#)).

j is the ultimate load factor of safety for all parts = 1.5 ([CS 27.303](#)).

K is an additional safety factor for textiles = 2.0 (see NOTE 1) ([CS 27.619](#)).

jf is an additional fitting factor = 1.33 applying to all joints, fittings, etc. ([CS 27.619](#)).

g is the acceleration due to gravity of 9.81 m/s².

The resulting values to ensure compliance with the CS-27 static strength requirements are:

UL_{min} for metallic elements with a fitting factor (needed for all joints and fittings): = 7 Mg.

(NOTE: To address fatigue, a value of 10 Mg may be required; see the section below on fatigue.)

UL_{min} for textiles (webbing, ropes, etc.) with fitting factor: = 14 Mg (see NOTE 1).

UL_{min} may be compared to the strength of the PCDS components already substantiated according to Directive 89/686/EEC, or Regulation (EU) 2016/425, as applicable, or subsequent revision and the corresponding EN Standards or Directive 2006/42/EC Annex I Point 6. Where UL_{min} is greater than that laid down in the Directives/EN requirements, a static test to not less than UL_{min} will be necessary. The test load must be sustained for 3 minutes. In addition, there should be no detrimental or permanent deformation of the metallic components at 3.5 Mg ([CS 27.305](#)).

NOTE 7: Directive 2006/42/EC Annex I Point 6 recommends a safety factor of 14 (2×7) for textiles applied to the working load (equivalent to 14 M above) for equipment lifting humans, whereas for a rescue harness, EN 1497 requires a static test load of not less than the greater of either 15 kN or 10 times the working load. Considering this difference, for each textile component within the PCDS certified to one of the following ENs, the value of K may be reduced, such that UL_{min} is not less than 10 M x g, where M is not more than 150 kg:

For harnesses, EN 361, EN 1497 or EN 12277A, EN 813 or EN 12277C apply; for belts or straps and for lanyards, EN 354 applies. This allowance is not applicable to ropes.

Furthermore, to allow this reduced value of UL_{min} and to address any potential deterioration of textiles due to environmental and other hidden damage, the ICA must include a life limitation of 5 years (or the life indicated by the PCDS manufacturer, if less) and an annual detailed inspection of the general condition of the harness.

Simple PCDS Fatigue

When the simple PCDS and the related attachment elements are limited to the carriage of HEC only, no further specific fatigue substantiation is necessary for each part of the simple PCDS that is either:

- (a) certified in accordance with an applicable EN that is referenced in this AMC for which the allowable working load is not exceeded by the mass M ; or
- (b) substantiated for static strength as described above with UL_{min} not less than 10 Mg

[Amdt No: 27/5]

MISCELLANEOUS

CS 27.871 Levelling marks

ED Decision 2003/15/RM

There must be reference marks for levelling the rotorcraft on the ground.

CS 27.873 Ballast provisions

ED Decision 2003/15/RM

Ballast provisions must be designed and constructed to prevent inadvertent shifting of ballast in flight.

SUBPART E – POWERPLANT

GENERAL

CS 27.901 Installation

ED Decision 2003/15/RM

- (a) For the purpose of this CS-27, the powerplant installation includes each part of the rotorcraft (other than the main and auxiliary rotor structures) that:
 - (1) Is necessary for propulsion;
 - (2) Affects the control of the major propulsive units; or
 - (3) Affects the safety of the major propulsive units between normal inspections or overhauls.
- (b) For each powerplant installation:
 - (1) Each component of the installation must be constructed, arranged, and installed to ensure its continued safe operation between normal inspections or overhauls for the range of temperature and altitude for which approval is requested;
 - (2) Accessibility must be provided to allow any inspection and maintenance necessary for continued airworthiness;
 - (3) Electrical interconnections must be provided to prevent differences of potential between major components of the installation and the rest of the rotorcraft;
 - (4) Axial and radial expansion of turbine engines may not affect the safety of the installation; and
 - (5) Design precautions must be taken to minimise the possibility of incorrect assembly of components and equipment essential to safe operation of the rotorcraft, except where operation with the incorrect assembly can be shown to be extremely improbable.
- (c) The installation must comply with:
 - (1) The installation instructions provided under CS-E; and
 - (2) The applicable provisions of this Subpart.

CS 27.903 Engines

ED Decision 2007/013/R

- (a) (Reserved)
- (b) *Engine or drive system cooling fan blade protection.*
 - (1) If an engine or rotor drive system cooling fan is installed, there must be means to protect the rotorcraft and allow a safe landing if a fan blade fails. This must be shown by showing that:
 - (i) The fan blades are contained in case of failure;
 - (ii) Each fan is located so that a failure will not jeopardise safety; or
 - (iii) Each fan blade can withstand an ultimate load of 1.5 times the centrifugal force resulting from operation limited by the following:

- (A) For fans driven directly by the engine:
 - (1) The terminal engine rpm under uncontrolled conditions; or
 - (2) An overspeed limiting device.
- (B) For fans driven by the rotor drive system the maximum rotor drive system rotational speed to be expected in service, including transients.
- (2) Unless a fatigue evaluation under [CS 27.571](#) is conducted, it must be shown that cooling fan blades are not operating at resonant conditions within the operating limits of the rotorcraft.
- (c) *Turbine engine installation.* For turbine engine installations, the powerplant systems associated with engine control devices, systems, and instrumentation must be designed to give reasonable assurance that those engine operating limitations that adversely affect turbine rotor structural integrity will not be exceeded in service.
- (d) *Restart capability.* A means to restart any engine in flight must be provided.
 - (1) Except for the in-flight shutdown of all engines, engine restart capability must be demonstrated throughout a flight envelope for the rotorcraft.
 - (2) Following the in-flight shutdown of all engines, in-flight engine restart capability must be provided.

[Amdt. No.: 27/1]

CS 27.907 Engine vibration

ED Decision 2003/15/RM

- (a) Each engine must be installed to prevent the harmful vibration of any part of the engine or rotorcraft.
- (b) The addition of the rotor and the rotor drive system to the engine may not subject the principal rotating parts of the engine to excessive vibration stresses. This must be shown by a vibration investigation.
- (c) No part of the rotor drive system may be subjected to excessive vibration stresses.

ROTOR DRIVE SYSTEM

CS 27.917 Design

ED Decision 2003/15/RM

- (a) Each rotor drive system must incorporate a unit for each engine to automatically disengage that engine from the main and auxiliary rotors if that engine fails.
- (b) Each rotor drive system must be arranged so that each rotor necessary for control in autorotation will continue to be driven by the main rotors after disengagement of the engine from the main and auxiliary rotors.
- (c) If a torque limiting device is used in the rotor drive system, it must be located so as to allow continued control of the rotorcraft when the device is operating.
- (d) The rotor drive system includes any part necessary to transmit power from the engines to the rotor hubs. This includes gear boxes, shafting, universal joints, couplings, rotor brake assemblies, clutches, supporting bearings for shafting, any attendant accessory pads or drives, and any cooling fans that are a part of, attached to, or mounted on the rotor drive system.

CS 27.921 Rotor brake

ED Decision 2003/15/RM

If there is a means to control the rotation of the rotor drive system independently of the engine, any limitations on the use of that means must be specified, and the control for that means must be guarded to prevent inadvertent operation.

CS 27.923 Rotor drive system and control mechanism tests

ED Decision 2003/15/RM

- (a) Each part tested as prescribed in this paragraph must be in a serviceable condition at the end of the tests. No intervening disassembly which might affect test results may be conducted.
- (b) Each rotor drive system and control mechanism must be tested for not less than 100 hours. The test must be conducted on the rotorcraft, and the torque must be absorbed by the rotors to be installed, except that other ground or flight test facilities with other appropriate methods of torque absorption may be used if the conditions of support and vibration closely simulate the conditions that would exist during a test on the rotorcraft.
- (c) A 60-hour part of the test prescribed in sub-paragraph (b) must be run at not less than maximum continuous torque and the maximum speed for use with maximum continuous torque. In this test, the main rotor controls must be set in the position that will give maximum longitudinal cyclic pitch change to simulate forward flight. The auxiliary rotor controls must be in the position for normal operation under the conditions of the test.
- (d) A 30-hour or, for rotorcraft for which the use of either 30-minute OEI power or continuous OEI power is requested, a 25-hour part of the test prescribed in sub-paragraph (b) must be run at not less than 75% of maximum continuous torque and the minimum speed for use with 75% of maximum continuous torque. The main and auxiliary rotor controls must be in the position for normal operation under the conditions of the test.
- (e) A 10-hour part of the test prescribed in sub-paragraph (b) must be run at not less than take-off torque and the maximum speed for use with take-off torque. The main and auxiliary rotor controls must be in the normal position for vertical ascent.

- (1) For multi-engine rotorcraft for which the use of 2½ minute OEI power is requested, 12 runs during the 10-hour test must be conducted as follows:
 - (i) Each run must consist of at least one period of 2½ minutes with take-off torque and the maximum speed for use with take-off torque on all engines.
 - (ii) Each run must consist of at least one period for each engine in sequence, during which that engine simulates a power failure and the remaining engines are run at 2½-minute OEI torque and the maximum speed for use with 2½-minute OEI torque for 2½ minutes.
- (2) For multi-engine turbine-powered rotorcraft for which the use of 30-second and 2-minute OEI power is requested, 10 runs must be conducted as follows:
 - (i) Immediately following a take-off run of at least 5 minutes, each power source must simulate a failure, in turn, and apply the maximum torque and the maximum speed for use with 30-second OEI power to the remaining affected drive system power inputs for not less than 30 seconds, followed by application of the maximum torque and the maximum speed for use with 2-minute OEI power for not less than 2 minutes. At least one run sequence must be conducted from a simulated 'flight idle' condition. When conducted on a bench test, the test sequence must be conducted following stabilisation at take-off power.
 - (ii) For the purpose of this paragraph, an affected power input includes all parts of the rotor drive system which can be adversely affected by the application of higher or asymmetric torque and speed prescribed by the test.
 - (iii) This test may be conducted on a representative bench test facility when engine limitations either preclude repeated use of this power or would result in premature engine removal during the test. The loads, the vibration frequency, and the methods of application to the affected rotor drive system components must be representative of rotorcraft conditions. Test components must be those used to show compliance with the remainder of this paragraph.
- (f) The parts of the test prescribed in subparagraphs (c) and (d) must be conducted in intervals of not less than 30 minutes and may be accomplished either on the ground or in flight. The part of the test prescribed in sub-paragraph (e) must be conducted in intervals of not less than 5 minutes.
- (g) At intervals of not more than five hours during the tests prescribed in sub-paragraphs (c),(d), and (e), the engine must be stopped rapidly enough to allow the engine and rotor drive to be automatically disengaged from the rotors.
- (h) Under the operating conditions specified in sub-paragraph (c), 500 complete cycles of lateral control, 500 complete cycles of longitudinal control of the main rotors, and 500 complete cycles of control of each auxiliary rotor must be accomplished. A 'complete cycle' involves movement of the controls from the neutral position, through both extreme positions, and back to the neutral position, except that control movements need not produce loads or flapping motions exceeding the maximum loads or motions encountered in flight. The cycling may be accomplished during the testing prescribed in sub-paragraph (c).
- (i) At least 200 start-up clutch engagements must be accomplished:
 - (1) So that the shaft on the driven side of the clutch is accelerated; and
 - (2) Using a speed and method selected by the applicant.

- (j) For multi-engine rotorcraft for which the use of 30-minute OEI power is requested, five runs must be made at 30-minute OEI torque and the maximum speed for use with 30-minute OEI torque, in which each engine, in sequence, is made inoperative and the remaining engine(s) is run for a 30-minute period.
- (k) For multi-engine rotorcraft for which the use of continuous OEI power is requested, five runs must be made at continuous OEI torque and the maximum speed for use with continuous OEI torque, in which each engine, in sequence, is made inoperative and the remaining engine(s) is run for a 1-hour period.

CS 27.927 Additional tests

ED Decision 2003/15/RM

- (a) Any additional dynamic, endurance, and operational tests, and vibratory investigations necessary to determine that the rotor drive mechanism is safe, must be performed.
- (b) If turbine engine torque output to the transmission can exceed the highest engine or transmission torque rating limit, and that output is not directly controlled by the pilot under normal operating conditions (such as where the primary engine power control is accomplished through the flight control), the following test must be made:
 - (1) Under conditions associated with all engines operating, make 200 applications, for 10 seconds each, of torque that is at least equal to the lesser of:
 - (i) The maximum torque used in meeting [CS 27.923](#) plus 10%; or
 - (ii) The maximum attainable torque output of the engines, assuming that torque limiting devices, if any, function properly.
 - (2) For multi-engine rotorcraft under conditions associated with each engine in turn becoming inoperative, apply to the remaining transmission torque inputs, the maximum torque attainable under probable operating conditions, assuming that torque limiting devices, if any, function properly. Each transmission input must be tested at this maximum torque for at least 15 minutes.
 - (3) The tests prescribed in this paragraph must be conducted on the rotorcraft at the maximum rotational speed intended for the power condition of the test and the torque must be absorbed by the rotors to be installed, except that other ground or flight test facilities with other appropriate methods of torque absorption may be used if the conditions of support and vibration closely simulate the conditions that would exist during a test on the rotorcraft.
- (c) It must be shown by tests that the rotor drive system is capable of operating under autorotative conditions for 15 minutes after the loss of pressure in the rotor drive primary oil system.

CS 27.931 Shafting critical speed

ED Decision 2003/15/RM

- (a) The critical speeds of any shafting must be determined by demonstration except that analytical methods may be used if reliable methods of analysis are available for the particular design.
- (b) If any critical speed lies within, or close to, the operating ranges for idling, power on, and autorotative conditions, the stresses occurring at that speed must be within safe limits. This must be shown by tests.

- (c) If analytical methods are used and show that no critical speed lies within the permissible operating ranges, the margins between the calculated critical speeds and the limits of the allowable operating ranges must be adequate to allow for possible variations between the computed and actual values.

CS 27.935 Shafting joints

ED Decision 2003/15/RM

Each universal joint, slip joint, and other shafting joints whose lubrication is necessary for operation must have provision for lubrication.

CS 27.939 Turbine engine operating characteristics

ED Decision 2003/15/RM

- (a) Turbine engine operating characteristics must be investigated in flight to determine that no adverse characteristics (such as stall, surge, or flameout) are present, to a hazardous degree, during normal and emergency operation within the range of operating limitations of the rotorcraft and of the engine.
- (b) The turbine engine air inlet system may not, as a result of airflow distortion during normal operation, cause vibration harmful to the engine.
- (c) For governor-controlled engines, it must be shown that there exists no hazardous torsional instability of the drive system associated with critical combinations of power, rotational speed, and control displacement.

FUEL SYSTEM

CS 27.951 General

ED Decision 2003/15/RM

- (a) Each fuel system must be constructed and arranged to ensure a flow of fuel at a rate and pressure established for proper engine functioning under any likely operating condition, including the manoeuvres for which certification is requested.
- (b) Each fuel system must be arranged so that –
 - (1) No fuel pump can draw fuel from more than one tank at a time; or
 - (2) There are means to prevent introducing air into the system.
- (c) Each fuel system for a turbine engine must be capable of sustained operation throughout its flow and pressure range with fuel initially saturated with water at 27°C (80°F) and having 0.198 cc of free water per litre (0.75 cc per US gallon) added and cooled to the most critical condition for icing likely to be encountered in operation.

CS 27.952 Fuel system crash resistance

ED Decision 2003/15/RM

Unless other means acceptable to the Agency are employed to minimise the hazard of fuel fires to occupants following an otherwise survivable impact (crash landing), the fuel systems must incorporate the design features of this paragraph. These systems must be shown to be capable of sustaining the static and dynamic deceleration loads of this paragraph, considered as ultimate loads acting alone, measured at the system component's centre of gravity without structural damage to system components, fuel tanks, or their attachments that would leak fuel to an ignition source.

- (a) *Drop test requirements.* Each tank, or the most critical tank, must be drop-tested as follows:
 - (1) The drop height must be at least 15.2 m (50 ft).
 - (2) The drop impact surface must be non-deforming.
 - (3) The tank must be filled with water to 80% of the normal, full capacity.
 - (4) The tank must be enclosed in a surrounding structure representative of the installation unless it can be established that the surrounding structure is free of projections or other design features likely to contribute to rupture of the tank.
 - (5) The tank must drop freely and impact in a horizontal position $\pm 10^\circ$.
 - (6) After the drop test there must be no leakage.
- (b) *Fuel tank load factors.* Except for fuel tanks located so that tank rupture with fuel release to either significant ignition sources, such as engines, heaters, and auxiliary power units, or occupants is extremely remote, each fuel tank must be designed and installed to retain its contents under the following ultimate inertial load factors, acting alone.
 - (1) For fuel tanks in the cabin:
 - (i) Upward – 4 g.
 - (ii) Forward – 16 g.
 - (iii) Sideward – 8 g.

- (iv) Downward – 20 g.
 - (2) For fuel tanks located above or behind the crew or passenger compartment that, if loosened, could injure an occupant in an emergency landing:
 - (i) Upward – 1.5 g.
 - (ii) Forward – 8 g.
 - (iii) Sideward – 2 g.
 - (iv) Downward – 4 g.
 - (3) For fuel tanks in other areas:
 - (i) Upward – 1.5 g.
 - (ii) Forward – 4 g.
 - (iii) Sideward – 2 g.
 - (iv) Downward – 4 g.
- (c) *Fuel line self-sealing breakaway couplings.* Self-sealing breakaway couplings must be installed unless hazardous relative motion of fuel system components to each other or to local rotorcraft structure is demonstrated to be extremely improbable or unless other means are provided. The couplings or equivalent devices must be installed at all fuel tank-to-fuel line connections, tank-to-tank interconnects, and at other points in the fuel system where local structural deformation could lead to release of fuel.
- (1) The design and construction of self-sealing breakaway couplings must incorporate the following design features:
 - (i) The load necessary to separate a breakaway coupling must be between 25 and 50% of the minimum ultimate failure load (ultimate strength) of the weakest component in the fluid-carrying line. The separation load must in no case be less than 1334 N (300 lb), regardless of the size of the fluid line.
 - (ii) A breakaway coupling must separate whenever its ultimate load (as defined in subparagraph (c)(1)(i)) is applied in the failure modes most likely to occur.
 - (iii) All breakaway couplings must incorporate design provisions to visually ascertain that the coupling is locked together (leak-free) and is open during normal installation and service.
 - (iv) All breakaway couplings must incorporate design provisions to prevent uncoupling or unintended closing due to operational shocks, vibrations, or accelerations.
 - (v) No breakaway coupling design may allow the release of fuel once the coupling has performed its intended function.
 - (2) All individual breakaway couplings, coupling fuel feed systems, or equivalent means must be designed, tested, installed and maintained so that inadvertent fuel shut-off in flight is improbable in accordance with [CS 27.955\(a\)](#) and must comply with the fatigue evaluation requirements of [CS 27.571](#) without leaking.
 - (3) Alternate, equivalent means to the use of breakaway couplings must not create a survivable impact-induced load on the fuel line to which it is installed greater than 25 to 50% of the ultimate load (strength) of the weakest component of the line and must comply with the fatigue requirements of [CS 27.571](#) without leaking.

- (d) *Frangible or deformable structural attachments.* Unless hazardous relative motion of fuel tanks and fuel system components to local rotorcraft structure is demonstrated to be extremely improbable in an otherwise survivable impact, frangible or locally deformable attachments of fuel tanks and fuel system components to local rotorcraft structure must be used. The attachment of fuel tanks and fuel system components to local rotorcraft structure, whether frangible or locally deformable, must be designed such that its separation or relative local deformation will occur without rupture or local tear-out of the fuel tank and fuel system components that will cause fuel leakage. The ultimate strength of frangible or deformable attachments must be as follows:
- (1) The load required to separate a frangible attachment from its support structure, or to deform a locally deformable attachment relative to its support structure, must be between 25 and 50% of the minimum ultimate load (ultimate strength) of the weakest component in the attached system. In no case may the load be less than 1330 N (300 lbs).
 - (2) A frangible or locally deformable attachment must separate or locally deform as intended whenever its ultimate load (as defined in sub-paragraph (d)(1)) is applied in the modes most likely to occur.
 - (3) All frangible or locally deformable attachments must comply with the fatigue requirements of [CS 27.571](#).
- (e) *Separation of fuel and ignition sources.* To provide maximum crash resistance, fuel must be located as far as practicable from all occupiable areas and from all potential ignition sources.
- (f) *Other basic mechanical design criteria.* Fuel tanks, fuel lines, electrical wires, and electrical devices must be designed, constructed and installed, as far as practicable, to be crash resistant.
- (g) *Rigid or semi-rigid fuel tanks.* Rigid or semi-rigid fuel tank or bladder walls must be impact and tear resistant.

CS 27.953 Fuel system independence

ED Decision 2003/15/RM

- (a) Each fuel system for multi-engine rotorcraft must allow fuel to be supplied to each engine through a system independent of those parts of each system supplying fuel to other engines. However, separate fuel tanks need not be provided for each engine.
- (b) If a single fuel tank is used on a multiengine rotorcraft, the following must be provided:
- (1) Independent tank outlets for each engine, each incorporating a shut-off valve at the tank. This shut-off valve may also serve as the firewall shut-off valve required by [CS 27.995](#) if the line between the valve and the engine compartment does not contain a hazardous amount of fuel that can drain into the engine compartment.
 - (2) At least two vents arranged to minimise the probability of both vents becoming obstructed simultaneously.
 - (3) Filler caps designed to minimise the probability of incorrect installation or inflight loss.
 - (4) A fuel system in which those parts of the system from each tank outlet to any engine are independent of each part of each system supplying fuel to other engines.

CS 27.954 Fuel system lightning protection

ED Decision 2003/15/RM

The fuel system must be designed and arranged to prevent the ignition of fuel vapour within the system by:

- (a) Direct lightning strikes to areas having a high probability of stroke attachment;
- (b) Swept lightning strokes to areas where swept strokes are highly probable; or
- (c) Corona and streamering at fuel vent outlets.

CS 27.955 Fuel flow

ED Decision 2003/15/RM

- (a) *General.* The fuel system for each engine must be shown to provide the engine with at least 100% of the fuel required under each operating and manoeuvring condition to be approved for the rotorcraft including, as applicable, the fuel required to operate the engine(s) under the test conditions required by [CS 27.927](#). Unless equivalent methods are used, compliance must be shown by test during which the following provisions are met except that combinations of conditions which are shown to be improbable need not be considered:
 - (1) The fuel pressure, corrected for critical accelerations, must be within the limits specified by the engine type certificate data sheet.
 - (2) The fuel level in the tank may not exceed that established as unusable fuel supply for the tank under [CS 27.959](#), plus the minimum additional fuel necessary to conduct the test.
 - (3) The fuel head between the tank outlet and the engine inlet must be critical with respect to rotorcraft flight attitudes.
 - (4) The critical fuel pump (for pumped systems) is installed to produce (by actual or simulated failure) the critical restriction to fuel flow to be expected from pump failure.
 - (5) Critical values of engine rotation speed, electrical power, or other sources of fuel pump motive power must be applied.
 - (6) Critical values of fuel properties which adversely affect fuel flow must be applied.
 - (7) The fuel filter required by [CS 27.997](#) must be blocked to the degree necessary to simulate the accumulation of fuel contamination required to activate the indicator required by [CS 27.1305\(q\)](#).
- (b) *Fuel transfer systems.* If normal operation of the fuel system requires fuel to be transferred to an engine feed tank, the transfer must occur automatically via a system which has been shown to maintain the fuel level in the engine feed tank within acceptable limits during flight or surface operation of the rotorcraft.
- (c) *Multiple fuel tanks.* If an engine can be supplied with fuel from more than one tank, the fuel systems must, in addition to having appropriate manual switching capability, be designed to prevent interruption of fuel flow to that engine, without attention by the flightcrew, when any tank supplying fuel to that engine is depleted of usable fuel during normal operation, and any other tank that normally supplies fuel to the engine alone contains usable fuel.

CS 27.959 Unusable fuel supply

ED Decision 2003/15/RM

The unusable fuel supply for each tank must be established as not less than the quantity at which the first evidence of malfunction occurs under the most adverse fuel feed condition occurring under any intended operations and flight manoeuvres involving that tank.

CS 27.961 Fuel system hot weather operation

ED Decision 2003/15/RM

Each suction lift fuel system and other fuel systems with features conducive to vapour formation must be shown by test to operate satisfactorily (within certification limits) when using fuel at a temperature of 43°C (110°F) under critical operating conditions including, if applicable, the engine operating conditions defined by [CS 27.927\(b\)\(1\) and \(b\)\(2\)](#).

CS 27.963 Fuel tanks: general

ED Decision 2003/15/RM

- (a) Each fuel tank must be able to withstand, without failure, the vibration, inertia, fluid, and structural loads to which it may be subjected in operation.
- (b) Each fuel tank of 38 litres (8.3 Imperial gallons/10 US gallons) or greater capacity must have internal baffles, or must have external support to resist surging.
- (c) Each fuel tank must be separated from the engine compartment by a firewall. At least one-half inch of clear airspace must be provided between the tank and the firewall.
- (d) Spaces adjacent to the surfaces of fuel tanks must be ventilated so that fumes cannot accumulate in the tank compartment in case of leakage. If two or more tanks have interconnected outlets, they must be considered as one tank, and the airspaces in those tanks must be interconnected to prevent the flow of fuel from one tank to another as a result of a difference in pressure between those airspaces.
- (e) The maximum exposed surface temperature of any component in the fuel tank must be less, by a safe margin, than the lowest expected auto-ignition temperature of the fuel or fuel vapour in the tank. Compliance with this requirement must be shown under all operating conditions and under all failure or malfunction conditions of all components inside the tank.
- (f) Each fuel tank installed in personnel compartments must be isolated by fume-proof and fuel-proof enclosures that are drained and vented to the exterior of the rotorcraft. The design and construction of the enclosures must provide necessary protection for the tank, must be crash resistant during a survivable impact in accordance with [CS 27.952](#) and must be adequate to withstand loads and abrasions to be expected in personnel compartments.
- (g) Each flexible fuel tank bladder or liner must be approved or shown to be suitable for the particular application and must be puncture resistant. Puncture resistance must be shown by meeting the ETSO-C80, paragraph 16.0, requirements using a minimum puncture force of 1646 N (370 lbs).
- (h) Each integral fuel tank must have provisions for inspection and repair of its interior.

CS 27.965 Fuel tank tests

ED Decision 2003/15/RM

- (a) Each fuel tank must be able to withstand the applicable pressure tests in this paragraph without failure or leakage. If practicable, test pressures may be applied in a manner simulating the pressure distribution in service.
- (b) Each conventional metal tank, non-metallic tank with walls that are not supported by the rotorcraft structure, and integral tank must be subjected to a pressure of 24 kPa (3.5 psi) unless the pressure developed during maximum limit acceleration or emergency deceleration with a full tank exceeds this value, in which case a hydrostatic head, or equivalent test, must be applied to duplicate the acceleration loads as far as possible. However, the pressure need not exceed 24 kPa (3.5 psi) on surfaces not exposed to the acceleration loading.
- (c) Each non-metallic tank with walls supported by the rotorcraft structure must be subjected to the following tests:
 - (1) A pressure test of at least 14 kPa (2.0 psi). This test may be conducted on the tank alone in conjunction with the test specified in sub-paragraph (c)(2).
 - (2) A pressure test, with the tank mounted in the rotorcraft structure, equal to the load developed by the reaction of the contents, with the tank full, during maximum limit acceleration or emergency deceleration. However, the pressure need not exceed 14 kPa (2.0 psi) on surfaces not exposed to the acceleration loading.
- (d) Each tank with large unsupported or unstiffened flat areas, or with other features whose failure or deformation could cause leakage, must be subjected to the following test or its equivalent:
 - (1) Each complete tank assembly and its support must be vibration tested while mounted to simulate the actual installation.
 - (2) The tank assembly must be vibrated for 25 hours while two-thirds full of any suitable fluid. The amplitude of vibration may not be less than 0.8 mm (1/32 inch), unless otherwise substantiated.
 - (3) The test frequency of vibration must be as follows:
 - (i) If no frequency of vibration resulting from any rpm within the normal operating range of engine or rotor system speeds is critical, the test frequency of vibration, in number of cycles per minute must, unless a frequency based on a more rational calculation is used, be the number obtained by averaging the maximum and minimum power-on engine speeds (rpm) for reciprocating engine-powered rotorcraft or 2000 cpm for turbine engine-powered rotorcraft.
 - (ii) If only one frequency of vibration resulting from any rpm within the normal operating range of engine or rotor system speeds is critical, that frequency of vibration must be the test frequency.
 - (iii) If more than one frequency of vibration resulting from any rpm within the normal operating range of engine or rotor system speeds is critical, the most critical of these frequencies must be the test frequency.
 - (4) Under sub-paragraphs (d)(3)(ii) and (iii), the time of test must be adjusted to accomplish the same number of vibration cycles as would be accomplished in 25 hours at the frequency specified in sub-paragraph (d)(3)(i).
 - (5) During the test, the tank assembly must be rocked at the rate of 16 to 20 complete cycles per minute through an angle of 15° on both sides of the horizontal (30° total), about the

most critical axis, for 25 hours. If motion about more than one axis is likely to be critical, the tank must be rocked about each critical axis for 12½ hours.

CS 27.967 Fuel tank installation

ED Decision 2003/15/RM

- (a) Each fuel tank must be supported so that tank loads are not concentrated on unsupported tank surfaces. In addition:
 - (1) There must be pads, if necessary, to prevent chafing between each tank and its supports;
 - (2) The padding must be nonabsorbent or treated to prevent the absorption of fuel;
 - (3) If flexible tank liners are used, they must be supported so that it is not necessary for them to withstand fluid loads; and
 - (4) Each interior surface of tank compartments must be smooth and free of projections that could cause wear of the liner unless –
 - (i) There are means for protection of the liner at those points; or
 - (ii) The construction of the liner itself provides such protection.
- (b) Any spaces adjacent to tank surfaces must be adequately ventilated to avoid accumulation of fuel or fumes in those spaces due to minor leakage. If the tank is in a sealed compartment, ventilation may be limited to drain holes that prevent clogging and excessive pressure resulting from altitude changes. If flexible tank liners are installed, the venting arrangement for the spaces between the liner and its container must maintain the proper relationship to tank vent pressures for any expected flight condition.
- (c) The location of each tank must meet the requirements of [CS 27.1185\(a\) and \(c\)](#).
- (d) No rotorcraft skin immediately adjacent to a major air outlet from the engine compartment may act as the wall of the integral tank.

CS 27.969 Fuel tank expansion space

ED Decision 2003/15/RM

Each fuel tank or each group of fuel tanks with interconnected vent systems must have an expansion space of not less than 2% of the tank capacity. It must be impossible to fill the fuel tank expansion space inadvertently with the rotorcraft in the normal ground attitude.

CS 27.971 Fuel tank sump

ED Decision 2003/15/RM

- (a) Each fuel tank must have a drainable sump with an effective capacity in any ground attitude to be expected in service of 0.25% of the tank capacity or 0.24 litres (0.05 Imperial gallons/one sixteenth US gallon), whichever is greater, unless:
 - (1) The fuel system has a sediment bowl or chamber that is accessible for preflight drainage and has a minimum capacity of 30 ml (1 ounce) for every 76 litres (16.7 Imperial gallons/20 US gallons) of fuel tank capacity; and
 - (2) Each fuel tank drain is located so that in any ground attitude to be expected in service, water will drain from all parts of the tank to the sediment bowl or chamber.

- (b) Each sump, sediment bowl, and sediment chamber drain required by the paragraph must comply with the drain provisions of [CS 27.999\(b\)](#).

CS 27.973 Fuel tank filler connection

ED Decision 2003/15/RM

- (a) Each fuel tank filler connection must prevent the entrance of fuel into any part of the rotorcraft other than the tank itself during normal operations and must be crash resistant during a survivable impact in accordance with [CS 27.952\(c\)](#). In addition:
- (1) Each filler must be marked as prescribed in [CS 27.1557\(c\)\(1\)](#);
 - (2) Each recessed filler connection that can retain any appreciable quantity of fuel must have a drain that discharges clear of the entire rotorcraft; and
 - (3) Each filler cap must provide a fuel-tight seal under the fluid pressure expected in normal operation and in a survivable impact.
- (b) Each filler cap or filler cap cover must warn when the cap is not fully locked or seated on the filler connection.

CS 27.975 Fuel tank vents

ED Decision 2003/15/RM

- (a) Each fuel tank must be vented from the top part of the expansion space so that venting is effective under all normal flight conditions. Each vent must minimise the probability of stoppage by dirt or ice.
- (b) The venting system must be designed to minimise spillage of fuel through the vents to an ignition source in the event of a rollover during landing, ground operation, or a survivable impact.

CS 27.977 Fuel tank outlet

ED Decision 2003/15/RM

- (a) There must be a fuel strainer for the fuel tank outlet or for the booster pump. This strainer must:
- (1) For reciprocating engine-powered rotorcraft have 3 to 6 meshes per cm (8 to 16 meshes per inch); and
 - (2) For turbine engine-powered rotorcraft, prevent the passage of any object that could restrict fuel flow or damage any fuel system component.
- (b) The clear area of each fuel tank outlet strainer must be at least 5 times the area of the outlet line.
- (c) The diameter of each strainer must be at least that of the fuel tank outlet.
- (d) Each finger strainer must be accessible for inspection and cleaning.

FUEL SYSTEM COMPONENTS

CS 27.991 Fuel pumps

ED Decision 2003/15/RM

Compliance with [CS 27.955](#) may not be jeopardised by failure of:

- (a) Any one pump except pumps that are approved and installed as parts of a type certificated engine; or
- (b) Any component required for pump operation except, for engine driven pumps, the engine served by that pump.

CS 27.993 Fuel system lines and fittings

ED Decision 2003/15/RM

- (a) Each fuel line must be installed and supported to prevent excessive vibration and to withstand loads due to fuel pressure and accelerated flight conditions.
- (b) Each fuel line connected to components of the rotorcraft between which relative motion could exist must have provisions for flexibility.
- (c) Flexible hose must be approved.
- (d) Each flexible connection in fuel lines that may be under pressure or subjected to axial loading must use flexible hose assemblies.
- (e) No flexible hose that might be adversely affected by high temperatures may be used where excessive temperatures will exist during operation or after engine shutdown.

CS 27.995 Fuel valves

ED Decision 2003/15/RM

- (a) There must be a positive, quick-acting valve to shut-off fuel to each engine individually.
- (b) The control for this valve must be within easy reach of appropriate crew members.
- (c) Where there is more than one source of fuel supply there must be means for independent feeding from each source.
- (d) No shut-off valve may be on the engine side of any firewall.

CS 27.997 Fuel strainer or filter

ED Decision 2003/15/RM

There must be a fuel strainer or filter between the fuel tank outlet and the inlet of the first fuel system component which is susceptible to fuel contamination, including but not limited to the fuel metering device or an engine positive displacement pump, whichever is nearer the fuel tank outlet. This fuel strainer or filter must:

- (a) Be accessible for draining and cleaning and must incorporate a screen or element which is easily removable;
- (b) Have a sediment trap and drain except that it need not have a drain if the strainer or filter is easily removable for drain purposes;

- (c) Be mounted so that its weight is not supported by the connecting lines or by the inlet or outlet connections of the strainer or filter itself, unless adequate strength margins under all loading conditions are provided in the lines and connections; and
- (d) Provide a means to remove from the fuel any contaminant which would jeopardise the flow of fuel through rotorcraft or engine fuel system components required for proper rotorcraft fuel system or engine fuel system operation.

CS 27.999 Fuel system drains

ED Decision 2003/15/RM

- (a) There must be at least one accessible drain at the lowest point in each fuel system to completely drain the system with the rotorcraft in any ground attitude to be expected in service.
- (b) Each drain required by sub-paragraph (a) must:
 - (1) Discharge clear of all parts of the rotorcraft;
 - (2) Have manual or automatic means to assure positive closure in the off position; and
 - (3) Have a drain valve:
 - (i) That is readily accessible and which can be easily opened and closed; and
 - (ii) That is either located or protected to prevent fuel spillage in the event of a landing with landing gear retracted.

OIL SYSTEM

CS 27.1011 Engines: general

ED Decision 2003/15/RM

- (a) Each engine must have an independent oil system that can supply it with an appropriate quantity of oil at a temperature not above that safe for continuous operation.
- (b) The usable oil capacity of each system may not be less than the product of the endurance of the rotorcraft under critical operating conditions and the maximum oil consumption of the engine under the same conditions, plus a suitable margin to ensure adequate circulation and cooling. Instead of a rational analysis of endurance and consumption, a usable oil capacity of 3.8 litres (0.83 Imperial gallon /1 US gallon) for each 151 litres (33.3 Imperial gallons/40 US gallons) of usable fuel may be used.
- (c) The oil cooling provisions for each engine must be able to maintain the oil inlet temperature to that engine at or below the maximum established value. This must be shown by flight tests.

CS 27.1013 Oil tanks

ED Decision 2003/15/RM

Each oil tank must be designed and installed so that –

- (a) It can withstand, without failure, each vibration, inertia, fluid, and structural load expected in operation;
- (b) (Reserved)
- (c) Where used with a reciprocating engine, it has an expansion space of not less than the greater of 10% of the tank capacity or 1.9 litre (0.42 Imperial gallon/0.5 US gallon), and where used with a turbine engine, it has an expansion space of not less than 10% of the tank capacity.
- (d) It is impossible to fill the tank expansion space inadvertently with the rotorcraft in the normal ground attitude;
- (e) Adequate venting is provided; and
- (f) There are means in the filler opening to prevent oil overflow from entering the oil tank compartment.

CS 27.1015 Oil tank tests

ED Decision 2003/15/RM

Each oil tank must be designed and installed so that it can withstand, without leakage, an internal pressure of 34 kPa (5 psi), except that each pressurised oil tank used with a turbine engine must be designed and installed so that it can withstand, without leakage, an internal pressure of 34 kPa (5 psi), plus the maximum operating pressure of the tank.

CS 27.1017 Oil lines and fittings

ED Decision 2003/15/RM

- (a) Each oil line must be supported to prevent excessive vibration.
- (b) Each oil line connected to components of the rotorcraft between which relative motion could exist must have provisions for flexibility.

- (c) Flexible hose must be approved.
- (d) Each oil line must have an inside diameter of not less than the inside diameter of the engine inlet or outlet. No line may have splices between connections.

CS 27.1019 Oil strainer or filter

ED Decision 2003/15/RM

- (a) Each turbine engine installation must incorporate an oil strainer or filter through which all of the engine oil flows and which meets the following requirements:
 - (1) Each oil strainer or filter that has a bypass must be constructed and installed so that oil will flow at the normal rate through the rest of the system with the strainer or filter completely blocked.
 - (2) The oil strainer or filter must have the capacity (with respect to operating limitations established for the engine) to ensure that engine oil system functioning is not impaired when the oil is contaminated to a degree (with respect to particle size and density) that is greater than that established for the engine under CS-E.
 - (3) The oil strainer or filter, unless it is installed at an oil tank outlet, must incorporate a means to indicate contamination before it reaches the capacity established in accordance with sub-paragraph (a)(2).
 - (4) The bypass of a strainer or filter must be constructed and installed so that the release of collected contaminants is minimised by appropriate location of the bypass to ensure that collected contaminants are not in the bypass flow path.
 - (5) An oil strainer or filter that has no bypass, except one that is installed at an oil tank outlet, must have a means to connect it to the warning system required in [CS 27.1305\(r\)](#).
- (b) Each oil strainer or filter in a powerplant installation using reciprocating engines must be constructed and installed so that oil will flow at the normal rate through the rest of the system with the strainer or filter element completely blocked.

CS 27.1021 Oil system drains

ED Decision 2003/15/RM

A drain (or drains) must be provided to allow safe drainage of the oil system. Each drain must:

- (a) Be accessible; and
- (b) Have manual or automatic means for positive locking in the closed position.

CS 27.1027 Transmissions and gearboxes: general

ED Decision 2003/15/RM

- (a) The lubrication system for components of the rotor drive system that require continuous lubrication must be sufficiently independent of the lubrication systems of the engine(s) to ensure lubrication during autorotation.
- (b) Pressure lubrication systems for transmissions and gear-boxes must comply with the engine oil system requirements of [CS 27.1013](#) (except sub-paragraph (c)), [CS 27.1015](#), [27.1017](#), [27.1021](#), and [27.1337\(d\)](#).

- (c) Each pressure lubrication system must have an oil strainer or filter through which all of the lubricant flows and must –
 - (1) Be designed to remove from the lubricant any contaminant which may damage transmission and drive system components or impede the flow of lubricant to a hazardous degree;
 - (2) Be equipped with a means to indicate collection of contaminants on the filter or strainer at or before opening of the bypass required by sub-paragraph (c)(3); and
 - (3) Be equipped with a bypass constructed and installed so that:
 - (i) The lubricant will flow at the normal rate through the rest of the system with the strainer or filter completely blocked; and
 - (ii) The release of collected contaminants is minimised by appropriate location of the bypass to ensure that collected contaminants are not in the bypass flow path.
- (d) For each lubricant tank or sump outlet supplying lubrication to rotor drive systems and rotor drive system components, a screen must be provided to prevent entrance into the lubrication system of any object that might obstruct the flow of lubricant from the outlet to the filter required by sub-paragraph (c). The requirements of sub-paragraph (c) do not apply to screens installed at lubricant tank or sump outlets.
- (e) Splash-type lubrication systems for rotor drive system gearboxes must comply with [CS 27.1021](#) and [27.1337\(d\)](#).

COOLING

CS 27.1041 General

ED Decision 2003/15/RM

- (a) Each powerplant cooling system must be able to maintain the temperatures of powerplant components within the limits established for these components under critical surface (ground or water) and flight operating conditions for which certification is required and after normal shutdown. Powerplant components to be considered include but may not be limited to engines, rotor drive system components, auxiliary power units, and the cooling or lubricating fluids used with these components.
- (b) Compliance with sub-paragraph (a) must be shown in tests conducted under the conditions prescribed in that paragraph.

CS 27.1043 Cooling tests

ED Decision 2003/15/RM

- (a) *General.* For the tests prescribed in [CS 27.1041\(b\)](#), the following apply:
 - (1) If the tests are conducted under conditions deviating from the maximum ambient atmospheric temperature specified in sub-paragraph (b), the recorded powerplant temperatures must be corrected under sub-paragraphs (c) and (d) unless a more rational correction method is applicable.
 - (2) No corrected temperature determined under sub-paragraph (a)(1) may exceed established limits.
 - (3) For reciprocating engines, the fuel used during the cooling tests must be of the minimum grade approved for the engines, and the mixture settings must be those normally used in the flight stages for which the cooling tests are conducted.
 - (4) The test procedures must be as prescribed in [CS 27.1045](#).
- (b) *Maximum ambient atmospheric temperature.* A maximum ambient atmospheric temperature corresponding to sea-level conditions of at least 38°C (100°F) must be established. The assumed temperature lapse rate is 1.98°C (3.6°F) per 305 m (1000 ft) of altitude above sea-level until a temperature of -56.5°C (-69.7°F) is reached, above which altitude the temperature is considered constant at -56.5°C (-69.7°F). However, for winterization installations, the applicant may select a maximum ambient atmospheric temperature corresponding to sea-level conditions of less than 38°C (100°F).
- (c) *Correction factor (except cylinder barrels).* Unless a more rational correction applies, temperatures of engine fluids and powerplant components (except cylinder barrels) for which temperature limits are established, must be corrected by adding to them the difference between the maximum ambient atmospheric temperature and the temperature of the ambient air at the time of the first occurrence of the maximum component or fluid temperature recorded during the cooling test.
- (d) *Correction factor for cylinder barrel temperatures.* Cylinder barrel temperatures must be corrected by adding to them 0.7 times the difference between the maximum ambient atmospheric temperature and the temperature of the ambient air at the time of the first occurrence of the maximum cylinder barrel temperature recorded during the cooling test.

CS 27.1045 Cooling test procedures

ED Decision 2003/15/RM

- (a) *General.* For each stage of flight, the cooling tests must be conducted with the rotorcraft:
 - (1) In the configuration most critical for cooling; and
 - (2) Under the conditions most critical for cooling.
- (b) *Temperature stabilisation.* For the purpose of the cooling tests, a temperature is ‘stabilised’ when its rate of change is less than 1°C (2°F) per minute. The following component and engine fluid temperature stabilisation rules apply:
 - (1) For each rotorcraft, and for each stage of flight:
 - (i) The temperatures must be stabilised under the conditions from which entry is made into the stage of flight being investigated; or
 - (ii) If the entry condition normally does not allow temperatures to stabilise, operation through the full entry condition must be conducted before entry into the stage of flight being investigated in order to allow the temperatures to attain their natural levels at the time of entry.
 - (2) For each helicopter during the take-off stage of flight the climb at take-off power must be preceded by a period of hover during which the temperatures are stabilised.
- (c) *Duration of test.* For each stage of flight the tests must be continued until:
 - (1) The temperatures stabilise or 5 minutes after the occurrence of the highest temperature recorded, as appropriate to the test condition;
 - (2) That stage of flight is completed; or
 - (3) An operating limitation is reached.

INDUCTION SYSTEM

CS 27.1091 Air induction

ED Decision 2003/15/RM

- (a) The air induction system for each engine must supply the air required by that engine under the operating conditions and manoeuvres for which certification is requested.
- (b) Each cold air induction system opening must be outside the cowling if backfire flames can emerge.
- (c) If fuel can accumulate in any air induction system, that system must have drains that discharge fuel –
 - (1) Clear of the rotorcraft; and
 - (2) Out of the path of exhaust flames.
- (d) For turbine engine-powered rotorcraft:
 - (1) There must be means to prevent hazardous quantities of fuel leakage or overflow from drains, vents, or other components of flammable fluid systems from entering the engine intake system; and
 - (2) The air inlet ducts must be located or protected so as to minimise the ingestion of foreign matter during take-off, landing, and taxiing.

CS 27.1093 Induction system icing protection

ED Decision 2003/15/RM

- (a) *Reciprocating engines.* Each reciprocating engine air induction system must have means to prevent and eliminate icing. Unless this is done by other means, it must be shown that, in air free of visible moisture at a temperature of -1°C (30°F) and with the engines at 75% of maximum continuous power:
 - (1) Each rotorcraft with sea-level engines using conventional venturi carburettors has a preheater that can provide a heat rise of 50°C (90°F);
 - (2) Each rotorcraft with sea-level engines using carburettors tending to prevent icing has a sheltered alternate source of air, and that the preheat supplied to the alternate air intake is not less than that provided by the engine cooling air downstream of the cylinders;
 - (3) Each rotorcraft with altitude engines using conventional venturi carburettors has a preheater capable of providing a heat rise of 67°C (120°F); and
 - (4) Each rotorcraft with altitude engines using carburettors tending to prevent icing has a preheater that can provide a heat rise of:
 - (i) 56°C (100°F); or
 - (ii) If a fluid de-icing system is used, at least 22°C (40°F).
- (b) *Turbine engines*
 - (1) It must be shown that each turbine engine and its air inlet system can operate throughout the flight power range of the engine (including idling):

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- (i) Without accumulating ice on engine or inlet system components that would adversely affect engine operation or cause a serious loss of power under the icing conditions specified in appendix C of CS-29; and
 - (ii) In snow, both falling and blowing, without adverse effect on engine operation, within the limitations established for the rotorcraft.
 - (2) Each turbine engine must idle for 30 minutes on the ground, with the air bleed available for engine icing protection at its critical condition, without adverse effect, in an atmosphere that is at a temperature between -9°C and -1°C (15° and 30°F) and has a liquid water content not less than 0.3 grams per cubic metre in the form of drops having a mean effective diameter of not less than 20 microns, followed by momentary operation at take-off power or thrust. During the 30 minutes of idle operation, the engine may be run up periodically to a moderate power or thrust setting in a manner acceptable to the Agency.
 - (c) *Supercharged reciprocating engines.* For each engine having superchargers to pressurise the air before it enters the carburettor, the heat rise in the air caused by that supercharging at any altitude may be utilised in determining compliance with sub-paragraph (a) if the heat rise utilised is that which will be available, automatically, for the applicable altitude and operating condition because of supercharging.

EXHAUST SYSTEM

CS 27.1121 General

ED Decision 2003/15/RM

For each exhaust system:

- (a) There must be means for thermal expansion of manifolds and pipes;
- (b) There must be means to prevent local hot spots;
- (c) Exhaust gases must discharge clear of the engine air intake, fuel system components, and drains;
- (d) Each exhaust system part with a surface hot enough to ignite flammable fluids or vapours must be located or shielded so that leakage from any system carrying flammable fluids or vapours will not result in a fire caused by impingement of the fluids or vapours on any part of the exhaust system including shields for the exhaust system;
- (e) Exhaust gases may not impair pilot vision at night due to glare;
- (f) If significant traps exist, each turbine engine exhaust system must have drains discharging clear of the rotorcraft, in any normal ground and flight attitudes, to prevent fuel accumulation after the failure of an attempted engine start; and
- (g) Each exhaust heat exchanger must incorporate means to prevent blockage of the exhaust port after any internal heat exchanger failure.

CS 27.1123 Exhaust piping

ED Decision 2003/15/RM

- (a) Exhaust piping must be heat and corrosion resistant and must have provisions to prevent failure due to expansion by operating temperatures.
- (b) Exhaust piping must be supported to withstand any vibration and inertia loads to which it would be subjected in operations.
- (c) Exhaust piping connected to components between which relative motion could exist must have provisions for flexibility.

POWERPLANT CONTROLS AND ACCESSORIES

CS 27.1141 Powerplant controls: general

ED Decision 2003/15/RM

- (a) Powerplant controls must be located and arranged under [CS 27.777](#) and marked under [CS 27.1555](#).
- (b) Each flexible powerplant control must be approved.
- (c) Each control must be able to maintain any set position without:
 - (1) Constant attention; or
 - (2) Tendency to creep due to control loads or vibration.
- (d) Controls of powerplant valves required for safety must have:
 - (1) For manual valves, positive stops or in the case of fuel valves suitable index provisions, in the open and closed position; and
 - (2) For power-assisted valves, a means to indicate to the flight crew when the valve:
 - (i) Is in the fully open or fully closed position; or
 - (ii) Is moving between the fully open and fully closed position.
- (e) For turbine-engine-powered rotorcraft, no single failure or malfunction, or probable combination thereof, in any powerplant control system may cause the failure of any powerplant function necessary for safety.

CS 27.1143 Engine controls

ED Decision 2003/15/RM

- (a) There must be a separate power control for each engine.
- (b) Power controls must be grouped and arranged to allow:
 - (1) Separate control of each engine; and
 - (2) Simultaneous control of all engines.
- (c) Each power control must provide a positive and immediately responsive means of controlling its engine.
- (d) If a power control incorporates a fuel shut-off feature, the control must have a means to prevent the inadvertent movement of the control into the shut-off position. The means must:
 - (1) Have a positive lock or stop at the idle position; and
 - (2) Require a separate and distinct operation to place the control in the shut-off position.
- (e) For rotorcraft to be certificated for a 30-second OEI power rating, a means must be provided to automatically activate and control the 30-second OEI power and prevent any engine from exceeding the installed engine limits associated with the 30-second OEI power rating approved for the rotorcraft.

CS 27.1145 Ignition switches

ED Decision 2003/15/RM

- (a) There must be means to quickly shut off all ignition by the grouping of switches or by a master ignition control.
- (b) Each group of ignition switches, except ignition switches for turbine engines for which continuous ignition is not required, and each master ignition control must have a means to prevent its inadvertent operation.

CS 27.1147 Mixture controls

ED Decision 2003/15/RM

If there are mixture controls, each engine must have a separate control and the controls must be arranged to allow:

- (a) Separate control of each engine; and
- (b) Simultaneous control of all engines.

CS 27.1151 Rotor brake controls

ED Decision 2003/15/RM

- (a) It must be impossible to apply the rotor brake inadvertently in flight.
- (b) There must be means to warn the crew if the rotor brake has not been completely released before take-off.

CS 27.1163 Powerplant accessories

ED Decision 2003/15/RM

- (a) Each engine-mounted accessory must:
 - (1) Be approved for mounting on the engine involved;
 - (2) Use the provisions on the engine for mounting; and
 - (3) Be sealed in such a way as to prevent contamination of the engine oil system and the accessory system.
- (b) Unless other means are provided, torque limiting means must be provided for accessory drives located on any component of the transmission and rotor drive system to prevent damage to these components from excessive accessory load.

POWERPLANT FIRE PROTECTION

CS 27.1183 Lines, fittings, and components

ED Decision 2003/15/RM

- (a) Except as provided in sub-paragraph (b), each line, fitting, and other component carrying flammable fluid in any area subject to engine fire conditions must be fire resistant, except that flammable fluid tanks and supports which are part of and attached to the engine must be fireproof or be enclosed by a fireproof shield unless damage by fire to any non-fireproof part will not cause leakage or spillage of flammable fluid. Components must be shielded or located so as to safeguard against the ignition of leaking flammable fluid. An integral oil sump of less than 24 litres (5.2 Imperial gallons/25 US quart) capacity on a reciprocating engine need not be fireproof nor be enclosed by a fireproof shield.
- (b) Sub-paragraph (a) does not apply to:
 - (1) Lines, fittings, and components which are already approved as part of a type certificated engine; and
 - (2) Vent and drain lines, and their fittings, whose failure will not result in, or add to, a fire hazard.
- (c) Each flammable fluid drain and vent must discharge clear of the induction system air inlet.

CS 27.1185 Flammable fluids

ED Decision 2003/15/RM

- (a) Each fuel tank must be isolated from the engines by a firewall or shroud.
- (b) Each tank or reservoir, other than a fuel tank, that is part of a system containing flammable fluids or gases must be isolated from the engine by a firewall or shroud unless the design of the system, the materials used in the tank and its supports, the shutoff means, and the connections, lines and controls provide a degree of safety equal to that which would exist if the tank or reservoir were isolated from the engines.
- (c) There must be at least 13 mm ($\frac{1}{2}$ in) of clear airspace between each tank and each firewall or shroud isolating that tank, unless equivalent means are used to prevent heat transfer from each engine compartment to the flammable fluid.
- (d) Absorbent materials close to flammable fluid system components that might leak must be covered or treated to prevent the absorption of hazardous quantities of fluids.

CS 27.1187 Ventilation and drainage

ED Decision 2003/15/RM

Each compartment containing any part of the powerplant installation must have provision for ventilation and drainage of flammable fluids. The drainage means must be:

- (a) Effective under conditions expected to prevail when drainage is needed; and
- (b) Arranged so that no discharged fluid will cause an additional fire hazard.

CS 27.1189 Shut-off means

ED Decision 2003/15/RM

- (a) There must be means to shut off each line carrying flammable fluids into the engine compartment, except:
 - (1) Lines, fittings, and components forming an integral part of an engine;
 - (2) For oil systems for which all components of the system, including oil tanks, are fireproof or located in areas not subject to engine fire conditions; and
 - (3) For reciprocating engine installations only, engine oil system lines in installations using engines of less than 8195 cm³ (500 cubic inches) displacement.
- (b) There must be means to guard against inadvertent operation of each shutoff, and to make it possible for the crew to reopen it in flight after it has been closed.
- (c) Each shut-off valve and its control must be designed, located, and protected to function properly under any condition likely to result from an engine fire.

CS 27.1191 Firewalls

ED Decision 2003/15/RM

- (a) Each engine, including the combustor, turbine, and tailpipe sections of turbine engines must be isolated by a firewall, shroud or equivalent means, from personnel compartments, structures, controls, rotor mechanisms, and other parts that are:
 - (1) Essential to a controlled landing; and
 - (2) Not protected under [CS 27.861](#).
- (b) Each auxiliary power unit and combustion heater, and any other combustion equipment to be used in flight, must be isolated from the rest of the rotorcraft by firewalls, shrouds, or equivalent means.
- (c) In meeting sub-paragraphs (a) and (b), account must be taken of the probable path of a fire as affected by the airflow in normal flight and in autorotation.
- (d) Each firewall and shroud must be constructed so that no hazardous quantity of air, fluids, or flame can pass from any engine compartment to other parts of the rotorcraft.
- (e) Each opening in the firewall or shroud must be sealed with close-fitting, fireproof grommets, bushings, or firewall fittings.
- (f) Each firewall and shroud must be fireproof and protected against corrosion.

CS 27.1193 Cowling and engine compartment covering

ED Decision 2003/15/RM

- (a) Each cowling and engine compartment covering must be constructed and supported so that it can resist the vibration, inertia, and air loads to which it may be subjected in operation.
- (b) There must be means for rapid and complete drainage of each part of the cowling or engine compartment in the normal ground and flight attitudes.
- (c) No drain may discharge where it might cause a fire hazard.
- (d) Each cowling and engine compartment covering must be at least fire resistant.

- (e) Each part of the cowling or engine compartment covering subject to high temperatures due to its nearness to exhaust system parts or exhaust gas impingement must be fireproof.
- (f) A means of retaining each openable or readily removable panel, cowling, or engine or rotor drive system covering must be provided to preclude hazardous damage to rotors or critical control components in the event of structural or mechanical failure of the normal retention means, unless such failure is extremely improbable.

CS 27.1194 Other surfaces

ED Decision 2003/15/RM

All surfaces aft of, and near, powerplant compartments, other than tail surfaces not subject to heat, flames, or sparks emanating from a powerplant compartment, must be at least fire resistant.

CS 27.1195 Fire detector systems

ED Decision 2003/15/RM

Each turbine engine-powered rotorcraft must have approved quick-acting fire detectors in numbers and locations insuring prompt detection of fire in the engine compartment which cannot be readily observed in flight by the pilot in the cockpit.

SUBPART F – EQUIPMENT

GENERAL

CS 27.1301 Function and installation

ED Decision 2003/15/RM

Each item of installed equipment must:

- (a) Be of a kind and design appropriate to its intended function;
- (b) Be labelled as to its identification, function, or operating limitations, or any applicable combination of these factors;
- (c) Be installed according to limitations specified for that equipment; and
- (d) Function properly when installed.

CS 27.1303 Flight and navigation instruments

ED Decision 2003/15/RM

The following are the required flight and navigation instruments:

- (a) An airspeed indicator.
- (b) An altimeter.
- (c) A magnetic direction indicator.

CS 27.1305 Powerplant instruments

ED Decision 2008/009/R

The following are the required powerplant instruments:

- (a) A carburettor air temperature indicator, for each engine having a pre-heater that can provide a heat rise in excess of 33°C (60°F).
- (b) A cylinder head temperature indicator, for each:
 - (1) Air cooled engine;
 - (2) Rotorcraft with cooling shutters; and
 - (3) Rotorcraft for which compliance with [CS 27.1043](#) is shown in any condition other than the most critical flight condition with respect to cooling.
- (c) A fuel pressure indicator, for each pump-fed engine.
- (d) A fuel quantity indicator, for each fuel tank.
- (e) A manifold pressure indicator, for each altitude engine.
- (f) An oil temperature warning device to indicate when the temperature exceeds a safe value in each main rotor drive gearbox (including any gearboxes essential to rotor phasing) having an oil system independent of the engine oil system.
- (g) An oil pressure warning device to indicate when the pressure falls below a safe value in each pressure-lubricated main rotor drive gearbox (including any gearboxes essential to rotor phasing) having an oil system independent of the engine oil system.

- (h) An oil pressure indicator for each engine.
- (i) An oil quantity indicator for each oil tank.
- (j) An oil temperature indicator for each engine.
- (k) At least one tachometer to indicate the rpm of each engine and, as applicable:
 - (1) The rpm of the single main rotor;
 - (2) The common rpm of any main rotors whose speeds cannot vary appreciably with respect to each other; or
 - (3) The rpm of each main rotor whose speed can vary appreciably with respect to that of another main rotor.
- (l) A low fuel warning device for each fuel tank which feeds an engine. This device must:
 - (1) Provide a warning to the flight crew when approximately 10 minutes of usable fuel remains in the tank; and
 - (2) Be independent of the normal fuel quantity indicating system.
- (m) Means to indicate to the flight crew the failure of any fuel pump installed to show compliance with [CS 27.955](#).
- (n) A gas temperature indicator for each turbine engine.
- (o) Means to enable the pilot to determine the torque of each turboshaft engine, if a torque limitation is established for that engine under [CS 27.1521\(e\)](#).
- (p) For each turbine engine, an indicator to indicate the functioning of the powerplant ice protection system.
- (q) An indicator for the fuel filter required by [CS 27.997](#) to indicate the occurrence of contamination of the filter at the degree established by the applicant in compliance with [CS 27.955](#).
- (r) For each turbine engine, a warning means for the oil strainer or filter required by [CS 27.1019](#), if it has no by-pass, to warn the pilot of the occurrence of contamination of the strainer or filter before it reaches the capacity established in accordance with [CS 27.1019\(a\)\(2\)](#).
- (s) An indicator to indicate the proper functioning of any selectable or controllable heater used to prevent ice clogging of fuel system components.
- (t) For rotorcraft for which a 30-second/2-minute OEI power rating is requested, a means must be provided to alert the pilot when the engine is at the 30-second and 2-minute OEI power levels, when the event begins, and when the time interval expires.
- (u) For each turbine engine utilising 30-second/2-minute OEI power, a device or system must be provided for use by ground personnel which:
 - (1) Automatically records each usage and duration of power in the 30-second and 2-minute OEI levels;
 - (2) Permits retrieval of the recorded data;
 - (3) Can be reset only by ground maintenance personnel; and
 - (4) Has a means to verify proper operation of the system or device.
- (v) Warning or caution devices to signal to the flight crew when ferromagnetic particles are detected by the chip detector required by [27.1337\(e\)](#).

[Amdt 27/2]

CS 27.1307 Miscellaneous equipment

ED Decision 2003/15/RM

The following is the required miscellaneous equipment:

- (a) An approved seat for each occupant.
- (b) An approved safety belt for each occupant.
- (c) A master switch arrangement.
- (d) An adequate source of electrical energy, where electrical energy is necessary for operation of the rotorcraft.
- (e) Electrical protective devices.

CS 27.1309 Equipment, systems, and installations

ED Decision 2016/024/R

- (a) The equipment, systems, and installations whose functioning is required by this CS-27 must be designed and installed to ensure that they perform their intended functions under any foreseeable operating condition.
- (b) The equipment, systems, and installations of a multi-engine rotorcraft must be designed to prevent hazards to the rotorcraft in the event of a probable malfunction or failure.
- (c) The equipment, systems, and installations of single-engine rotorcraft must be designed to minimise hazards to the rotorcraft in the event of a probable malfunction or failure.

[Amdt 27/4]

CS 27.1316 Electrical and electronic system lightning protection

ED Decision 2016/024/R

- (a) Each electrical and electronic system that performs a function whose failure would prevent the continued safe flight and landing of the rotorcraft must be designed and installed in a way that:
 - (1) the function is not adversely affected during and after the rotorcraft's exposure to lightning; and
 - (2) the system automatically recovers normal operation of that function in a timely manner after the rotorcraft's exposure to lightning unless the system's recovery conflicts with other operational or functional requirements of the system that would prevent continued safe flight and landing of the rotorcraft.
- (b) For rotorcraft approved for instrument flight rules operation, each electrical and electronic system that performs a function whose failure would reduce the capability of the rotorcraft or the ability of the flight crew to respond to an adverse operating condition must be designed and installed in a way that the function recovers normal operation in a timely manner after the rotorcraft's exposure to lightning.

[Amdt 27/4]

CS 27.1317 High-Intensity Radiated Fields (HIRF) protection

ED Decision 2016/024/R

- (a) Each electrical and electronic system that performs a function whose failure would prevent the continued safe flight and landing of the rotorcraft must be designed and installed in a way that:
- (1) the function is not adversely affected during and after the rotorcraft's exposure to HIRF environment I as described in [Appendix D](#);
 - (2) the system automatically recovers normal operation of that function in a timely manner after the rotorcraft's exposure to HIRF environment I as described in [Appendix D](#) unless the system's recovery conflicts with other operational or functional requirements of the system that would prevent continued safe flight and landing of the rotorcraft;
 - (3) the system is not adversely affected during and after the rotorcraft's exposure to HIRF environment II as described in [Appendix D](#); and
 - (4) each function required during operation under visual flight rules is not adversely affected during and after the rotorcraft's exposure to HIRF environment III as described in [Appendix D](#).
- (b) Each electrical and electronic system that performs a function whose failure would significantly reduce the capability of the rotorcraft or the ability of the flight crew to respond to an adverse operating condition must be designed and installed in a way that the system is not adversely affected when the equipment providing the function is exposed to equipment HIRF test level 1 or 2 as described in [Appendix D](#).
- (c) Each electrical and electronic system that performs a function whose failure would reduce the capability of the rotorcraft or the ability of the flight crew to respond to an adverse operating condition must be designed and installed in a way that the system is not adversely affected when the equipment providing the function is exposed to equipment HIRF test level 3 as described in [Appendix D](#).

[Amdt 27/4]

Appendix D – HIRF Environments and Equipment HIRF Test Levels

ED Decision 2016/024/R

This Appendix specifies the HIRF environments and equipment HIRF test levels for electrical and electronic systems under [CS 27.1317](#). The field strength values for the HIRF environments and equipment HIRF test levels are expressed in root-mean-square units measured during the peak of the modulation cycle.

- (a) HIRF environment I is specified in the following table:

Table I – HIRF Environment I

FREQUENCY	FIELD STRENGTH (V/m)	
	PEAK	AVERAGE
10 kHz–2 MHz	50	50
2–30 MHz	100	100
100–400 MHz	100	100
400–700 MHz	700	50
700 MHz–1 GHz	700	100
1–2 GHz	2000	200
2–6 GHz	3000	200
6–8 GHz	1000	200
8–12 GHz	3000	300
12–18 GHz	2000	200
18–40 GHz	600	200

In this table, the higher field strength applies to the frequency band edges.

- (b) HIRF environment II is specified in the following table:

Table II – HIRF Environment II

FREQUENCY	FIELD STRENGTH (V/m)	
	PEAK	AVERAGE
10–500 kHz	20	20
500 kHz–2 MHz	30	30
2–30 MHz	100	100
30–100 MHz	10	10
100–200 MHz	30	10
200–400 MHz	10	10
400 MHz–1 GHz	700	40
1–2 GHz	1300	160
2–4 GHz	3000	120
4–6 GHz	3000	160
6–8 GHz	400	170
8–12 GHz	1230	230
12–18 GHz	730	190
18–40 GHz	600	150

In this table, the higher field strength applies to the frequency band edges.

- (c) HIRF environment III is specified in the following table:

Table III — HIRF Environment III

FREQUENCY	FIELD STRENGTH (V/m)	
	PEAK	AVERAGE
10–100 kHz	150	150
100 kHz–400 MHz	200	200
400–700 MHz	730	200
700 MHz–1 GHz	1400	240
1–2 GHz	5000	250
2–4 GHz	6000	490
4–6 GHz	7200	400
6–8 GHz	1100	170
8–12 GHz	5000	330
12–18 GHz	2000	330
18–40 GHz	1000	420

In this table, the higher field strength applies at the frequency band edges.

- (d) Equipment HIRF Test Level 1

- (1) From 10 kilohertz (kHz) to 400 megahertz (MHz), use conducted susceptibility tests with continuous wave (CW) and 1 kHz square wave modulation with 90 % depth or greater. The conducted susceptibility current must start at a minimum of 0.6 milliamperes (mA) at 10 kHz, increasing 20 decibels (dB) per frequency decade to a minimum of 30 mA at 500 kHz.
- (2) From 500 kHz to 40 MHz, the conducted susceptibility current must be at least 30 mA.
- (3) From 40 MHz to 400 MHz, use conducted susceptibility tests, starting at a minimum of 30 mA at 40 MHz, decreasing 20 dB per frequency decade to a minimum of 3 mA at 400 MHz.
- (4) From 100 MHz to 400 MHz, use radiated susceptibility tests at a minimum of 20 volts per meter (V/m) peak with CW and 1 kHz square wave modulation with 90 % depth or greater.
- (5) From 400 MHz to 8 gigahertz (GHz), use radiated susceptibility tests at a minimum of 150 V/m peak with pulse modulation of 4 % duty cycle with 1 kHz pulse repetition frequency. This signal must be switched on and off at a rate of 1 Hz with a duty cycle of 50 %.

- (e) Equipment HIRF Test Level 2. Equipment HIRF Test Level 2 is HIRF environment II in Table II of this Appendix reduced by acceptable aircraft transfer function and attenuation curves. Testing must cover the frequency band of 10 kHz to 8 GHz.

- (f) Equipment HIRF Test Level 3

- (1) From 10 kHz to 400 MHz, use conducted susceptibility tests, starting at a minimum of 0.15 mA at 10 kHz, increasing 20 dB per frequency decade to a minimum of 7.5 mA at 500 kHz.
- (2) From 500 kHz to 40 MHz, use conducted susceptibility tests at a minimum of 7.5 mA.

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- (3) From 40 MHz to 400 MHz, use conducted susceptibility tests, starting at a minimum of 7.5 mA at 40 MHz, decreasing 20 dB per frequency decade to a minimum of 0.75 mA at 400 MHz.
 - (4) From 100 MHz to 8 GHz, use radiated susceptibility tests at a minimum of 5 V/m.

[Amdt 27/4]

INSTRUMENTS: INSTALLATION

CS 27.1321 Arrangement and visibility

ED Decision 2003/15/RM

- (a) Each flight, navigation, and powerplant instrument for use by any pilot must be easily visible to him.
- (b) For each multi-engine rotorcraft, identical powerplant instruments must be located so as to prevent confusion as to which engine each instrument relates.
- (c) Instrument panel vibration may not damage, or impair the readability or accuracy of, any instrument.
- (d) If a visual indicator is provided to indicate malfunction of an instrument, it must be effective under all probable cockpit lighting conditions.

CS 27.1322 Warning, caution, and advisory lights

ED Decision 2003/15/RM

If warning, caution or advisory lights are installed in the cockpit, they must, unless otherwise approved the Agency, be:

- (a) Red, for warning lights (lights indicating a hazard which may require immediate corrective action);
- (b) Amber, for caution lights (lights indicating possible need for future corrective action);
- (c) Green, for safe operation lights; and
- (d) Any other colour, including white, for lights not described in sub-paragraphs (a) to (c), provided the colour differs sufficiently from the colours prescribed in sub-paragraphs (a) to (c) to avoid possible confusion.

CS 27.1323 Airspeed indicating system

ED Decision 2003/15/RM

- (a) Each airspeed indicating instrument must be calibrated to indicate true airspeed (at sea-level with a standard atmosphere) with a minimum practicable instrument calibration error when the corresponding pitot and static pressures are applied.
- (b) The airspeed indicating system must be calibrated in flight at forward speeds of 37 km/h (20 knots) and over.
- (c) At each forward speed above 80% of the climbout speed, the airspeed indicator must indicate true airspeed, at sea-level with a standard atmosphere, to within an allowable installation error of not more than the greater of:
 - (1) $\pm 3\%$ of the calibrated airspeed; or
 - (2) 9.3 km/h (5 knots).

CS 27.1325 Static pressure systems

ED Decision 2003/15/RM

- (a) Each instrument with static air case connections must be vented so that the influence of rotorcraft speed, the opening and closing of windows, airflow variation, and moisture or other foreign matter does not seriously affect its accuracy.
- (b) Each static pressure port must be designed and located in such a manner that the correlation between air pressure in the static pressure system and true ambient atmospheric static pressure is not altered when the rotorcraft encounters icing conditions. An anti-icing means or an alternate source of static pressure may be used in showing compliance with this requirement. If the reading of the altimeter, when on the alternate static pressure system, differs from the reading of the altimeter when on the primary static system by more than 15 m (50 feet), a correction card must be provided for the alternate static system.
- (c) Except as provided in sub-paragraph (d), if the static pressure system incorporates both a primary and an alternate static pressure source, the means for selecting one or the other source must be designed so that:
 - (1) When either source is selected, the other is blocked off, and
 - (2) Both sources cannot be blocked off simultaneously.
- (d) For unpressurised rotorcraft, sub-paragraph (c)(1) does not apply if it can be demonstrated that the static pressure system calibration, when either static pressure source is selected is not changed by the other static pressure source being open or blocked.

CS 27.1327 Magnetic direction indicator

ED Decision 2003/15/RM

- (a) Except as provided in sub-paragraph (b):
 - (1) Each magnetic direction indicator must be installed so that its accuracy is not excessively affected by the rotorcraft's vibration or magnetic fields; and
 - (2) The compensated installation may not have a deviation, in level flight, greater than 10° on any heading.
- (b) A magnetic non-stabilised direction indicator may deviate more than 10° due to the operation of electrically powered systems such as electrically heated windshields if either a magnetic stabilised direction indicator, which does not have a deviation in level flight greater than 10° on any heading, or a gyroscopic direction indicator, is installed. Deviations of a magnetic non-stabilised direction indicator of more than 10° must be placarded in accordance with [CS 27.1547\(e\)](#).

CS 27.1329 Automatic pilot system

ED Decision 2003/15/RM

- (a) Each automatic pilot system must be designed so that the automatic pilot can:
 - (1) Be sufficiently overpowered by one pilot to allow control of the rotorcraft; and
 - (2) Be readily and positively disengaged by each pilot to prevent it from interfering with control of the rotorcraft.
- (b) Unless there is automatic synchronisation, each system must have a means to readily indicate to the pilot the alignment of the actuating device in relation to the control system it operates.

- (c) Each manually operated control for the system's operation must be readily accessible to the pilots.
- (d) The system must be designed and adjusted so that, within the range of adjustment available to the pilot, it cannot produce hazardous loads on the rotorcraft or create hazardous deviations in the flight path under any flight condition appropriate to its use, either during normal operation or in the event of a malfunction, assuming that corrective action begins within a reasonable period of time.
- (e) If the automatic pilot integrates signals from auxiliary controls or furnishes signals for operation of other equipment, there must be positive interlocks and sequencing of engagement to prevent improper operation.
- (f) If the automatic pilot system can be coupled to airborne navigation equipment, means must be provided to indicate to the pilots the current mode of operation. Selector switch position is not acceptable as a means of indication.

CS 27.1335 Flight director systems

ED Decision 2003/15/RM

If a flight director system is installed, means must be provided to indicate to the flight crew its current mode of operation. Selector switch position is not acceptable as a means of indication.

CS 27.1337 Powerplant instruments

ED Decision 2003/15/RM

- (a) *Instruments and instrument lines*
 - (1) Each powerplant instrument line must meet the requirements of [CS 27.961](#) and [27.993](#).
 - (2) Each line carrying flammable fluids under pressure must:
 - (i) Have restricting orifices or other safety devices at the source of pressure to prevent the escape of excessive fluid if the line fails; and
 - (ii) Be installed and located so that the escape of fluids would not create a hazard.
 - (3) Each powerplant instrument that utilises flammable fluids must be installed and located so that the escape of fluid would not create a hazard.
- (b) *Fuel quantity indicator.* Each fuel quantity indicator must be installed to clearly indicate to the flight crew the quantity of fuel in each tank in flight. In addition:
 - (1) Each fuel quantity indicator must be calibrated to read 'zero' during level flight when the quantity of fuel remaining in the tank is equal to the unusable fuel supply determined under [CS 27.959](#);
 - (2) When two or more tanks are closely interconnected by a gravity feed system and vented, and when it is impossible to feed from each tank separately, at least one fuel quantity indicator must be installed; and
 - (3) Each exposed sight gauge used as a fuel quantity indicator must be protected against damage.
- (c) *Fuel flow meter system.* If a fuel flow meter system is installed, each metering component must have a means for bypassing the fuel supply if malfunction of that component severely restricts fuel flow.

- (d) *Oil quantity indicator.* There must be means to indicate the quantity of oil in each tank:
 - (1) On the ground (including during the filling of each tank); and
 - (2) In flight, if there is an oil transfer system or reserve oil supply system.
- (e) Rotor drive system transmissions and gearboxes utilising ferromagnetic materials must be equipped with chip detectors designed to indicate the presence of ferromagnetic particles resulting from damage or excessive wear. Chip detectors must:
 - (1) be designed to provide a signal to the indicator required by [27.1305\(v\)](#); and
 - (2) be provided with a means to allow crew members to check, in flight, the function of each detector electrical circuit and signal.

ELECTRICAL SYSTEMS AND EQUIPMENT

CS 27.1351 General

ED Decision 2003/15/RM

- (a) *Electrical system capacity.* Electrical equipment must be adequate for its intended use. In addition:
- (1) Electric power sources, their transmission cables, and their associated control and protective devices must be able to furnish the required power at the proper voltage to each load circuit essential for safe operation; and
 - (2) Compliance with paragraph (a)(1) must be shown by an electrical load analysis, or by electrical measurements that take into account the electrical loads applied to the electrical system, in probable combinations and for probable durations.
- (b) *Function.* For each electrical system the following apply:
- (1) Each system, when installed, must be:
 - (i) Free from hazards in itself, in its method of operation, and in its effects on other parts of the rotorcraft; and
 - (ii) Protected from fuel, oil, water, other detrimental substances, and mechanical damage.
 - (2) Electric power sources must function properly when connected in combination or independently.
 - (3) No failure or malfunction of any source may impair the ability of any remaining source to supply load circuits essential for safe operation.
 - (4) Each electric power source control must allow the independent operation of each source.
- (c) *Generating system.* There must be at least one generator if the system supplies power to load circuits essential for safe operation. In addition:
- (1) Each generator must be able to deliver its continuous rated power;
 - (2) Generator voltage control equipment must be able to dependably regulate each generator output within rated limits;
 - (3) Each generator must have a reverse current cut-out designed to disconnect the generator from the battery and from the other generators when enough reverse current exists to damage that generator; and
 - (4) Each generator must have an over voltage control designed and installed to prevent damage to the electrical system, or to equipment supplied by the electrical system, that could result if that generator were to develop an over voltage condition.
- (d) *Instruments.* There must be means to indicate to appropriate crew members the electric power system quantities essential for safe operation of the system. In addition –
- (1) For direct current systems, an ammeter that can be switched into each generator feeder may be used; and
 - (2) If there is only one generator, the ammeter may be in the battery feeder.

- (e) *External power.* If provisions are made for connecting external power to the rotorcraft, and that external power can be electrically connected to equipment other than that used for engine starting, means must be provided to ensure that no external power supply having a reverse polarity, or a reverse phase sequence, can supply power to the rotorcraft's electrical system.

CS 27.1353 Storage battery design and installation

ED Decision 2003/15/RM

- (a) Each storage battery must be designed and installed as prescribed in this paragraph.
- (b) Safe cell temperatures and pressures must be maintained during any probable charging and discharging condition. No uncontrolled increase in cell temperature may result when the battery is recharged (after previous complete discharge):
- (1) At maximum regulated voltage or power;
 - (2) During a flight of maximum duration; and
 - (3) Under the most adverse cooling condition likely to occur in service.
- (c) Compliance with sub-paragraph (b) must be shown by test unless experience with similar batteries and installations has shown that maintaining safe cell temperatures and pressures presents no problem.
- (d) No explosive or toxic gases emitted by any battery in normal operation, or as the result of any probable malfunction in the charging system or battery installation, may accumulate in hazardous quantities within the rotorcraft.
- (e) No corrosive fluids or gases that may escape from the battery may damage surrounding structures or adjacent essential equipment.
- (f) Each nickel cadmium battery installation capable of being used to start an engine or auxiliary power unit must have provisions to prevent any hazardous effect on structure or essential systems that may be caused by the maximum amount of heat the battery can generate during a short circuit of the battery or of its individual cells.
- (g) Nickel cadmium battery installations capable of being used to start an engine or auxiliary power unit must have:
- (1) A system to control the charging rate of the battery automatically so as to prevent battery overheating;
 - (2) A battery temperature sensing and over-temperature warning system with a means for disconnecting the battery from its charging source in the event of an over-temperature condition; or
 - (3) A battery failure sensing and warning system with a means for disconnecting the battery from its charging source in the event of battery failure.

CS 27.1357 Circuit protective devices

ED Decision 2003/15/RM

- (a) Protective devices, such as fuses or circuit breakers, must be installed in each electrical circuit other than:
- (1) The main circuits of starter motors; and
 - (2) Circuits in which no hazard is presented by their omission.

- (b) A protective device for a circuit essential to flight safety may not be used to protect any other circuit.
- (c) Each resettable circuit protective device ('trip free' device in which the tripping mechanism cannot be overridden by the operating control) must be designed so that:
 - (1) A manual operation is required to restore service after tripping; and
 - (2) If an overload or circuit fault exists, the device will open the circuit regardless of the position of the operating control.
- (d) If the ability to reset a circuit breaker or replace a fuse is essential to safety in flight, that circuit breaker or fuse must be located and identified so that it can be readily reset or replaced in flight.
- (e) If fuses are used, there must be one spare of each rating, or 50% spare fuses of each rating, whichever is greater.

CS 27.1361 Master switch

ED Decision 2003/15/RM

- (a) There must be a master switch arrangement to allow ready disconnection of each electric power source from the main bus. The point of disconnection must be adjacent to the sources controlled by the switch.
- (b) Load circuits may be connected so that they remain energised after the switch is opened, if they are protected by circuit protective devices, rated at five amperes or less, adjacent to the electric power source.
- (c) The master switch or its controls must be installed so that the switch is easily discernible and accessible to a crew member in flight.

CS 27.1365 Electric cables

ED Decision 2003/15/RM

- (a) Each electric connecting cable must be of adequate capacity.
- (b) Each cable that would overheat in the event of circuit overload or fault must be at least flame resistant and may not emit dangerous quantities of toxic fumes.
- (c) Insulation on electrical wire and cable installed in the rotorcraft must be self-extinguishing when tested in accordance with CS-25, appendix F, part I (a)(3).

CS 27.1367 Switches

ED Decision 2003/15/RM

Each switch must be:

- (a) Able to carry its rated current;
- (b) Accessible to the crew; and
- (c) Labelled as to operation and the circuit controlled.

LIGHTS

CS 27.1381 Instrument lights

ED Decision 2003/15/RM

The instrument lights must:

- (a) Make each instrument, switch, and other devices for which they are provided easily readable; and
- (b) Be installed so that:
 - (1) Their direct rays are shielded from the pilot's eyes; and
 - (2) No objectionable reflections are visible to the pilot.

CS 27.1383 Landing lights

ED Decision 2003/15/RM

- (a) Each required landing or hovering light must be approved.
- (b) Each landing light must be installed so that:
 - (1) No objectionable glare is visible to the pilot;
 - (2) The pilot is not adversely affected by halation; and
 - (3) It provides enough light for night operation, including hovering and landing.
- (c) At least one separate switch must be provided, as applicable:
 - (1) For each separately installed landing light; and
 - (2) For each group of landing lights installed at a common location.

CS 27.1385 Position light system installation

ED Decision 2003/15/RM

- (a) *General.* Each part of each position light system must meet the applicable requirements of this paragraph, and each system as a whole must meet the requirements of [CS 27.1387](#) to [27.1397](#).
- (b) *Forward position lights.* Forward position lights must consist of a red and a green light spaced laterally as far apart as practicable and installed forward on the rotorcraft so that, with the rotorcraft in the normal flying position, the red light is on the left side and the green light is on the right side. Each light must be approved.
- (c) *Rear position light.* The rear position light must be a white light mounted as far aft as practicable, and must be approved.
- (d) *Circuit.* The two forward position lights and the rear position light must make a single circuit.
- (e) *Light covers and colour filters.* Each light cover or colour filter must be at least flame resistant and may not change colour or shape or lose any appreciable light transmission during normal use.

CS 27.1387 Position light system dihedral angles

ED Decision 2003/15/RM

- (a) Except as provided in sub-paragraph (e), each forward and rear position light must, as installed, show unbroken light within the dihedral angles described in this paragraph.
- (b) Dihedral angle L (left) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the rotorcraft, and the other at 110° to the left of the first, as viewed when looking forward along the longitudinal axis.
- (c) Dihedral angle R (right) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the rotorcraft, and the other at 110° to the right of the first, as viewed when looking forward along the longitudinal axis.
- (d) Dihedral angle A (aft) is formed by two intersecting vertical planes making angles of 70° to the right and to the left, respectively, to a vertical plane passing through the longitudinal axis, as viewed when looking aft along the longitudinal axis.
- (e) If the rear position light, when mounted as far aft as practicable in accordance with [27.1385\(c\)](#), cannot show unbroken light within dihedral angle A (as defined in sub-paragraph (d)), a solid angle or angles of obstructed visibility totalling not more than 0.04 steradians is allowable within that dihedral angle, if such solid angle is within a cone whose apex is at the rear position light and whose elements make an angle of 30° with a vertical line passing through the rear position light.

CS 27.1389 Position light distribution and intensities

ED Decision 2003/15/RM

- (a) *General.* The intensities prescribed in this paragraph must be provided by new equipment with light covers and colour filters in place. Intensities must be determined with the light source operating at a steady value equal to the average luminous output of the source at the normal operating voltage of the rotorcraft. The light distribution and intensity of each position light must meet the requirements of sub-paragraph (b).
- (b) *Forward and rear position lights.* The light distribution and intensities of forward and rear position lights must be expressed in terms of minimum intensities in the horizontal plane, minimum intensities in any vertical plane, and maximum intensities in overlapping beams, within dihedral angles L, R, and A, and must meet the following requirements:
 - (1) *Intensities in the horizontal plane.* Each intensity in the horizontal plane (the plane containing the longitudinal axis of the rotorcraft and perpendicular to the plane of symmetry of the rotorcraft) must equal or exceed the values in [CS 27.1391](#).
 - (2) *Intensities in any vertical plane.* Each intensity in any vertical plane (the plane perpendicular to the horizontal plane) must equal or exceed the appropriate value in [CS 27.1393](#), where *I* is the minimum intensity prescribed in [CS 27.1391](#) for the corresponding angles in the horizontal plane.
 - (3) *Intensities in overlaps between adjacent signals.* No intensity in any overlap between adjacent signals may exceed the values in [CS 27.1395](#), except that higher intensities in overlaps may be used with main beam intensities substantially greater than the minima specified in [CS 27.1391](#) and [27.1393](#), if the overlap intensities in relation to the main beam intensities do not adversely affect signal clarity. When the peak intensity of the forward position lights is greater than 100 candelas, the maximum overlap intensities between them may exceed the values in [CS 27.1395](#) if the overlap intensity in Area A is

not more than 10% of peak position light intensity and the overlap intensity in Area B is not more than 2.5% of peak position light intensity.

CS 27.1391 Minimum intensities in the horizontal plane of forward and rear position lights

ED Decision 2003/15/RM

Each position light intensity must equal or exceed the applicable values in the following table:

Dihedral angle (light included)	Angle from right or left of longitudinal axis, measured from dead ahead	Intensity (candelas)
L and R (forward red and green)	0° to 10°	40
	10° to 20°	30
A (rear white)	20° to 110°	5
	110° to 180°	20

CS 27.1393 Minimum intensities in any vertical plane of forward and rear position lights

ED Decision 2003/15/RM

Each position light intensity must equal or exceed the applicable values in the following table:

Angle above or below the horizontal plane –	Intensity
0	1.0 I
0° to 5°	0.90 I
5° to 10°	0.80 I
10° to 15°	0.70 I
15° to 20°	0.50 I
20° to 30°	0.30 I
30° to 40°	0.10 I
40° to 90°	0.05 I

CS 27.1395 Maximum intensities in overlapping beams of forward and rear position lights

ED Decision 2003/15/RM

No position light intensity may exceed the applicable values in the following table, except as provided in [CS 27.1389\(b\)\(3\)](#):

Overlaps	Maximum intensity	
	Area A (candelas)	Area B (candelas)
Green in dihedral angle L	10	1
Red in dihedral angle R	10	1
Green in dihedral angle A	5	1
Red in dihedral angle A	5	1
Rear white in dihedral angle L	5	1
Rear white in dihedral angle R	5	1

Where:

- (a) Area A includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 10° but less than 20°; and
- (b) Area B includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 20°.

CS 27.1397 Colour specifications

ED Decision 2003/15/RM

Each position light colour must have the applicable International Commission on Illumination chromaticity co-ordinates as follows:

- (a) *Aviation red*: 'y' is not greater than 0.335; and
'z' is not greater than 0.002.
- (b) *Aviation green*:
'x' is not greater than 0.440–0.320y;
'x' is not greater than y–0.170; and
'y' is not less than 0.390–0.170x.
- (c) *Aviation white*:
'x' is not less than 0.300 and not greater than 0.540;
'y' is not less than 'x–0.040' or 'y_o–0.010', whichever is the smaller; and
'y' is not greater than 'x + 0.020' nor '0.636–0.400x';
Where 'y_o' is the 'y' co-ordinate of the Planckian radiator for the value of 'x' considered.

CS 27.1399 Riding light

ED Decision 2003/15/RM

- (a) Each riding light required for water operation must be installed so that it can:
 - (1) Show a white light for at least 3.7 km (two nautical miles) at night under clear atmospheric conditions; and
 - (2) Show a maximum practicable unbroken light with the rotorcraft on the water.
- (b) Externally hung lights may be used.

CS 27.1401 Anti-collision light system

ED Decision 2003/15/RM

- (a) *General*. If certification for night operation is requested, the rotorcraft must have an anti-collision light system that:
 - (1) Consists of one or more approved anti-collision lights located so that their emitted light will not impair the crew's vision or detract from the conspicuity of the position lights; and
 - (2) Meets the requirements of sub-paragraphs (b) to (f).

- (b) *Field of coverage.* The system must consist of enough lights to illuminate the vital areas around the rotorcraft, considering the physical configuration and flight characteristics of the rotorcraft. The field of coverage must extend in each direction within at least 30° above and 30° below the horizontal plane of the rotorcraft, except that there may be solid angles of obstructed visibility totalling not more than 0.5 steradians.
- (c) *Flashing characteristics.* The arrangement of the system, that is, the number of light sources, beam width, speed of rotation, and other characteristics, must give an effective flash frequency of not less than 40, nor more than 100, cycles per minute. The effective flash frequency is the frequency at which the rotorcraft's complete anti-collision light system is observed from a distance, and applies to each sector of light including any overlaps that exist when the system consists of more than one light source. In overlaps, flash frequencies may exceed 100, but not 180, cycles per minute.
- (d) *Colour.* Each anti-collision light must be aviation red and must meet the applicable requirements of [CS 27.1397](#).
- (e) *Light intensity.* The minimum light intensities in any vertical plane, measured with the red filter (if used) and expressed in terms of 'effective' intensities, must meet the requirements of subparagraph (f). The following relation must be assumed:

$$I_e = \frac{\int_{t_1}^{t_2} I(t) dt}{0.2 + (t_2 - t_1)}$$

where:

I_e = effective intensity (candelas).

$I(t)$ = instantaneous intensity as a function of time.

$t_2 - t_1$ = flash time interval (seconds).

Normally, the maximum value of effective intensity is obtained when t_2 and t_1 are chosen so that the effective intensity is equal to the instantaneous intensity at t_2 and t_1 .

- (f) *Minimum effective intensities for anticollision light.* Each anti-collision light effective intensity must equal or exceed the applicable values in the following table:

Angle above or below the horizontal plane	Effective intensity (candelas)
0° to 5°	150
5° to 10°	90
10° to 20°	30
20° to 30°	15

SAFETY EQUIPMENT

CS 27.1411 General

ED Decision 2018/007/R

- (a) *Accessibility.* Required safety equipment to be used by the crew in an emergency must be readily accessible.
- (b) *Stowage provisions.* Stowage provisions for required safety equipment must be furnished and must:
 - (1) be arranged so that the equipment is directly accessible and its location is obvious; and
 - (2) protect the safety equipment from inadvertent damage.

[Amdt No: 27/5]

AMC 27.1411 Safety equipment – General

ED Decision 2018/007/R

This AMC replaces FAA AC 27.1411.

(a) Explanation

CS-27 Amendment 5 introduced changes related to ditching and associated equipment. In particular, it defined a standard set of terminology, it simplified [CS 27.1411](#) in line with it being a general certification specification for safety equipment, reorganised [CS 27.1415](#) specifically for ditching equipment, and created a new [CS 27.1470](#) on the installation and carriage of emergency locator transmitters (ELTs). All requirements relating to life raft installations are now co-located in [CS 27.1415](#).

- (1) The safety equipment should be accessible and appropriately stowed, and it should be ensured that:
 - (i) locations for stowage of all required safety equipment have been provided;
 - (ii) safety equipment is readily accessible to both crew members and passengers, as appropriate, during any reasonably probable emergency situation;
 - (iii) stowage locations for all required safety equipment will adequately protect such equipment from inadvertent damage during normal operations; and
 - (iv) safety equipment stowage provisions will protect the equipment from damage during emergency landings when subjected to the inertia loads specified in [CS 27.561](#).

(b) Procedures

- (1) A cockpit evaluation should be conducted to demonstrate that all required emergency equipment to be used by the flight crew will be readily accessible during any probable emergency situation. This evaluation should include, for example, emergency flotation equipment actuation devices, remote life raft releases, door jettison handles, handheld fire extinguishers, and protective breathing equipment.
- (2) Stowage provisions for safety equipment shown to be compatible with the vehicle configuration presented for certification should be provided and identified so that:
 - (i) equipment is readily accessible regardless of the operational configuration;

- (ii) stowed equipment is free from inadvertent damage from passengers and handling; and
- (iii) stowed equipment is adequately restrained to withstand the inertia forces specified in [CS 27.561\(b\)\(3\)](#) without sustaining damage.

[Amdt No: 27/5]

CS 27.1413 Safety belts

ED Decision 2003/15/RM

Each safety belt must be equipped with a metal to metal latching device.

CS 27.1415 Ditching equipment

ED Decision 2018/007/R

If certification with ditching provisions or emergency flotation provisions is requested by the applicant, the additional safety equipment required by any applicable operating rule must meet the requirements of this CS.

- (a) All equipment must be approved.
- (b) Life rafts.
 - (1) Required life raft(s) must be remotely deployable for use in an emergency. Remote controls capable of deploying the life raft(s) must be located within easy reach of the flight crew, occupants of the passenger cabin and survivors in the water, with the rotorcraft in the upright floating or capsized position. It must be substantiated that life rafts sufficient to accommodate all rotorcraft occupants, without exceeding the rated capacity of any life raft, can be reliably deployed with the rotorcraft in any reasonably foreseeable floating attitude, including capsized, and in the sea conditions chosen for showing compliance with [CS 27.801\(e\)](#).
 - (2) Each life raft must be attached to the rotorcraft by a short retaining line to keep it alongside the rotorcraft and a long retaining line designed to keep it attached to the rotorcraft. Both retaining lines must be weak enough to break before submerging the empty life raft to which they are attached. The long retaining line must be of sufficient length that a drifting life raft will not be drawn towards any part of the rotorcraft that would pose a danger to the life raft itself or the persons on board.
 - (3) Each life raft must be substantiated as suitable for use in all sea conditions covered by the certification with ditching or emergency flotation provisions.
- (c) Life preservers.

If the applicable operating rule allows for life preservers not to be worn at all times, stowage provisions must be provided that accommodate one life preserver for each occupant for which certification with ditching or emergency flotation provisions.

[Amdt No: 27/5]

AMC 27.1415 Ditching equipment

ED Decision 2018/007/R

This AMC replaces FAA AC 27.1415.

(a) Explanation

- (1) Additional safety equipment is not required for all rotorcraft overwater operations. However, if such equipment is required by the applicable operating rule, the equipment supplied should satisfy this AMC.

NOTE: Although the term ‘ditching’ is most commonly associated with the design standards related to [CS 27.801](#) (ditching approval), a rotorcraft equipped to the less demanding requirements of [CS 27.802](#) (emergency flotation approval), when performing an emergency landing on to water, would nevertheless be commonly described as carrying out the process of ditching. The term ‘ditching equipment’ is therefore to be considered to apply to any safety equipment required by operational rule for operation over water.

It is a frequent practice for the rotorcraft manufacturer to provide the substantiation for only those portions of the ditching requirements relating to rotorcraft flotation and emergency exits. Completion of the ditching certification to include the safety equipment installation and stowage provisions is then left to the affected operator to arrange via a modifier so that those aspects can best be adopted to the selected cabin interior. In such cases, the ‘Limitations’ section of the rotorcraft flight manual (RFM) should identify the substantiations yet to be provided in order to justify the full certification with ditching provisions. The modifier performing these final installations is then concerned directly with the details of this AMC. Any issues arising from aspects of the basic rotorcraft flotation and emergency exits certification that are not compatible with the modifier’s proposed safety equipment provisions should be resolved between the type certificate (TC) holder and the modifier prior to the certifying authority’s certification with ditching provisions (see [AMC 27.801\(b\)\(12\)](#) and [AMC 27.1415\(a\)\(2\)\(ii\)](#)).

- (2) Compliance with the requirements of [CS 27.801](#) for rotorcraft ditching requires compliance with the safety equipment stowage requirements and ditching equipment requirements of [CS 27.1411](#) and [CS 27.1415](#), respectively.
- (i) Ditching equipment, installed to complete ditching certification, or required by the applicable operating rule, should be compatible with the basic rotorcraft configuration presented for ditching certification. It is satisfactory if the operating equipment is not incorporated at the time of the original rotorcraft type certification provided that suitable information is included in the ‘Limitations’ section of the rotorcraft flight manual (RFM) to identify the extent of ditching certification not yet completed.
- (ii) When ditching equipment is being installed by a person other than the applicant who provided the rotorcraft flotation system and emergency exits, special care should be taken to avoid degrading the functioning of those items, and to make the ditching equipment compatible with them (see [AMC 27.801\(b\)\(12\)](#) and [AMC 27.1411\(a\)\(2\)](#)).

(b) Procedures

All ditching equipment, including life rafts, life preservers, immersion suits, emergency breathing systems etc., should be of an approved type. Life rafts should be chosen to be suitable for use in all sea conditions covered by the certification with ditching provisions.

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- (1) Life rafts
- (i) Life rafts are rated during their approval according to the number of people that can be carried under normal conditions and the number that can be accommodated in an overload condition. Only the normal rating may be used in relation to the number of occupants permitted to fly in the rotorcraft.
 - (ii) Where two life rafts are installed, they should deploy on opposite sides of the rotorcraft in order to minimise the probability that both will be damaged during water entry/impact, and to provide the maximum likelihood that at least one raft will be useable in any wind condition.
 - (iii) Successful deployment of life raft installations should be demonstrated in representative orientations. Testing should be performed, including underwater deployment, if applicable, to demonstrate that life rafts sufficient to accommodate all rotorcraft occupants, without exceeding the rated capacity of any life raft, will deploy reliably with the rotorcraft in any reasonably foreseeable floating attitude, including capsized. It should also be substantiated that reliable deployment will not be compromised by inertial effects from the rolling/pitching/heaving of the rotorcraft in the sea conditions chosen for the demonstration of compliance with the flotation/trim requirements of [CS 27.801\(e\)](#), or by intermittent submerging of the stowed raft location (if applicable) and the effects of wind. This substantiation should also consider all reasonably foreseeable rotorcraft floating attitudes, including capsized. Reasonably foreseeable floating attitudes are considered to be, as a minimum, upright, with and without loss of the critical emergency flotation system (EFS) compartment, and capsized, also with and without loss of the critical EFS compartment. Consideration should also be given towards maximising, where practicable, the likelihood of life raft deployment for other cases of EFS damage.
 - (iv) Rotorcraft fuselage attachments for the life raft retaining lines should be provided.
 - (A) Each life raft must be equipped with two retaining lines to be used for securing the life raft to the rotorcraft. The short retaining line should be of such a length as to hold the raft at a point next to an upright floating rotorcraft such that the occupants can enter the life raft directly without entering the water. If the design of the rotorcraft is such that the flight crew cannot enter the passenger cabin, it is acceptable that they would need to take a more indirect route when boarding the life raft. After life raft boarding is completed, the short retaining line may be cut and the life raft then remain attached to the rotorcraft by means of the long retaining line.
 - (B) Attachments on the rotorcraft for the retaining lines should not be susceptible to damage when the rotorcraft is subjected to the maximum water entry loads established by [CS 27.563](#).
 - (C) Attachments on the rotorcraft for the retaining lines should be structurally adequate to restrain a fully loaded life raft.
 - (D) Life rafts should be attached to the rotorcraft by the required retaining lines after deployment without further action from the crew or passengers.
 - (E) It should be verified that the length of the long retaining line will not result in the life raft taking up a position which could create a potential puncture risk or hazard to the occupants, such as directly under the tail boom, tail rotor or main rotor disc.

- (v) Life raft stowage provisions should be sufficient to accommodate rafts for the maximum number of occupants for which certification for ditching is requested by the applicant.

- (vi) Life raft activation

The following should be provided for each life raft:

- (A) primary activation: manual activation control(s), readily accessible to each pilot on the flight deck whilst seated;
- (B) secondary activation: manual activation control(s) accessible from the passenger cabin; if any control is located within the cabin, it should be protected from inadvertent operation; and
- (C) tertiary activation: manual activation control(s) accessible to a person in the water, with the rotorcraft in all foreseeable floating attitudes, including capsized.

It is acceptable for two or more of the above functions to be incorporated into one control.

Automatic life raft activation is not prohibited (e.g. it could be triggered by water immersion). However, if such a capability is provided, it should be in addition to the above manual activation controls, not instead of them, and issues such as inadvertent deployment in flight and the potential for damage from turning rotors during deployment on the water should be mitigated.

Placards should be installed, of appropriate size, number and location, to highlight the location of each of the above life raft activation controls. All reasonably foreseeable rotorcraft floating attitudes should be considered.

- (vii) Protection of life rafts from damage

Service experience has shown that following deployment, life rafts are susceptible to damage while in the water adjacent to the rotorcraft due to projections on the exterior of the rotorcraft such as antennas, overboard vents, unprotected split-pin tails, guttering, etc. and any projections sharper than a three-dimensional right angled corner. Projections likely to cause damage to a deployed life raft should be avoided by design, or suitably protected to minimise the likelihood of their causing damage to a deployed life raft. In general, projections on the exterior surface of the helicopter, that are located in a zone delineated by boundaries that are 1.22 m (4 ft) above and 0.61 m (2 ft) below the established static water line should be assessed. Relevant maintenance information should also provide procedures for maintaining such protection for rotorcraft equipped with life rafts. Furthermore, due account should be taken of the likely damage that may occur (e.g. disintegration of carbon-fibre panels or structure) during water entry and its potential hazard to deployed life rafts.

- (2) Life preservers.

No provision for the stowage of life preservers is necessary if the applicable operating rule mandates the need for constant-wear life preservers.

- (3) Emergency signalling equipment.

Emergency signalling equipment required by the applicable operating rule should be free from hazards in its operation, and operable using either bare or gloved hands. Required

signalling equipment should be easily accessible to the passengers or crew and located near an emergency exit or included in the survival equipment attached to the life rafts.

[Amdt No: 27/5]

CS 27.1419 Ice protection

ED Decision 2003/15/RM

- (a) To obtain certification for flight into icing conditions, compliance with this paragraph must be shown.
- (b) It must be demonstrated that the rotorcraft can be safely operated in the continuous maximum and intermittent maximum icing conditions determined under appendix C of CS-29 within the rotorcraft altitude envelope. An analysis must be performed to establish, on the basis of the rotorcraft's operational needs, the adequacy of the ice protection system for the various components of the rotorcraft.
- (c) In addition to the analysis and physical evaluation prescribed in sub-paragraph (b), the effectiveness of the ice protection system and its components must be shown by flight tests of the rotorcraft or its components in measured atmospheric icing conditions and by one or more of the following tests as found necessary to determine the adequacy of the ice protection system:
 - (1) Laboratory dry air or simulated icing tests, or a combination of both, of the components or models of the components.
 - (2) Flight dry air tests of the ice protection system as a whole, or its individual components.
 - (3) Flight tests of the rotorcraft or its components in measured simulated icing conditions.
- (d) The ice protection provisions of this paragraph are considered to be applicable primarily to the airframe. Powerplant installation requirements are contained in Subpart E of this CS-27.
- (e) A means must be identified or provided for determining the formation of ice on critical parts of the rotorcraft. Unless otherwise restricted, the means must be available for night-time as well as daytime operation. The rotorcraft flight manual must describe the means of determining ice formation and must contain information necessary for safe operation of the rotorcraft in icing conditions.

CS 27.1435 Hydraulic systems

ED Decision 2003/15/RM

- (a) *Design.* Each hydraulic system and its elements must withstand, without yielding, any structural loads expected in addition to hydraulic loads.
- (b) *Tests.* Each system must be substantiated by proof pressure tests. When proof tested, no part of any system may fail, malfunction, or experience a permanent set. The proof load of each system must at least 1.5 times the maximum operating pressure of that system.
- (c) *Accumulators.* No hydraulic accumulator or pressurised reservoir may be installed on the engine side of any firewall unless it is an integral part of an engine.

CS 27.1457 Cockpit voice recorders

ED Decision 2003/15/RM

- (a) Each cockpit voice recorder required by the applicable operating rules must be approved, and must be installed so that it will record the following:
- (1) Voice communications transmitted from or received in the rotorcraft by radio.
 - (2) Voice communications of flight-crew members on the flight deck.
 - (3) Voice communications of flight-crew members on the flight deck, using the rotorcraft's interphone system.
 - (4) Voice or audio signals identifying navigation or approach aids introduced into a headset or speaker.
 - (5) Voice communications of flight-crew members using the passenger loudspeaker system, if there is such a system, and if the fourth channel is available in accordance with the requirements of sub-paragraph (c)(4)(ii).
- (b) The recording requirements of subparagraph (a)(2) may be met:
- (1) By installing a cockpit-mounted area microphone located in the best position for recording voice communications originating at the first and second pilot stations and voice communications of other crew members on the flight deck when directed to those stations; or
 - (2) By installing a continually energised or voice-activated lip microphone at the first and second pilot stations. The microphone specified in this paragraph must be so located and if necessary, the preamplifiers and filters of the recorder must be adjusted or supplemented so that the recorded communications are intelligible when recorded under flight cockpit noise conditions and played back. The level of intelligibility must be approved by the Agency. Repeated aural or visual playback of the record may be used in evaluating intelligibility.
- (c) Each cockpit voice recorder must be installed so that the part of the communication or audio signals specified in sub-paragraph (a) obtained from each of the following sources is recorded on a separate channel:
- (1) For the first channel, from each microphone, headset, or speaker used at the first pilot station.
 - (2) For the second channel, from each microphone, headset, or speaker used at the second pilot station.
 - (3) For the third channel, from the cockpit-mounted area microphone, or the continually energised or voice-activated lip microphone at the first and second pilot stations.
 - (4) For the fourth channel, from:
 - (i) Each microphone, headset, or speaker used at the stations for the third and fourth crew members; or
 - (ii) If the stations specified in sub-paragraph (c)(4)(i) are not required or if the signal at such a station is picked up by another channel, each microphone on the flight deck that is used with the passenger loud-speaker system if its signals are not picked up by another channel.

- (iii) Each microphone on the flight deck that is used with the rotorcraft's loudspeaker system if its signals are not picked up by another channel.
- (d) Each cockpit voice recorder must be installed so that:
 - (1) It receives its electric power from the bus that provides the maximum reliability for operation of the cockpit voice recorder without jeopardising service to essential or emergency loads;
 - (2) There is an automatic means to simultaneously stop the recorder and prevent each erasure feature from functioning, within 10 minutes after crash impact; and
 - (3) There is an aural or visual means for pre-flight checking of the recorder for proper operation.
- (e) The record container must be located and mounted to minimise the probability of rupture of the container as a result of crash impact and consequent heat damage to the record from fire.
- (f) If the cockpit voice recorder has a bulk erasure device, the installation must be designed to minimise the probability of inadvertent operation and actuation of the device during crash impact.
- (g) Each recorder container must be either bright orange or bright yellow.

CS 27.1459 Flight recorders

ED Decision 2003/15/RM

- (a) Each flight recorder required by the applicable operating rules must be installed so that:
 - (1) It is supplied with airspeed, altitude, and directional data obtained from sources that meet the accuracy requirements of [CS 27.1323](#), [CS 27.1325](#), and [27.1327](#), as applicable;
 - (2) The vertical acceleration sensor is rigidly attached, and located longitudinally within the approved centre of gravity limits of the rotorcraft;
 - (3) It receives its electrical power from the bus that provides the maximum reliability for operation of the flight recorder without jeopardising service to essential or emergency loads;
 - (4) There is an aural or visual means for pre-flight checking of the recorder for proper recording of data in the storage medium;
 - (5) Except for recorders powered solely by the engine-driven electrical generator system, there is an automatic means to simultaneously stop a recorder that has a data erasure feature and prevent each erasure feature from functioning, within 10 minutes after any crash impact; and
- (b) Each non-ejectable recorder container must be located and mounted so as to minimise the probability of container rupture resulting from crash impact and subsequent damage to the record from fire.
- (c) A correlation must be established between the flight recorder readings of airspeed, altitude, and heading and the corresponding readings (taking into account correction factors) of the first pilot's instruments. This correlation must cover the airspeed range over which the aircraft is to be operated, the range of altitude to which the aircraft is limited, and 360° of heading. Correlation may be established on the ground as appropriate.

- (d) Each recorder container must:
- (1) Be either bright orange or bright yellow;
 - (2) Have a reflective tape affixed to its external surface to facilitate its location underwater; and
 - (3) Have an underwater locating device, when required by the applicable operating rules, on or adjacent to the container which is secured in such a manner that they are not likely to be separated during crash impact.

CS 27.1461 Equipment containing high energy rotors

ED Decision 2003/15/RM

- (a) Equipment containing high energy rotors must meet sub-paragraphs (b), (c), or (d).
- (b) High energy rotors contained in equipment must be able to withstand damage caused by malfunctions, vibration, abnormal speeds, and abnormal temperatures. In addition:
- (1) Auxiliary rotor cases must be able to contain damage caused by the failure of high energy rotor blades; and
 - (2) Equipment control devices, systems, and instrumentation must reasonably ensure that no operating limitations affecting the integrity of high energy rotors will be exceeded in service.
- (c) It must be shown by test that equipment containing high energy rotors can contain any failure of a high energy rotor that occurs at the highest speed obtainable with the normal speed control devices inoperative.
- (d) Equipment containing high energy rotors must be located where rotor failure will neither endanger the occupants nor adversely affect continued safe flight.

CS 27.1470 Emergency locator transmitter (ELT)

ED Decision 2018/007/R

Each emergency locator transmitter, including sensors and antennae, required by the applicable operating rule, must be installed so as to minimise damage that would prevent its functioning following an accident or incident.

[Amdt No: 27/5]

AMC 27.1470 Emergency locator transmitters (ELTs)

ED Decision 2018/007/R

- (a) Explanation

The purpose of this AMC is to provide specific guidance for compliance with [CS 27.1301](#), [CS 27.1309](#), [CS 27.1470](#), [CS 27.1529](#) and [CS 27.1581](#) regarding emergency locator transmitters (ELT) and their installation.

An ELT is considered to be a passive and dormant device whose status is unknown until it is required to perform its intended function. As such, its performance is highly dependent on proper installation and post-installation testing.

(b) References

Further guidance on this subject can be found in the following references:

- (1) ETSO-C126b 406 and 121.5 MHz Emergency Locator Transmitter;
- (2) ETSO-C126b 406 MHz Emergency Locator Transmitter;
- (3) FAA TSO-C126b 406 MHz Emergency Locator Transmitter (ELT);
- (4) EUROCAE ED-62A MOPS for aircraft emergency locator transmitters (406 MHz and 121.5 MHz (optional 243 MHz));
- (5) RTCA DO-182 Emergency Locator Transmitter (ELT) Equipment Installation and Performance; and
- (6) RTCA DO-204A Minimum Operational Performance Standards for 406 MHz Emergency Locator Transmitters (ELTs).

(c) Definitions

- (1) ELT (AF): an ELT (automatic fixed) is intended to be permanently attached to the rotorcraft before and after a crash, is automatically activated by the shock of the crash, and is designed to aid search and rescue (SAR) teams in locating a crash site.
- (2) ELT (AP): an ELT (automatic portable) is intended to be rigidly attached to the rotorcraft before a crash and is automatically activated by the shock of the crash, but is readily removable from the rotorcraft after a crash. It functions as an ELT (AF) during the crash sequence. If the ELT does not employ an integral antenna, the rotorcraft-mounted antenna may be disconnected and an auxiliary antenna (stowed in the ELT case) connected in its place. The ELT can be tethered to a survivor or a life raft. This type of ELT is intended to assist SAR teams in locating the crash site or survivor(s).
- (3) ELT (S): an ELT (survival) should survive the crash forces, be capable of transmitting a signal, and have an aural or visual indication (or both) that power is on. Activation of an ELT (S) usually occurs by manual means but automatic activation (e.g. activation by water) may also apply.
 - (i) ELT (S) Class A (buoyant): this type of ELT is intended to be removed from the rotorcraft, deployed and activated by survivors of a crash. It can be tethered to a life raft or a survivor. The equipment should be buoyant and it should be designed to operate when floating in fresh or salt water, and should be self-righting to establish the antenna in its nominal position in calm conditions.
 - (ii) ELT (S) Class B (non-buoyant): this type of ELT should be integral to a buoyant device in the rotorcraft, deployed and activated by the survivors of a crash.
- (4) ELT (AD) or automatically deployable emergency locator transmitter (ADELT): this type of automatically deployable ELT is intended to be rigidly attached to the rotorcraft before a crash and automatically deployed after the crash sensor determines that a crash has occurred or after activation by a hydrostatic sensor. This type of ELT should float in water and is intended to aid SAR teams in locating the crash site.
- (5) A crash acceleration sensor (CAS) is a device that detects an acceleration and initiates the transmission of emergency signals when the acceleration exceeds a predefined threshold (Gth). It is also often referred to as 'g switch'.

(d) Procedures

(1) Installation aspects of ELTs.

The installation of the equipment should be designed in accordance with the ELT manufacturer's instructions.

(i) Installation of the ELT transmitter unit and crash acceleration sensors

The location of the ELT should be chosen to minimise the potential for inadvertent activation or damage by impact, fire, or contact with passengers, baggage or cargo.

The ELT transmitter unit should ideally be mounted on primary rotorcraft load-carrying structures such as trusses, bulkheads, longerons, spars, or floor beams (not rotorcraft skin). Alternatively, the structure should meet the requirements of the test specified in 6.1.8 of ED-62A. For convenience, the requirements of this test are reproduced here, as follows:

'The mounts shall have a maximum static local deflection no greater than 2.5 mm when a force of 450 Newtons (100 lbf) is applied to the mount in the most flexible direction. Deflection measurements shall be made with reference to another part of the airframe not less than 0.3 m or more than 1.0 m from the mounting location.'

However, this does not apply to an ELT (S), which should be installed or stowed in a location that is conspicuously marked and readily accessible, or should be integral to a buoyant device such as a life raft, depending on whether it is of Class A or B.

A poorly designed crash acceleration sensor installation can be a source of problems such as nuisance triggers, failures to trigger and failures to deploy.

Nuisance triggers can occur when the crash acceleration sensor does not work as expected or is installed in a way that exposes it to shocks or vibration levels outside those assumed during equipment qualification. This can also occur as a result of improper handling and installation practices.

A failure to trigger can occur when an operational ELT is installed such that the crash sensor is prevented from sensing the relevant crash accelerations.

Particular attention should be paid to the installation orientation of the crash acceleration sensor. If the equipment contains a crash sensor with particular installation orientation needs, the part of the equipment containing the crash sensor will be clearly marked by the ELT manufacturer to indicate the correct installation orientation(s).

The design of the installation should follow the instructions contained in the installation manual provided by the equipment manufacturer. In the absence of an installation manual, in general, in the case of a helicopter installation, if the equipment has been designed to be installed on fixed-wing aircraft, it may nevertheless be acceptable for a rotorcraft application. In such cases, guidance should be sought from the equipment manufacturer. This has typically resulted in a recommendation to install the ELT with a different orientation, e.g. 45 degrees with respect to the main longitudinal axis (versus zero degrees for a fixed wing application). This may help the sensor to detect forces in directions other than the main longitudinal axis, since, during a helicopter crash, the direction of the impact may differ appreciably from the main aircraft axis. However, some ELTs are designed specifically for helicopters or designed to sense forces in several axes.

(ii) Use of hook and loop style fasteners

In several recent aircraft accidents, ELTs mounted with hook and loop style fasteners, commonly known from the brand name Velcro®, have detached from their aircraft mountings. The separation of the ELT from its mount could cause the antenna connection to be severed, rendering the ELT ineffective.

Inconsistent installation and reinstallation practices can lead to the hook and loop style fastener not having the necessary strength to perform its intended function. Furthermore, the retention capability of the hook and loop style fastener may degrade over time, due to wear and environmental factors such as vibration, temperature, or contamination. The safety concern about these attachments increases when the ELT manufacturer's instructions for continued airworthiness (ICA) do not contain specific instructions for regularly inspecting the hook and loop style fasteners, or a replacement interval (e.g. Velcro life limit). This concern applies, regardless of how the hook and loop style fastener is installed in the aircraft.

Separation of ELTs has occurred, even though the associated hook and loop style fastener design was tested during initial European Technical Standard Order (ETSO) compliance verification against crash shock requirements.

Therefore, it is recommended that when designing an ELT installation, the ELT manufacturer's ICA is reviewed and it is ensured that the ICA for the rotorcraft (or the modification, as applicable) appropriately addresses the in-service handling of hook and loop style fasteners.

It is to be noted that ETSO/TSO-C126b states that the use of hook and loop fasteners is not an acceptable means of attachment for automatic fixed (AF) and automatic portable (AP) ELTs.

(iii) ELT antenna installation

This section does not apply to the ELT(S) or ELT (AD) types of ELT. The most recurrent issue found during accident investigations concerning ELTs is the detachment of the antenna (coaxial cable), causing the transmission of the ELT unit to be completely ineffective.

Chapter 6 of ED-62A addresses the installation of an external antenna and provides guidance, in particular, on:

- (A) the location of the antenna;
- (B) the position of the antenna relative to the ELT transmission unit;
- (C) the characteristics of coaxial-cables; and
- (D) the installation of coaxial-cables.

Any ELT antenna should be located away from other antennas to avoid disruption of the antenna radiation patterns. In any case, during installation of the antenna, it should be ensured that the antenna has a free line of sight to the orbiting COSPAS-SARSAT satellites at most times when the aircraft is in the normal flight attitude.

Ideally, for the 121.5 MHz ELT antenna, a separation of 2.5 metres from antennas receiving very high frequency (VHF) communications and navigation data is sufficient to minimise unwanted interference. The 406 MHz ELT antenna should be

positioned at least 0.8 metres from antennas receiving VHF communications and navigation data to minimise interference.

External antennas which have been shown to be compatible with a particular ELT will either be part of the ETSO/TSO-approved ELT or will be identified in the ELT manufacturer's installation instructions. Recommended methods for installing antennas are outlined in FAA AC 43.13-2B.

The antenna should be mounted as close to the respective ELT as practicable. Provision should be taken to protect coaxial cables from disconnection or from being cut. Therefore, installation of the external antenna close to the ELT unit is recommended. Coaxial cables connecting the antenna to the ELT unit should not cross rotorcraft production breaks.

In the case of an external antenna installation, ED-62A recommends that its mounting surface should be able to withstand a static load equal to 100 times the antenna's weight applied at the antenna mounting base along the longitudinal axis of the rotorcraft. This strength can be substantiated by either test or conservative analysis.

If the antenna is installed within a fin cap, the fin cap should be made of an RF-transparent material that will not severely attenuate the radiated transmission or adversely affect the antenna radiation pattern shape.

In the case of an internal antenna location, the antenna should be installed as close to the ELT unit as practicable, insulated from metal window casings and restrained from movement within the cabin area. The antenna should be located such that its vertical extension is exposed to an RF-transparent window. The antenna's proximity to the vertical sides of the window and to the window pane and casing as well as the minimum acceptable window dimensions should be in accordance with the equipment manufacturer's instructions.

The voltage standing wave ratio (VSWR) of the installed external antenna should be checked at all working frequencies, according to the test equipment manufacturer's recommendations, during the first certification exercise for installation on a particular rotorcraft type.

Coaxial cables between the antenna and the ELT unit should be provided on each end with an RF connector that is suitable for the vibration environment of the particular installation application. When the coaxial cable is installed and the connectors mated, each end should have some slack in the cable, and the cable should be secured to rotorcraft structures for support and protection.

In order to withstand exposure to fire or flames, the use of fire-resistant coaxial cables or the use of fire sleeves compliant to SAE AS1072 is recommended.

(2) Deployment aspects of ELTs

Automatically deployable emergency locator transmitters (ADELTs) have particularities in their designs and installations that need to be addressed independently of the general recommendations.

The location of an ADELT and its manner of installation should minimise the risk of injury to persons or damage to the rotorcraft in the event of its inadvertent deployment. The means to manually deploy the ADELT should be located in the cockpit, and be guarded, such that the risk of inadvertent manual deployment is minimised.

Automatically deployable ELTs should be located so as to minimise any damage to the structure and surfaces of the rotorcraft during their deployment. The deployment trajectory of the ELT should be demonstrated to be clear of interference from the airframe or any other parts of the rotorcraft, or from the rotor in the case of helicopters. The installation should not compromise the operation of emergency exits or of any other safety features.

In some helicopters, where an ADELt is installed aft of the transport joint in the tail boom, any disruption of the tail rotor drive shaft has the potential to disrupt or disconnect the ADELt wiring. From accident investigations, it can be seen that if a tail boom becomes detached, an ADELt that is installed there, aft of the transport joint, will also become detached before signals from sensors that trigger its deployment can be received.

Therefore, it is recommended to install the ADELt forward of the transport joint of the tail boom. Alternatively, it should be assured that ELT system operation will not be impacted by the detachment of the structural part on which it is installed.

The hydrostatic sensor used for automatic deployment should be installed in a location shown to be immersed in water within a short time following a ditching or water impact, but not subject to water exposure in the expected rotorcraft operations. This assessment should include the most probable rotorcraft attitude when crashed, i.e. its capability to keep an upright position after a ditching or a crash into water.

The installation supporting the deployment feature should be demonstrated to be robust to immersion. Assuming a crash over water or a ditching, water may immerse not only the beacon and the hydrostatic sensor, which is designed for this, but also any electronic component, wires and the source of power used for the deployment.

(3) Additional considerations

(i) Human factors (HF)

The ELT controls should be designed and installed so that they are not activated unintentionally. These considerations should address the control panel locations, which should be clear from normal flight crew movements when getting into and out of the cockpit and when operating the rotorcraft, and the control itself. The means for manually activating the ELT should be guarded in order to avoid unintentional activation.

(ii) The rotorcraft flight manual (RFM) should document the operation of the ELT, and in particular, any feature specific to the installed model.

(iii) Batteries

An ELT operates using its own power source. The ELT manufacturer indicates the useful life and expiration date of the batteries by means of a dedicated label. The installation of the ELT should be such that the label indicating the battery expiration date is clearly visible without requiring the removal of the ELT or other LRU from the rotorcraft.

(4) Maintenance and inspection aspects

This Chapter provides guidance for the applicant to produce ICA related to ELT systems. The guidance is based on Chapter 7 of ED-62A.

- (i) The ICA should explicitly mention that:
 - (A) The self-test function should be performed according to the manufacturer's recommendation but no less than once every 6 months. Regulation at the place of operation should be considered when performing self-tests, as national aviation authorities (NAAs) may have established specific procedures to perform self-tests.
 - (B) As a minimum, a periodic inspection should occur at every battery replacement unless an inspection is required more frequently by the airworthiness authorities or the manufacturer.
- (ii) Each inspection should include:
 - (A) the removal of all interconnections to the ELT antenna, and inspection of the cables and terminals;
 - (B) the removal of the ELT unit, and inspection of the mounting;
 - (C) access to the battery to check that there is no corrosion;
 - (D) a check of all the sensors as recommended by Chapter 7.6 of ED-62A — Periodic inspection; and
 - (E) measurement of the transmission frequencies and the power output.
- (5) Rotorcraft flight manual (RFM)/Rotorcraft flight manual supplement (RFMS)

The rotorcraft flight manual (RFM) or supplement (RFMS), as appropriate, should contain all the pertinent information related to the operation of the ELT, including the use of the remote control panel in the cockpit. If there are any limitations on its use, these should be declared in the 'Limitations' section.

Detailed instructions for pre-flight and post-flight checks should be provided. As a pre-flight check, the ELT remote control should be checked to ensure that it is in the armed position. Post-flight, the ELT should be checked to ensure that it does not transmit, by activating the indicator on the remote control or monitoring 121.5 MHz.

Information on the location and deactivation of ELTs should also be provided. Indeed, accident investigations have shown that following aircraft ground impact, the remote control switch on the instrument panel may become inoperative, and extensive fuselage disruption may render the localisation of, and the access to, the ELT unit difficult. As a consequence, in the absence of information available to the accident investigators and first responders, this has led to situations where the ELT transmitted for a long time before being shut down, thus blocking the SAR channel for an extended time period. It is therefore recommended that information explaining how to disarm or shut down the ELT after an accident, including when the remote control switch is inoperative, should be included.

[Amdt No: 27/5]

SUBPART G – OPERATING LIMITATIONS AND INFORMATION

GENERAL

CS 27.1501 General

ED Decision 2016/024/R

- (a) Each operating limitation specified in [CS 27.1503](#) to [27.1525](#) and other limitations and information necessary for safe operation must be established.
- (b) The operating limitations and other information necessary for safe operation must be made available to the crew members as prescribed in [CS 27.1541](#) to [27.1593](#).

[Amdt 27/4]

OPERATING LIMITATIONS

CS 27.1503 Airspeed limitations: general

ED Decision 2003/15/RM

- (a) An operating speed range must be established.
- (b) When airspeed limitations are a function of weight, weight distribution, altitude, rotor speed, power, or other factors, airspeed limitations corresponding with the critical combinations of these factors must be established.

CS 27.1505 Never-exceed speed

ED Decision 2003/15/RM

- (a) The never-exceed speed, V_{NE} , must be established so that it is:
 - (1) Not less than 74 km/h (40 knots) (CAS); and
 - (2) Not more than the lesser of:
 - (i) 0.9 times the maximum forward speeds established under [CS 27.309](#);
 - (ii) 0.9 times the maximum speed shown under [CS 27.251](#) and [27.629](#); or
 - (iii) 0.9 times the maximum speed substantiated for advancing blade tip mach number effects.
- (b) V_{NE} may vary with altitude, rpm, temperature, and weight, if:
 - (1) No more than two of these variables (or no more than two instruments integrating more than one of these variables) are used at one time; and
 - (2) The ranges of these variables (or of the indications on instruments integrating more than one of these variables) are large enough to allow an operationally practical and safe variation of V_{NE} .
- (c) For helicopters, a stabilised power-off V_{NE} denoted as V_{NE} (power-off) may be established at a speed less than V_{NE} established pursuant to sub-paragraph (a), if the following conditions are met:
 - (1) V_{NE} (power-off) is not less than a speed midway between the power-on V_{NE} and the speed used in meeting the requirements of:
 - (i) [CS 27.65\(b\)](#) for single engine helicopters; and
 - (ii) [CS 27.67](#) for multi-engine helicopters.
 - (2) V_{NE} (power-off) is –
 - (i) A constant airspeed;
 - (ii) A constant amount less than power-on V_{NE} ; or
 - (iii) A constant airspeed for a portion of the altitude range for which certification is requested, and a constant amount less than power-on V_{NE} for the remainder of the altitude range.

CS 27.1509 Rotor speed

ED Decision 2003/15/RM

- (a) *Maximum power-off (autorotation).* The maximum power-off rotor speed must be established so that it does not exceed 95% of the lesser of –
 - (1) The maximum design rpm determined under [CS 27.309\(b\)](#); and
 - (2) The maximum rpm shown during the type tests.
- (b) *Minimum power-off.* The minimum power-off rotor speed must be established so that it is not less than 105% of the greater of:
 - (1) The minimum shown during the type tests; and
 - (2) The minimum determined by design substantiation.
- (c) *Minimum power-on.* The minimum power-on rotor speed must be established so that it is:
 - (1) Not less than the greater of:
 - (i) The minimum shown during the type tests; and
 - (ii) The minimum determined by design substantiation; and
 - (2) Not more than a value determined under [CS 27.33\(a\)\(1\) and \(b\)\(1\)](#).

CS 27.1519 Weight and centre of gravity

ED Decision 2003/15/RM

The weight and centre of gravity limitations determined under [CS 27.25](#) and [27.27](#), respectively, must be established as operating limitations.

CS 27.1521 Powerplant limitations

ED Decision 2012/021/R

- (a) *General.* The powerplant limitations prescribed in this paragraph must be established so that they do not exceed the corresponding limits for which the engines are type certificated.
- (b) *Take-off operation.* The powerplant take-off operation must be limited by:
 - (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value shown during the type tests;
 - (2) The maximum allowable manifold pressure (for reciprocating engines);
 - (3) The time limit for the use of the power corresponding to the limitations established in sub-paragraphs (b)(1) and (2);
 - (4) If the time limit in sub-paragraph (b)(3) exceeds 2 minutes, the maximum allowable cylinder head, coolant outlet, or oil temperatures;
 - (5) The gas temperature limits for turbine engines over the range of operating and atmospheric conditions for which certification is requested.
- (c) *Continuous operation.* The continuous operation must be limited by:
 - (1) The maximum rotational speed which may not be greater than:

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- (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value shown during the type tests;
 - (2) The minimum rotational speed shown under the rotor speed requirements in [CS 27.1509\(c\)](#); and
 - (3) The gas temperature limits for turbine engines over the range of operating and atmospheric conditions for which certification is requested.
- (d) *Fuel grade or designation.* The minimum fuel grade (for reciprocating engines), or fuel designation (for turbine engines), must be established so that it is not less than that required for operation of the engines within the limitations in sub-paragraphs (b) and (c).
- (e) *Turboshaft engine torque.* For rotorcraft with main rotors driven by turboshaft engines, and that do not have a torque limiting device in the transmission system, the following apply:
- (1) A limit engine torque must be established if the maximum torque that the engine can exert is greater than:
 - (i) The torque that the rotor drive system is designed to transmit; or
 - (ii) The torque that the main rotor assembly is designed to withstand in showing compliance with [CS 27.547\(d\)](#).
 - (2) The limit engine torque established under sub-paragraph (e)(1) may not exceed either torque specified in sub-paragraph (e)(1)(i) or (ii).
- (f) *Ambient temperature.* For turbine engines, ambient temperature limitations (including limitations for winterization installations, if applicable) must be established as the maximum ambient atmospheric temperature at which compliance with the cooling provisions of [CS 27.1041](#) to [27.1045](#) is shown.
- (g) *Two and one-half minute OEI power operation.* Unless otherwise authorised, the use of 2½-minute OEI power must be limited to engine failure operation of multi-engine, turbine-powered rotorcraft for not longer than 2½ minutes after failure of an engine. The use of 2½-minute OEI power must also be limited by:
- (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum demonstrated during the type tests;
 - (2) The maximum allowable gas temperature; and
 - (3) The maximum allowable torque.
- (h) *Thirty-minute OEI power operation.*
- Unless otherwise authorised, the use of 30-minute OEI power must be limited to multi-engine, turbine-powered rotorcraft for not longer than 30 minutes after failure of an engine. The use of 30-minute OEI power must also be limited by:
- (1) The maximum rotational speed which may not be greater than –
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value demonstrated during the type tests;
 - (2) The maximum allowable gas temperature; and
 - (3) The maximum allowable torque.

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- (i) *Continuous OEI power operation.* Unless otherwise authorised, the use of continuous OEI power must be limited to multi-engine, turbine-powered rotorcraft for continued flight after failure of an engine. The use of continuous OEI power must also be limited by:
- (1) The maximum rotational speed, which may not be greater than –
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value demonstrated during the type tests;
 - (2) The maximum allowable gas temperature; and
 - (3) The maximum allowable torque.
- (j) *Rated 30-second OEI power operation.* Rated 30-second OEI power is permitted only on multi-engine, turbine-powered rotorcraft, also certificated for the use of rated 2-minute OEI power, and can only be used for continued operation of the remaining engine(s) after a failure or precautionary shutdown of an engine. It must be shown that following application of 30-second OEI power, any damage will be readily detectable by the applicable inspections and other related procedures furnished in accordance with paragraph A27.4 of [Appendix A](#) of this CS-27. The use of 30-second OEI power must be limited to not more than 30 seconds for any period in which that power is used, and by:
- (1) The maximum rotational speed which may not be greater than:
 - (i) The maximum value determined by the rotor design: or
 - (ii) The maximum value demonstrated during the type tests:
 - (2) The maximum allowable gas temperature; and
 - (3) The maximum allowable torque.
- (k) *Rated 2-minute OEI power operation.* Rated 2-minute OEI power is permitted only on multi-engine, turbine-powered rotorcraft, also certificated for the use of rated 30-second OEI power, and can only be used for continued operation of the remaining engine(s) after a failure or precautionary shutdown of an engine. It must be shown that following application of 2-minute OEI power, any damage will be readily detectable by the applicable inspections and other related procedures furnished in accordance with A27.4 of [appendix A](#) of this CS-27. The use of 2-minute OEI power must be limited to not more than 2 minutes for any period in which that power is used, and by:
- (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value demonstrated during the type tests;
 - (2) The maximum allowable gas temperature; and
 - (3) The maximum allowable torque.

[Amdt 27/3]

CS 27.1523 Minimum flight crew

ED Decision 2003/15/RM

The minimum flight crew must be established so that it is sufficient for safe operation, considering:

- (a) The workload on individual crew members;
- (b) The accessibility and ease of operation of necessary controls by the appropriate crew member; and
- (c) The kinds of operation authorised under [CS 27.1525](#).

CS 27.1525 Kinds of operations

ED Decision 2003/15/RM

The kinds of operations (such as VFR, IFR, day, night, or icing) for which the rotorcraft is approved are established by demonstrated compliance with the applicable certification requirements and by the installed equipment.

CS 27.1527 Maximum operating altitude

ED Decision 2003/15/RM

The maximum altitude up to which operation is allowed, as limited by flight, structural, powerplant, functional, or equipment characteristics, must be established.

CS 27.1529 Instructions for Continued Airworthiness

ED Decision 2003/15/RM

Instructions for Continued Airworthiness in accordance with [Appendix A](#) must be prepared.

Appendix A – Instructions for Continued Airworthiness

ED Decision 2012/021/R

A27.1 General

- (a) This appendix specifies requirements for the preparation of instructions for continued airworthiness as required by [CS 27.1529](#).
- (b) The instructions for continued airworthiness for each rotorcraft must include the instructions for continued airworthiness for each engine and rotor (hereinafter designated 'products'), for each appliance required by any applicable CS or operating rule, and any required information relating to the interface of those appliances and products with the rotorcraft. If instructions for continued airworthiness are not supplied by the manufacturer of an appliance or product installed in the rotorcraft the instructions for continued airworthiness for the rotorcraft must include the information essential to the continued airworthiness of the rotorcraft.

A27.2 Format

- (a) The instructions for continued airworthiness must be in the form of a manual or manuals as appropriate for the quantity of data to be provided.
- (b) The format of the manual or manuals must provide for a practical arrangement.

A27.3 Content

The contents of the manual or manuals must be prepared in a language acceptable to the Agency. The instructions for continued airworthiness must contain the following manuals or sections, as appropriate, and information:

- (a) Rotorcraft maintenance manual or section
 - (1) Introduction information that includes an explanation of the rotorcraft's features and data to the extent necessary for maintenance or preventive maintenance.
 - (2) A description of the rotorcraft and its systems and installations including its engines, rotors, and appliances.
 - (3) Basic control and operation information describing how the rotorcraft components and systems are controlled and how they operate, including any special procedures and limitations that apply.
 - (4) Servicing information that covers details regarding servicing points, capacities of tanks, reservoirs, types of fluids to be used, pressures applicable to the various systems, location of access panels for inspection and servicing, locations of lubrication points, the lubricants to be used, equipment required for servicing, tow instructions and limitations, mooring, jacking, and levelling information.
- (b) Maintenance instructions
 - (1) Scheduling information for each part of the rotorcraft and its engines, auxiliary power units, rotors, accessories, instruments and equipment that provides the recommended periods at which they should be cleaned, inspected, adjusted, tested, and lubricated, and the degree of inspection, the applicable wear tolerances, and work recommended at these periods. However, it is allowed to refer to an accessory, instrument, or equipment manufacturer as the source of this information if it is shown that the item has an exceptionally high degree of complexity requiring specialised maintenance techniques, test equipment, or expertise. The recommended overhaul periods and necessary cross references to the Airworthiness Limitations section of the manual must also be included. In addition an inspection program that includes the frequency and extent of the inspections necessary to provide for the continued airworthiness of the rotorcraft must be included.
 - (2) Troubleshooting information describing probable malfunctions, how to recognise those malfunctions, and the remedial action for those malfunctions.
 - (3) Information describing the order and method of removing and replacing products and parts with any necessary precautions to be taken.
 - (4) Other general procedural instructions including procedures for system testing during ground running, symmetry checks, weighing and determining the centre of gravity, lifting and shoring, and storage limitations.
- (c) Diagrams of structural access plates and information needed to gain access for inspections when access plates are not provided.
- (d) Details for the application of special inspection techniques including radiographic and ultrasonic testing where such processes are specified.
- (e) Information needed to apply protective treatments to the structure after inspection.

- (f) All data relative to structural fasteners such as identification, discard recommendations, and torque values.
- (g) A list of special tools needed.

A27.4 Airworthiness Limitations Section

The instructions for continued airworthiness must contain a section titled airworthiness limitations, that is segregated and clearly distinguishable from the rest of the document. This section must set forth each mandatory replacement time, structural inspection interval, and related structural inspection procedure required for type-certification. If the instructions for continued airworthiness consist of multiple documents, the section required by this paragraph must be included in the principal manual. This section must contain a legible statement in a prominent location that reads: ‘the airworthiness limitations section is approved and variations must also be approved.’

[Amdt 27/2]

[Amdt 27/3]

MARKINGS AND PLACARDS

CS 27.1541 General

ED Decision 2003/15/RM

- (a) The rotorcraft must contain:
 - (1) The markings and placards specified in [CS 27.1545](#) to [27.1565](#), and
 - (2) Any additional information, instrument markings, and placards required for the safe operation of rotorcraft with unusual design, operating or handling characteristics.
- (b) Each marking and placard prescribed in sub-paragraph (a):
 - (1) Must be displayed in a conspicuous place; and
 - (2) May not be easily erased, disfigured, or obscured.

CS 27.1543 Instrument markings: general

ED Decision 2003/15/RM

For each instrument:

- (a) When markings are on the cover glass of the instrument, there must be means to maintain the correct alignment of the glass cover with the face of the dial; and
- (b) Each arc and line must be wide enough, and located, to be clearly visible to the pilot.

CS 27.1545 Airspeed indicator

ED Decision 2003/15/RM

- (a) Each airspeed indicator must be marked as specified in sub-paragraph (b), with the marks located at the corresponding indicated airspeeds.
- (b) The following markings must be made:
 - (1) A red radial line:
 - (i) For rotorcraft other than helicopters, at V_{NE} ; and
 - (ii) For helicopters at V_{NE} (power-on).
 - (2) A red cross-hatched radial line at V_{NE} (power-off) for helicopters, if V_{NE} (power-off) is less than V_{NE} (power-on).
 - (3) For the caution range, a yellow arc.
 - (4) For the safe operating range, a green arc.

CS 27.1547 Magnetic direction indicator

ED Decision 2003/15/RM

- (a) A placard meeting the requirements of this paragraph must be installed on or near the magnetic direction indicator.
- (b) The placard must show the calibration of the instrument in level flight with the engines operating.

- (c) The placard must state whether the calibration was made with radio receivers on or off.
- (d) Each calibration reading must be in terms of magnetic heading in not more than 45° increments.
- (e) If a magnetic non-stabilised direction indicator can have a deviation of more than 10° caused by the operation of electrical equipment, the placard must state which electrical loads, or combination of loads, would cause a deviation of more than 10° when turned on.

CS 27.1549 Powerplant instruments

ED Decision 2003/15/RM

For each required powerplant instrument, as appropriate to the type of instrument:

- (a) Each maximum and, if applicable, minimum safe operating limit must be marked with a red radial or a red line;
- (b) Each normal operating range must be marked with a green arc or green line, not extending beyond the maximum and minimum safe limits;
- (c) Each take-off and precautionary range must be marked with a yellow arc or yellow line;
- (d) Each engine or propeller range that is restricted because of excessive vibration stresses must be marked with red arcs or red lines; and
- (e) Each OEI limit or approved operating range must be marked to be clearly differentiated from the markings of sub-paragraphs (a) to (d) except that no marking is normally required for the 30-second OEI limit.

CS 27.1551 Oil quantity indicator

ED Decision 2003/15/RM

Each oil quantity indicator must be marked with enough increments to indicate readily and accurately the quantity of oil.

CS 27.1553 Fuel quantity indicator

ED Decision 2003/15/RM

If the unusable fuel supply for any tank exceeds 3.8 litres (0.8 Imperial gallon/1 US gallon), or 5% of the tank capacity, whichever is greater, a red arc must be marked on its indicator extending from the calibrated zero reading to the lowest reading obtainable in level flight.

CS 27.1555 Control markings

ED Decision 2018/007/R

- (a) Each cockpit control, other than primary flight controls or control whose function is obvious, must be plainly marked as to its function and method of operation.
- (b) For powerplant fuel controls:
 - (1) Each fuel tank selector control must be marked to indicate the position corresponding to each tank and to each existing cross feed position;
 - (2) If safe operation requires the use of any tanks in a specific sequence, that sequence must be marked on, or adjacent to, the selector for those tanks; and

- (3) Each valve control for any engine of a multi-engine rotorcraft must be marked to indicate the position corresponding to each engine controlled.
- (c) Usable fuel capacity must be marked as follows:
 - (1) For fuel systems having no selector controls, the usable fuel capacity of the system must be indicated at the fuel quantity indicator.
 - (2) For fuel systems having selector controls, the usable fuel capacity available at each selector control position must be indicated near the selector control.
- (d) For accessory, auxiliary, and emergency controls –
 - (1) each essential visual position indicator, such as those showing rotor pitch or landing gear position, must be marked so that each crew member can determine at any time the position of the unit to which it relates; and
 - (2) each emergency control must be marked as to method of operation and be red unless it may need to be operated underwater, in which case it must be marked with yellow and black stripes.
- (e) For rotorcraft incorporating retractable landing gear, the maximum landing gear operating speed must be displayed in clear view of the pilot.

[Amdt No: 27/5]

AMC 27.1555 Control markings

ED Decision 2018/007/R

This AMC supplements FAA AC 27.1555.

- (a) Explanation

CS-27 Amendment 5 introduced the need to mark emergency controls for use following a ditching or water impact with black and yellow stripes, instead of red, to make them more conspicuous when viewed underwater.
- (b) Procedures
 - (1) Any emergency control that may be required to be operated underwater (e.g. an emergency flotation system deployment switch, a life raft deployment switch or handle) should be coloured with black and yellow stripes.
 - (2) Black and yellow markings should consist of at least two bands of each colour of approximately equal widths.

[Amdt No: 27/5]

CS 27.1557 Miscellaneous markings and placards

ED Decision 2018/007/R

- (a) *Baggage and cargo compartments, and ballast location.* Each baggage and cargo compartment and each ballast location must have a placard stating any limitations on contents, including weight, that are necessary under the loading requirements.
- (b) *Seats.* If the maximum allowable weight to be carried in a seat is less than 77 kg (170 lbs), a placard stating the lesser weight must be permanently attached to the seat structure.

- (c) *Fuel and oil filler openings.* The following apply:
- (1) Fuel filler openings must be marked at or near the filler cover with:
 - (i) The word 'fuel';
 - (ii) For reciprocating enginepowered rotorcraft, the minimum fuel grade;
 - (iii) For turbine engine-powered rotorcraft, the permissible fuel designations; and
 - (iv) For pressure fuelling systems, the maximum permissible fuelling supply pressure and the maximum permissible defuelling pressure.
 - (2) Oil filler openings must be marked at or near the filler cover with the word 'oil'.
- (d) *Emergency exit placards.* Each placard and operating control for each emergency exit must differ in colour from the surrounding fuselage. A placard must be near each emergency exit control and must clearly indicate the location of that exit and its method of operation.

[Amdt No: 27/5]

CS 27.1559 Limitations placard

ED Decision 2003/15/RM

There must be a placard in clear view of the pilot that specifies the kinds of operations (such as VFR, IFR, day, night or icing) for which the rotorcraft is approved.

CS 27.1561 Safety equipment

ED Decision 2018/007/R

- (a) Each safety equipment control to be operated by the crew or passenger in an emergency must be plainly marked with its identification and its method of operation.
- (b) Each location, such as a locker or compartment that carries any fire extinguishing, signalling, or other safety equipment, must be appropriately marked in order to identify the contents and if necessary indicate how to remove the equipment.
- (c) Each item of safety equipment carried must be marked with its identification and must have obviously marked operating instructions.

[Amdt No: 27/5]

AMC 27.1561 Safety equipment

ED Decision 2018/007/R

This AMC supplements FAA AC 27.1561.

- (a) Explanation

[CS 27.1561](#) requires each safety equipment control that can be operated by a crew member or passenger to be plainly marked to identify its function and method of operation. (Note that the marking of safety equipment controls located within the cockpit and intended for use by the flight crew is addressed in [CS 27.1555](#).)

In addition, a location marking for each item of stowed safety equipment should be provided that identifies the contents and how to remove them. All safety equipment, including ditching and survival equipment, should be clearly identifiable and provided with operating instructions.

Markings and placards should be conspicuous and durable as per [CS 27.1541](#). Both passengers and crew should be able to easily identify and then use the safety equipment.

(b) Procedures

- (1) Release devices such as levers or latch handles for life rafts and other safety equipment should be plainly marked to identify their function and method of operation. The method of operation should be also marked. Stencils, permanent decals, placards, or other permanent labels or instructions may be used.
- (2) Lockers, compartments, or pouches used to contain safety equipment such as life vests, etc., should be marked to identify the equipment therein and to also identify, if not obvious, the method or means of accessing or releasing the equipment.
- (3) Safety equipment should be labelled and provided with instructions for its use or operation.
- (4) Locating signs for safety equipment should be legible in daylight from the furthest seated point in the cabin or recognisable from a distance equal to the width of the cabin. Letters, 2.5 cm (1 in) high, should be acceptable to satisfy the recommendation. Operating instructions should be legible from a distance of 76 cm (30 in). These recommendations are based on the exit requirements of CS 27.811(b) and (e)(1).
- (5) As prescribed, each life raft and its installed equipment should be provided with clear operating instruction markings that cannot be easily erased or disfigured and are readable at low levels of illumination.
- (6) Easily recognised or identified and easily accessible safety equipment located in sight of the occupants, such as a passenger compartment fire extinguisher that all passengers can see, may not require locating signs, stencils, or decals. However, operating instructions are required.

[Amdt No: 27/5]

CS 27.1565 Tail rotor

ED Decision 2003/15/RM

Each tail rotor must be marked so that its disc is conspicuous under normal daylight ground conditions.

ROTORCRAFT FLIGHT MANUAL AND APPROVED MANUAL MATERIAL

CS 27.1581 General

ED Decision 2003/15/RM

- (a) *Furnishing information.* A rotorcraft flight manual must be furnished with each rotorcraft, and it must contain the following:
 - (1) Information required by [CS 27.1583](#) to [27.1589](#).
 - (2) Other information that is necessary for safe operation because of design, operating, or handling characteristics.
- (b) *Approved information.* Each part of the manual listed in [CS 27.1583](#) to [27.1589](#), that is appropriate to the rotorcraft, must be furnished, verified, and approved, and must be segregated, identified, and clearly distinguished from each unapproved part of that manual.
- (c) (Reserved).
- (d) *Table of contents.* Each rotorcraft flight manual must include a table of contents if the complexity of the manual indicates a need for it.

CS 27.1583 Operating limitations

ED Decision 2003/15/RM

- (a) *Airspeed and rotor limitations.* Information necessary for the marking of airspeed and rotor limitations on, or near, their respective indicators must be furnished. The significance of each limitation and of the colour coding must be explained.
- (b) *Powerplant limitations.* The following information must be furnished:
 - (1) Limitations required by [CS 27.1521](#).
 - (2) Explanation of the limitations, when appropriate.
 - (3) Information necessary for marking the instruments required by [CS 27.1549](#) to [27.1553](#).
- (c) *Weight and loading distribution.* The weight and centre of gravity limits required by [CS 27.25](#) and [27.27](#), respectively, must be furnished. If the variety of possible loading conditions warrants, instructions must be included to allow ready observance of the limitations.
- (d) *Flight crew.* When a flight crew of more than one is required, the number and functions of the minimum flight crew determined under [CS 27.1523](#) must be furnished.
- (e) *Kinds of operation.* Each kind of operation for which the rotorcraft and its equipment installations are approved must be listed.
- (f) (Reserved)
- (g) *Altitude.* The altitude established under [CS 27.1527](#) and an explanation of the limiting factors must be furnished.

CS 27.1585 Operating procedures

ED Decision 2003/15/RM

- (a) Parts of the manual containing operating procedures must have information concerning any normal and emergency procedures and other information necessary for safe operation, including take-off and landing procedures and associated airspeeds. The manual must contain any pertinent information including:
 - (1) The kind of take-off surface used in the tests and each appropriate climb out speed; and
 - (2) The kind of landing surface used in the tests and appropriate approach and glide airspeeds.
- (b) For multi-engine rotorcraft, information identifying each operating condition in which the fuel system independence prescribed in [CS 27.953](#) is necessary for safety must be furnished, together with instructions for placing the fuel system in a configuration used to show compliance with that paragraph.
- (c) For helicopters for which a V_{NE} (power-off) is established under [CS 27.1505\(c\)](#), information must be furnished to explain the V_{NE} (power-off) and the procedures for reducing airspeed to not more than the V_{NE} (power-off) following failure of all engines.
- (d) For each rotorcraft showing compliance with [CS 27.1353\(g\)\(2\) or \(g\)\(3\)](#), the operating procedures for disconnecting the battery from its charging source must be furnished.
- (e) If the unusable fuel supply in any tank exceeds 5% of the tank capacity, or 3.8 litres (0.8 Imperial gallon/1 US gallon), whichever is greater, information must be furnished which indicates that when the fuel quantity indicator reads 'zero' in level flight, any fuel remaining in the fuel tank cannot be used safely in flight.
- (f) Information on the total quantity of usable fuel for each fuel tank must be furnished.
- (g) The airspeeds and rotor speeds for minimum rate of descent and best glide angle as prescribed in [CS 27.71](#) must be provided.

CS 27.1587 Performance information

ED Decision 2018/007/R

- (a) The rotorcraft flight manual (RFM) must contain the following information, determined in accordance with [CS 27.49](#) through [CS 27.79](#) and [CS 27.143\(c\) and \(d\)](#):
 - (1) Enough information to determine the limiting height-speed envelope.
 - (2) Information relative to:
 - (i) The steady rates of climb and descent, in-ground effect and out-of-ground effect hovering ceilings, together with the corresponding airspeeds and other pertinent information including the calculated effects of altitude and temperatures;
 - (ii) The maximum weight for each altitude and temperature condition at which the rotorcraft can safely hover in-ground effect and out-of-ground effect in winds of not less than 31 km/h (17 knots) from all azimuths. This data must be clearly referenced to the appropriate hover charts. In addition, if there are other combinations of weight, altitude, and temperature for which performance information is provided and at which the rotorcraft cannot land and take-off safely

with the maximum wind value, those portions of the operating envelope and the appropriate safe wind conditions must be stated in the Rotorcraft Flight Manual;

- (iii) For reciprocating engine-powered rotorcraft, the maximum atmospheric temperature at which compliance with the cooling provisions of [CS 27.1041](#) to [27.1045](#) is shown; and
- (iv) Glide distance as a function of altitude when autorotating at the speeds and conditions for minimum rate of descent and best glide as determined in [CS 27.71](#).

(b) The RFM must contain:

- (1) In its performance information section any pertinent information concerning the take-off weights and altitudes used in compliance with [CS 27.51](#);
- (2) The horizontal take-off distance determined in accordance with [CS 27.65\(a\)\(2\)\(i\)](#), and
- (3) The substantiated sea conditions and any associated information relating to the certification obtained with ditching or emergency flotation provisions.

[Amdt No: 27/1]

[Amdt No: 27/5]

AMC 27.1587(b)(3) Performance Information

ED Decision 2018/007/R

a. Explanation

The rotorcraft flight manual (RFM) is an important element in the certification process of the rotorcraft for approval with ditching or emergency flotation provisions. The material may be presented in the form of a supplement or a revision to the basic manual. This material should include:

- (1) A statement in the 'Limitations' section stating that the rotorcraft is approved for ditching or emergency flotation, as appropriate.

If certification with ditching provisions is obtained in a segmented fashion (i.e. one applicant performing the safety equipment installation and operations portion and another designing and substantiating the safety equipment's performance and deployment facilities), the RFM limitations should state that the ditching provisions are not approved until all the segments are completed. The outstanding ditching provisions for a complete certification should be identified in the 'Limitations' section.

- (2) Procedures and limitations for the inflation of a flotation device.
- (3) A statement in the performance information section of the RFM, identifying the substantiated sea conditions and any other pertinent information. If substantiation was performed using the default North Sea wave climate (JONSWAP), the maximum substantiated significant wave height (H_s) should be stated. If extended testing was performed in accordance with the [AMC to 27.801\(e\)](#) and [27.802\(c\)](#) to demonstrate that the target level of capsizing probability can be reached without any operational limitations, this should also be stated. If substantiation was performed for other sea conditions, the maximum substantiated significant wave height (H_s) and the limits of the geographical area represented should be stated.

- (4) Recommended rotorcraft water entry attitude and speed.
- (5) Procedures for the use of safety equipment.
- (6) Egress and life raft entry procedures.

[Amdt No: 27/5]

CS 27.1589 Loading information

ED Decision 2003/15/RM

There must be loading instructions for each possible loading condition between the maximum and minimum weights determined under [CS 27.25](#) that can result in a centre of gravity beyond any extreme prescribed in [CS 27.27](#), assuming any probable occupant weights.

CS 27.1593 Exposure to volcanic cloud hazards

ED Decision 2016/024/R

If required by an operating rule, the susceptibility of rotorcraft features to the effects of volcanic cloud hazards must be established.

[Amdt 27/4]

AMC 27.1593 Exposure to volcanic cloud hazards

ED Decision 2016/024/R

The aim of [CS 27.1593](#) is to support commercial and non-commercial operators operating complex motor-powered rotorcraft by identifying and assessing airworthiness hazards associated with operations in contaminated airspace. Providing such data to operators will enable those hazards to be properly managed as part of an established management system.

Acceptable means of establishing the susceptibility of rotorcraft features to the effects of volcanic clouds should include a combination of experience, studies, analysis, and/or testing of parts or sub-assemblies.

Information necessary for safe operation should be contained in the unapproved part of the flight manual or other appropriate manual, and should be readily usable by operators in preparing a safety risk assessment as part of their overall management system.

A volcanic cloud comprises volcanic ash together with gases and other chemicals. Although the primary hazard is volcanic ash itself, other elements of the volcanic cloud may also be undesirable to operate through, thus their effect on airworthiness should be assessed.

In determining the susceptibility of rotorcraft features to the effects of volcanic clouds as well as the necessary information to be provided to operators, the following points should be considered:

- (a) Identify the features of the rotorcraft that are susceptible to airworthiness effects of volcanic clouds. These may include but are not limited to the following:
 - (1) malfunction or failure of one or more engines, leading not only to reduction or complete loss of thrust but also to failures of electrical, pneumatic and hydraulic systems;
 - (2) blockage of pitot and static sensors, resulting in unreliable airspeed indications and erroneous warnings;

- (3) windscreen abrasion, resulting in windscreens rendered partially or completely opaque;
 - (4) fuel contamination;
 - (5) volcanic-ash and/or toxic chemical contamination of cabin air-conditioning packs, possibly leading to loss of cabin pressurisation or noxious fumes in the cockpit and/or cabin;
 - (6) erosion, blockage or malfunction of external and internal rotorcraft components;
 - (7) volcanic-cloud static discharge, leading to prolonged loss of communications; and
 - (8) reduced cooling efficiency of electronic components, leading to a wide range of rotorcraft system failures.
- (b) The nature and severity of effects.
- (c) Details of any device or system installed on the rotorcraft that can detect the presence of volcanic cloud hazards (e.g. volcanic ash (particulate) sensors or volcanic gas sensors).
- (d) The effect of volcanic ash on operations arriving to or departing from contaminated aerodromes.
- (e) The related pre-flight, in-flight and post-flight precautions to be taken by the operator including any necessary amendments to Aircraft Operating Manuals, Aircraft Maintenance Manuals, Master Minimum Equipment List/Dispatch Deviation or equivalents, required to support the operator. Pre-flight precautions should include clearly defined procedures for the removal of any volcanic ash detected on parked rotorcraft.
- (f) The recommended continuing-airworthiness inspections associated with operations in airspace contaminated by (a) volcanic cloud(s) and arriving to or departing from aerodromes contaminated by volcanic ash; this may take the form of Instructions for Continued Airworthiness (ICA) or other advice.

[Amdt 27/4]

MISCELLANEOUS GUIDANCE

MG4 Full Authority Digital Electronic Controls (FADEC)

ED Decision 2008/009/R

Note: Certification procedures identified in MG4 refer specifically to the FAA regulatory system. For guidance on EASA procedures, reference should be made to Commission Regulation (EC) No 1702/2003 (as amended) (Part-21), AMC-20 (and specifically AMC 20-1 and 20-3) and to EASA internal working procedures, all of which are available on EASA's web site: <http://www.easa.europa.eu/>

[Amdt 27/2]

MG5 Agricultural dispensing equipment installation

ED Decision 2016/024/R

Certification procedures identified in MG5 refer specifically to the FAA regulatory system and are not fully applicable to the EASA regulatory system due to the different applicability of restricted certification. The EASA regulatory system does not encompass a restricted certification category for design changes or Supplemental Type Certificates.

The certification basis of design changes or Supplemental Type Certificates for agricultural dispensing is to be established in accordance with 21.A.101 of Annex I to Regulation (EU) No 748/2012, on a case-by-case basis through compliance with the applicable airworthiness requirements contained in MG5, supplemented by any special conditions in accordance with 21.A.16B of Regulation (EU) No 748/2012 that are appropriate to the application and specific operating limitations and conditions. If appropriate to the proposed design, compliance with the above could be achieved through the provisions contained in 21.A.103(a)2(ii) or 21.A.115(b)2 of Regulation (EU) No 748/2012.

[Amdt 27/4]

MG6 Emergency Medical Service (EMS) systems installations, including interior arrangements, equipment, Helicopter Terrain Awareness and Warning System (HTAWS), radio altimeter, and Flight Data Monitoring System (FDMS)

ED Decision 2016/024/R

This AMC provides further guidance and acceptable means of compliance to supplement the FAA AC 27-1B Change 4 MG6 which is the EASA acceptable means of compliance, as provided for in AMC 27 General. Specifically, this AMC addresses aspects where the FAA AC has been deemed by EASA as being at variance with the EASA's interpretation or regulatory system. These aspects are as follows and the remaining paragraphs of FAA AC 27-1B MG6 that are not referenced below are considered to be EASA acceptable means of compliance:

- a. Explanation. This AMC pertains to EMS configurations and associated rotorcraft airworthiness standards. EMS configurations are usually unique interior arrangements that are subject to the appropriate airworthiness standards (CS-27 or other applicable standards) to which the rotorcraft was certified. No relief from the standards is intended except through the procedures contained in Regulation (EU) No 748/2012 (namely Part-21 point 21.A.21(c)). EMS configurations are seldom, if ever, done by the original manufacturer.

- (1) Regulation (EU) No 965/2012 specifies the minimum equipment required to operate as a helicopter air ambulance service provider. This equipment, as well as all other equipment presented for evaluation and approval, is subject to compliance with airworthiness standards. Any equipment not essential to the safe operation of the rotorcraft may be approved provided the use, operation, and possible failure modes of the equipment are not hazardous to the rotorcraft Safe flight, safe landing, and prompt evacuation of the rotorcraft, in the event of a minor crash landing, for any reason, are the objectives of the EASA's evaluation of interiors and equipment unique to EMS.
 - i. For example, a rotorcraft equipped only for transportation of a non-ambulatory person (e.g. a police rotorcraft with one litter) as well as a rotorcraft equipped with multiple litters and complete life support systems and two or more attendants or medical personnel may be submitted for approval. These configurations will be evaluated to the airworthiness standards appropriate to the rotorcraft certification basis.
 - ii. Small category rotorcraft should comply with flight crew and passenger safety standards, which will result in the need to re-evaluate certain features of the baseline existing type certified rotorcraft related to the EMS arrangement, such as doors and emergency exits, and occupant protection. Compliance with airworthiness standards results in the following features that should be retained as part of the rotorcraft's baseline type design: an emergency interior lighting system, placards or markings for doors and exits, exit size, exit quantity and location, exit access, safety belts and possibly shoulder harnesses or other restraint or passenger protection means. The features, placards, markings, and 'emergency' systems required as part of the rotorcraft's baseline type design should be retained unless specific replacements or alternate designs are necessary for the EMS configuration to comply with airworthiness standards.
- (2) Many EMS configurations of small rotorcraft are typically equipped with the following:
 - i. attendant and medical personnel seats, which may swivel;
 - ii. multiple litters, some of which may tilt;
 - iii. medical equipment stowage compartments;
 - iv. life support and other complex medical equipment;
 - v. human infant incubator ('isolette');
 - vi. curtains or other interior light shielding for the flight crew compartment;
 - vii. external loudspeakers and search lights;
 - viii. special internal and external communication radio equipment;
 - ix. FDMS;
 - x. radio altimeter;
 - xi. HTAWS.
- (3) All helicopter air ambulance service providers are required to operate at all times in accordance with Regulation (EU) No 965/2012, which also defines the equipment required for an operational approval to be obtained.

b. Procedures**(2) Evacuation and interior arrangements**

- iii. When an evacuation demonstration is determined to be appropriate for compliance, 90 seconds should be used as the time interval for evacuation of the rotorcraft. Attendants and flight crew, trained in the evacuation procedures, may be used to remove the litter patient(s). It is preferable for the patient(s) to remain in the litter; however, the patient(s) may be removed from the litter to facilitate rapid evacuation through the exit. The patient(s) is (are) not ambulatory during the demonstration. Evacuation procedures should be included if isolettes are part of the interior. The demonstration may be conducted in daylight with the dark of the night simulated and the rotorcraft in a normal attitude with the landing gear extended. For the purpose of the demonstration, exits on one side (critical side) should be used. Exits on the opposite side are blocked and not accessible for the demonstration.

(3) Restraint of occupants and equipment

The emergency landing conditions specified in [CS 27.561\(b\)](#) dictate the design load conditions. See FAA AC 27-1, sections 27.561 and 27.785, for further information.

- i. Whether seated or recumbent, the occupants must be protected from serious injury as prescribed in [CS 27.785](#). Swivel seats and tilt litters may be used provided they are substantiated for the appropriate loads for the position selected for approval. Placards or markings may be used to ensure proper orientation for flight, take-off, or landing and emergency landing conditions. The seats and litters should be listed in the type design data for the configuration. See paragraph b.(17) for substitutions.

(6) Interior or 'medical' lights

The view of the flight crew must be free from glare and reflections that could cause interference. Curtains that meet flammability standards may be used. Complete partition or separation of the flight crew and passenger compartment is not prudent. Means for visual and verbal communication are usually necessary. Refer to FAA AC 27-1, section 27.773, which addresses pilot visibility aspects.

[Amdt 27/4]